

# FAA APPROVED AIRPLANE FLIGHT MANUAL

for the CIRRUS VISION SF50

with Cirrus Perspective Touch+ Avionics System  
and FL310 Maximum Operating Altitude



FAA Approved in Normal Category based on FAR 23. This document must be carried in the airplane at all times and be kept within the reach of the pilot during all flight operations.

THIS MANUAL INCLUDES THE MATERIAL REQUIRED TO BE FURNISHED TO THE PILOT BY FAR PART 23 AND ADDITIONAL INFORMATION PROVIDED BY CIRRUS AIRCRAFT AND CONSTITUTES THE FAA APPROVED AIRPLANE FLIGHT MANUAL.

Model - Serial #:

Registration #:

FAA Approved

A handwritten signature in black ink, appearing to read 'C. Roberts', is written over a horizontal line.

Nov 20, 2018  
Date

Manager, Southwest Flight Test Section, AIR-713  
Federal Aviation Administration  
Ft. Worth, TX



**CIRRUS**  
A I R C R A F T

Copyright © 2016 - All Rights Reserved  
Cirrus Aircraft  
4515 Taylor Circle  
Duluth, MN 55811

## List of Effective Pages

Use this page to determine the current effective date for each page in the AFM. Supplements are issued individually and are controlled by the Log of Supplements Page in Section 9.

Dates of original issue and revised pages are:

Original Issue..... 20 Nov 2018  
Revision 1 ..... 20 Aug 2020  
Revision 2 ..... 04 Feb 2022

| Page          | Status         | Page | Status         | Page | Status         |
|---------------|----------------|------|----------------|------|----------------|
| LOEP-1        | Revision 2     | 2-17 | Original Issue | 3-5  | Original Issue |
| LOEP-2        | Revision 2     | 2-18 | Original Issue | 3-6  | Revision 1     |
| LOEP-3        | Revision 2     | 2-19 | Original Issue | 3-7  | Revision 1     |
| LOEP-4        | Revision 2     | 2-20 | Revision 1     | 3-8  | Revision 1     |
| Frontmatter-1 | Revision 1     | 2-21 | Revision 1     | 3-9  | Revision 1     |
| Frontmatter-2 | Revision 1     | 2-22 | Revision 1     | 3-10 | Revision 1     |
| Frontmatter-3 | Revision 2     | 2-23 | Revision 1     | 3-11 | Revision 1     |
| Frontmatter-4 | Revision 1     | 2-24 | Revision 1     | 3-12 | Revision 1     |
| 1-1           | Original Issue | 2-25 | Revision 1     | 3-13 | Revision 1     |
| 1-2           | Original Issue | 2-26 | Revision 1     | 3-14 | Revision 1     |
| 1-3           | Original Issue | 2-27 | Revision 1     | 3-15 | Revision 1     |
| 1-4           | Original Issue | 2-28 | Revision 1     | 3-16 | Revision 1     |
| 1-5           | Original Issue | 2-29 | Revision 1     | 3-17 | Revision 1     |
| 1-6           | Revision 1     | 2-30 | Revision 2     | 3-18 | Revision 1     |
| 1-7           | Revision 1     | 2-31 | Revision 2     | 3-19 | Revision 1     |
| 1-8           | Revision 1     | 2-32 | Revision 2     | 3-20 | Revision 1     |
| 1-9           | Revision 1     | 2-33 | Revision 2     | 3-21 | Revision 1     |
| 1-10          | Original Issue | 2-34 | Revision 2     | 3-22 | Revision 1     |
| 1-11          | Original Issue | 2-35 | Revision 2     | 3-23 | Revision 1     |
| 1-12          | Revision 1     | 2-36 | Revision 2     | 3-24 | Revision 1     |
| 1-13          | Original Issue | 2-37 | Revision 2     | 3-25 | Revision 1     |
| 1-14          | Original Issue | 2-38 | Revision 2     | 3-26 | Revision 1     |
| 1-15          | Original Issue | 2-39 | Revision 2     | 3-27 | Revision 1     |
| 1-16          | Original Issue | 2-40 | Revision 2     | 3-28 | Revision 1     |
| 2-1           | Revision 2     | 2-41 | Revision 2     | 3-29 | Revision 1     |
| 2-2           | Revision 2     | 2-42 | Revision 2     | 3-30 | Revision 1     |
| 2-3           | Revision 2     | 2-43 | Revision 2     | 3-31 | Revision 1     |
| 2-4           | Revision 2     | 2-44 | Revision 2     | 3-32 | Revision 1     |
| 2-5           | Revision 1     | 2-45 | Revision 2     | 3-33 | Revision 2     |
| 2-6           | Revision 1     | 2-46 | Revision 2     | 3-34 | Revision 2     |
| 2-7           | Original Issue | 2-47 | Revision 2     | 3-35 | Revision 2     |
| 2-8           | Revision 1     | 2-48 | Revision 2     | 3-36 | Revision 2     |
| 2-9           | Revision 1     | 2-49 | Revision 2     | 3-37 | Revision 2     |
| 2-10          | Revision 1     | 2-50 | Revision 2     | 3-38 | Revision 2     |
| 2-11          | Original Issue | 2-51 | Revision 2     | 3-39 | Revision 2     |
| 2-12          | Original Issue | 2-52 | Revision 2     | 3-40 | Revision 2     |
| 2-13          | Original Issue | 3-1  | Revision 2     | 3-41 | Revision 2     |
| 2-14          | Revision 1     | 3-2  | Revision 2     | 3-42 | Revision 2     |
| 2-15          | Original Issue | 3-3  | Revision 2     | 3-43 | Revision 2     |
| 2-16          | Revision 1     | 3-4  | Revision 2     | 3-44 | Revision 2     |

FAA Approved

*Addison P. Tower*

Digitally signed by ADDISON P TOWER  
Date: 2022.03.22 11:12:21 -05'00'

for Date 22 Mar 2022

Manager, Southwest Flight Test Section, AIR-713  
Federal Aviation Administration  
Ft. Worth, TX

## *List of Effective Pages (Cont.)*

| Page  | Status         | Page  | Status     | Page | Status         |
|-------|----------------|-------|------------|------|----------------|
| 3-45  | Revision 2     | 3A-34 | Revision 1 | 4-3  | Original Issue |
| 3-46  | Revision 2     | 3A-35 | Revision 1 | 4-4  | Original Issue |
| 3-47  | Revision 2     | 3A-36 | Revision 1 | 4-5  | Revision 1     |
| 3-48  | Revision 2     | 3A-37 | Revision 1 | 4-6  | Revision 1     |
| 3-49  | Revision 2     | 3A-38 | Revision 1 | 4-7  | Revision 1     |
| 3-50  | Revision 2     | 3A-39 | Revision 1 | 4-8  | Original Issue |
| 3-51  | Revision 2     | 3A-40 | Revision 1 | 4-9  | Revision 1     |
| 3-52  | Revision 2     | 3A-41 | Revision 1 | 4-10 | Original Issue |
| 3-53  | Revision 2     | 3A-42 | Revision 1 | 4-11 | Revision 1     |
| 3-54  | Revision 2     | 3A-43 | Revision 1 | 4-12 | Revision 1     |
| 3-55  | Revision 2     | 3A-44 | Revision 1 | 4-13 | Revision 2     |
| 3-56  | Revision 2     | 3A-45 | Revision 1 | 4-14 | Revision 1     |
| 3-57  | Revision 2     | 3A-46 | Revision 1 | 4-15 | Revision 2     |
| 3-58  | Revision 2     | 3A-47 | Revision 1 | 4-16 | Revision 1     |
| 3-59  | Revision 2     | 3A-48 | Revision 1 | 4-17 | Revision 1     |
| 3-60  | Revision 2     | 3A-49 | Revision 1 | 4-18 | Revision 1     |
| 3-61  | Revision 2     | 3A-50 | Revision 1 | 4-19 | Revision 2     |
| 3-62  | Revision 2     | 3A-51 | Revision 1 | 4-20 | Revision 1     |
| 3-63  | Revision 2     | 3A-52 | Revision 1 | 4-21 | Revision 1     |
| 3-64  | Revision 2     | 3A-53 | Revision 1 | 4-22 | Revision 1     |
| 3-65  | Revision 2     | 3A-54 | Revision 1 | 4-23 | Revision 1     |
| 3-66  | Revision 2     | 3A-55 | Revision 2 | 4-24 | Revision 1     |
| 3A-1  | Revision 2     | 3A-56 | Revision 1 | 5-1  | Revision 2     |
| 3A-2  | Revision 2     | 3A-57 | Revision 1 | 5-2  | Revision 2     |
| 3A-3  | Revision 2     | 3A-58 | Revision 1 | 5-3  | Revision 2     |
| 3A-4  | Revision 2     | 3A-59 | Revision 2 | 5-4  | Revision 2     |
| 3A-5  | Original Issue | 3A-60 | Revision 1 | 5-5  | Revision 1     |
| 3A-6  | Original Issue | 3A-61 | Revision 1 | 5-6  | Original Issue |
| 3A-7  | Original Issue | 3A-62 | Revision 1 | 5-7  | Original Issue |
| 3A-8  | Revision 1     | 3A-63 | Revision 1 | 5-8  | Original Issue |
| 3A-9  | Revision 1     | 3A-64 | Revision 1 | 5-9  | Original Issue |
| 3A-10 | Revision 1     | 3A-65 | Revision 1 | 5-10 | Original Issue |
| 3A-11 | Revision 2     | 3A-66 | Revision 1 | 5-11 | Original Issue |
| 3A-12 | Revision 1     | 3A-67 | Revision 1 | 5-12 | Original Issue |
| 3A-13 | Revision 1     | 3A-68 | Revision 1 | 5-13 | Original Issue |
| 3A-14 | Revision 1     | 3A-69 | Revision 1 | 5-14 | Original Issue |
| 3A-15 | Revision 1     | 3A-70 | Revision 1 | 5-15 | Original Issue |
| 3A-16 | Revision 1     | 3A-71 | Revision 1 | 5-16 | Original Issue |
| 3A-17 | Revision 1     | 3A-72 | Revision 1 | 5-17 | Original Issue |
| 3A-18 | Revision 1     | 3A-73 | Revision 1 | 5-18 | Original Issue |
| 3A-19 | Revision 1     | 3A-74 | Revision 1 | 5-19 | Original Issue |
| 3A-20 | Revision 1     | 3A-75 | Revision 1 | 5-20 | Original Issue |
| 3A-21 | Revision 1     | 3A-76 | Revision 1 | 5-21 | Revision 2     |
| 3A-22 | Revision 1     | 3A-77 | Revision 1 | 5-22 | Revision 2     |
| 3A-23 | Revision 1     | 3A-78 | Revision 1 | 5-23 | Revision 2     |
| 3A-24 | Revision 1     | 3A-79 | Revision 1 | 5-24 | Revision 2     |
| 3A-25 | Revision 1     | 3A-80 | Revision 1 | 5-25 | Revision 2     |
| 3A-26 | Revision 1     | 3A-81 | Revision 1 | 5-26 | Revision 2     |
| 3A-27 | Revision 1     | 3A-82 | Revision 1 | 5-27 | Revision 2     |
| 3A-28 | Revision 1     | 3A-83 | Revision 2 | 5-28 | Revision 2     |
| 3A-29 | Revision 1     | 3A-84 | Revision 2 | 5-29 | Revision 2     |
| 3A-30 | Revision 1     | 3A-85 | Revision 1 | 5-30 | Revision 2     |
| 3A-31 | Revision 1     | 3A-86 | Revision 1 | 5-31 | Revision 2     |
| 3A-32 | Revision 1     | 4-1   | Revision 2 | 5-32 | Revision 2     |
| 3A-33 | Revision 1     | 4-2   | Revision 2 | 5-33 | Revision 2     |

## ***List of Effective Pages (Cont.)***

| Page | Status     | Page  | Status     | Page  | Status         |
|------|------------|-------|------------|-------|----------------|
| 5-34 | Revision 2 | 5-89  | Revision 2 | 5-144 | Revision 2     |
| 5-35 | Revision 2 | 5-90  | Revision 2 | 5-145 | Revision 2     |
| 5-36 | Revision 2 | 5-91  | Revision 2 | 5-146 | Revision 2     |
| 5-37 | Revision 2 | 5-92  | Revision 2 | 5-147 | Revision 2     |
| 5-38 | Revision 2 | 5-93  | Revision 2 | 5-148 | Revision 2     |
| 5-39 | Revision 2 | 5-94  | Revision 2 | 5-149 | Revision 2     |
| 5-40 | Revision 2 | 5-95  | Revision 2 | 5-150 | Revision 2     |
| 5-41 | Revision 2 | 5-96  | Revision 2 | 5-151 | Revision 2     |
| 5-42 | Revision 2 | 5-97  | Revision 2 | 5-152 | Revision 2     |
| 5-43 | Revision 2 | 5-98  | Revision 2 | 5-153 | Revision 2     |
| 5-44 | Revision 2 | 5-99  | Revision 2 | 5-154 | Revision 2     |
| 5-45 | Revision 2 | 5-100 | Revision 2 | 5-155 | Revision 2     |
| 5-46 | Revision 2 | 5-101 | Revision 2 | 5-156 | Revision 2     |
| 5-47 | Revision 2 | 5-102 | Revision 2 | 5-157 | Revision 2     |
| 5-48 | Revision 2 | 5-103 | Revision 2 | 5-158 | Revision 2     |
| 5-49 | Revision 2 | 5-104 | Revision 2 | 5-159 | Revision 2     |
| 5-50 | Revision 2 | 5-105 | Revision 2 | 5-160 | Revision 2     |
| 5-51 | Revision 2 | 5-106 | Revision 2 | 5-161 | Revision 2     |
| 5-52 | Revision 2 | 5-107 | Revision 2 | 5-162 | Revision 2     |
| 5-53 | Revision 2 | 5-108 | Revision 2 | 5-163 | Revision 2     |
| 5-54 | Revision 2 | 5-109 | Revision 2 | 5-164 | Revision 2     |
| 5-55 | Revision 2 | 5-110 | Revision 2 | 5-165 | Revision 2     |
| 5-56 | Revision 2 | 5-111 | Revision 2 | 5-166 | Revision 2     |
| 5-57 | Revision 2 | 5-112 | Revision 2 | 5-167 | Revision 2     |
| 5-58 | Revision 2 | 5-113 | Revision 2 | 5-168 | Revision 2     |
| 5-59 | Revision 2 | 5-114 | Revision 2 | 5-169 | Revision 2     |
| 5-60 | Revision 2 | 5-115 | Revision 2 | 5-170 | Revision 2     |
| 5-61 | Revision 2 | 5-116 | Revision 2 | 5-171 | Revision 2     |
| 5-62 | Revision 2 | 5-117 | Revision 2 | 5-172 | Revision 2     |
| 5-63 | Revision 2 | 5-118 | Revision 2 | 5-173 | Revision 2     |
| 5-64 | Revision 2 | 5-119 | Revision 2 | 5-174 | Revision 2     |
| 5-65 | Revision 2 | 5-120 | Revision 2 | 5-175 | Revision 2     |
| 5-66 | Revision 2 | 5-121 | Revision 2 | 5-176 | Revision 2     |
| 5-67 | Revision 2 | 5-122 | Revision 2 | 5-177 | Revision 2     |
| 5-68 | Revision 2 | 5-123 | Revision 2 | 5-178 | Revision 2     |
| 5-69 | Revision 2 | 5-124 | Revision 2 | 5-179 | Revision 2     |
| 5-70 | Revision 2 | 5-125 | Revision 2 | 5-180 | Revision 2     |
| 5-71 | Revision 2 | 5-126 | Revision 2 | 5-181 | Revision 2     |
| 5-72 | Revision 2 | 5-127 | Revision 2 | 5-182 | Revision 2     |
| 5-73 | Revision 2 | 5-128 | Revision 2 | 5-183 | Revision 2     |
| 5-74 | Revision 2 | 5-129 | Revision 2 | 5-184 | Revision 2     |
| 5-75 | Revision 2 | 5-130 | Revision 2 | 5-185 | Revision 2     |
| 5-76 | Revision 2 | 5-131 | Revision 2 | 5-186 | Revision 2     |
| 5-77 | Revision 2 | 5-132 | Revision 2 | 5-187 | Revision 2     |
| 5-78 | Revision 2 | 5-133 | Revision 2 | 5-188 | Revision 2     |
| 5-79 | Revision 2 | 5-134 | Revision 2 | 6-1   | Original Issue |
| 5-80 | Revision 2 | 5-135 | Revision 2 | 6-2   | Original Issue |
| 5-81 | Revision 2 | 5-136 | Revision 2 | 6-3   | Original Issue |
| 5-82 | Revision 2 | 5-137 | Revision 2 | 6-4   | Revision 1     |
| 5-83 | Revision 2 | 5-138 | Revision 2 | 6-5   | Original Issue |
| 5-84 | Revision 2 | 5-139 | Revision 2 | 6-6   | Revision 1     |
| 5-85 | Revision 2 | 5-140 | Revision 2 | 6-7   | Revision 1     |
| 5-86 | Revision 2 | 5-141 | Revision 2 | 6-8   | Revision 1     |
| 5-87 | Revision 2 | 5-142 | Revision 2 | 6-9   | Revision 1     |
| 5-88 | Revision 2 | 5-143 | Revision 2 | 6-10  | Original Issue |

## *List of Effective Pages (Cont.)*

| Page | Status         | Page  | Status         | Page | Status |
|------|----------------|-------|----------------|------|--------|
| 6-11 | Original Issue | 10-2  | Revision 2     |      |        |
| 6-12 | Original Issue | 10-3  | Original Issue |      |        |
| 7-1  | Original Issue | 10-4  | Original Issue |      |        |
| 7-2  | Original Issue | 10-5  | Revision 1     |      |        |
| 8-1  | Revision 2     | 10-6  | Revision 2     |      |        |
| 8-2  | Revision 2     | 10-7  | Revision 2     |      |        |
| 8-3  | Revision 1     | 10-8  | Revision 2     |      |        |
| 8-4  | Revision 1     | 10-9  | Revision 2     |      |        |
| 8-5  | Revision 1     | 10-10 | Revision 2     |      |        |
| 8-6  | Revision 1     | 10-11 | Revision 2     |      |        |
| 8-7  | Revision 1     | 10-12 | Revision 2     |      |        |
| 8-8  | Revision 1     |       |                |      |        |
| 8-9  | Revision 1     |       |                |      |        |
| 8-10 | Original Issue |       |                |      |        |
| 8-11 | Original Issue |       |                |      |        |
| 8-12 | Original Issue |       |                |      |        |
| 8-13 | Revision 1     |       |                |      |        |
| 8-14 | Revision 1     |       |                |      |        |
| 8-15 | Revision 1     |       |                |      |        |
| 8-16 | Revision 1     |       |                |      |        |
| 8-17 | Revision 2     |       |                |      |        |
| 8-18 | Revision 2     |       |                |      |        |
| 8-19 | Revision 2     |       |                |      |        |
| 8-20 | Revision 2     |       |                |      |        |
| 8-21 | Revision 2     |       |                |      |        |
| 8-22 | Revision 2     |       |                |      |        |
| 8-23 | Revision 2     |       |                |      |        |
| 8-24 | Revision 2     |       |                |      |        |
| 8-25 | Revision 2     |       |                |      |        |
| 8-26 | Revision 2     |       |                |      |        |
| 8-27 | Revision 2     |       |                |      |        |
| 8-28 | Revision 2     |       |                |      |        |
| 8-29 | Revision 2     |       |                |      |        |
| 8-30 | Revision 2     |       |                |      |        |
| 8-31 | Revision 2     |       |                |      |        |
| 8-32 | Revision 2     |       |                |      |        |
| 8-33 | Revision 2     |       |                |      |        |
| 8-34 | Revision 2     |       |                |      |        |
| 8-35 | Revision 2     |       |                |      |        |
| 8-36 | Revision 2     |       |                |      |        |
| 8-37 | Revision 2     |       |                |      |        |
| 8-38 | Revision 2     |       |                |      |        |
| 8-39 | Revision 2     |       |                |      |        |
| 8-40 | Revision 2     |       |                |      |        |
| 8-41 | Revision 2     |       |                |      |        |
| 8-42 | Revision 2     |       |                |      |        |
| 8-43 | Revision 2     |       |                |      |        |
| 8-44 | Revision 2     |       |                |      |        |
| 8-45 | Revision 2     |       |                |      |        |
| 8-46 | Revision 2     |       |                |      |        |
| 8-47 | Revision 2     |       |                |      |        |
| 8-48 | Revision 2     |       |                |      |        |
| 9-1  | Original Issue |       |                |      |        |
| 9-2  | Original Issue |       |                |      |        |
| 10-1 | Revision 2     |       |                |      |        |

## **Foreword**

This Airplane Flight Manual (AFM) has been prepared by Cirrus Aircraft to familiarize operators with the aircraft. Read this AFM carefully. It provides operational procedures that will assure the operator obtains the performance published in the AFM, data designed to allow the most efficient and safe use of the airplane, and basic information to assist in maintaining the airplane in airworthy condition.

### **• NOTE •**

All limitations, procedures, maintenance & servicing requirements, and performance data contained in this AFM are mandatory for compliance with Federal Aviation Administration (FAA) operating rules and for continued airworthiness of the airplane.

This AFM includes the material required to be furnished to the pilot by the Code of Federal Regulations (CFR) and additional information provided by Cirrus Aircraft and constitutes the FAA Approved AFM for the aircraft.

## **The Airplane Flight Manual**

This AFM has been prepared using GAMA Specification #1 for Pilot's Operating Handbook, Revision 2, dated 18 October 1996 as the content model and format guide. However, some deviations from this specification were made for clarity. The AFM is presented in loose-leaf form for ease in inserting revisions and is sized for convenient storage. Tabbed dividers throughout the AFM allow quick reference to each section. Logical and convenient Tables of Contents are located at the beginning of each section to aid in locating specific data within that section. The AFM is divided into ten sections as follows:

|                  |                        |
|------------------|------------------------|
| Section 1 .....  | General                |
| Section 2 .....  | Limitations            |
| Section 3 .....  | Emergency Procedures   |
| Section 3A ..... | Abnormal Procedures    |
| Section 4 .....  | Normal Procedures      |
| Section 5 .....  | Performance Data       |
| Section 6 .....  | Weight and Balance     |
| Section 7 .....  | Systems Description    |
| Section 8 .....  | Handling and Servicing |
| Section 9 .....  | Log of Supplements     |
| Section 10 ..... | Safety Information     |

The data presented in this AFM is the result of extensive flight tests and is approved by the FAA. However, as new procedures or performance data are developed, the AFM will be revised.

• NOTE •

It is the responsibility of the owner to ensure that the AFM is current at all times. Therefore, it is very important that all revisions be properly incorporated into this AFM as soon as they become available.

## Revising the Airplane Flight Manual

Two types of revisions may be issued for this AFM: Temporary and Numbered.

Temporary revisions are printed on yellow paper, normally cover only one topic or procedure, and are issued to provide safety related information in a timely manner. All the information needed to properly file a temporary revision is included on the revision itself. Typically, a temporary revision is superseded and replaced by the next numbered revision.

Numbered revisions are printed on white paper, normally cover several subjects, and are issued as general updates to the AFM. Each numbered revision includes an "Instruction Sheet," a "List of Effective Pages", and a "Revision Highlights" page. The "Instruction Sheet" is intended to assist the manual holder in removing superseded pages and inserting new or superseding pages. The "List of Effective Pages" shows the issue or revision status of all pages in the AFM. The "Revision Highlights" page gives a brief description of changes made to each page in the current revision.

## Identifying Revised Material

Each page in the AFM has revision identification at the lower inside corner opposite the page number. Original issue pages will be identified by the words "Original Issue" at this location. In the event that the majority of pages in the AFM are revised, Cirrus Aircraft may determine that it is more effective to reissue the AFM. Reissued pages will be identified by the word "Reissue" followed by a letter indicating the reissue level; for example, "Reissue A" Revised pages will be identified by the word "Revision" followed by the revision number at this location; for example, "Revision 2" (Original Issue, Revision 2) or "Revision B1" (Reissue B, Revision 1).

Revised material on a page can be identified by a change bar located at the outside page margin. Revision bars are not used at reissues of the AFM.



## Revisions to the Airplane Flight Manual

AFM revisions, temporary revisions, and supplements can be downloaded from Cirrus Aircraft at [www.cirrusaircraft.com](http://www.cirrusaircraft.com), or from the Authorized Service Center website.

Paper copies of AFM revisions and supplements can be purchased from Cirrus Direct at [www.buycirrusdirect.com](http://www.buycirrusdirect.com).

## Checklists

*Serials without Safe Return Autoland:* Electronic Checklist P/N 39723-001 applies to this AFM.

*Serials with Safe Return Autoland:* Electronic Checklist P/N 39723-003 applies to this AFM.

## Supplements

The Supplements section (Section 9) of this AFM contains FAA Approved Supplements necessary to safely and efficiently operate the airplane when equipped with optional equipment not provided with the standard airplane or not included in the AFM. Supplements are essentially “mini-hand-books” and may contain data corresponding to most sections of the AFM. Data in a supplement either adds to, supersedes, or replaces similar data in the basic AFM.

Section 9 includes a “Log of Supplements” page preceding all Cirrus Aircraft Supplements produced for this airplane. The “Log of Supplements” page can be utilized as a “Table of Contents” for Section 9. If the airplane is modified at a non-Cirrus Aircraft facility through an STC or other approval method, it is the owner’s responsibility to ensure that the proper supplement, if applicable, is installed in the AFM and that the supplement is properly recorded on the “Log of Supplements” page.

FAA Approved AFM Supplements must be in the airplane for flight operations when the subject optional equipment is installed or the special operations are to be performed.

## Retention of Data

In the event a new title page is issued, the weight and balance data changes, the equipment list changes, or the “Log of Supplements” is replaced, the owner must ensure that all information applicable to the airplane is transferred to the new pages and the aircraft records are current. It is not a requirement that owners retain information, such as supplements, that is not applicable to their airplane.

In the event a new AFM is purchased, the owner must ensure that all information applicable to the airplane is transferred to the new AFM and the aircraft records are current.

## **Warnings, Cautions, and Notes**

Warnings, Cautions, and Notes are used throughout this AFM to focus attention on special conditions or procedures as follows:

**• WARNING •**

**Warnings are used to call attention to operating procedures which, if not strictly observed, may result in personal injury or loss of life.**

**• CAUTION •**

Cautions are used to call attention to operating procedures which, if not strictly observed, may result in damage to equipment.

**• NOTE •**

Notes are used to highlight specific operating conditions or steps of a procedure.

# ***Section 1: General***

## **Table of Contents**

|                                                  |    |
|--------------------------------------------------|----|
| Introduction.....                                | 3  |
| The Airplane .....                               | 7  |
| Engine .....                                     | 7  |
| Fuel .....                                       | 7  |
| Approved Fuel Grades .....                       | 7  |
| Oil .....                                        | 7  |
| Oil Capacity .....                               | 7  |
| Landing Gear Hydraulic Fluid & Brake Fluid ..... | 8  |
| Ice Protection System (IPS) Fluid .....          | 8  |
| IPS Fluid Capacity .....                         | 8  |
| Maximum Certificated Weights.....                | 8  |
| Noise Characteristics .....                      | 9  |
| Terminology.....                                 | 10 |

Intentionally Left Blank

## **Introduction**

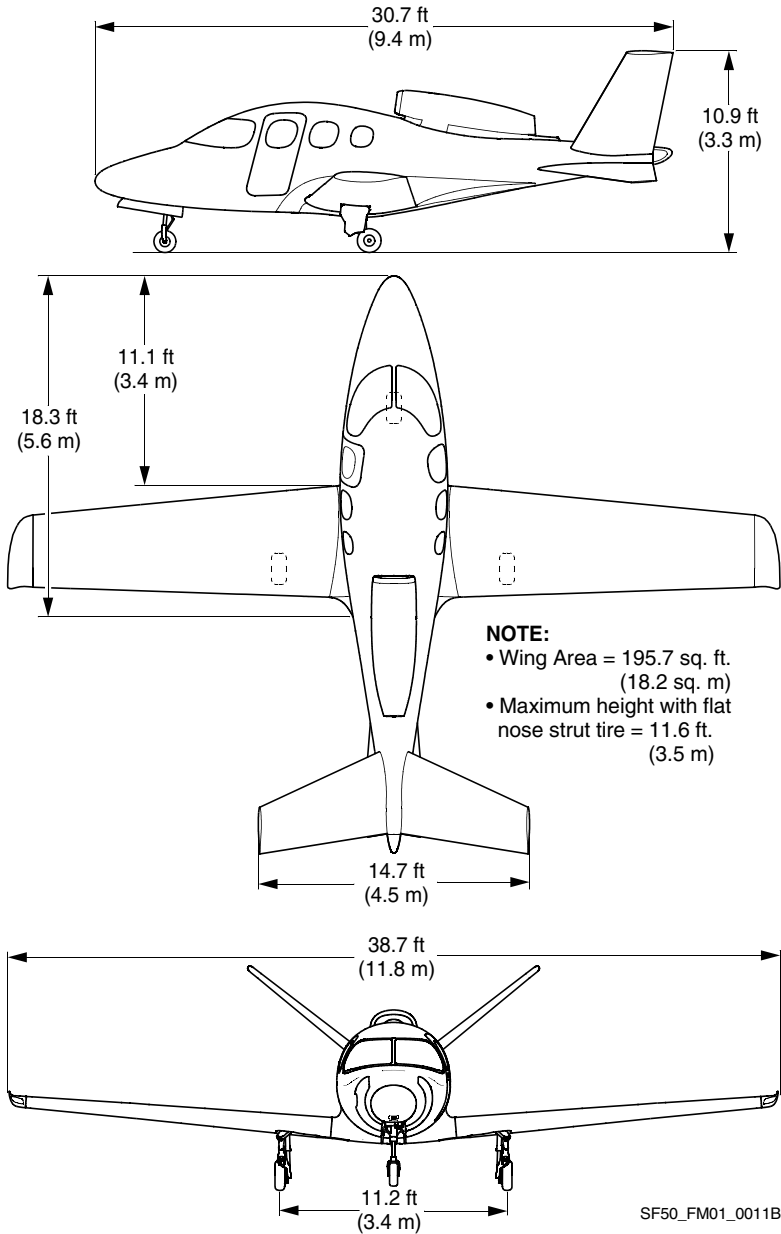
This section contains information of general interest to pilots and owners. You will find the information useful in acquainting yourself with the airplane, as well as in loading, fueling, sheltering, and handling the airplane during ground operations. Additionally, this section contains definitions or explanations of symbols, abbreviations, and terminology used throughout this AFM.

### **• NOTE •**

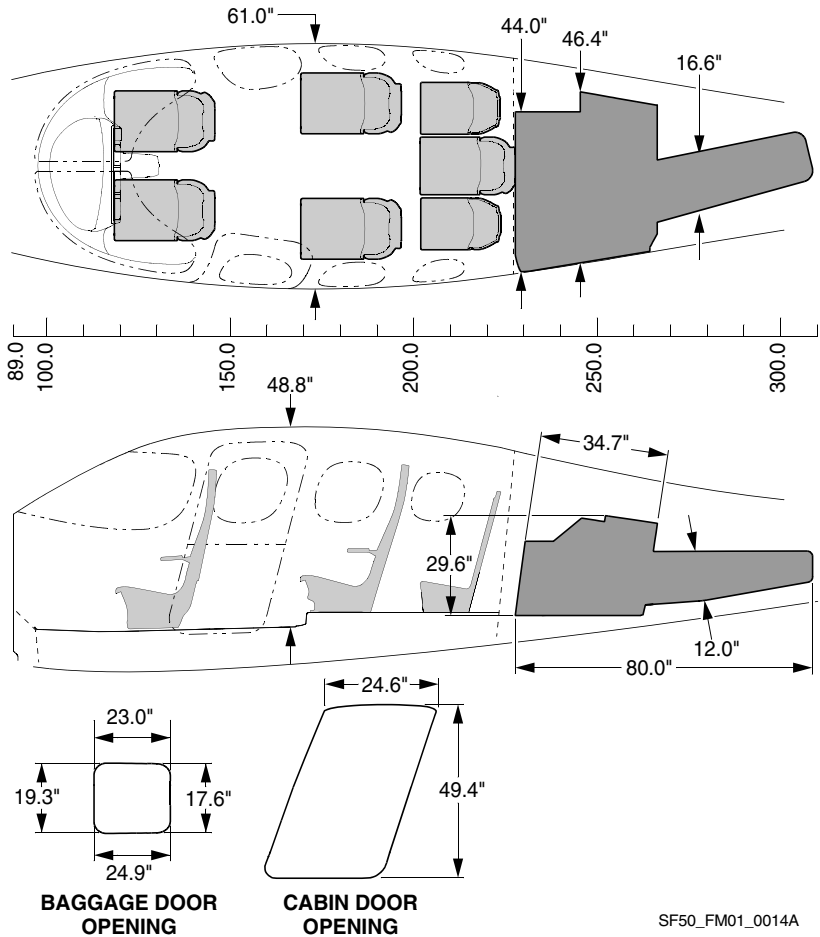
For specific information regarding the organization of this AFM, revisions, supplements, and procedures to be used to obtain publications, see the “Foreword” section.

All liquid volumes referenced in this publication are expressed in United States Customary Units, e.g., U.S. Gallons.

**Figure 1-1: Airplane Three View**

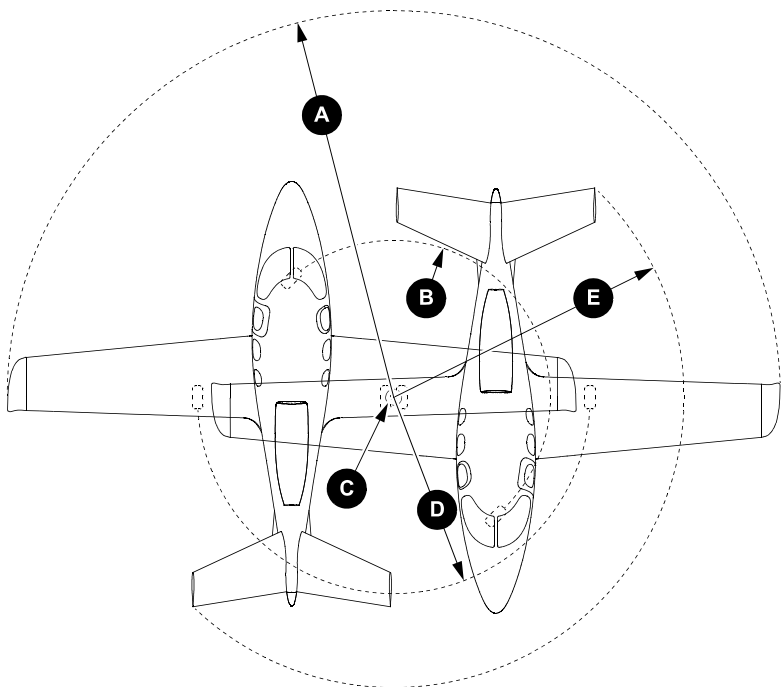


**Figure 1-2: Airplane Interior Dimensions**



| Location                 | Length | Width | Height | Volume      |
|--------------------------|--------|-------|--------|-------------|
| Cabin                    | 137.8" | 61.0" | 48.8"  | 170.0 cu ft |
| Main Baggage             | 35.0"  | 44.0" | 28.0"  | 23.5 cu ft  |
| Cargo X-Tend™ (optional) |        | 16.6" | 12.0"  | 3.9 cu ft   |

**Figure 1-3: Turning Radius**



**GROUND TURNING CLEARANCE**

|          |                                   |       |          |         |
|----------|-----------------------------------|-------|----------|---------|
| <b>A</b> | RADIUS FOR WING TIP               | ----- | 26.1 ft. | (7.9 m) |
| <b>B</b> | RADIUS FOR NOSE GEAR              | ----- | 11.6 ft. | (3.5 m) |
| <b>C</b> | RADIUS FOR INSIDE GEAR            | ----- | 0.5 ft.  | (0.2 m) |
| <b>D</b> | RADIUS FOR OUTSIDE GEAR           | ----- | 11.7 ft. | (3.6 m) |
| <b>E</b> | RADIUS FOR STABILIZER/RUDDERVATOR | ----- | 20.7 ft. | (6.3 m) |

TURNING RADII ARE CALCULATED USING ONE BRAKE AND PARTIAL POWER. ACTUAL TURNING RADIUS MAY VARY AS MUCH AS 3.0 FEET (0.9 METER).

SF50\_FM01\_0012A



## **The Airplane**

### **Engine**

|                           |                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Engines .....   | 1                                                                                                                                                                                                                                                                           |
| Engine Manufacturer ..... | Williams International                                                                                                                                                                                                                                                      |
| Engine Model .....        | FJ33-5A                                                                                                                                                                                                                                                                     |
| Engine Type.....          | Twin spool, turbofan engine with a single-stage fan and a three-stage axial compressor direct-driven by a two-stage turbine, a single-stage centrifugal compressor driven by a single-stage turbine, an annular combustor, a full length bypass duct, and an exhaust mixer. |
| Thrust Rating .....       | 1846 lb                                                                                                                                                                                                                                                                     |

### **Fuel**

|                      |                        |
|----------------------|------------------------|
| Total Capacity ..... | 298.5 US gal (2018 lb) |
| Usable .....         | 296 US gal (2001 lb)   |
| Unusable .....       | 2.5 US gal (17 lb)     |

**• NOTE •**

Fuel mass provided for reference assuming nominal 6.76 lb/gallon  
at 59 °F (15 °C).

### **Approved Fuel Grades**

|                                  |               |
|----------------------------------|---------------|
| Jet A Specification Fuel .....   | ASTM D1655    |
| Jet A-1 Specification Fuel ..... | ASTM D1655    |
| JP-8 Specification Fuel .....    | MIL-DTL-83133 |

### **Oil**

#### **Oil Capacity**

|                             |                    |
|-----------------------------|--------------------|
| Oil Tank Total Volume.....  | 3 Quarts (2.8 L)   |
| Oil Tank FULL Volume.....   | 2.2 Quarts (2.1 L) |
| Oil Tank ADD Volume .....   | 1.6 Quarts (1.5 L) |
| Oil Tank Usable Volume..... | 1.5 Quarts (1.4 L) |

The oil tank total volume represents the volume at which the filler port will overflow. The oil tank “FULL” and “ADD” correspond to marking on the sight glass. Usable volume represents the volume the engine can consume between the sight glass “FULL” level and the onset of oil pressure fluctuations.

All engine oil level readings should be made at least 5 minutes after engine shutdown.

For a list of approved engine lubricants, refer to Section 2: Limitations, “Oil”.

## Landing Gear Hydraulic Fluid & Brake Fluid

- MIL-PRF-87257
- Fire resistant, synthetic-based

## Ice Protection System (IPS) Fluid

- IPS Fluid (AL-5) DTD-406B

### ***IPS Fluid Capacity***

|                           |                           |
|---------------------------|---------------------------|
| Total Tank Capacity ..... | 3.1 U.S. gallons (11.7 L) |
| Usable .....              | 3.0 U.S. gallons (11.4 L) |
| Unusable .....            | 0.1 U.S. gallon (0.4 L)   |

## Maximum Certificated Weights

|                                        |                              |
|----------------------------------------|------------------------------|
| Max. Takeoff Gross Wt. (MTOGW) .....   | 6000 lb (2722 kg)            |
| Maximum Landing Weight (MLW) .....     | 5550 lb (2517 kg)            |
| Maximum Ramp Weight (MRW) .....        | 6040 lb (2740 kg)            |
| Max. Zero Fuel Weight (MZFW) .....     | 4900 lb (2223 kg)            |
| Maximum Aft Baggage Compartment .....  | 300 lb (136 kg) <sup>1</sup> |
| Maximum Cargo X-Tend Compartment ..... | 40 lb (18 kg) <sup>2</sup>   |

### **• NOTE •**

Load limit for cargo and baggage on all floor surfaces is 50 lb/ft<sup>2</sup>  
(244 kg/m<sup>2</sup>).

- 
1. Includes Cargo X-Tend weight limit.
  2. Cargo X-Tend is an optional storage space that extends the main baggage area.

## Noise Characteristics

The following noise levels comply with 14 CFR Part 36, Appendix B, Stage 4 maximum noise level requirements and were obtained by analysis of approved data from noise tests conducted under the provisions of Part 36, Amendment 36-28. The noise measurement and evaluation procedures used to obtain these noise levels are considered by the FAA to be equivalent to the Chapter 4 noise level required by the International Civil Aviation Organization (ICAO) in Annex 16, Volume I, Appendix 2, Amendment 7, effective March 21, 2002.

| NOISE<br>REFERENCE | WEIGHT<br>(LBS.) | FLAP SETTING | NOISE LEVELS<br>(EPNdB) |
|--------------------|------------------|--------------|-------------------------|
| LATERAL            | 6000             | 50%          | 79.6                    |
| FLYOVER            | 6000             | 50%          | 70.9                    |
| APPROACH           | 5550             | 100%         | 80.3                    |
| Stage 4 Margin     |                  |              | 50.2                    |

Compliance with 14 CFR, Part 36 flyover and lateral noise levels was demonstrated using rated takeoff power with landing gear up, 50% flaps, and a climb speed of 122 KIAS. For flyover noise, a thrust cutback was initiated at 1558 ft AGL from takeoff N1 to 83.5% N1. Approach noise was obtained at 5550 lb (2495 kg), landing gear down, and 100% flaps at 94 KIAS.

No determination has been made by the FAA that the noise levels of this aircraft are, or should be, acceptable or unacceptable for operation at, into, or out of, any airport.

## Terminology

**Table 1:** General Airspeed Terminology

| General Airspeed Terminology |                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Terminology                  | Definition                                                                                                                               |
| KCAS                         | Knots Calibrated Airspeed is the indicated airspeed corrected for position and instrument error.                                         |
| KIAS                         | Knots Indicated Airspeed is the speed shown on the airspeed indicator. The IAS values published in this AFM assume no instrument error.  |
| KTAS                         | Knots True Airspeed is the airspeed expressed in knots relative to undisturbed air which is KCAS corrected for altitude and temperature. |
| M                            | Mach Number is the ratio of true airspeed to the speed of sound.                                                                         |
| M <sub>MO</sub>              | Maximum Operating Mach Number is the highest Mach at which the aircraft may be safely operated.                                          |
| V <sub>R</sub>               | Rotation Speed is the speed at which the rotation of the aircraft should be initiated to takeoff attitude.                               |
| V <sub>MIN ICING</sub>       | Minimum airspeed for flight in icing conditions with flaps and gear UP (excluding climbs, approach, and landing).                        |
| V <sub>O</sub>               | Operating Maneuvering Speed is the maximum speed at which application of full control movement will not overstress the airplane.         |
| V <sub>LO_EXT</sub>          | Maximum Landing Gear Operating Speed (Extension) is the maximum speed at which it is safe to extend the landing gear.                    |
| V <sub>LO_RET</sub>          | Maximum Landing Gear Operating Speed (Retraction) is the maximum speed at which it is safe to retract the landing gear.                  |
| V <sub>LE</sub>              | Maximum Landing Gear Extended Speed is the maximum speed at which it is safe to fly with the landing gear extended.                      |

| <b>General Airspeed Terminology (Continued)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                              | <b>Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $V_{FE_{50\%}}$                                 | Maximum Flap Extended Speed (50%) is the highest speed permissible with wing flaps extended to the 50% position (typical of takeoff and approach).                                                                                                                                                                                                                                                                                                                                                            |
| $V_{FE_{100\%}}$                                | Maximum Flap Extended Speed (100%) is the highest speed permissible with wing flaps extended to the 100% position (typical of landing).                                                                                                                                                                                                                                                                                                                                                                       |
| $V_{MO}$                                        | Maximum Operating Speed is the highest speed at which the aircraft may be safely operated.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $V_{REF}$                                       | <p>Minimum final approach speed is the airspeed equal to the landing 50-foot point speed with flaps 100% (IPS OFF) or with flaps 50% and landing gear extended.</p> <p>For emergency and abnormal procedures, it is the airspeed equal to the 50-foot point speed with flaps in landing position as defined and landing gear extended.</p> <p>The green donut provides approximate <math>V_{REF}</math> and indicates an airspeed that is 1.3 times the stall speed for the given aircraft configuration.</p> |
| $V_S$                                           | Stalling Speed is the minimum steady flight speed at which the aircraft is controllable.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $V_{S_{50\%}}$                                  | Stalling Speed is the minimum steady flight speed at which the aircraft is controllable with 50% flaps.                                                                                                                                                                                                                                                                                                                                                                                                       |
| $V_{SO}$                                        | Stalling Speed is the minimum steady flight speed at which the aircraft is controllable in the landing configuration (100% flaps) at the most unfavorable weight and balance.                                                                                                                                                                                                                                                                                                                                 |
| $V_X$                                           | Best Angle of Climb Speed is the speed at which the airplane will obtain the highest altitude in a given horizontal distance. The best angle of climb speed normally increases slightly with altitude.                                                                                                                                                                                                                                                                                                        |

**Table 2:** Meteorological Terminology

| <b>Meteorological Terminology</b> |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                | <b>Definition</b>                                                                                                                                                                                                                                                                                                                            |
| IMC                               | Instrument Meteorological Conditions are meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling less than the minima for visual flight defined in 14 CFR 91.155.                                                                                                                                       |
| ISA                               | International Standard Atmosphere (standard day) is an atmosphere in which (1) the air is a dry perfect gas, (2) the temperature at sea level is 59 °F (15 °C), and (3) the pressure at sea level is 29.92 in.Hg (1013.2 millibars)                                                                                                          |
| MSL                               | Mean Sea Level is the average height of the surface of the sea for all stages of tide. In this AFM, altitude given as MSL is the altitude above the mean sea level. It is the altitude read from the altimeter when the altimeter's barometric adjustment has been set to the altimeter setting obtained from ground meteorological sources. |
| TAT                               | Total Air Temperature is the temperature measurement of air in motion. The total of static temperature plus temperature rise due to ram effect.                                                                                                                                                                                              |
| OAT                               | Outside Air Temperature is the free air static temperature obtained from in-flight temperature indications or from ground meteorological sources. It is expressed in either °C or °F.                                                                                                                                                        |
| PA                                | Pressure Altitude is the altitude read from the altimeter when the altimeter's barometric adjustment has been set to 29.92 inHg (1013.21 mb) corrected for position and instrument error. In this AFM, altimeter instrument errors are assumed to be zero.                                                                                   |

| <b>Meteorological Terminology (Continued)</b> |                                                                                                                                                                                                                                                                              |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                            | <b>Definition</b>                                                                                                                                                                                                                                                            |
| Standard Temperature                          | Standard Temperature is the temperature that would be found at a given pressure altitude in the standard atmosphere. It is 59 °F (15 °C) at sea level pressure altitude and decreases approximately 4 °F (2 °C) for each 1000 feet of altitude increase. See ISA definition. |

**Table 3:** Engine Power Terminology

| <b>Engine Power Terminology</b> |                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>              | <b>Definition</b>                                                                                                                                                                                                                                                                                                                             |
| TLA                             | Thrust Lever Angle is the position of the thrust lever.                                                                                                                                                                                                                                                                                       |
| T/O                             | Takeoff thrust is the maximum installed thrust that the engine will produce. This is obtained when Thrust Lever Angle (TLA) is in Takeoff (T/O) position.                                                                                                                                                                                     |
| MCT                             | Maximum Continuous Thrust is the maximum power that can be used continuously. This is obtained when TLA is at the MCT detent.                                                                                                                                                                                                                 |
| FI                              | Flight Idle is the minimum steady engine speed attainable in flight.                                                                                                                                                                                                                                                                          |
| GI                              | Ground Idle is the minimum steady engine speed attainable on the ground.                                                                                                                                                                                                                                                                      |
| N1                              | N1 is the rotational speed (RPM) of the low pressure (LP) rotary group, which consists of a single-stage fan, followed by a three-stage axial intermediate pressure (IP) compressor in the gas generator flow path. The fan and IP compressor are directly driven by two LP turbines. N1 speed is sensed from a toothed ring on the LP shaft. |

| <b>Engine Power Terminology (Continued)</b> |                                                                                                                                                                                                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                          | <b>Definition</b>                                                                                                                                                                                                      |
| N2                                          | N2 is the rotational speed (RPM) of the high pressure (HP) rotary group, which consists of a single-stage compressor driven by a single-stage turbine. N2 speed is sensed in the gearbox via the accessory drive gear. |
| ITT                                         | Inter-Turbine Temperature is the exhaust temperature measured between the HP and LP turbines.                                                                                                                          |

**Table 4:** Performance and Flight Planning Terminology

| <b>Performance and Flight Planning Terminology</b> |                                                                                                                                                                                                                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                                 | <b>Definition</b>                                                                                                                                                                                                                                               |
| g                                                  | One “g” is a quantity of acceleration equal to that of earth’s gravity.                                                                                                                                                                                         |
| Demonstrated Crosswind Velocity                    | Demonstrated Crosswind Velocity is the velocity of the crosswind component for which adequate control of the airplane during taxi, takeoff, and landing was demonstrated during certification testing. Demonstrated crosswind is not considered to be limiting. |
| GPH                                                | Gallons Per Hour is the amount of fuel (in gallons) consumed by the aircraft per hour.                                                                                                                                                                          |
| NMPG                                               | Nautical Miles Per Gallon is the distance (in nautical miles) which can be expected per gallon of fuel consumed at a specific engine power setting and/or flight configuration.                                                                                 |
| Unusable Fuel                                      | Unusable Fuel is the quantity of fuel that cannot be safely used in flight.                                                                                                                                                                                     |
| Usable Fuel                                        | Usable Fuel is the fuel available for flight planning.                                                                                                                                                                                                          |



**Table 5:** Weight and Balance Terminology

| <b>Weight and Balance Terminology</b> |                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminology</b>                    | <b>Definition</b>                                                                                                                                                                                                      |
| Reference Datum                       | Reference Datum is an imaginary vertical plane from which all horizontal distances are measured for balance purposes.                                                                                                  |
| Fuselage Station                      | Fuselage Station (FS) is a location along the airplane fuselage measured in inches from the reference datum and expressed as a number. For example: A point 123 inches aft of the reference datum is FS 123.           |
| CG                                    | Center of Gravity is the point at which an airplane would balance if suspended. Its distance from the reference datum is found by dividing the total moment by the total weight of the airplane.                       |
| Arm                                   | Arm is the horizontal distance from the reference datum to the center of gravity (CG) of an item. The airplane's arm is obtained by adding the airplane's individual moments and dividing the sum by the total weight. |
| Moment                                | Moment is the product of the weight of an item multiplied by its arm.                                                                                                                                                  |
| Basic Empty Weight                    | Basic Empty Weight is the actual weight of the airplane including all operating equipment that has a fixed location in the airplane. The basic empty weight includes the weight of unusable fuel and full oil.         |
| Maximum Ramp Weight                   | Maximum Ramp Weight is the maximum weight approved for ground maneuvering and includes the weight of the fuel used for startup and taxi.                                                                               |
| Maximum Gross Weight                  | Maximum Gross Weight is the maximum permissible weight of the airplane and its contents as listed in the aircraft specifications.                                                                                      |

| Weight and Balance Terminology (Continued) |                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminology                                | Definition                                                                                                                                                          |
| Maximum Takeoff Weight                     | Maximum Takeoff Weight is the maximum weight approved for the start of the takeoff run.                                                                             |
| Maximum Zero Fuel Weight                   | Maximum Zero Fuel Weight is the maximum permissible weight of the airplane and its contents minus the total weight of the fuel onboard.                             |
| Useful Load                                | Useful Load is the basic empty weight subtracted from the maximum ramp weight. It is the maximum allowable combined weight of pilot, passengers, fuel, and baggage. |
| Maximum Landing Weight                     | Maximum Landing Weight is the maximum weight approved for the landing touchdown.                                                                                    |

## ***Section 2: Limitations***

### **Table of Contents**

|                                                    |    |
|----------------------------------------------------|----|
| Introduction.....                                  | 5  |
| Certification Status .....                         | 5  |
| RVSM .....                                         | 5  |
| Taxiing, Takeoff, and Landing Limitations .....    | 6  |
| Operational Limits .....                           | 6  |
| Airspeed Limitations .....                         | 6  |
| Operating Speeds .....                             | 6  |
| Minimum Maneuvering Speeds .....                   | 7  |
| Flap Speeds .....                                  | 7  |
| Landing Gear Speeds .....                          | 7  |
| Airspeed Indicator Markings .....                  | 8  |
| Powerplant Limitations .....                       | 8  |
| Operational Limits .....                           | 8  |
| Engine Limit Exceedances .....                     | 10 |
| Fuel .....                                         | 10 |
| Oil .....                                          | 11 |
| Landing Gear Hydraulic Fluid and Brake Fluid ..... | 11 |
| Engine Start.....                                  | 11 |
| Engine Start (General) .....                       | 11 |
| Engine Start (Ground) .....                        | 11 |
| Start Abort Criteria.....                          | 12 |
| Engine Start (In flight).....                      | 12 |
| Starter Generator Limits (Start Mode) .....        | 12 |
| Engine Instrument Markings & Annunciations.....    | 14 |
| Powerplant .....                                   | 14 |
| Fuel .....                                         | 16 |
| Electrical.....                                    | 17 |
| Weight Limits.....                                 | 18 |
| Center of Gravity Limits.....                      | 18 |
| Maneuver Limits .....                              | 20 |
| Configuration Limitations.....                     | 20 |
| Maximum demonstrated crosswind .....               | 20 |
| Flight Load Factor Limits .....                    | 20 |
| Kinds of Operation.....                            | 20 |
| Maximum Operating Altitude Limit .....             | 20 |
| Outside Air Temperature Limits.....                | 21 |

---

|                                                               |    |
|---------------------------------------------------------------|----|
| Takeoff Temperatures .....                                    | 21 |
| Operating Temperatures .....                                  | 21 |
| Minimum Crew Requirements .....                               | 23 |
| Maximum Passenger Seating Limits .....                        | 23 |
| Child Restraint System Limits.....                            | 23 |
| Seat Belt Limits.....                                         | 23 |
| Seat Handling Limitations.....                                | 23 |
| Systems and Equipment Limits .....                            | 24 |
| Flight Controls .....                                         | 24 |
| Flaps.....                                                    | 24 |
| Landing Gear.....                                             | 24 |
| Warning .....                                                 | 24 |
| Stall Warning and Protection .....                            | 24 |
| Traffic Alert and Collision Avoidance System (TCAS-1).....    | 24 |
| Electrical System .....                                       | 24 |
| Maximum Generator Loads .....                                 | 25 |
| Icing Conditions .....                                        | 25 |
| Severe Icing .....                                            | 25 |
| Operation .....                                               | 26 |
| Tactile Inspection.....                                       | 27 |
| IPS Activation .....                                          | 27 |
| Ice Protection System (IPS) Fluid.....                        | 27 |
| Use of Autopilot in Icing Conditions .....                    | 28 |
| Autopilot.....                                                | 28 |
| Safe Return Autoland System (if installed).....               | 30 |
| Navigation and Communication Equipment .....                  | 31 |
| Attitude and Heading Reference System (AHRS) .....            | 31 |
| Cirrus Perspective Touch+ by Garmin Avionics System .....     | 31 |
| Cirrus Perspective Touch+ by Garmin GPS Navigation System ... | 32 |
| Enhanced Vision System (Optional).....                        | 33 |
| Weather .....                                                 | 33 |
| Garmin GSR 56 Iridium Datalink .....                          | 33 |
| Portable Electronic Devices.....                              | 34 |
| Other Limitations .....                                       | 34 |
| Oxygen .....                                                  | 34 |
| Fill Pressure vs. Temperature .....                           | 35 |
| Altitude vs. Minutes of Available Oxygen .....                | 36 |
| Pilot Oxygen Mask Mode: NORM.....                             | 36 |
| Pilot Oxygen Mask Mode: 100% .....                            | 37 |
| Crew Headset.....                                             | 38 |
| Smoking.....                                                  | 38 |
| Brakes .....                                                  | 39 |

---

|                                                            |    |
|------------------------------------------------------------|----|
| Tire Inflation Pressure (nominal) .....                    | 39 |
| Circuit Breakers .....                                     | 39 |
| In-Flight Entertainment .....                              | 39 |
| Exterior Windows .....                                     | 40 |
| Placards .....                                             | 40 |
| Interior Placards .....                                    | 40 |
| Safe Return Autoland Activation Panel (if installed) ..... | 49 |
| Exterior Placards .....                                    | 49 |
| Safe Return Autoland Equipped (if installed) .....         | 51 |
| Electronic Placards.....                                   | 52 |

Intentionally Left Blank

## **Introduction**

The limitations included in this Section of the AFM are approved by the FAA.

This section provides operating limitations, instrument markings, and basic placards required by regulation and necessary for the safe operation of the aircraft and its standard systems and equipment.

### **• NOTE •**

Compliance with the operating limitations in this section and in Section 9 is required by the Code of Federal Regulations.

For installed equipment described in an FAA Approved AFM Supplement, refer to [Section 9: Log of Supplements](#) of this AFM for amended operating limitations.

## **Certification Status**

The aircraft is certificated under the requirements of 14 CFR, Part 23, Airworthiness Standards: Normal Category, Part 34, Fuel Venting and Exhaust Emissions, Part 36, Noise Standards, and Special Conditions prescribed by the Administrator.

## **RVSM**

This airplane is eligible for operations in Reduced Vertical Separation Minimum (RVSM) airspace when the following equipment is installed and operating normally upon entering RVSM airspace:

- ADC1 and ADC2
- Autopilot
- Altitude Miscompare Alert
- ATCRBS Transponder

This equipment must be maintained in accordance with the Airplane Maintenance Manual (AMM).

This does not constitute operational approval. Each operator must ensure compliance with required crew training and operating practices and procedures. Individual airplane and operational approval must be obtained in accordance with the applicable operating rules.

## Taxiing, Takeoff, and Landing Limitations

### Operational Limits

For taxiing operations, the lower cabin door must be closed.

Takeoffs and landings are limited to:

- Smooth surfaces, and
- Paved, concrete, or dry hard packed (dirt/clay) surfaces, and
- Surfaces free from loose dirt, rocks, or gravel

Maximum Tailwind for Takeoff and Landing..... 10 knots

### Airspeed Limitations

#### Operating Speeds

Operating Maneuvering Speed ( $V_O$ ) ..... 150 KIAS

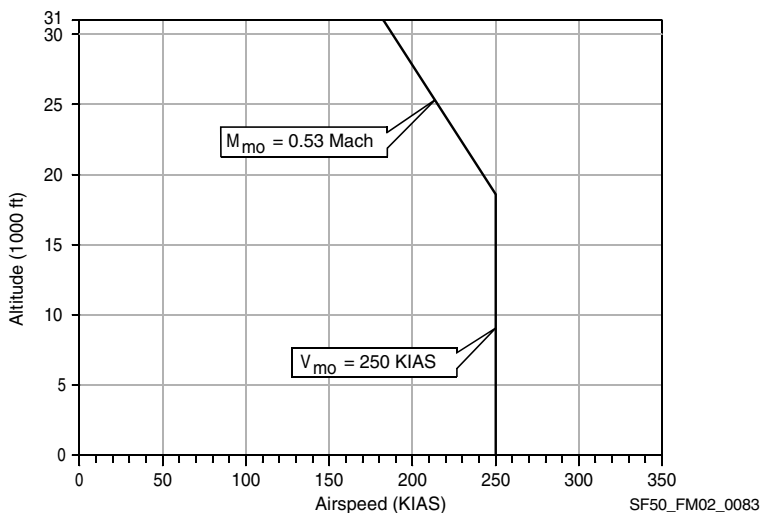
Max. Operating Speed ( $V_{MO}$ ) ..... 250 KIAS

Max. Operating Mach Number ( $M_{MO}$ ).....0.53 Mach

#### • WARNING •

**Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw, as they may result in loss of control in flight and/or structural failures at any speed, including below  $V_O$ .**

**Figure 2-1: Maximum Operating Speed ( $V_{MO}/M_{MO}$ )**





## Minimum Maneuvering Speeds

| FLAP CONFIGURATION | SPEED        |
|--------------------|--------------|
| UP                 | $V_{REF}+20$ |
| 50%                | $V_{REF}+15$ |
| 100%               | $V_{REF}+10$ |

### • NOTE •

These speeds are based on up to 40° angle of bank and are independent of thrust settings.

Flaps UP is an abnormal landing configuration, approved for use in the event of an emergency.

## Flap Speeds

Maximum flap extended speed, 50%/Takeoff/Approach  
( $V_{FE_{50\%}}$ ) ..... 190 KIAS  
Maximum flap extended speed, 100%/Land ( $V_{FE_{100\%}}$ ) ..... 150 KIAS

## Landing Gear Speeds

Maximum landing gear operation speed, Extension  
( $V_{LO_{EXT}}$ ) ..... 210 KIAS  
Maximum landing gear operation speed, Retraction  
( $V_{LO_{RET}}$ ) ..... 150 KIAS  
Max. landing gear extended speed ( $V_{LE}$ ) ..... 210 KIAS

### • Note •

Landing gear extension / retraction times significantly increase with cold temps, high airspeeds, yaw angle, and increased acceleration.

## Airspeed Indicator Markings

| Marking    | Value (KIAS/Mach)                   | Remarks                                                                                   |
|------------|-------------------------------------|-------------------------------------------------------------------------------------------|
| Red Line   | 250 KIAS/<br>0.53 Mach              | Maximum Operating Speed ( $V_{MO}$ ) and<br>Maximum Operating Mach Number<br>( $M_{MO}$ ) |
|            | > 210 KIAS                          | Landing Gear: DOWN                                                                        |
|            | > 190 KIAS                          | $V_{FE\_50\%}$                                                                            |
|            | > 150 KIAS                          | $V_{FE\_100\%}$                                                                           |
| Green Band | ≤ Red Line                          | Based on aircraft configuration                                                           |
|            | ≥ Low Speed<br>Awareness<br>Marking | Dynamically based on sensed AOA                                                           |

• Note •

The low speed color marking on the airspeed tape is computed dynamically based on the sensed AOA.

## Powerplant Limitations

### Operational Limits

The Cirrus Perspective Touch+™ by Garmin® avionics system provides dynamic gauge markings for engine instruments. Markings are automatically determined based on current conditions and mode of operation (start, climb, etc.).

Automated markings are designed to reduce pilot workload and eliminate pilot error; operators are recommended to rely on the indicated instrument markings for limitations, not memorization.

| Operating Mode | ITT (°C)                                                                 | N1 (%)                                                         | N2 (%)                                                  | Oil Pressure (psig)                                                                        | Oil Temperature (°C)                                                            |
|----------------|--------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Start          | ≤ 836<br><br><u>Transients:</u><br>1000<br>(15 sec.)<br>900<br>(30 sec.) | -                                                              | -                                                       | -                                                                                          | -40 – 135                                                                       |
| Idle           | 836                                                                      | -                                                              | < 80%<br>until oil temp.<br>>10 °C                      | <u>N2 &lt; 80%</u><br>35 - 120<br><br><u>N2 ≥ 80%</u><br>45 - 120<br><br><u>Transient:</u> | 10 – 135<br><br><u>Transient:</u><br>-40<br>with N2 < 80%<br>149<br>(5 minutes) |
| Takeoff (T/O)  | 877<br>(10 sec.)<br>862<br>(5 minutes)                                   | 104.7%<br>max.<br><br><u>Transient:</u><br>105.7%<br>(30 sec.) | 100% max.<br><br><u>Transient:</u><br>101%<br>(30 sec.) | 23<br>(5 minutes)<br>130<br>(5 minutes)                                                    | 10 – 135<br><br><u>Transient:</u><br>149<br>(5 minutes)                         |
| MCT            | 836                                                                      |                                                                |                                                         |                                                                                            |                                                                                 |

• NOTE •

High oil temperatures and associated low oil pressures can occur when returning to ground idle from high power. Therefore, ground operation at or near Takeoff thrust should be limited in duration to avoid oil temperatures rising above 194 °F (90 °C) or perform a partial power cooldown prior to returning to ground idle.

Airflow disturbances across engine intake, particularly during gusty or crosswind conditions, may result in N1 or N2 speed fluctuations. This is normal at high power settings during static or low speed operations.

For maximum occupant comfort, avoid operations for more than 20 minutes at or near 78% N1.

It is highly recommended that while aircraft is unattended, inlet and exhaust covers are installed to prevent excessive windmilling and Foreign Object Damage (FOD).

Avoid unnecessary rapid movements of the thrust lever.

# Engine Limit Exceedances

In the event of ITT, N1, N2, oil temperature, and/or oil pressure exceedances, the pilot should record each instance of the exceedance, along with the duration and degree of excess. This information should then be reported to Cirrus Aircraft.

Long term changes in N1 and N2 or ITT readings can be useful in tracking trends in engine health. Therefore, the pilot should monitor engine trend data and notify Cirrus Aircraft of any sudden or gradual changes over time.

## Fuel

• NOTE •

Fuel mass values below provided for reference assuming nominal 6.76 lb/gallon at 59 °F (15 °C).

|                                     |                            |
|-------------------------------------|----------------------------|
| Jet A Specification Fuel .....      | ASTM D1655                 |
| Jet A-1 Specification Fuel.....     | ASTM D1655                 |
| JP-8 Specification Fuel .....       | MIL-DTL-83133              |
| Total fuel capacity .....           | 298.5 US Gallons (2018 lb) |
| Usable fuel .....                   | 296 US Gallons (2001 lb)   |
| Max. allowable fuel imbalance ..... | 15 US Gallons (101 lb)     |
| Minimum fuel temperature .....      | -40 °F (-40 °C)            |
| Maximum fuel temperature .....      | 122 °F (50 °C)             |

• NOTE •

Recommended fueling best practice is to alternate between wings in 75-gallon (284-liter) increments when fueling from a single nozzle. Larger fuel imbalance may lead to fuel venting from low wing if aircraft “leans” towards a heavy wing as fuel reaches maximum tank capacity.

| Fuel           | Left                         | Right                        | Total                       |
|----------------|------------------------------|------------------------------|-----------------------------|
| Total Capacity | 149.25 U.S. gal<br>(1009 lb) | 149.25 U.S. gal<br>(1009 lb) | 298.5 U.S. gal<br>(2018 lb) |
| Usable         | 148 U.S. gal<br>(1000 lb)    | 148 U.S. gal<br>(1000 lb)    | 296 U.S. gal<br>(2001 lb)   |
| Unusable       | 1.25 U.S. gal<br>(8 lb)      | 1.25 U.S. gal<br>(8 lb)      | 2.5 U.S. gal<br>(17 lb)     |

## Oil

Oil quantity..... Adequate for intended flight

Only the following approved lubrications shall be used for the engine:

Mobil Jet II .....(MIL-L-23699)

Mobil 254 .....(MIL-L-23699)

Mixing of approved oils is permissible.

## Landing Gear Hydraulic Fluid and Brake Fluid

Only the following approved hydraulic fluid shall be used for the landing gear and brakes:

Specification .....MIL-PRF-87257

## Engine Start

### Engine Start (General)

Minimum oil temperature.....-40 °F (-40 °C)

Minimum voltage for external power unit start..... 28 VDC

Minimum current for external power unit start ..... 800 A

Maximum current for external power unit start..... 1000 A

Max. duration for Takeoff thrust .....5 minutes (max.)<sup>1</sup>

### Engine Start (Ground)

Maximum tailwind for start.....15 knots<sup>2</sup>

Maximum crosswind for start .....25 knots

1. 2 minutes recommended for achieving rated engine life.

2. Tailwinds will increase the time required for start, and increase the likelihood of a Hot Start. The aircraft should be positioned into the wind whenever possible prior to start.

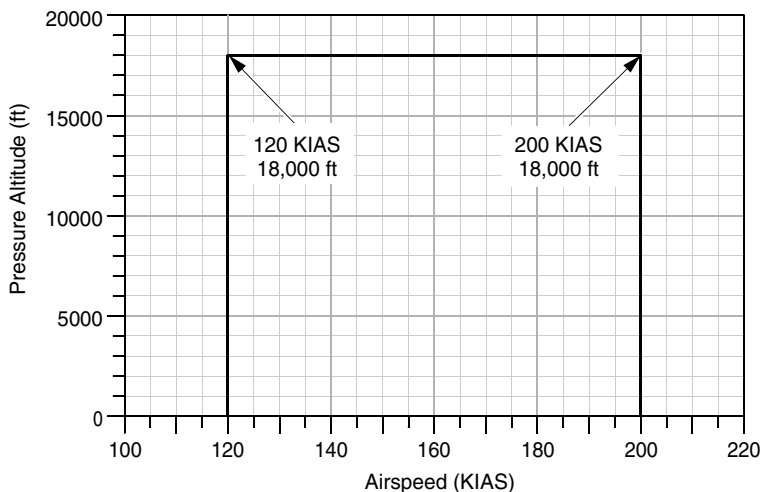
## Start Abort Criteria

Abort start if any of the following are observed:

- Failure to accelerate after 30 seconds.
- No N1 indication at 25% N2.
- Low oil pressure after start.
- ITT exceeds limits (red line).

## Engine Start (In flight)

**Figure 2-2:** Engine Start (In flight)



SF50\_FM02\_0007

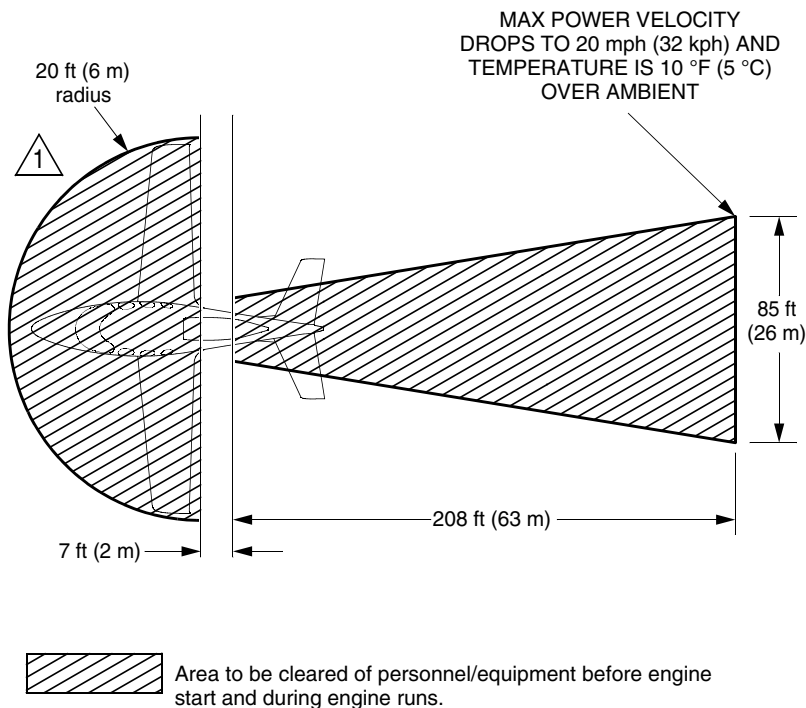
## Starter Generator Limits (Start Mode)

The starter duty cycle is limited as follows:

1. 30-second maximum first start attempt followed by a 30-second minimum spool down and cool down period.
2. 30-second maximum second start attempt followed by a 5-minute minimum spool down and cool down period.
3. 30-second maximum third start attempt followed by a 30-minute minimum spool down and cool down period.

Repeat cycle (Steps 1 through 3 above), if required.

**Figure 2-3: Danger Areas During Start and Ground Operation**



**NOTE**

1 Aircraft Static - Sea Level I.S.A. - No Wind.

**• WARNING •**

**When operating on the ground, check danger areas for personnel, equipment, or Foreign Object Debris (FOD) before starting engine.**

**• CAUTION •**

Point the nose of the aircraft into the wind. If this is not possible, and if the speed of the tailwind component is greater than 10 knots, do not operate engine N1 speed above 60% until the aircraft rolling speed is greater than the tailwind component. Running the engine with a tailwind can either cause the engine to surge or result in high ITT.

SF50\_FM02\_0008C

## **Engine Instrument Markings & Annunciations**

The following describes the engine instrument markings. Associated Warning and Caution annunciations are shown in capitalized text.

### **Powerplant**

The Cirrus Perspective Touch+ by Garmin avionics system provides dynamic gauge markings for engine instruments. Markings are automatically determined based on current conditions and mode of operation (start, climb, etc.).

Automated markings are designed to reduce pilot workload and eliminate pilot error; operators are recommended to rely on the indicated instrument markings for limitations, not memorization.



| Instrument<br>(Range &<br>Units)        | Low Limit<br>(Red<br>Line)                                                            | Low Operating<br>Range<br>(Green Arc<br>Bottom)                                       | High<br>Operating<br>Range<br>(Green Arc<br>Top)          | High<br>Limit<br>(Red Line)          |
|-----------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------|
| Oil Pressure<br>(0 – 130 psi)           | <u>N2 &lt; 80%</u><br>23 psi<br>(5<br>minutes)<br>35 psi<br><u>N2 ≥ 80%</u><br>45 psi | <u>N2 &lt; 80%</u><br>35 psi<br><u>N2 ≥ 80%</u><br>45 psi                             | 120 psi                                                   | 130 psi<br>(5<br>minutes)<br>120 psi |
| Oil<br>Temperature<br>(-40 –<br>135 °C) | <u>N2 &lt; 80%</u><br>-40 °C<br><u>N2 ≥ 80%</u><br>10 °C                              | 10 °C                                                                                 | 135 °C                                                    | 149 °C<br>(5<br>minutes)<br>135 °C   |
| ITT<br>(0 –<br>1150 °C)                 | -                                                                                     | 0 °C                                                                                  | 836 °C                                                    | See Note <sup>a</sup>                |
| N1 (0 –<br>110%)                        | -                                                                                     | 0%                                                                                    | MCT <sup>b</sup> (Blue "T"<br>Bug at T/O) <sup>b</sup>    | 106%<br>(30 sec.)<br>105%            |
| N2 (0 –<br>110%)                        | -                                                                                     | <u>On Ground:</u><br>53%<br><br><u>In Flight:</u><br>61%<br>70% (AI ON <sup>c</sup> ) | 100%                                                      | 101%<br>(30 sec.)<br>100%            |
| Thrust<br>(0 – 110%)                    | -                                                                                     | 0%                                                                                    | MCT <sup>b</sup><br>(Blue "T" Bug at<br>T/O) <sup>b</sup> | -                                    |

- ITT Limitations (Engine Start: 1000 °C (15 sec.), 900 °C (30 sec.), 836 °C (unlimited); Takeoff: 877 °C (10 sec.), 862 °C (5 minutes); Continuous: 836 °C (unlimited).
- Top of %N1 and % Thrust green arc depict Maximum Continuous Thrust (MCT); a value automatically determined by FADEC and avionics system based on current conditions. Takeoff Target %N1 and %Thrust are also automatically determined by FADEC and displayed by blue "T" bugs.
- Minimum flight idle %N2 with Engine anti ice ON increases to 70% in cruise configuration, and decreases to 61% with gear extended to facilitate landing thrust.

Fuel

| Instrument<br>(Range &<br>Units)                    | Red<br>Arc/Bar | Yellow<br>Arc/Bar        | Green<br>Arc/Bar | Yellow<br>Arc/Bar        | Red<br>Arc/Bar |
|-----------------------------------------------------|----------------|--------------------------|------------------|--------------------------|----------------|
|                                                     | Min.           | Min.<br>Caution<br>Range | Normal<br>Range  | Max.<br>Caution<br>Range | Max.           |
| Fuel Flow<br>(0 – 250 U.S.<br>Gal./Hr.)             | --             | --                       | 0 – 175          | --                       | --             |
| Fuel Quantity<br>Gauge<br>(0 – 150 U.S.<br>Gallons) | 0              | 1 – 25                   | 26 – 150         | --                       | --             |

## Electrical

| Instrument<br>(Range &<br>Units)                                                            | Red<br>Arc/Bar        | Yellow<br>Arc/Bar                          | Green Arc/Bar                                                                             | Yellow<br>Arc/Bar        | Red<br>Arc/Bar                 |
|---------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------|--------------------------------|
|                                                                                             | Min.                  | Min.<br>Caution<br>Range                   | Normal<br>Range                                                                           | Max.<br>Caution<br>Range | Max.                           |
| Ess. Bus<br>Voltage<br>Synoptic Only<br>(0 – 35.0 V)                                        | --                    | < 26.5 V                                   | $26.5 \leq V \leq 32.0$                                                                   | --                       | > 32.0 V                       |
| Main Bus<br>Voltage<br>Synoptic Only<br>(0 – 35.0 V)                                        | --                    | < 25.5 V<br>or<br>< 26.5<br>V <sup>a</sup> | $25.5 \leq V \leq 32.0^a$ or<br>$26.5 \leq V \leq 32.0^b$                                 | --                       | > 32.0 V                       |
| Emer. Bus<br>Voltage<br>(0 – 35.0 V)                                                        | < 26.5 V <sup>b</sup> | < 25.0<br>V <sup>b</sup>                   | $25.0 \leq V \leq 32.0^c$ or<br>$26.5 \leq V \leq 32.0^b$                                 | --                       | > 32.0 V                       |
| Gen. 1 Current<br>Digital<br>Display:<br>(-380 – 380 A)<br>Analog Scale:<br>(0 – 275 A)     | --                    | Digital<br>Display:<br>< 5 A <sup>b</sup>  | Digital Display:<br>$5 \leq A \leq 275^b$ or $\leq 275 A^c$<br>Analog Scale:<br>0 - 275 A | --                       | Digital<br>Display:<br>> 275 A |
| Gen. 2 Current<br>Digital<br>Display:<br>(0 – 200 A)<br>Analog Scale:<br>(0 – 75 A)         | --                    | Digital<br>Display:<br>< 4 A <sup>b</sup>  | Digital Display:<br>$4 \leq A \leq 75^b$ or $\leq 75 A^c$<br>Analog Scale:<br>0 - 75 A    | --                       | Digital<br>Display:<br>> 75 A  |
| Batt. 1 Current<br>Digital<br>Display:<br>(-245 – 245 A)<br>Analog Scale:<br>(-100 – 100 A) | --                    | < -5 A                                     | $\geq 5 A$                                                                                | --                       | --                             |
| Batt. 2 Current<br>Digital<br>Display:<br>(-99 – 100 A)<br>Analog Scale:<br>(-50 – 100 A)   | --                    | < -4 A                                     | $\geq 4 A$                                                                                | --                       | --                             |

a. Only if the engine has been running for 15 seconds, or while in the air.

b. Only if on ground with the engine OFF, or during an engine start.

## **Weight Limits**

• NOTE •

The airplane must be operated in accordance with the approved loading schedule.

|                                        |                              |
|----------------------------------------|------------------------------|
| Max. Takeoff Gross Wt. (MTOGW) .....   | 6000 lb (2722 kg)            |
| Maximum Landing Weight (MLW) .....     | 5550 lb (2517 kg)            |
| Maximum Ramp Weight (MRW) .....        | 6040 lb (2740 kg)            |
| Max. Zero Fuel Weight (MZFW) .....     | 4900 lb (2223 kg)            |
| Maximum Aft Baggage Compartment .....  | 300 lb (136 kg) <sup>1</sup> |
| Maximum Cargo X-Tend Compartment ..... | 40 lb (18 kg)                |

## **Center of Gravity Limits**

| CG Envelope          | Weight (lb) | FS (inches) |
|----------------------|-------------|-------------|
| Forward Light        | 4000        | 189.44      |
| Forward Intermediate | 4700        | 189.44      |
| Forward Gross        | 6000        | 192.55      |
| Aft Gross            | 6000        | 198.15      |
| Aft Intermediate     | 5200        | 198.15      |
| Aft Light            | 3600        | 196.28      |

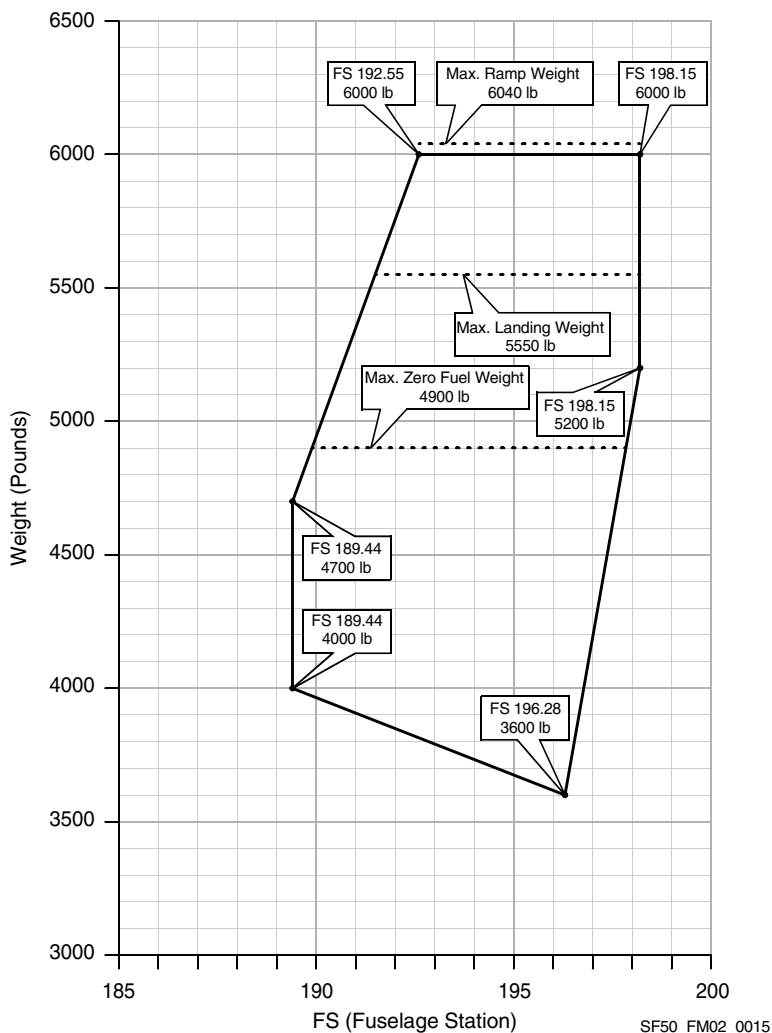
• Note •

Specifications for weight and balance data are defined in detail in [Section 6: Weight and Balance, "Weight and Balance Data"](#).

---

1. Includes Cargo X-Tend weight limit.

**Figure 2-4: Center of Gravity Envelope**



## **Maneuver Limits**

Acrobatic maneuvers are strictly prohibited.

Spins are strictly prohibited.

This airplane is certified in the normal category and is not designed for acrobatic operations. Only those operations incidental to normal flight are approved. These operations include normal stalls, chandelles, lazy eights, and turns in which the angle of bank is limited to 60°.

A normal stall is a non-accelerated stall normally performed at a deceleration of approximately 1 kt per second. It does not include whip stalls, cross controlled stalls, etc.

The stall speed for any configuration is the speed at which the stick pusher activates.

## **Configuration Limitations**

### ***Maximum demonstrated crosswind***

Flaps 50% ..... 18 knots

Flaps 100% ..... 16 knots

## **Flight Load Factor Limits**

Flaps UP, any weight ..... +3.6g, -1.44g

Flaps 50%, any weight ..... +2.0g, 0g

Flaps 100%, any weight ..... +2.0g, 0g

Landing Gear Up or Extended ..... +3.6g, -1.44g

Landing Gear Operating ..... +2.0g, 0g

## **Kinds of Operation**

This airplane is approved for the following types of operations when the required equipment listed in the Minimum Equipment List (MEL) is installed and operational:

- VFR Day and Night
- IFR Day and Night
- Flight Into Known Icing (FIKI)

Acrobatic maneuvers, including spins, are prohibited.

## **Maximum Operating Altitude Limit**

Maximum Airport Elevation ..... 10,000 ft MSL

Maximum Operating Altitude ..... FL310

Minimum Airport Elevation ..... -1000 ft MSL<sup>1</sup>

Maximum Positive Cabin Differential Pressure ..... 7.1 psid

## **Outside Air Temperature Limits**

### **Takeoff Temperatures**

Minimum Takeoff Temperature .....-40 °F (-40 °C)

Maximum Takeoff Temperature.....122 °F (50 °C)

### **Operating Temperatures**

Min. Operating Temp. (up to 20,000 ft).....-65 °F (-54 °C)

Min. Operating Temp. (up to 31,000 ft).....-82 °F (-63 °C)

Min. Landing Gear Operating Temp. ....-40 °F (-40 °C)

Min. Wing/Stab IPS Operating Temp. ....-40 °F (-40 °C)

Max. Engine IPS Operating Temperature.....50 °F (10 °C)

Maximum Operating Temperature .....122 °F (50 °C)

#### **• CAUTION •**

Power-on ground operations exceeding 30 minutes are prohibited either with ambient temperatures above 115 °F (46 °C) or with the Air Conditioning System (ACS) not operating and ambient temperatures above 100 °F (38 °C). Failure to comply could lead to overheating which may damage avionics Line Replaceable Units (LRUs).

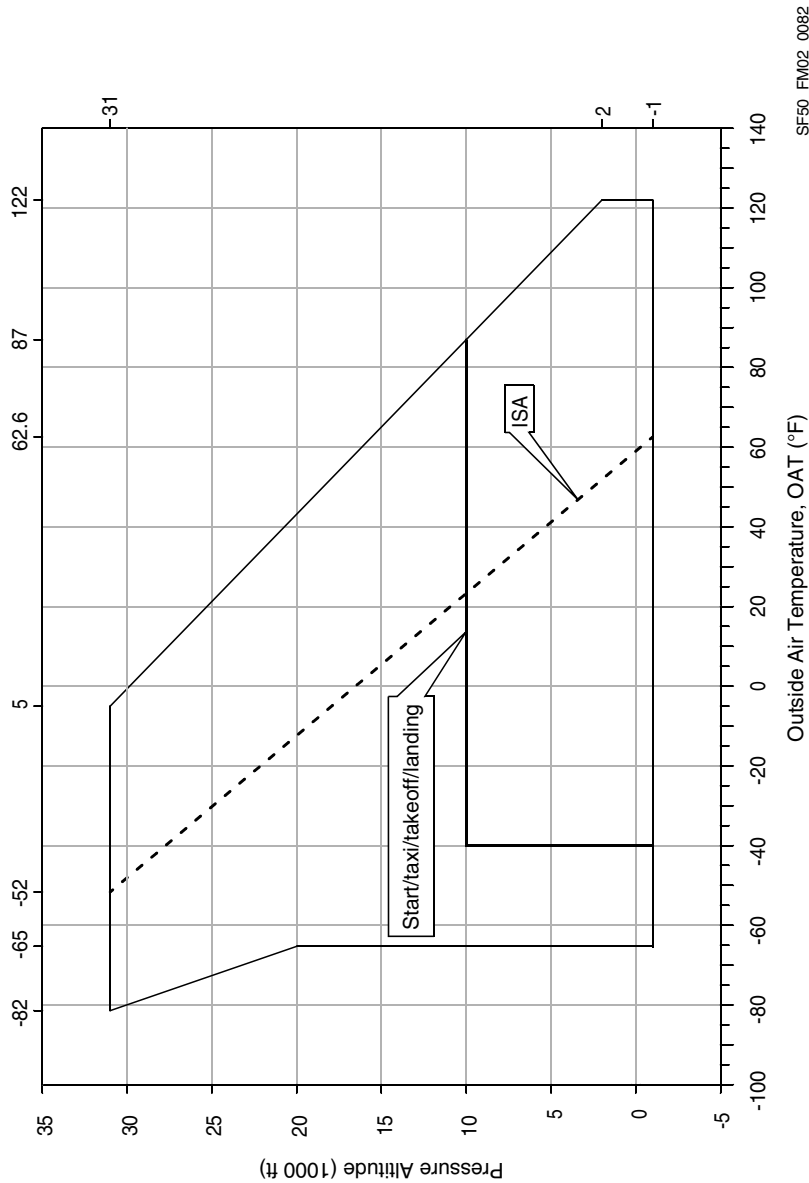
#### **• NOTE •**

Windshield IPS performance is degraded below -22 °F (-30 °C). Avionics that have been cold soaked for two or more hours at an ambient temperature of -40 °F (-40 °C) or lower must be allowed to warm up for at least 20 minutes after power on prior to takeoff.

---

1. For operations at altitudes below sea level, use Mean Sea Level (MSL) values.

**Figure 2-5:** Outside Air Temperature Limits





## **Minimum Crew Requirements**

The minimum crew requirement for all operations is one appropriately rated pilot.

## **Maximum Passenger Seating Limits**

Occupancy of this airplane is limited to seven persons - the pilot and six passengers. The two optional outboard XC seats are weight-limited to 90 lb (40 kg). If this weight restriction for the XC seats cannot be met, or if the XC seats are not installed, these seats cannot be used and occupancy is thus limited to five persons.

When the XC seats are occupied, seat 4 (if installed) must be in the aft position for all ground and air segments of the flight to facilitate cabin emergency egress.

If the wider executive seats are installed, seat 4 will not fit if installed in the forward position (aligned with seats 3 and 5).

When the convenience console is installed between seats 3 and 5, blocking access to the back row, seats 4, 6, and 7 may be installed but not occupied. Passengers in the third row must have an unobstructed egress path at all times.

• NOTE •

Refer to PIM Chapter 7 for seat layout.

## **Child Restraint System Limits**

Rear seat configuration for LATCH-compliant child seats is limited to Seats 3, 4, and 5.

Installation of child seats in Seats 1, 2, 6, and 7 is prohibited.

• NOTE •

Refer to PIM Chapter 7 for seat layout.

## **Seat Belt Limits**

Seat belts must be worn and tightened snug and low on the waist.

## **Seat Handling Limitations**

• WARNING •

**Do not move (or remove) the passenger seat while the aircraft is moving; do so when the aircraft is stationary.**

## **Systems and Equipment Limits**

### **Flight Controls**

#### ***Flaps***

|                                             |               |
|---------------------------------------------|---------------|
| Approved Takeoff Settings .....             | 50%           |
| Approved Approach and Landing Settings..... | UP, 50%, 100% |

• NOTE •

Flaps UP is an abnormal landing configuration, approved for use in the event of flap failure.

Use of flaps above 17,500 feet is prohibited.

### **Landing Gear**

- 8 in-flight cycles per 32 minutes (Maximum)
  - 1 cycle consists of UP-Down-UP or Down-UP-Down
- 10 seconds rest between cycles (Minimum)

### **Warning**

#### ***Stall Warning and Protection***

Ensure stall warning and protection system are tested prior to each flight.

• NOTE •

Refer to [Section 4: Normal Procedures, "Preflight Inspection"](#).

#### ***Traffic Alert and Collision Avoidance System (TCAS-1)***

TCAS-1 is for advisory use only to aid in visually acquiring traffic.

TCAS-1 traffic information must not be the only basis for avoidance maneuvers. The pilot is responsible to see and avoid traffic without TCAS-1 reliance.

### **Electrical System**

The Cirrus Perspective Touch+ by Garmin avionics system provides dynamic gauge markings for engine instruments. Markings are automatically determined based on current conditions and mode of operation (start, climb, etc.).

Automated markings are designed to reduce pilot workload and eliminate pilot error; operators are recommended to rely on the indicated instrument markings for limitations, not memorization.

| Electrical Bus | Normal Range (V) | Outside Normal Range Indications |
|----------------|------------------|----------------------------------|
| Main           | 26.5 – 28.5      | Yellow                           |
| Essential      | 27.3 – 29.5      | Yellow                           |
| Emergency      | 26.5 – 29.5      | Red                              |

### **Maximum Generator Loads**

|                   |       |
|-------------------|-------|
| Generator 1 ..... | 270 A |
| Generator 2 ..... | 76 A  |

### **Icing Conditions**

Icing conditions are defined as visible moisture and OAT  $\leq 41$  °F (5 °C) or visible ice accumulations on the airframe.

Visible moisture is defined as any cloud, fog with visibility of one mile (1.6 km) or less, mist, rain, sleet, or snow.

Icing conditions may exist whenever Outside Air Temperature (OAT) on the ground, or for takeoff, is below 41 °F (5 °C) and visible moisture in any form is present.

Icing conditions may also exist when OAT on the ground and for takeoff is below 41 °F (5 °C) during operations where snow, ice, standing water, or slush may be encountered.

SAE Type 1 deicing fluid is approved for ground deicing.

### **Severe Icing**

The airplane is prohibited from operating in severe icing conditions. Severe icing conditions are defined as any freezing drizzle, any freezing rain, Supercooled Large Droplets (SLD), or any icing conditions that overwhelm the ice protection systems. If the airplane encounters such conditions, the pilot must (i) immediately exit icing conditions by changing altitude or course, and (ii) remain clear of icing conditions for the remainder of the flight.

Severe icing conditions may be identified by the following:

- Unusually extensive ice accumulation on the airframe or windshield in areas not normally observed to collect ice.
- Accumulation of ice on the wing surface, aft of the deice boots.
- Inability of the airplane to maintain the published ice-contaminated performance specifications listed in [Section 5: Performance Data](#).
- Repeated use of MAX mode to clear the windshield.

The following weather conditions may be conducive to severe icing:

- Visible rain at temperatures colder than 32 °F (0 °C) static air temperature.
- Droplets that splash or splatter at temperatures colder than 32 °F (0 °C) static air temperature.

**• CAUTION •**

Delaying the use of the IPS may result in ice ingestion and possible engine damage or flameout.

## Operation

In icing conditions, the airplane must be operated, and its ice protection systems used, as described in the operating procedures section of this AFM. Where specific operational speeds and performance information have been established for such conditions, this information must be used.

Takeoff is prohibited with any frost (polished or not), ice, snow, or slush adhering to the wings, stabilizers, control surfaces, engine inlet, or fuselage forward of the engine inlet.

When holding in icing conditions, higher speed will result in increased ice accretion rate.

Autopilot altitude holding performance has not been demonstrated in RVSM airspace (above FL280) with wing/stab ice protection on.

Minimum airspeed for flight in icing conditions with flaps and gear UP  
( $V_{MIN\ ICING}$ ) .....160 KIAS<sup>1</sup>  
Flaps (when holding in icing conditions)..... UP  
Maximum flap deflection with ice accretions .....50%  
Max. duty cycle (WINDSHLD IPS MAX Mode) .....50%,  
with maximum of 10 seconds ON.

**• WARNING •**

**Exit icing conditions immediately if any grinding or sticking is felt in the controls.**

**• CAUTION •**

When OAT is above 41 °F (5 °C), probe heat may only be ON for 5 minutes (continuous) during ground operations. Extended use in warmer temperatures may damage the composite structure adjacent to the probe.

1. Excluding climbs, approach, and landing.

## ***Tactile Inspection***

A tactile (hands on surface) check of the wing leading edge and the upper surface must be accomplished no more than 5 minutes prior to takeoff when operating in ground icing conditions.

This check must be performed whenever the OAT is below 41 °F (5 °C), or when the wing fuel temperature is not above 32 °F (0 °C), and visible moisture is present, or the airplane was exposed to visible moisture since the previous landing, or the difference between the dew point temperature and the OAT is 6 °F (3 °C) or less, or water is present in any form on the wing.

### **• WARNING •**

**On ground, do not rely on visual icing evidence to turn on the Engine IPS. Use the temperature and visual moisture criteria as specified earlier. Visual evidence of icing may not precede actual engine or engine inlet icing. Excessive ice accretions on the inlet or engine may cause engine damage if ice is ingested by the engine. The resulting loss of thrust may lead to reduced takeoff performance or an off-airport landing, either of which could cause serious injury or loss of life.**

## ***IPS Activation***

The IPS must be activated:

- When OAT  $\leq$  41 °F (5 °C) with visible moisture or visibility < 1 mile (1.6 km)  
OR
- At first sign of ice.

### **• WARNING •**

**At first sign of ice, the WING/STAB IPS switch must be selected to the ON position and remain ON until the entire wing is free of ice accretion and ice is no longer accreting on the aircraft.**

**Stall Warning offset must remain ON until entire airframe is free of ice.**

### **• NOTE •**

The left wing leading edge is the primary reference area for determining if ice is accumulating on the aircraft. Ice may be visible on other areas of the aircraft as well, such as the forward windshield.

## ***Ice Protection System (IPS) Fluid***

### **Minimum Dispatch Quantity**

IPS Fluid Minimum Dispatch Quantity..... 1.0 U.S. gal (4 L)

## Endurance Limits (at minimum dispatch quantity)

|                |            |
|----------------|------------|
| NORM Mode..... | 90 minutes |
| HIGH Mode..... | 45 minutes |

## Use of Autopilot in Icing Conditions

In light-to-moderate icing conditions, autopilot use with periodic checks (disconnect and hand fly) is permitted. However, autopilot use is prohibited in the following conditions:

- Severe icing;
- Any unusually small or large control forces, or control deflections, to move flight controls when the autopilot is disconnected periodically for checking purposes; or
- Indications of frequent autopilot re-trimming during straight and level flight.

## Autopilot

Autopilot must be on above FL290. Pilot must notify ATC of non-RVSM status if autopilot fails.

The Garmin GFC 700 Automatic Flight Control System (AFCS) has the following limitations:

- Minimum Autopilot Speed

| Flaps | Minimum Autopilot Speed |
|-------|-------------------------|
| 100%  | 85 KIAS                 |
| 50%   | 95 KIAS                 |
| UP    | 105 KIAS                |

- Maximum Autopilot Speed

| Flaps             | Maximum Autopilot Speed |
|-------------------|-------------------------|
| 100%              | 150 KIAS                |
| 50%               | 190 KIAS                |
| Landing Gear DOWN | 210 KIAS                |
| Clean             | 250 KIAS / 0.53 Mach    |

- Autopilot Minimum-Use Height:

• NOTE •

Transient descent below published minimum use height is permitted while transitioning to climb using coupled go around (GA) mode.

- Takeoff and Climb..... 400 ft AGL
- Enroute and Descent..... 1000 ft AGL
- Approach (GP or GS Mode) .....Higher of 200 ft AGL or Approach MDA, DA, DH
- Approach (IAS, VS, PIT or ALT Mode) ..... Higher of 400 ft AGL or Approach MDA
- The autopilot cannot be engaged beyond the Engagement Limits. If the autopilot is engaged beyond the command limits (up to engagement limits), it will be rolled or pitched to within the command limits, and an altitude loss of 1000 feet or more can be expected while attitude is established in the selected mode.

| Axis  | Autopilot Engagement Limit |
|-------|----------------------------|
| Pitch | $\pm 50^\circ$             |
| Roll  | $\pm 75^\circ$             |

- The autopilot and Flight Director (FD) will not command pitch or roll beyond the Command Limits.

| Axis                    | Autopilot Command Limit |
|-------------------------|-------------------------|
| FD Pitch Command Limits | +30°, -25°              |
| FD Roll Command Limits  | $\pm 30^\circ$          |

- Use of VNAV is not supported during an approach with a teardrop course reversal. VNAV will be disabled at the beginning of the teardrop.
- If Stall Warning is inoperative, Autopilot Underspeed Protection (USP) will not be provided in Altitude Critical Modes (ALT, GS, GP, T/O and GA).
- The autothrottle pilot controllable functions (if installed) are part of the autopilot, and are subject to autopilot limitations (minimum use height, speed, etc.).
- The autothrottle pilot controllable functions (if installed) cannot be engaged unless the autopilot is engaged.

## Safe Return Autoland System (if installed)

The Safe Return Autoland system is for emergency situations only and should not be used when the pilot is able to land the aircraft. It is intended to be used in the event that the pilot in command is unable to operate the aircraft, such as if the pilot is impaired or incapacitated.

Once activated, Safe Return Autoland declares the emergency condition, then maneuvers the aircraft to a suitable airport for landing, selecting the airport based upon a variety of parameters, including weather, terrain, available approach, and runway length.

### • NOTE •

Safe Return Autoland cannot account for GPS outages or runway conditions (NOTAMs) which could degrade ability to safely land the airplane.

Safe Return Autoland assumes a fully functional airplane in order to function properly, and is not designed to mitigate system failures. Safe Return Autoland activation will be inhibited if there are missing or corrupt terrain, navigation, or obstacle databases. Other failures including (but not limited to) GPS, landing gear, flaps, wheel speed sensors, AFCS/pitch trim servos, or radar altimeter, will degrade Safe Return Autoland performance. This may result in damage to the airframe and possibly injury to the occupants.

A red slashed circle (⊗) illuminated on the Autoland activation panel to the left of the activation button indicates when a system failure may have caused Autoland to be degraded or unavailable, and is illuminated at all times when the aircraft is below 600 feet AGL. A green landing airplane icon illuminated on the Autoland activation panel to the right of the activation button indicates when Autoland is activated.



## Navigation and Communication Equipment

### ***Attitude and Heading Reference System (AHRS)***

The airplane must not operate in the regions listed below. Flying in these regions may lead to loss of attitude or heading information, and possibly even loss of aircraft control. Exceeding aircraft limitations may cause the aircraft to break up while in flight.

| <b>Magnetic Cut-out Regions</b> | <b>Latitude</b> | <b>Longitude</b>                                                          |
|---------------------------------|-----------------|---------------------------------------------------------------------------|
| North                           | North of 72° N  | All longitudes                                                            |
|                                 | North of 65° N  | Between 75° W and 120° W.<br>(Northern Canada)                            |
|                                 | North of 70° N  | Between 70° W and 128° W.<br>(Northern Canada)                            |
|                                 | North of 70° N  | Between 85° E and 114° E.<br>(Northern Russia)                            |
| South                           | South of 70° S  | All longitudes                                                            |
|                                 | South of 55° S  | Between 120° E and 165° E.<br>(Region south of Australia and New Zealand) |

• Note •

Alternative procedures must be established for dispatch if the indication GEO LIMITS is displayed.

### ***Cirrus Perspective Touch+ by Garmin Avionics System***

The appropriate revision of the Cirrus Perspective Touch+ Cockpit Reference Guide (p/n 190-02471-XX, where X can be any digit from 0 to 9) must be immediately available to the pilot during flight. The system software version stated in the reference guide must be appropriate for the system software version displayed on the equipment.

The avionics system has the following limitations:

- Use of VNAV is prohibited during the intermediate segment of an approach that includes a teardrop course reversal.
- Dead Reckoning (DR) Mode use is allowed only in Enroute (ENR) or Oceanic (OCN) phases of flight. The estimated navigation data supplied by the system in DR Mode must not be used as a sole means of navigation.

- The fuel quantity, fuel required, fuel remaining, and gross weight estimate functions presented in the AUX-WEIGHT PLANNING and AUX-TRIP PLANNING pages of the MFD are supplemental information only and must be confirmed by the flight crew prior to use.

### ***Cirrus Perspective Touch+ by Garmin GPS Navigation System***

The GPS receivers are approved under TSO C145a Class 3. The system has been demonstrated accurate for the following operations receiving usable navigation data.

- GPS and GPS/SBAS navigation for enroute and terminal navigation per AC 20-138.
- Non-precision instrument approach operations using GPS and Wide Area Augmentation System (WAAS), and approach procedures with vertical guidance (including “LNAV/VNAV”, “LNAV + V” for AC 90-105, and “LPV” for AC 90-107) in accordance with the latest revision of AC 20-138.
- Barometric VNAV is approved to enroute and terminal descents, as per AC 20-129. Guidance is provided up to the FAF waypoint when there is no procedure for vertical guidance following the FAF. Guidance is provided up to the waypoint preceding the FAF (FAF-1) when there are provisions for vertical guidance (ILS or GPS WAAS) following the FAF.
- Oceanic/Remote/MNPS/RNP-10 (per FAA AC 20-138A and FAA Order 8400-12A). Both GPS receivers are required to be operating and receiving usable signals except for routes requiring only one Long Range Navigation (LRN) sensor.

#### **• NOTE •**

For Oceanic/Remote operations, GPS Fault Detection/Exclusion (FDE) availability must be verified prior to flight with either the on board RAIM prediction tool or other approved FDE software.

- Enroute and Terminal including RNP-5/BRNAV and PRNAV (RNP-1) in accordance with JAA TGL-10, ACJ 20X4, and AC 90-105 provided the FMS is receiving usable navigation information from one or more GPS receivers.
- RNAV and RNP Terminal (RNP 1 or RNP 2, RNAV-1 or RNAV-2) per AC 20-138B and AC 90-105.
- VHF Navigation: Maximum demonstrated VHF VOR reception abeam aircraft position is 100 NM. Refer to FAA FAR AIM (1-1-8) for NAVAID service volumes.

The GPS navigation system has the following limitations:

- GPS-based IFR enroute, oceanic, and terminal navigation is prohibited unless the pilot verifies the currency of the database or verifies each selected waypoint for accuracy by reference to current approved data.
- RNAV/GPS instrument approaches must be accomplished in accordance with approved instrument approach procedures that are retrieved from the navigation database.

The database must incorporate the current update cycle.

• Note •

Not all the published approaches are in the navigation database. The flight crew must ensure that the planned approach is in the database.

- Receiver Autonomous Integrity Monitoring (RAIM) must be available when conducting instrument approaches utilizing the GPS receiver.
- When using GPS/NAV mode to intercept the final approach course, monitor the flight director transition to LOC/GS. If the transition does not occur automatically, manually select HDG mode and CDI source to LOC in order to intercept the final approach course.
- Use of the GPS receiver to accomplish ILS, LOC, LOC-BC, LDA, SDF, MLS or any other type of approach not approved for GPS overlay is not authorized.
- Operation in airspace referenced to a datum other than WGS-84 or NAD-83 is prohibited.
- Dispatch is not authorized with inoperative GPS, predicted degraded GPS (RAIM prediction), or known GPS outage (NOTAM) expected during the planned flight.

### ***Enhanced Vision System (Optional)***

The Enhanced Vision System (EVS) must not be used for flight guidance, navigation, traffic avoidance, or terrain avoidance. Maneuvering the airplane in any phase of flight such as taxi, takeoff, approach, landing, or roll out shall not be predicated on EVS imagery. The EVS must only be used as an aid to assist the flight crew with visually acquiring objects normally viewed through the cockpit windows.

## **Weather**

### ***Garmin GSR 56 Iridium Datalink***

System functionality at cabin temperatures below -4 °F (-20 °C) should not be expected.

## **Portable Electronic Devices**

The SF50 navigation and communication systems have been demonstrated to be tolerant to common Transmitting Portable Electronic Devices (examples include but are not limited to: mobile phones, laptop computers, tablet computers, and hand-held gaming devices) through all phases of flight in accordance with AC 20-164 (noted in AC 91.21-1D).

## **Other Limitations**

### **Oxygen**

The oxygen system is serviced with Oxygen, Aviator's Breathing, Liquid And Gas (MIL-O-27210, Type I - Gaseous). It is prohibited to use technical/medical oxygen.

The RH crew seat is equipped with a quick-don mask. During Preflight Inspection, the passenger must be briefed on its use. The second quick-don oxygen mask shall be considered a passenger oxygen mask, and operated as such, unless acting as pilot in command or required otherwise.

An additional passenger oxygen mask has been installed in the passenger compartment to accommodate multiple seat 4 positions. In the event of emergency oxygen use, all passenger masks will automatically deploy and user will have to pull pin to initiate flow.

In addition to the requirements of applicable operating rules, the following aircraft certification rules must be observed:

- Oxygen system is not approved for extended use of auto deploy passenger oxygen masks above 25,000 feet cabin altitude.
- To qualify as a quick-donning oxygen mask, comply with proper stowage of the pressure demand crew oxygen masks in their containers.

The oxygen endurance assumes the:

- Single pilot quick-don oxygen mask should be used in 100% mode above 20,000 feet.
- Single pilot quick-don oxygen mask should be used in NORMAL mode below 20,000 feet.
- Passenger oxygen is not used at 15,000 feet and below (including passenger using second quick-don mask).
- Passenger oxygen use is limited to 10 minutes or less.

• **WARNING** •

**Ensure that the crew oxygen mask can be donned quickly without any interferences (i.e. eyewear, headsets, hats, hairstyles).**

**Ensure that the passenger oxygen masks are donned during an emergency descent to an altitude not requiring supplemental oxygen.**

• **NOTE** •

Passenger oxygen is automatically made available at cabin altitudes exceeding 15,000 feet or if manually selected.

## Fill Pressure vs. Temperature

| <b>OXYGEN FILL PRESSURE VERSUS TEMPERATURE</b> |                |                    |
|------------------------------------------------|----------------|--------------------|
| <b>TEMP °F</b>                                 | <b>TEMP °C</b> | <b>FILL TO PSI</b> |
| -40                                            | -40            | 1321               |
| -30                                            | -34            | 1370               |
| -20                                            | -29            | 1418               |
| -10                                            | -23            | 1466               |
| 0                                              | -18            | 1515               |
| 10                                             | -12            | 1563               |
| 20                                             | -7             | 1611               |
| 30                                             | -1             | 1659               |
| 40                                             | 4              | 1707               |
| 50                                             | 10             | 1755               |
| 60                                             | 16             | 1802               |
| 70                                             | 21             | 1850               |
| 80                                             | 27             | 1898               |
| 90                                             | 32             | 1945               |
| 100                                            | 38             | 1993               |
| 110                                            | 43             | 2040               |
| 120                                            | 49             | 2088               |

## Altitude vs. Minutes of Available Oxygen

### Pilot Oxygen Mask Mode: NORM

The information in the tables below is based on the following:

- The quick-don (QD) masks set to NORM, and
- The bottle being filled to pressure noted in the following tables.

#### 22 cu ft Oxygen Bottle

| 22 cu ft: 1850 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| ALT<br>(FT.)       | M I N U T E S      |                    |                    |                    |                    |                    |                    |
|                    | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | 0 Pax              | +                  | +                  | +                  | +                  | +                  | +                  |
|                    |                    | 0 Pax              | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |
| 10,000             | 463                | 232                | 71                 | 42                 | 30                 | 23                 | 19                 |
| 15,000             | 438                | 219                | 71                 | 42                 | 30                 | 23                 | 19                 |
| 20,000             | 344                | 172                | 66                 | 41                 | 29                 | 23                 | 19                 |
| 25,000             | 222                | 111                | 55                 | 36                 | 27                 | 22                 | 18                 |
| 30,000             | 250                | 125                | 58                 | 38                 | 28                 | 22                 | 19                 |
| 31,000             | 264                | 132                | 60                 | 39                 | 29                 | 23                 | 19                 |

| 22 cu ft: 1600 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| ALT<br>(FT.)       | M I N U T E S      |                    |                    |                    |                    |                    |                    |
|                    | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | 0 Pax              | +                  | +                  | +                  | +                  | +                  | +                  |
|                    |                    | 0 Pax              | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |
| 10,000             | 390                | 195                | 60                 | 35                 | 25                 | 19                 | 16                 |
| 15,000             | 368                | 184                | 60                 | 35                 | 25                 | 20                 | 16                 |
| 20,000             | 289                | 144                | 55                 | 34                 | 25                 | 19                 | 16                 |
| 25,000             | 187                | 94                 | 46                 | 31                 | 23                 | 18                 | 15                 |
| 30,000             | 210                | 105                | 49                 | 32                 | 24                 | 19                 | 16                 |
| 31,000             | 222                | 111                | 50                 | 33                 | 24                 | 19                 | 16                 |

| 22 cu ft: 1000 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| M I N U T E S      |                    |                    |                    |                    |                    |                    |                    |
| ALT<br>(FT.)       | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              |
|                    |                    | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |                    |
| 10,000             | 213                | 107                | 33                 | 19                 | 14                 | 11                 | 9                  |
| 15,000             | 202                | 101                | 33                 | 19                 | 14                 | 11                 | 9                  |
| 20,000             | 158                | 79                 | 30                 | 19                 | 14                 | 11                 | 9                  |
| 25,000             | 102                | 51                 | 25                 | 17                 | 13                 | 10                 | 8                  |
| 30,000             | 115                | 58                 | 27                 | 18                 | 13                 | 10                 | 9                  |
| 31,000             | 122                | 61                 | 28                 | 18                 | 13                 | 10                 | 9                  |

### Pilot Oxygen Mask Mode: 100%

#### • CAUTION •

With 100% flow, endurance can be substantially lower than with diluted flow.

The information in the tables below is based on the following:

- The quick-don (QD) masks set to 100%, and
- The bottle being filled to pressures noted in the following tables.

### 22 cu ft Oxygen Bottle

| 22 cu ft: 1850 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| M I N U T E S      |                    |                    |                    |                    |                    |                    |                    |
| ALT<br>(FT.)       | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              | 0 Pax              |
|                    |                    | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |                    |
| 10,000             | 47                 | 24                 | 19                 | 16                 | 14                 | 12                 | 11                 |
| 15,000             | 59                 | 29                 | 23                 | 19                 | 16                 | 14                 | 12                 |
| 20,000             | 74                 | 37                 | 27                 | 22                 | 18                 | 15                 | 14                 |
| 25,000             | 95                 | 47                 | 33                 | 25                 | 20                 | 17                 | 15                 |
| 30,000             | 125                | 62                 | 40                 | 29                 | 23                 | 19                 | 16                 |
| 31,000             | 134                | 67                 | 42                 | 30                 | 24                 | 19                 | 17                 |

| 22 cu ft: 1600 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| ALT<br>(FT.)       | M I N U T E S      |                    |                    |                    |                    |                    |                    |
|                    | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 0 Pax              | 0 Pax              | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |
| 10,000             | 40                 | 20                 | 16                 | 14                 | 12                 | 10                 | 9                  |
| 15,000             | 49                 | 25                 | 19                 | 16                 | 13                 | 12                 | 10                 |
| 20,000             | 62                 | 31                 | 23                 | 18                 | 15                 | 13                 | 11                 |
| 25,000             | 80                 | 40                 | 28                 | 21                 | 17                 | 14                 | 12                 |
| 30,000             | 105                | 53                 | 33                 | 25                 | 19                 | 16                 | 14                 |
| 31,000             | 112                | 56                 | 35                 | 25                 | 20                 | 16                 | 14                 |

| 22 cu ft: 1000 PSI |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| ALT<br>(FT.)       | M I N U T E S      |                    |                    |                    |                    |                    |                    |
|                    | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             | 1 Crew             |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD | 2 <sup>nd</sup> QD |
|                    | +                  | +                  | +                  | +                  | +                  | +                  | +                  |
|                    | 0 Pax              | 0 Pax              | 1 Pax              | 2 Pax              | 3 Pax              | 4 Pax              | 5 Pax              |
| 10,000             | 22                 | 11                 | 9                  | 7                  | 6                  | 6                  | 5                  |
| 15,000             | 27                 | 13                 | 11                 | 9                  | 7                  | 6                  | 6                  |
| 20,000             | 34                 | 17                 | 13                 | 10                 | 8                  | 7                  | 6                  |
| 25,000             | 44                 | 22                 | 15                 | 12                 | 9                  | 8                  | 7                  |
| 30,000             | 58                 | 29                 | 18                 | 13                 | 11                 | 9                  | 7                  |
| 31,000             | 62                 | 31                 | 19                 | 14                 | 11                 | 9                  | 8                  |

## Crew Headset

One headset with a boom microphone is required for pilot use.

## Smoking

Smoking is prohibited in this airplane.



## Brakes

### • WARNING •

**Brake assemblies, discs, and pads can be VERY HOT. Approach slowly and cautiously to detect the amount of radiant heat before proceeding with tactile inspection (hands on surface).**

A cool down period and brake overheat inspection are required after all high-energy braking events. The minimum cool down period is 30 minutes, or until the brake discs have cooled enough to allow for tactile inspection, whichever occurs first.

High-energy braking events include:

- Rejected Takeoffs (RTOs) or engine emergencies on takeoff.
- High-speed landings (Landings with 0% flaps, ice contaminated airframe, or high-density altitude).

The cool down period is necessary because a high-energy use of the brakes leads to elevated brake temperatures, significant pad wear, as well as possible damage to discs and pads.

After each cool down period, inspect for damaged or overheated components.

### • NOTE •

Refer to [Section 8: Handling and Servicing, "Brake Overheat Inspection"](#).

## Tire Inflation Pressure (nominal)

|                                 |                        |
|---------------------------------|------------------------|
| Nose Tire 5.00 x 5 (10 PR)..... | 88±5 psi (607±34 kPa)  |
| Main Tires 18 x 5.5 (8 PR)..... | 105±5 psi (724±34 kPa) |

## Circuit Breakers

Intentional pulling of circuit breakers during flight, other than as required in specific procedures, is prohibited.

## In-Flight Entertainment

The In-Flight Entertainment (IFE) display screen must be stowed during ground operations, takeoff, and landing.

Exterior Windows

Adhering foreign items to exterior windows is prohibited. This includes but is not limited to: suction cups, films, window tint, stickers, decals, etc. This does not include the interior passenger window frost panes, which may have clear suction cup shades applied.

• CAUTION •

Objects in close contact with the windows, especially dark or black, may result in temperature damage or distortion of the window. Objects with dyes may transfer dye into the window acrylic resulting in damage to the aircraft window.

Placards

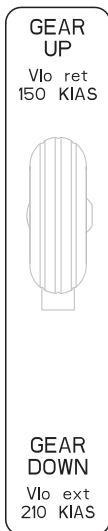
Interior Placards

Figure 2-6: Oxygen Fill Chart

| OXYGEN FILL CHART |                          |                   |
|-------------------|--------------------------|-------------------|
| TEMPERATURE<br>°F | FILL<br>PRESSURE<br>PSIG | TEMPERATURE<br>°C |
| -40               | 1321                     | -40.0             |
| -30               | 1370                     | -34.4             |
| -20               | 1418                     | -28.9             |
| -10               | 1466                     | -23.3             |
| 0                 | 1515                     | -17.8             |
| 10                | 1563                     | -12.2             |
| 20                | 1611                     | -6.7              |
| 30                | 1659                     | -1.1              |
| 40                | 1707                     | 4.4               |
| 50                | 1755                     | 10.0              |
| 60                | 1802                     | 15.6              |
| 70                | 1850                     | 21.1              |
| 80                | 1898                     | 26.7              |
| 90                | 1945                     | 32.2              |
| 100               | 1993                     | 37.8              |
| 110               | 2040                     | 43.3              |
| 120               | 2088                     | 48.9              |

SF50\_FM02\_0013

**Figure 2-7:** Landing Gear  $V_{LO}$  Placard



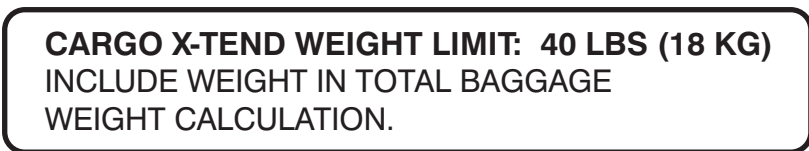
SF50\_FM02\_0025

**Figure 2-8:** Fasten Seat Belts/No Smoking



SF50\_FM02\_0026

**Figure 2-9:** Cargo X-Tend (if installed)



SF50\_FM02\_0027

**Figure 2-10: Baggage Compartment Limitations**

**TOTAL BAGGAGE COMPARTMENT  
WEIGHT LIMIT: 300 LBS (136 KG)**

**50 LB/FT<sup>2</sup> MAXIMUM (244 KG/M<sup>2</sup>)**

**LOAD IN ACCORDANCE WITH AFM.  
REFER TO AFM FOR BAGGAGE TIE-DOWN,  
WEIGHT AND BALANCE INFORMATION.**

SF50\_FM02\_0028

**Figure 2-11: Seats 3, 4, 5 (if installed)**

- Locate seat per AFM.
- Do not block egress path.
- Lock seat in position before flight.

-This seat weighs:  lbs.

*\* Actual seat weight to be recorded on each seat placard.*

SF50\_FM02\_0087

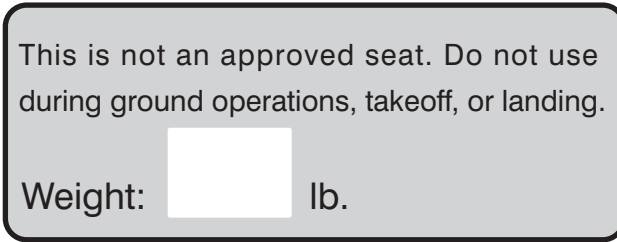
**Figure 2-12: Child Seats (if installed)**

- Locate seat per AFM.
- Lock seat in position before flight.
- Occupant weight limit: 90 lb.

-This seat weighs:  lb.

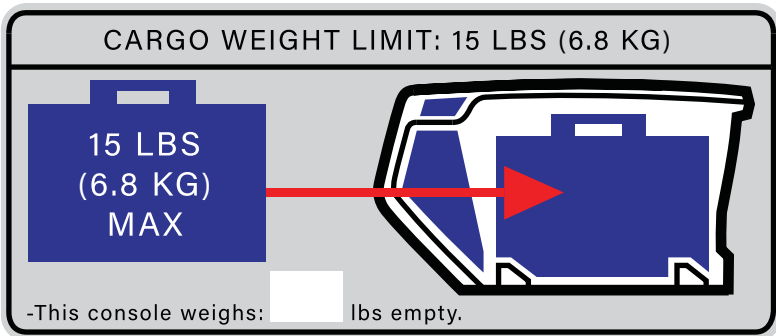
SF50\_FM02\_0032

**Figure 2-13:** Lavatory (if installed)



SF50\_FM02\_0040A

**Figure 2-14:** Convenience Console Weight Limit (if installed)



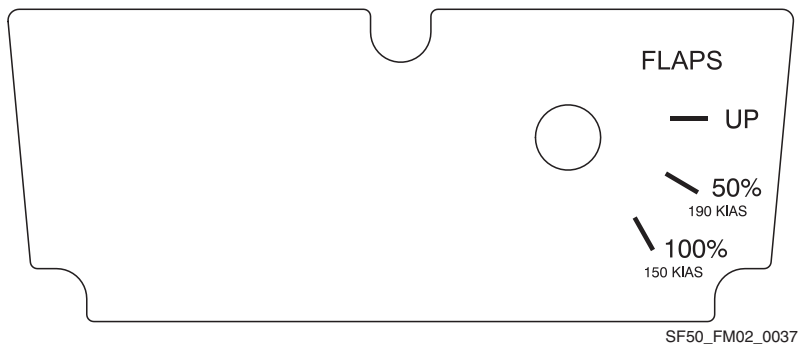
SF50\_FM02\_0088

**Figure 2-15:** Convenience Console (if installed)



SF50\_FM02\_0089

**Figure 2-16: Flap Positions/Airspeeds**



**Figure 2-17: Pitch Trim Direction**



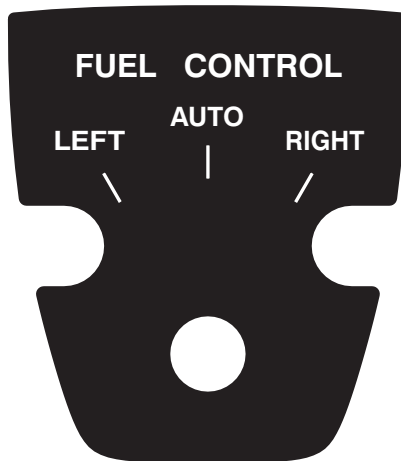
SF50\_FM02\_0035

**Figure 2-18:** Thrust Position



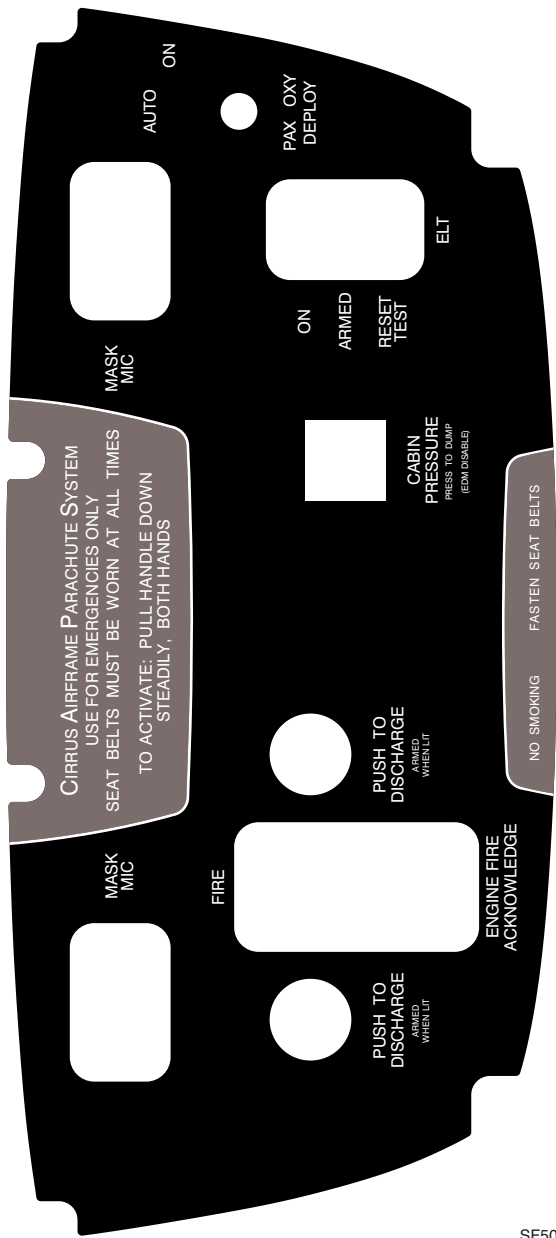
SF50\_FM02\_0036

**Figure 2-19:** Fuel Selector



SF50\_FM02\_0039

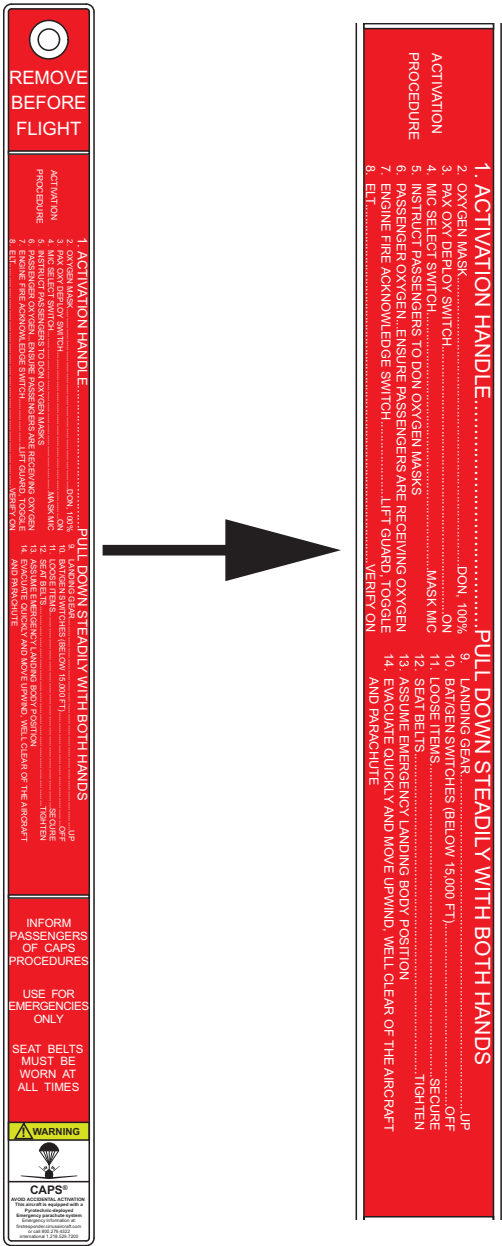
Figure 2-20: Emergency Panel



SF50\_FM02\_0086



Figure 2-21: CAPS Safety Pin Flag



SF50\_FM02\_0090A

**Figure 2-22: Emergency Exit**



SF50\_FM02\_0042

**Figure 2-23: Door Operation, Upper**



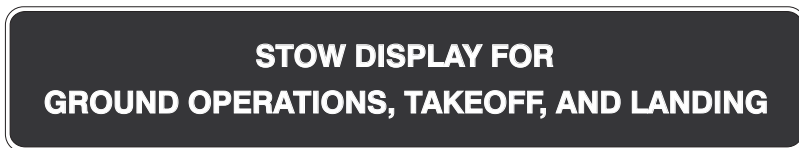
SF50\_FM02\_0044

**Figure 2-24: Door Operation, Lower**



SF50\_FM02\_0045

**Figure 2-25: IFE Screen Stowage (if installed)**



SF50\_FM02\_0046A

**Figure 2-26:** Egress



SF50\_FM02\_0081

**Figure 2-27:** Safe Return Autoland Brake Release (if installed)

**TO RELEASE BRAKES AFTER EMERGENCY AUTOLAND:**

1. SHUT DOWN ENGINE.  
LEFT COCKPIT ENGINE PANEL.  
ROTATE KNOB TO OFF AND PRESS BUTTON.
2. TURN OFF ALL ELECTRICAL SWITCHES (BELOW DISPLAYS).
3. SECURE AIRCRAFT (KEEP FROM ROLLING).
4. PRESS AND HOLD GREEN "BRAKE RESET" BUTTON.  
HOLD FOR APPROXIMATELY 5 SECONDS.
5. AIRCRAFT IS NOW FREE TO MOVE.

SF50\_FM02\_0162

**Safe Return Autoland Activation Panel (if installed)**

Placard stating "EMERGENCY USE ONLY" must appear adjacent to activation pushbutton.

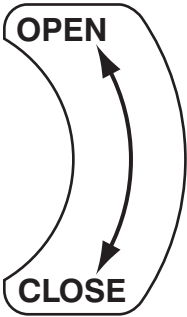
**Exterior Placards**

**Figure 2-28:** Emergency Exit



SF50\_FM02\_0041

Figure 2-29: Door Operation



SF50\_FM02\_0043

Figure 2-30: External Power



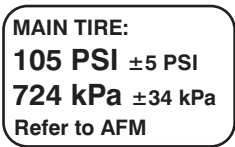
SF50\_FM02\_0052

Figure 2-31: Tire Pressure, Nose



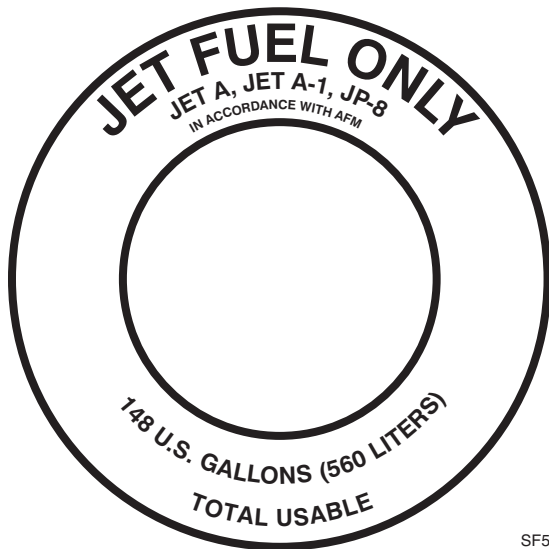
SF50\_FM02\_0022

Figure 2-32: Tire Pressure, Main



SF50\_FM02\_0023

**Figure 2-33:** Fuel Tank Capacity and Fuel Type



SF50\_FM02\_0021

**Figure 2-34:** TKS Fluid



SF50\_FM02\_0024A

### **Safe Return Autoland Equipped (if installed)**

Placard stating “See inside pilot footwell for brake reset and aircraft movement instructions following Safe Return Emergency Autoland” must appear adjacent to cabin door.

## ***Electronic Placards***

Normal category airplane

No acrobatic maneuvers, including spins, approved.

This aircraft is certified for the following flight operations: DAY – NIGHT  
– VFR – IFR, Flight in known icing (with required equipment).

Operate per Airplane Flight Manual (AFM).

Maximum flap position 50% if airframe is ice contaminated.

Maneuvering Speed:  $V_o$  150 KIAS

Seats must be in normal upright position for takeoff and landing.

Completely lock all seats in position before flight.

Aircraft is equipped with: Cirrus Airframe Parachute System (CAPS),  
Garmin Electronic Stability and Protection (ESP), Safe Return Emergency  
Automatic Landing System (if installed).

# ***Section 3: Emergency Procedures***

## **Table of Contents**

|                                                          |    |
|----------------------------------------------------------|----|
| Introduction.....                                        | 5  |
| Crew Alert System (CAS) Messaging .....                  | 5  |
| Warnings .....                                           | 5  |
| Preflight Planning .....                                 | 5  |
| Preflight Inspections/Maintenance .....                  | 6  |
| Methodology .....                                        | 6  |
| Maintain Aircraft Control .....                          | 6  |
| Analyze the Situation .....                              | 6  |
| Take Appropriate Action .....                            | 6  |
| Land as Soon as Conditions Permit .....                  | 7  |
| Circuit Breakers .....                                   | 7  |
| Memory Items .....                                       | 7  |
| Procedure Division Symbols .....                         | 8  |
| Landing Guidance .....                                   | 8  |
| Land as soon as practicable .....                        | 8  |
| Land as soon as possible .....                           | 8  |
| Airspeeds for Emergency Operations .....                 | 8  |
| Maneuvering Speed .....                                  | 8  |
| Emergency Landing .....                                  | 8  |
| Best Glide (Gear and Flaps: UP) .....                    | 9  |
| Glide .....                                              | 9  |
| Emergency Procedures .....                               | 10 |
| Automatic Flight Control Malfunction .....               | 10 |
| Engine Emergency On Takeoff - Speed Below VR .....       | 11 |
| Engine Fire On Takeoff - Low Altitude .....              | 12 |
| Engine Failure On Takeoff - Low Altitude .....           | 13 |
| Rejected Takeoff .....                                   | 14 |
| Engine Failure In Flight .....                           | 14 |
| Uncontained Engine Failure (Rotor Burst in Flight) ..... | 16 |
| Loss of All Flight Displays .....                        | 17 |
| Loss of Thrust Control .....                             | 17 |
| Loss of Reliable Airspeed Indication .....               | 18 |
| Abnormal Engine Start .....                              | 19 |
| Air Start .....                                          | 20 |
| Smoke Removal .....                                      | 21 |
| Cabin Fire In Flight .....                               | 22 |

---

|                                                   |    |
|---------------------------------------------------|----|
| Ditching.....                                     | 23 |
| Emergency Descent .....                           | 24 |
| Emergency Landing w/o Power .....                 | 25 |
| Flameout / Precautionary Approach .....           | 27 |
| Landing Without Aileron Control .....             | 28 |
| Landing Without Elevator Control .....            | 28 |
| Landing Without Rudder Control .....              | 28 |
| CAPS Activation .....                             | 29 |
| Emergency Ground Egress .....                     | 31 |
| Pitch Trim Runaway / Failure .....                | 31 |
| Roll Trim Runaway / Failure .....                 | 31 |
| Jammed Elevator Trim Tab .....                    | 32 |
| AFCS Alerts .....                                 | 32 |
| Emergency CAS Procedures.....                     | 33 |
| AUTO DESCENT Warning .....                        | 33 |
| CABIN ALTITUDE HIGH Warning .....                 | 33 |
| CABIN DIFF PRESSURE HIGH Warning .....            | 34 |
| CABIN DUMP Warning.....                           | 35 |
| CAPS ACTIVATED Warning .....                      | 36 |
| DOOR OPEN BAGGAGE Warning .....                   | 36 |
| DOOR OPEN MAIN Warning .....                      | 37 |
| EMER AUTOLAND ACTIVATING Warning.....             | 38 |
| EMER AUTOLAND BRAKES Warning.....                 | 39 |
| EMER BUS VOLTS Warning.....                       | 39 |
| ENGINE FIRE Warning - In Flight .....             | 43 |
| ENGINE FIRE Warning - On Ground .....             | 44 |
| ENGINE START N1 Warning.....                      | 45 |
| EXTERNAL POWER CONNECTED Warning - On Ground..... | 45 |
| FADEC CTRL DEGRADED Warning .....                 | 46 |
| FADEC NO DISPATCH Warning - On Ground .....       | 47 |
| FLAPS CAL MODE Warning - On Ground .....          | 47 |
| FLAPS ICE Warning.....                            | 47 |
| FUEL FILTER BYPASS Warning .....                  | 48 |
| FUEL IMBALANCE Warning .....                      | 48 |
| FUEL LOW LEFT/RIGHT/TOTAL Warning .....           | 49 |
| FUEL PRESSURE LOW Warning .....                   | 50 |
| FUEL TEMP LOW Warning.....                        | 51 |
| FUEL VALVES BOTH FAIL Warning .....               | 52 |
| GENERATOR 1 CURRENT Warning.....                  | 53 |
| GENERATOR 1 OVERVOLTAGE Warning .....             | 54 |
| GENERATOR 2 CURRENT Warning.....                  | 57 |
| GENERATOR 2 OVERVOLTAGE Warning .....             | 57 |



---

|                                                |    |
|------------------------------------------------|----|
| ITT Warning.....                               | 58 |
| LDG GEAR IS UP Warning.....                    | 59 |
| LDG GEAR SELECTED UP Warning .....             | 60 |
| N1 Warning .....                               | 60 |
| N2 Warning .....                               | 61 |
| OIL CHIP DETECT Warning - On Ground .....      | 61 |
| OIL FILTER BYPASS Warning - On Ground .....    | 61 |
| OIL PRESSURE Warning .....                     | 62 |
| OIL TEMPERATURE Warning.....                   | 62 |
| STALL WARNING .....                            | 63 |
| STICK PUSHER Warning .....                     | 63 |
| TAKEOFF PITCH TRIM Warning - On Ground.....    | 65 |
| THR LEVER CALIBRATION Warning - On Ground..... | 65 |
| TT2 OVERHEAT Warning.....                      | 65 |

Intentionally Left Blank

## **Introduction**

This section provides procedures for handling emergencies and critical flight situations that may occur while operating the aircraft. Although emergencies caused by airplane, systems, or engine malfunctions are rare, the guidelines described in this section should be considered and applied as necessary should an emergency arise.

Emergency procedures associated with optional equipment are not described in this section.

Although this section provides procedures for handling most emergencies and critical flight situations that could arise in the aircraft, it is not a substitute for proper flight training, thorough knowledge of the airplane, and recognized piloting techniques and standards. A thorough study of the information in this AFM while on the ground will help you prepare for time-critical situations in the air.

• NOTE •

Refer to [Section 9: Log of Supplements](#) for optional equipment Emergency Procedures.

## **Crew Alert System (CAS) Messaging**

### ***Warnings***

Displayed in red against a black background, Warning CAS messages arise during emergency situations that require immediate flight crew awareness and immediate flight crew response.

- A flashing Warning CAS message with an accompanying aural alert requires immediate action.
- A Warning CAS message with no accompanying aural alert requires attention, dependent on workload. It may also require performing maintenance or taking corrective action prior to next flight. Warnings with no aural alert typically occur while on ground.

## **Preflight Planning**

Enroute emergencies caused by weather can be minimized or eliminated by careful flight planning and good judgment when unexpected weather is encountered.

## Preflight Inspections/Maintenance

In-flight mechanical problems in the aircraft will be extremely rare if proper preflight inspections and maintenance are practiced. Always perform a thorough walk-around inspection before any flight to ensure that no damage occurred during the previous flight or while the airplane was on the ground. Pay special attention to any oil leaks or fuel stains that could indicate engine problems.

### • NOTE •

Refer to [Section 4: Normal Procedures, "Preflight Inspection"](#) for more information.

## Methodology

Aircraft emergencies are dynamic events. Because of this, it is impossible to enumerate every action a pilot should properly undertake in response to a particular situation. However, four basic actions can be applied to any emergency. They are:

### ***Maintain Aircraft Control***

#### • CAUTION •

Press and hold red Master Autopilot Disconnect Button (AP DISC) on the side stick if the aircraft and/or automatic flight control system are causing an unexpected or uncommanded aircraft response.

Many minor aircraft emergencies turn into major ones when the pilot fails to maintain aircraft control. Do not panic and do not fixate on a particular problem. Over-attention to a warning light during an instrument approach can lead to a pilot-induced unusual attitude, and possibly worse. To avoid this, even in an emergency: always aviate, navigate, and communicate, in that order. Never let anything interfere with your control of the airplane. Never stop flying.

### ***Analyze the Situation***

Once you are able to maintain control of the aircraft, assess the situation. Read all warning and caution messages. Evaluate the engine parameters. Consider all aircraft operational information at your disposal.

### ***Take Appropriate Action***

In many situations, the procedures listed in this section will either correct or mitigate the aircraft problem or allow safe recovery of the aircraft. Follow them and use good pilot judgment.

## ***Land as Soon as Conditions Permit***

Once you have evaluated and responded to the emergency, assess your next move. Perform any non-critical “clean-up” items in the checklist and land as soon as practicable. Even if the airplane appears to be in sound condition, it may not be.

• NOTE •

Refer to [Section 3: Emergency Procedures, "Landing Guidance"](#) for factors that determine landing criticality.

## **Circuit Breakers**

Some procedures involve manipulating circuit breakers (CBs). The following criteria should be followed during “Circuit Breaker” steps:

- Intentional pulling of circuit breakers during flight, other than as required in specific procedures, is prohibited.
- When instructed to “SET”, the appropriate circuit breaker should be checked for normal condition. If the circuit breaker is not “SET”, it may be reset only once. If the circuit breaker opens again, do not reset.
- When instructed to “PULL”, the appropriate circuit breaker should only be pulled and not reset.
- When instructed to “CYCLE”, the appropriate circuit breaker should be pulled, delayed for several seconds, and reset only once. Allow sufficient cooling time for circuit breakers that are reset through a “CYCLE” procedure.

## **Memory Items**

Checklist steps emphasized by a rectangular enclosure, such as the example below, should be memorized for accomplishment without reference to the procedure, due to the nature of their urgency.

|                                     |
|-------------------------------------|
| 1. Best Glide Speed ..... ESTABLISH |
|-------------------------------------|

If the rectangle is open at the top or bottom of the page, additional steps related to the memory item(s) precede or follow those listed on the current page.

## Procedure Division Symbols

For procedures requiring pilot decision, conditional steps are indented with a symbol to designate sub-sections within the procedure. On condition, the pilot makes a decision to identify the applicable sub-section. Following the initial decision, a further sub-division of the procedure may occur. In that event, one or more additional conditions guides the pilot through the remaining decisions. Once the applicable condition(s) are identified, the pilot follows the remaining steps until the indication “Procedure Complete” is reached.

The procedure symbol levels are:

- ◆ First Level
- Second Level
- Third Level

## Landing Guidance

### ***Land as soon as practicable***

Extended flight is not recommended and the pilot should proceed to land the plane, but not necessarily at the first or closest landing location. The landing location and duration of flight remains at the discretion of the pilot in command, after considering all factors related to the safety of the flight. For example, landing may be deferred to an airfield which could provide post-landing maintenance procedures or malfunction investigation. The choice of airfield may be determined by the pilot with consideration to weather, runway dimensions, surface conditions, and services available. However, in no case shall a suitable landing airfield be discounted in the name of convenience.

### ***Land as soon as possible***

Land without delay at the first site at which a safe landing can be made. Continued flight beyond an available airfield at which a safe landing can be made will result in increased risk and shall not be attempted. The nearest suitable landing site shall be determined by the pilot with consideration given to weather, surface dimensions and conditions, and pilot experience.

## **Airspeeds for Emergency Operations**

### **Maneuvering Speed**

6000 lb (2721 kg)..... 150 KIAS

### **Emergency Landing**

Flaps UP ..... 100 KIAS  
Flaps 50% ..... 95 KIAS  
Flaps 100% ..... 90 KIAS

## Best Glide (Gear and Flaps: UP)

|                         |          |
|-------------------------|----------|
| 6000 lb (2721 kg) ..... | 114 KIAS |
| 5000 lb (2268 kg) ..... | 104 KIAS |
| 4000 lb (1814 kg) ..... | 93 KIAS  |

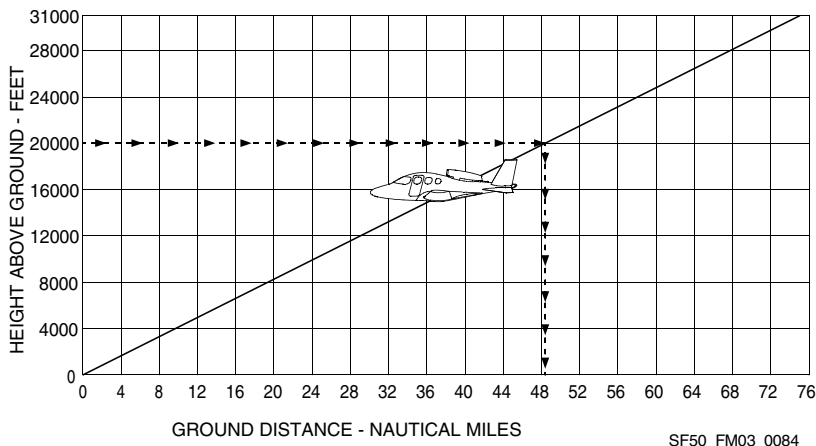
• NOTE •

In the clean configuration, the  $V_{REF}$  green donut indicates the speed for best glide.

## Glide

| Conditions |      | Example        |                |
|------------|------|----------------|----------------|
| Power      | OFF  | Altitude       | 20,000 ft. AGL |
| Flaps      | UP   | Glide Distance | 48.4 NM        |
| Wind       | Zero |                |                |
| Gear       | UP   |                |                |

**Figure 3-1:** Maximum Glide Ratio ~ 14.7: 1



## **Emergency Procedures**

### **Automatic Flight Control Malfunction (Stick Pusher, ESP, USP, Autopilot)**

1. Grip side stick firmly.
2. AP DISC Button ..... PRESS AND HOLD
3. Airplane Control ..... MAINTAIN
4. AP SERVOS CB (A3) ..... PULL
5. AP DISC Button ..... RELEASE

◆ If in RVSM airspace:

- a. Airplane Control ..... MAINTAIN
- b. Exit RVSM airspace.

• NOTE •

Refer to [Section 3A: Abnormal Procedures, "Autopilot Failure"](#).

6. Land as soon as practicable.

Procedure Complete

• NOTE •

If erroneous AOA indications are observed, green donut airspeed reference will be unavailable or unreliable. Low speed awareness indications on the airspeed tape should be considered unreliable and should not be referenced. Refer to [Section 3A: Abnormal Procedures, "STICK PUSHER FAIL Caution"](#).

Expect CAPS AUTOPILOT INOP CAS. Refer to [Section 3A: Abnormal Procedures, "CAPS AUTOPILOT INOP Advisory"](#).



## Engine Emergency On Takeoff - Speed Below $V_R$

• NOTE •

Includes emergencies such as engine failure or fire.

- |                      |                               |
|----------------------|-------------------------------|
| 1. Brakes .....      | MAX PILOT EFFORT W/O SKIDDING |
| 2. Thrust Lever..... | IDLE                          |
3. Engine Knob ..... OFF  
4. Engine Button ..... PRESS (MOMENTARY)

◆ If ENGINE FIRE Warning occurs:

- ENGINE FIRE ACKNOWLEDGE Switch.....LIFT GUARD, TOGGLE
- Left or Right PUSH TO DISCHARGE Button .....PUSH
- CABIN PRESSURE DUMP Button .....PRESS
- GEN 1 and GEN 2 Switches ..... OFF
- BAT 1 and BAT 2 Switches ..... OFF
- Evacuate airplane.

◆ If no indication of engine fire:

- Parking Brake.....DO NOT SET
- Exit aircraft normally when safe to do so.

After airplane comes to a complete stop:

5. Brakes ..... COOL DOWN

Procedure Complete

**(Continued on next page)**

**(Continued)**

**• CAUTION •**

For maximum brake effectiveness, retract flaps, hold side stick control full back, and bring the airplane to a stop by smooth, even application of the brakes.

A cool down period and brake overheat inspection are required after high-energy braking events.

Refer to [Section 2: Limitations, "Brakes"](#).

**• NOTE •**

Refer to [Section 4: Normal Procedures, "After Landing"](#).

## **Engine Fire On Takeoff - Low Altitude**

1. Land as soon as possible.

◆ If immediate landing is not possible:

- a. Perform [ENGINE FIRE Warning - In Flight](#) checklist.

Procedure Complete

**• WARNING •**

**If the engine fails immediately after becoming airborne, abort on the runway if possible. In most cases, when the engine fails below 500 feet AGL, the landing should be made straight ahead, turning only to avoid obstructions. In such a case, lower the nose to maintain flying airspeed and establish a glide attitude. After establishing a glide for landing, perform as many of the checklist items as time permits.**

**• NOTE •**

When determined that a forced landing is the safest option, a suitable landing site should be chosen as early as possible so that maximum time will be available to plan and execute the forced landing. While gliding, be aware that use of flaps will reduce glide distance. Flaps should not be selected until landing is assured. Land on the main gear and hold the nose wheel off the ground as long as possible.

## Engine Failure On Takeoff - Low Altitude

- |                                      |          |
|--------------------------------------|----------|
| 1. Best Glide or Landing Speed ..... | 100 KIAS |
| 2. Landing Gear .....                | A/R      |
| 3. Flaps .....                       | A/R      |
| 4. Land straight ahead.              |          |

◆ If time permits:

- a. Engine Knob..... OFF
- b. Engine Button ..... PRESS (MOMENTARY)
- c. CABIN PRESSURE DUMP Button ..... PRESS
- d. GEN 1 and GEN 2 Switches..... OFF
- e. BAT 1 and BAT 2 Switches ..... OFF

Procedure Complete

• WARNING •

**If engine failure is accompanied by fuel fumes in the cockpit, or if internal engine damage is suspected, lift the guard on the ENGINE FIRE ACKNOWLEDGE switch, toggle the switch, and shut down the engine. Do not attempt a restart.**

**If the engine fails immediately after becoming airborne, abort on the runway if possible. In most cases, when the engine fails below 500 feet AGL, the landing should be made straight ahead, turning only to avoid obstructions. In such a case, lower the nose to maintain flying airspeed and establish a glide attitude. After establishing a glide for landing, perform as many of the checklist items as time permits.**

• NOTE •

When determined that a forced landing is the safest option, a suitable landing site should be chosen as early as possible so that maximum time will be available to plan and execute the forced landing. While gliding, be aware that use of flaps will reduce glide distance. Flaps should not be selected until landing is assured. Land on the main gear and hold the nose wheel off the ground as long as possible.

## Rejected Takeoff

- |                      |                                |
|----------------------|--------------------------------|
| 1. Brakes.....       | MAX. PILOT EFFORT W/O SKIDDING |
| 2. Thrust Lever..... | IDLE                           |

After airplane comes to a complete stop:

3. Brakes.....COOL DOWN

Procedure Complete

### • CAUTION •

For maximum brake effectiveness, retract flaps, hold side stick control full back, and bring the airplane to a stop by smooth, even application of the brakes.

Do not set the parking brake following a Rejected Takeoff.

A cool down period and brake overheat inspection are required after high-energy braking events.

Refer to [Section 2: Limitations, Brakes](#).

### • NOTE •

Use as much of the remaining runway as needed to safely bring airplane to a stop or to slow airplane sufficiently to turn off runway.

Refer to [Section 4: Normal Procedures, After Landing](#).

## Engine Failure In Flight

- |                                          |                         |
|------------------------------------------|-------------------------|
| 1. Pitch for air start.....              | AIRSPEED 120 - 200 KIAS |
| 2. Turn toward possible landing site(s). |                         |
| 3. Oxygen Mask .....                     | DON, 100%               |
4. MIC SELECT Switch ..... MASK MIC

### • NOTE •

Engine can be considered safe to attempt restart if no abnormal, loud noises or vibrations were noted prior to or during shutdown, N1 rotation is observed, and no exceedances were noted prior to shutdown.

(Continued on next page)

**(Continued)**

- ◆ If engine appears safe to attempt a restart:

- a. Perform **Air Start** checklist.

Procedure Complete

- ◆ If engine is unsafe to restart:

- a. Engine Knob..... OFF
  - b. Engine Button .....PRESS (MOMENTARY)
  - c. Perform **Emergency Landing w/o Power** checklist.

Procedure Complete

- ◆ If engine failure is accompanied by fuel fumes in the cockpit, or if internal engine damage is suspected:

- a. ENGINE FIRE ACKNOWLEDGE Switch.....LIFT GUARD, TOGGLE
  - b. Shut down engine. Do not attempt a restart.
  - c. Perform **Emergency Landing w/o Power** checklist.

Procedure Complete

**• WARNING •**

**If all attempts to restart the engine fail and a forced landing is imminent, select a suitable landing site and prepare for landing. As forced landings on unimproved surfaces are not recommended, CAPS activation may be the safest option.**

**Refer to Section 3: Emergency Procedures, "CAPS Activation" checklist.**

**Refer to Section 10: Safety Information, "Cirrus Airframe Parachute System (CAPS)" for CAPS deployment scenarios and landing considerations.**

**Refer to Section 2: Limitations, "Oxygen" for oxygen requirements.**

## Uncontained Engine Failure (Rotor Burst in Flight)

1. Pitch for 120 KIAS (APPROX.)
2. Turn toward possible landing site(s).
3. Oxygen Mask ..... DON, 100%
4. MIC SELECT Switch ..... MASK MIC
5. ENGINE FIRE ACKNOWLEDGE Switch ..... LIFT GUARD, TOGGLE
6. Left or Right PUSH TO DISCHARGE Button ....PUSH (REGARDLESS OF ARMED LIGHT)
7. Engine Knob .....OFF
8. Engine Button.....PRESS (MOMENTARY)
9. GEN 1 and GEN 2 Switches.....OFF
10. Perform [Emergency Landing w/o Power](#) checklist.

Procedure Complete

• **NOTE** •

This condition is identified by complete loss of thrust accompanied by the following:

1. Loud bangs, knocks, vibration, or other noise, or
2. Engine Fire Indication (CAS Warning and/or FIRE light), or
3. Fire Detector Fail (CAS Caution), or
4. Erratic engine indications, such as the following:
  - Indication failure/loss of indication
  - Low or no N1 or N2
  - Low fuel pressure
  - Low or no oil pressure
  - High ITT, fuel flow, or oil temperature

• **NOTE** •

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## Loss of All Flight Displays

1. BAT 1 and BAT 2 Switches ..... VERIFY ON
2. GEN 1 and GEN 2 Switches ..... VERIFY ON
3. Land as soon as possible.

Procedure Complete

• NOTE •

BAT 2 is designed to power the Emergency Bus for at least 60 minutes.

FADEC NO DISPATCH faults will be displayed once normal power is restored. Faults can be cleared using the FADEC fault reset button.

FUEL VALVE #2 FAIL Caution will be displayed during normal tank AUTO selection since the secondary valve is not powered by emergency power.

## Loss of Thrust Control

1. Shut down engine, if required.
2. Flaps ..... A/R
3. Landing Gear ..... A/R
4. Land as soon as practicable.

◆ If the thrust lever is stuck at, or near, the T/O detent:

- a. Turn toward nearest practical landing site.
- b. Engine Knob ..... OFF
- c. Engine Button ..... PRESS (MOMENTARY)
- d. Fly a [Flameout / Precautionary Approach](#).
- e. If necessary, consider [CAPS Activation](#).

Procedure Complete

**(Continued on next page)**

**(Continued)**

◆ If the thrust lever is stuck at, or near, the IDLE position, and straight and level flight cannot be maintained:

- a. Pitch for glide speed.
- b. Turn toward nearest practical landing site.
- c. Fly a [Flameout / Precautionary Approach](#).
- d. If necessary, consider [CAPS Activation](#).

Procedure Complete

**• NOTE •**

Refer to [Section 3: Emergency Procedures, "Flameout / Precautionary Approach"](#).

**Loss of Reliable Airspeed Indication**

1. Autopilot ..... DISENGAGE
2. AOA Display ..... ON
3. Use following guidance pertaining to phase of flight.

◆ Climb:

- a. AOA ..... 0.25

**• NOTE •**

Airspeed is approximately 160 KIAS.

◆ Cruise:

- a. Power ..... SET N1% PER  
[Section 5: Performance Data, "Cruise Performance"](#)

◆ Descent:

- a. AOA ..... 0.25

**• NOTE •**

Airspeed is approximately 160 KIAS.

**(Continued on next page)**



**(Continued)**

◆ Approach:

○ Gear UP:

(1) AOA .....0.35

• NOTE •

Airspeed is approximately 120 KIAS.

○ Gear DOWN, Flaps DOWN:

(1) AOA .....0.53

• NOTE •

Refer to  $V_{REF}$  Speeds table.

◆ Go Around:

a. AOA .....0.53

• NOTE •

Refer to  $V_{REF}$  Speeds table.

4. Land as soon as practicable.

Procedure Complete

## Abnormal Engine Start

1. Engine Knob ..... OFF

2. Engine Button ..... PRESS AND HOLD 15 - 30 SECONDS

3. See [Section 2: Limitations, "Powerplant Limitations"](#) and [Section 2: Limitations, "Outside Air Temperature Limits"](#) in for corrective actions.

Procedure Complete

• NOTE •

An Abnormal Engine Start occurs when:

- there is a failure to accelerate after 30 seconds, or
- there is no N1 indication at 25% N2, or
- there is low oil pressure after start, or
- ITT exceeds limits (red line).

## Air Start

1. Airspeed..... 120 - 200 KIAS
2. Altitude..... < 18,000 FT
3. Thrust Lever..... IDLE

Engine reset:

4. Engine Knob ..... OFF
5. Engine Button ..... PRESS (MOMENTARY)
6. N2 ..... WAIT UNTIL < 24%

◆ If time permits:

- a. N2..... WAIT UNTIL < 16%

Engine start:

7. Engine Knob ..... RUN
8. Engine Button ..... PRESS (MOMENTARY)

◆ If air start is successful:

- a. GEN 1 and GEN 2 Switches ..... VERIFY ON
- b. Engine Instruments ..... MONITOR
- c. Land as soon as practicable.

Procedure Complete

◆ If air start is unsuccessful:

- a. Engine Knob ..... OFF
- b. Engine Button..... PRESS (MOMENTARY)
- c. GEN 1 and GEN 2 Switches ..... OFF
- d. Perform [Emergency Landing w/o Power](#) checklist.

Procedure Complete

### • CAUTION •

Probability of engine restart is greater when N2 < 16%.

Do not attempt a restart until N2 < 24%.

See [Section 2: Limitations](#), "Engine Start (In flight)" for Engine Air Start envelope.

## Smoke Removal

1. Oxygen Mask..... DON, EMER
2. Smoke Goggles (if equipped) ..... DON
3. PAX OXY DEPLOY Switch..... ON

4. MIC SELECT Switch .....MASK MIC
5. Instruct passenger(s) to don mask(s).
6. DEFOG Button ..... ON
7. CABIN PRESSURE DUMP Button..... PRESS

**• NOTE •**

Autopilot Emergency Descent Mode (EDM) is disabled when the cabin dump switch is activated.

8. Perform **Emergency Descent** checklist as required.
9. BLEED Switch (below 14,000 feet) ..... FRESH
10. ECS DISABLE ..... SELECT
11. FIRE EXTINGUISHER..... A/R
12. All Other Switches (**except BAT 2**) ..... OFF

◆ If setting switches off eliminates fire / fumes source and airplane is in night or IFR conditions:

- a. BAT 1 Switch..... ON
- b. GEN 1 and GEN 2 Switches ..... A/R
- c. Required Systems..... ACTIVATE ONE AT A TIME
- d. Temperature Select..... COLD
- e. Panel Air Outlets ..... OPEN
- f. Land as soon as possible.
- g. Oxygen Masks ..... ADJUST TO NORM IF ABLE AND BELOW 20,000 FEET

Procedure Complete

13. Land as soon as possible.
14. Oxygen Masks ..... ADJUST TO NORM IF ABLE AND BELOW 20,000 FEET

Procedure Complete

**(Continued on next page)**

**(Continued)**

**• WARNING •**

**Regardless of smoke evidence, if there is no visual confirmation that fire has been extinguished following fire suppression and/or smoke evacuation, land as soon as possible.**

**The fire extinguisher should only be used in 2-second blasts, and additional shots should only be used if necessary to extinguish the fire. Inhalation of the complete discharge of the fire extinguisher may result in cardiotoxicity and/or anesthetic effects to the occupants. Occupants shall use emergency oxygen masks during and after fire extinguisher use.**

**• NOTE •**

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## **Cabin Fire In Flight**

- |                                      |           |
|--------------------------------------|-----------|
| 1. Oxygen Mask .....                 | DON, EMER |
| 2. Smoke Goggles (if equipped) ..... | DON       |
| 3. PAX OXY DEPLOY Switch .....       | ON        |
- 
- |                                   |                                              |
|-----------------------------------|----------------------------------------------|
| 4. MIC SELECT Switch .....        | MASK MIC                                     |
| 5. BAT 1 Switch .....             | OFF                                          |
| 6. GEN 1 and GEN 2 Switches ..... | OFF                                          |
| 7. Fire Extinguisher .....        | A/R                                          |
| 8. Oxygen Mask .....              | ADJUST TO NORM IF ABLE AND BELOW 20,000 FEET |

◆ If airflow is not sufficient to clear smoke or fumes:

- Perform [Smoke Removal](#) checklist.

Procedure Complete

**(Continued on next page)**

**(Continued)**

**• WARNING •**

**The fire extinguisher should only be used in 2-second blasts, and additional shots should only be used if necessary to extinguish the fire. Inhalation of the complete discharge of the fire extinguisher may result in cardiotoxicity and/or anesthetic effects to the occupants. Occupants shall use emergency oxygen masks during and after fire extinguisher use.**

**• NOTE •**

With BAT and GEN switches OFF, engine will continue to run.  
However, no electrical power will be available.

If the airplane is in IMC, turn GEN 1, GEN 2, and BAT 1 switches OFF. Power from Battery 2 will power all the loads on the Emergency Bus for at least 60 minutes. If airplane is in day VFR conditions and turning off the GEN switches eliminated the fire situation, leave the GEN switches OFF.

If the cause of the fire is readily apparent and accessible, use the fire extinguisher to extinguish flames and land as soon as possible.

If required to re-activate systems, pause several seconds between activating each system to isolate malfunctioning system. Continue flight to earliest possible landing with malfunctioning system OFF. Activate only the minimum amount of equipment necessary to complete a safe landing.

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## Ditching

1. Radio..... TRANSMIT (ATC OR 121.5 MHZ) MAYDAY WITH LOCATION & INTENTIONS
2. Transponder .....SQUAWK 7700
3. ELT..... ON
4. Landing Gear ..... UP
5. Flaps..... 100% (RECOMMENDED)
6. CABIN PRESSURE DUMP Button.....PRESS
7. Airspeed ..... $V_{REF}$
8. Rate of Descent..... 200 - 300 FEET PER MINUTE

**(Continued on next page)**

**(Continued)**

9. Align approach parallel to swells and attempt to touchdown along a wave crest or behind it.
  - ◆ If surface winds are strong, or waves rough and irregular:
    - a. Ditch into the wind on the backside of a wave.
10. Pitch attitude at water contact.....SLIGHTLY HIGHER THAN NORMAL LANDING ATTITUDE
11. Airspeed & Rate of Descent..... REDUCE TO MINIMUM, ABOVE STALL WARNING
12. Thrust Lever ..... IDLE
13. Water Touchdown .... CONTACT ON CREST OF SWELL, PARALLEL TO MAJOR SWELL
14. Airplane .....EVACUATE

**• WARNING •**

**Evacuate via either egress window or main cabin door. However, main cabin door evacuation shall be done only via the upper cabin door. To avoid flooding the airplane, do not open lower cabin door.**

15. Flotation Devices (if equipped) ..... INFLATE WHEN CLEAR OF AIRPLANE

Procedure Complete

**• NOTE •**

If available, life preservers should be donned and the life raft prepared for immediate evacuation upon touchdown.

## **Emergency Descent**

1. AP DISC .....PRESS AND RELEASE
2. Thrust Lever ..... IDLE
3. Pitch ..... -20° (APPROX.)
4. Airspeed.....INCREASE TO  $V_{MO}$  (250 KIAS) /  $M_{MO}$  (0.53 Mach) MAXIMUM

Procedure Complete

**• CAUTION •**

If significant turbulence is expected, do not descend at indicated airspeeds greater than  $V_O$ .

## Emergency Landing w/o Power

1. Airplane Control .....MAINTAIN
2. Pitch for glide speed.
3. Turn toward nearest practical landing site.
4. Radio..... TRANSMIT (ATC OR 121.5 MHZ) MAYDAY WITH LOCATION & INTENTIONS
5. Transponder .....SQUAWK 7700

◆ If landing site is improved:

- a. Landing Gear .....DOWN
- b. Flaps.....A/R
- c. Seat Belt(s) .....SECURED
- d. CABIN PRESSURE DUMP Button .....PRESS
- e. Touchdown .....NORMAL TECHNIQUE

### • WARNING •

Refer to [Section 3: Emergency Procedures, "Flameout / Precautionary Approach" checklist.](#)

**If a safe landing is not assured by the High-Key/Low-Key approach, consider CAPS procedures.**

After landing:

- f. BAT 1 and BAT 2 Switches .....OFF
- g. Evacuate airplane.

Procedure Complete

**(Continued on next page)**

**(Continued)**

◆ If landing site is unimproved:

- a. Landing Gear ..... A/R
- b. Flaps ..... A/R WITHIN GLIDING DISTANCE
- c. Seat Belt(s) ..... SECURED
- d. CABIN PRESSURE DUMP Button ..... PRESS
- e. Touchdown ..... MINIMUM CONTROLLABLE AIRSPEED

After landing:

- f. BAT 1 and BAT 2 Switches ..... OFF
- g. Evacuate airplane.

Procedure Complete

**• WARNING •**

**If all attempts to restart the engine fail and a forced landing is imminent, select a suitable landing site and prepare for landing. As forced landings on unimproved surfaces are not recommended, CAPS activation may be the safest option.**

**Refer to [Section 3: Emergency Procedures, "CAPS Activation"](#) checklist.**

**Refer to [Section 10: Safety Information, "Cirrus Airframe Parachute System \(CAPS\)"](#) for CAPS deployment scenarios and landing considerations.**

**When determined that a forced landing is the safest option, a suitable landing site should be chosen as early as possible so that maximum time will be available to plan and execute the forced landing. While gliding, be aware that use of flaps will reduce glide distance. Flaps should not be selected until landing is assured. Land on the main gear and hold the nose wheel off the ground as long as possible.**



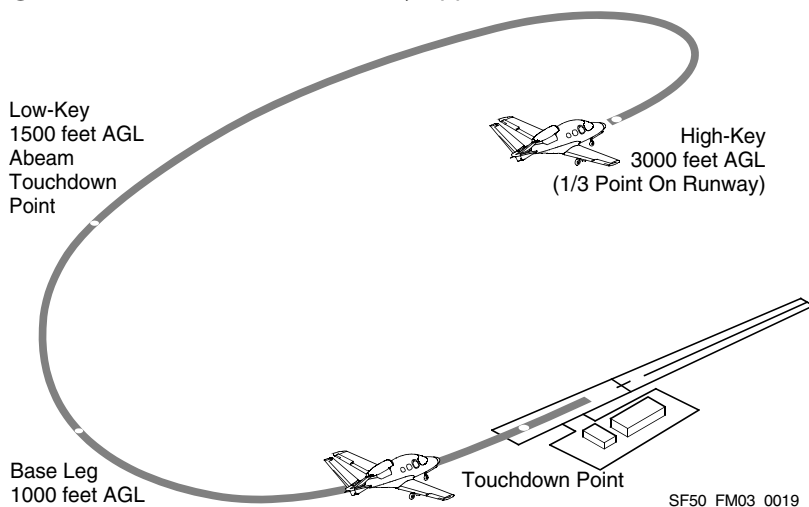
## Flameout / Precautionary Approach

| POSITION | ALTITUDE    | AIRSPEED     |
|----------|-------------|--------------|
| High-Key | 3000 ft AGL | 115-130 KIAS |
| Low-Key  | 1500 ft AGL | 120 KIAS     |
| Base Leg | 1000 ft AGL | 120 KIAS     |

• NOTE •

Gear down and flaps when landing is assured.  
Airspeed 110 KIAS (minimum) until short final.

**Figure 3-2:** Flameout / Precautionary Approach



## Landing Without Aileron Control

1. Flaps ..... 50%
2. Trim .....  $V_{REF}$
3. Thrust Lever ..... A/R
4. Use rudder as necessary to align the aircraft with the runway.

Procedure Complete

## Landing Without Elevator Control

1. Use manual trim wheel to control pitch attitude.
2. Flaps ..... 50%
3. Make small pitch and power changes and prepare for landing configuration early.

After touchdown and after the nose wheel is on the ground:

4. Brakes ..... A/R

Procedure Complete

## Landing Without Rudder Control

### • NOTE •

Avoid crosswind conditions if possible.

1. Touchdown ..... ACCOMPLISH
2. Maintain directional control on ground, using brakes and ailerons as required.

Procedure Complete

## **CAPS Activation**

### **• WARNING •**

The recommended CAPS activation speed is one that is less than 135 KIAS and 145 KTAS, whichever is lower, which will result in immediate parachute deployment once the CAPS handle is pulled. It is important for the operator to make all reasonable efforts to slow to the minimum possible airspeed prior to CAPS activation. This is necessary to minimize loads on the airframe and aircraft occupants and increase the chances of a successful outcome.

If CAPS is activated at speeds higher than 135 KIAS and 145 KTAS, whichever is lower, the system is designed to delay deployment of the parachute until the autopilot slows the airplane to the parachute deployment envelope. Do not attempt to overpower or disconnect the autopilot. The time required for the autopilot to slow the airplane from 250 KIAS to the parachute deployment airspeed has been demonstrated to be approximately 15-24 seconds. Flying at a slower airspeed when CAPS is activated will reduce this time.

If the airspeed is still above the maximum deployment speed approximately 30 seconds after CAPS activation, the system will automatically deploy the parachute.

CAPS deployment will likely result in damage or loss to the airframe and possible injury to the aircraft occupants. Seat belts must be worn at all times.

Jerking or rapidly pulling the CAPS activation handle will greatly increase the pull forces required to activate the rocket. Use a firm and steady pulling motion – a “chin-up” type pull ensures successful activation.

- |                                                            |
|------------------------------------------------------------|
| 1. Activation Handle....PULL DOWN STEADILY WITH BOTH HANDS |
|------------------------------------------------------------|

### **• NOTE •**

Wait for plane to stabilize beneath canopy before proceeding.

2. Oxygen Mask (above 15,000 feet)..... DON, 100%
3. PAX OXY DEPLOY Switch..... ON
4. MIC SELECT Switch .....MASK MIC
5. Instruct passengers to don oxygen masks.

**(Continued on next page)**

**(Continued)**

6. Passenger Oxygen ..... ENSURE PASSENGERS ARE RECEIVING OXYGEN
7. ENGINE FIRE ACKNOWLEDGE Switch ..... LIFT GUARD, TOGGLE
8. ELT ..... VERIFY ON
9. Landing Gear ..... UP

Turn the BAT/GEN Switches OFF after completing any necessary radio communications.

10. BAT/GEN Switches (below 15,000 ft) ..... OFF
11. Loose Items ..... SECURE
12. Seat Belts ..... TIGHTEN
13. Assume emergency landing body position.

After airplane comes to a complete stop:

14. Evacuate quickly and move upwind, well clear of the aircraft and parachute.

Procedure Complete

**• NOTE •**

In the event the cabin remains pressurized at touchdown, the egress hatch may not be removable.

After touchdown with the landing gear up, the lower half of the clamshell cabin door will not completely open with the belly of the aircraft resting on the ground.

Refer to [Section 2: Limitations](#), "Oxygen" for oxygen requirements.

## Emergency Ground Egress

1. Engine Knob ..... OFF
2. Engine Button..... PRESS (MOMENTARY)
3. BAT 1 and BAT 2 Switches ..... OFF
4. Parking Brake .....SET
5. Egress aircraft through exits.

Procedure Complete

### • WARNING •

**While exiting the airplane, make sure evacuation path is clear of aircraft and other hazards.**

### • NOTE •

In instances when the engine cannot be shut off, and has to be left running, set the parking brake prior to evacuating the airplane.

## Pitch Trim Runaway / Failure

Red PTRM annunciator on PFD.

1. Grip side stick firmly.
2. AP DISC Button..... PRESS AND HOLD
3. Pitch Trim ..... A/R (MANUAL TRIM WHEEL)
4. PITCH TRIM CB (A1) ..... PULL
5. AP DISC Button..... RELEASE

Procedure Complete

## Roll Trim Runaway / Failure

1. Grip side stick firmly.
2. AP DISC Button..... PRESS AND HOLD
3. Airspeed ..... REDUCE (A/R TO MINIMIZE FORCE)
4. ROLL TRIM CB (A2) ..... PULL
5. AP DISC Button..... RELEASE

Procedure Complete

## Jammed Elevator Trim Tab

### ◆ During Cruise:

- a. Thrust Lever..... A/R
- b. Landing Gear ..... A/R
- c. Airspeed .....MAINTAIN LAST TRIMMED AIRSPEED OR A/R
- d. Flaps ..... UP OR A/R FOR CONDITIONS
- e. Land as soon as practicable.

Procedure Complete

### ◆ During Takeoff or Go-Around:

- a. Pitch and Thrust ...MAINTAIN ALTITUDE AND FLYING SPEED  
AS REQUIRED

#### • NOTE •

Expect extremely high stick forces.

When safely airborne and clear of obstacles:

- b. Flaps .....PLACE IN POSITION THAT JAM OCCURRED

#### • NOTE •

100% flaps may alleviate excessive nose-up tendencies following a normal takeoff.

- c. Landing Gear ..... DO NOT RETRACT
- d. Land as soon as practicable.

Procedure Complete

## AFCS Alerts

For more information on AFCS alerts, refer to SF50 Garmin Cockpit Reference Guide.

## Emergency CAS Procedures

### AUTO DESCENT Warning

#### AUTO DESCENT

Aircraft descending to 14,000 ft.

• NOTE •

Aircraft will descend to 14,000 feet, level off for 4 minutes, then descend to 12,500 feet.

***Serials with Safe Return Autoland:*** Unless canceled, Safe Return Autoland will activate descending through 15,000 feet pressure altitude and change route to an emergency landing airport. Refer to [Section 10: Safety Information, "Safe Return Autoland System \(if installed\)"](#).

To cancel auto descent, disconnect the autopilot:

- a. AP DISC Button .....PRESS AND RELEASE

Procedure Complete

### CABIN ALTITUDE HIGH Warning

#### CABIN ALTITUDE HIGH

Cabin high altitude detected.

1. Oxygen Mask ..... DON, 100%

2. BLEED Switch ..... VERIFY BLEED

3. MIC SELECT Switch .....MASK MIC

4. Perform [Emergency Descent](#) checklist as required.

5. Passenger Oxygen ..... ENSURE PASSENGERS ARE RECEIVING OXYGEN

◆ If passenger oxygen masks do not deploy:

- a. PAX OXY DEPLOY Switch ..... ON

6. Transponder .....SQUAWK 7700

7. CABIN PRESSURE DUMP Button.....PRESS

Procedure Complete

• NOTE •

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

(Continued on next page)

**(Continued)**

**• NOTE •**

Refer to Section 2: Limitations, "Oxygen" for oxygen requirements.

If the CABIN PRESSURE DUMP is not activated, Emergency Descent Mode (EDM) will activate if the aircraft is at or above 14,900 feet pressure altitude and the cabin altitude is greater than or equal to 15,000 feet. An EDM activation can lead to an automatic activation of Safe Return Autoland. EDM and Safe Return Autoland will not automatically activate if CABIN PRESSURE DUMP is activated. Manual activation of Safe Return Autoland can still be accomplished following CABIN PRESSURE DUMP activation.

## CABIN DIFF PRESSURE HIGH Warning

### CABIN DIFF PRESSURE HIGH

Cabin differential pressure is too high.

1. Oxygen Mask ..... DON, 100%
2. BLEED Switch.....FRESH
3. MIC SELECT Switch ..... MASK MIC
4. Perform [Emergency Descent](#) checklist as required.

◆ If message persists:

- a. Thrust Lever..... REDUCE
- b. CABIN PRESSURE DUMP Button.....PRESS

Procedure Complete

**• NOTE •**

Autopilot Emergency Descent Mode (EDM) is disabled when the cabin dump switch is activated.

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.



## CABIN DUMP Warning

### CABIN DUMP

Cabin pressure dump activated.

• *NOTE* •

If the CABIN PRESSURE DUMP Button was inadvertently activated, it should be deactivated immediately.

Autopilot Emergency Descent Mode (EDM) is disabled when the cabin dump switch is activated.

◆ If message is accompanied by CABIN ALTITUDE HIGH Warning:

- a. Perform [CABIN ALTITUDE HIGH Warning](#) checklist.

Procedure Complete

## CAPS ACTIVATED Warning

### CAPS ACTIVATED

CAPS sequence has been activated.

◆ If message is displayed due to CAPS activation (pulling of CAPS Activation Handle):

1. Refer to [CAPS Activation](#) procedure.

Procedure Complete

#### • WARNING •

**Do not press and hold AP DISC Button if CAPS handle has been pulled; if handle has been pulled, parachute will deploy regardless and pressing AP DISC will prevent the aircraft from adequately slowing prior to deployment.**

◆ If message is displayed but CAPS handle is NOT pulled:

- a. AP DISC Button.....PRESS AND HOLD 5 SECONDS

Procedure Complete

#### • WARNING •

**CAPS AP mode will be unavailable for remainder of flight until next power cycle. If CAPS activation is necessary, the pilot must manually reduce airspeed below 135 KIAS and 145 KTAS, whichever is lower, to deploy the parachute.**

Refer to [Section 3: Emergency Procedures, "CAPS Activation"](#) procedure.

## DOOR OPEN BAGGAGE Warning

### DOOR OPEN BAGGAGE

Baggage door is open.

◆ If in flight:

- a. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Verify baggage door is closed.

Procedure Complete

## DOOR OPEN MAIN Warning

### DOOR OPEN MAIN

Main door is open.

◆ If in flight:

- a. Airplane Control.....MAINTAIN
- b. Airspeed ..... LIMIT TO 150 KIAS
- c. Altitude ..... AS LOW AS PRACTICAL
- d. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Verify main door is closed.

Procedure Complete

• WARNING •

**Stay clear of door if malfunction is believed to have occurred. Ensure seat belts are tightened and secured in order to avoid extraction from aircraft in the event of rapid door opening/decompression.**

**Maneuver with care, avoiding unnecessary accelerations and abrupt maneuvers.**

• NOTE •

If open door is discovered on takeoff roll, abort takeoff if sufficient runway remains. If already airborne, do not allow efforts to close the door interfere with the primary task of maintaining control of the airplane.

## EMER AUTOLAND ACTIVATING Warning

### EMER AUTOLAND ACTIVATING

Emergency Autoland is activating.

◆ If Safe Return Autoland activation is not desired:

- a. AP DISC Button.....PRESS AND HOLD 1 SECOND

○ If Safe Return Autoland activation has proceeded beyond 10-second EMER AUTOLAND ACTIVATING period:

• NOTE •

Safe Return Autoland will immediately squawk 7700, change the flight plan, and may change the altimeter setting.

- (1) Transponder..... SET  
(2) FMS ..... A/R  
(3) Altimeter ..... SET

□ If system has deployed flaps:

- (a) Flaps ..... 50%  
(b) Flaps ..... A/R

□ If system has deployed landing gear:

- (a) Landing Gear .....DOWN  
(b) Landing Gear ..... A/R

Procedure Complete

○ If malfunction of Safe Return Autoland activation is suspected:

- (1) EMER AUTOLAND CB (E11) ..... PULL  
(2) Perform Automatic Flight Control Malfunction (Stick Pusher, ESP, USP, Autopilot) checklist.

Procedure Complete

◆ If Safe Return Autoland activation is desired:

- a. No further action is required.

• NOTE •

Refer to [Section 8: Handling and Servicing, "Safe Return Autoland Runway Recovery"](#).

Procedure Complete

## EMER AUTOLAND BRAKES Warning

### EMER AUTOLAND BRAKES

Do not dispatch, autoland brakes in test mode.

1. Do not dispatch.

Procedure Complete

## EMER BUS VOLTS Warning

### EMER BUS VOLTS

Operating on emergency power.

◆ If GENERATOR 1 CURRENT OR GENERATOR 2 CURRENT CAS is also displayed:

- a. Perform GENERATOR 1 CURRENT OR GENERATOR 2 CURRENT procedure prior to performing remainder of EMER BUS VOLTS procedure.

◆ If GEN 1 and GEN 2 are not within limits:

- a. GEN 1 Switch ..... OFF
- b. GEN 2 Switch ..... OFF
- c. Interior Lights ..... OFF
- d. BAT 1 Switch..... OFF
- e. BAT 2 Switch..... VERIFY ON

• NOTE •

With GEN 1, GEN 2, and BAT 1 OFF, the autopilot and the displays on the MFD and GTC 3 will be unavailable.

- f. Land as soon as possible.

Procedure Complete

**(Continued on next page)**

**(Continued)**

**• NOTE •**

BAT 2 is designed to power the Emergency Bus for at least 60 minutes.

FADEC NO DISPATCH faults will be displayed once normal power is restored. Faults can be cleared using the FADEC fault reset button.

FUEL VALVE #2 FAIL Caution will be displayed during normal tank AUTO selection since the secondary valve is not powered by emergency power.

**• NOTE •**

With Battery 1, Generator 1, and Generator 2 switched off, power will be removed from the entire main bus, including the hydraulic powerpack. The following equipment will also be inoperative:

- Stick Pusher (Perform [STICK PUSHER FAIL Caution](#) checklist)
- Landing gear control (To lower landing gear, perform [Alternate Gear Extension](#) checklist)
- Flaps control (Perform [Flaps Inoperative Approach/Landing](#) checklist)
- Cabin pressure control (Perform [CABIN PRESSURE CTRL FAIL Caution](#) checklist)
- Engine start/shutdown control (To shutdown engine, perform [Abnormal Engine Shutdown On Ground](#) checklist)
- Yaw SAS (Perform YAW SAS SERVO Caution checklist)
- Bleed air shutoff and ECS (such as AC Compressor and Fans)
- Redundant equipment (such as Fuel Valve 2, AHRS 2 / ADC 2, Right Pitot Heat, COM 2, and GPS 2)
- Non-essential equipment (such as Electric Fuel Pump, Stick Shaker, Avionics Fan, Anti-collision Lights, Landing Lights, IFE USB, Transponder1, Navigation Lights, Fuel Level, Oil Pressure, Intercom, and Passenger Oxygen)
- Optional equipment (such as Traffic/TCAS, Transponder2, Radar, EVS, ADS-B/UAT, 115VAC PWR, Iridium WIFI Data, DME, XM Data, and Yaw Damper)
- Autopilot Servos, Roll Trim, and electric Pitch Trim
- Ice Protection (Engine, Windshield, WING/STAB, Ice Light, and AOA Heat)

## Emergency Bus Circuit Breakers

| CIRCUIT BREAKER                              | CIRCUIT                                     |
|----------------------------------------------|---------------------------------------------|
| <b>BATT1 HOT BUS</b>                         |                                             |
| CONV LIGHTS                                  | Interior Lights Controller                  |
| CAPS A                                       | Primary CAPS                                |
| <b>COPILOT CIRCUIT BREAKER PANEL (ROW D)</b> |                                             |
| FUEL VALVE 1 (D5)                            | Fuel Control Valve 1                        |
| LANDING GEAR IND (D6)                        | L MLG WOW Sensor                            |
|                                              | R MLG WOW Sensor                            |
|                                              | L MLG Down Sensor                           |
|                                              | L MLG Locked Sensor                         |
|                                              | R MLG Down Sensor                           |
|                                              | R MLG Locked Sensor                         |
|                                              | NLG Down Sensor                             |
|                                              | NLG Locked Sensor                           |
| STANDBY ATT (D2)                             | ADAHRS 3 (MD302)                            |
| GTC 1A (D3)                                  | GTC 1 (pwr A)                               |
| GEA (D4)                                     | RDM                                         |
|                                              | GEA 71                                      |
| CAPS B (D1)                                  | Secondary CAPS                              |
| <b>PILOT CIRCUIT BREAKER PANEL (ROW A)</b>   |                                             |
| CABIN PRESS 2 (A10)                          | Manual Dump SW & Controller<br>Solenoid Pwr |
| AP CTRL (A7)                                 | AP Controller (GMC 707)                     |
| FIRE PROT 1 (A9)                             | Fire Bottle 1                               |
|                                              | Fire Prot. CCA LED                          |
| CREW LIGHTS (A11)                            | Task Light, Copilot                         |
|                                              | Task Light, Pilot                           |
| PROBE HT LEFT (A8)                           | Left Pitot Heat                             |

| CIRCUIT BREAKER                     | CIRCUIT            |
|-------------------------------------|--------------------|
| PILOT CIRCUIT BREAKER PANEL (ROW B) |                    |
| FIRE PROT 2 (B9)                    | Fire Bottle 2      |
|                                     | Fuel Shutoff Valve |
| AHRS 1 (B10)                        | AHRS 1 (GRS79)     |
| ADC 1 (B11)                         | GDC 1              |
| GTC 2A (B5)                         | GTC 2 (pwr A)      |
| GPS NAV<br>GIA 1 (B6)               | GIA 1 MAIN         |
| PFD A (B8)                          | PFD (pwr A)        |
| COM 1 (B7)                          | GIA 1 COM          |



## ENGINE FIRE Warning - In Flight

### ENGINE FIRE

Engine fire detected.

1. Thrust Lever..... IDLE
2. Pitch for 120 KIAS (APPROX.)
3. Wait 15 seconds.

◆ If ENGINE FIRE Warning remains on, or secondary fire indications are present – probable engine fire:

- a. ENGINE FIRE ACKNOWLEDGE Switch.....LIFT GUARD, TOGGLE
- b. Left or Right PUSH TO DISCHARGE Button.....PUSH
- c. Oxygen Mask..... DON, 100%
- d. MIC SELECT Switch.....MASK MIC
- e. Engine Knob..... OFF
- f. Engine Button .....PRESS (MOMENTARY)
- g. GEN 1 and GEN 2 Switches..... OFF
- h. Perform [Emergency Landing w/o Power](#) checklist.

Procedure Complete

◆ If ENGINE FIRE Warning extinguishes and secondary fire indications are not present – probable bleed leak:

- a. Thrust Lever ..... REDUCE (MINIMUM REQUIRED)
- b. Land as soon as possible.

If time permits:

- c. BLEED Switch ..... FRESH
- d. Oxygen Masks ..... DON, 100%
- e. L / R MIC Switches.....MASK MIC
- f. Inlet Anti-Ice ..... OFF
- g. Exit icing conditions.
- h. Prepare for Emergency Descent, Loss of Cabin Pressure.

Procedure Complete

**(Continued on next page)**

**(Continued)**

**• NOTE •**

An actual engine fire may be accompanied by abnormal or failing engine indications, erratic or rough engine operation, visible flames, or smoke trail from aircraft.

Secondary fire indications include N1 / N2 fluctuation, generator failure, or erratic or failing engine indications.

A sustained fire may cause the structure to be compromised.

Structural failure may cause significant injury or loss of life.

Limiting speed, if possible, and avoiding high maneuvering loads may reduce the possibility of structural failure.

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## ENGINE FIRE Warning - On Ground

### ENGINE FIRE

Engine fire detected.

1. ENGINE FIRE ACKNOWLEDGE Switch ..... LIFT GUARD, TOGGLE
2. Left or Right PUSH TO DISCHARGE Button ..... PUSH
3. Engine Knob ..... OFF
4. Engine Button ..... PRESS (MOMENTARY)
5. Evacuate airplane.

Procedure Complete

**• NOTE •**

Engine Knob OFF inhibits Electric Fuel Pump.

## ENGINE START N1 Warning

### ENGINE START N1

N1 is not indicating rotation.

#### 1. ENGINE START N1 ..... ABORT START

◆ If in flight:

- a. Increase airspeed and retry start procedure.

Procedure Complete

◆ If on ground:

- b. Contact Cirrus for disposition.

Procedure Complete

## EXTERNAL POWER CONNECTED Warning - On Ground

### EXTERNAL POWER CONNECTED

External power is connected, set park brake.

1. Parking Brake or Brakes..... SET OR HOLD
2. Disconnect external power.

Procedure Complete

• *NOTE* •

Setting parking brake will extinguish CAS warning.

## FADEC CTRL DEGRADED Warning

### FADEC CTRL DEGRADED

Reference FADEC Fault Code Messages.

1. FADEC Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the FADEC Stale Fault Reset Button.

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

#### • WARNING •

**Possible inaccurate automatic power setting. The engine may exceed or go below allowable engine operating parameters which could ultimately result in loss of thrust. Avoid fast thrust lever movement or rapid thrust setting changes.**

○ If in flight:

(1) ENG. IPS Switch ..... ON

- (a) Allow sufficient time for ice to clear from inlet probes.

(2) FADEC Reset via MFD Controller:

- (a) From Home, touch Aircraft Systems.
- (b) Touch the FADEC Stale Fault Reset Button.

(3) Land as soon as possible.

Procedure Complete

○ If on ground:

(1) Do not dispatch.

Procedure Complete

## FADEC NO DISPATCH Warning - On Ground

### FADEC NO DISPATCH

Reference FADEC Fault Code Messages.

1. FADEC Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the FADEC Stale Fault Reset Button.

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

- a. Do not dispatch.

Procedure Complete

## FLAPS CAL MODE Warning - On Ground

### FLAPS CAL MODE

Do not dispatch, flaps system needs service.

1. Do not dispatch.

Procedure Complete

## FLAPS ICE Warning

### FLAPS ICE

Full flaps prohibited in icing conditions.

1. Flaps ..... SET UP OR 50%

Procedure Complete

• WARNING •

**Maximum flap deflection in icing conditions is limited to 50%.**

• NOTE •

If pilot can verify airframe is not ice contaminated, then IPS stall offset reset may be used to allow extension of flaps beyond 50%.

## FUEL FILTER BYPASS Warning

### FUEL FILTER BYPASS

Fuel filter diff press too high, bypass impending.

1. Thrust Lever.....REDUCE (MINIMUM REQUIRED)
2. Land as soon as possible.

Procedure Complete

#### • WARNING •

**Plan for possible imminent engine failure.**

**Refer to Section 3: Emergency Procedures, "Engine Failure In Flight" procedure for more information.**

## FUEL IMBALANCE Warning

### FUEL IMBALANCE

Fuel imbalance is  $\geq 50$  gallons.

#### • NOTE •

If no roll trim imbalance exists when autopilot is disconnected, fuel imbalance or fuel quantity indication may be erroneous.

1. Fuel Quantity/Fuel Remaining.....COMPARE
  - a. Refer to EIS for fuel quantity (based on level sensors).
2. Fuel Control Switch ..... FULLEST TANK
3. Airspeed..... MAINTAIN FOR ADEQUATE ROLL CONTROL

#### ◆ If message extinguishes or imbalance decreases:

- a. Continue flight.
- b. Fuel Control Switch ..... AUTO

Procedure Complete

#### ◆ If message persists or imbalance increases:

- a. Land as soon as possible.

Procedure Complete

## FUEL LOW LEFT/RIGHT/TOTAL Warning

| FUEL LOW LEFT  |
|----------------|
| FUEL LOW RIGHT |
| FUEL LOW TOTAL |

Left fuel tank is nearly empty, OR  
Right fuel tank is nearly empty, OR  
Total fuel quantity is low.

1. Fuel Quantity ..... CHECK
2. Fuel Control Switch ..... FULLEST TANK
3. Land as soon as possible.
4. Avoid excessive pitch attitudes and lateral acceleration to prevent uncovering the fuel inlet lines and possible fuel flow interruption.

Procedure Complete

### • NOTE •

Warning occurs when total fuel quantity  $\leq$  25 U.S. gallons (95 L),  
OR Left or Right fuel tank  $\leq$  5 U.S. gallons (19 L).

## FUEL PRESSURE LOW Warning

### FUEL PRESSURE LOW

FDU fuel inlet pressure is low.

1. Fuel Control Switch ..... SWITCH TO OTHER TANK
  - a. Refer to EIS for current tank selection (AUTO selection shown in white; LEFT/RIGHT selection shown in cyan).

◆ If message extinguishes:

- a. Continue flight.
- b. Fuel Quantity ..... MAINTAIN BALANCE

Procedure Complete

◆ If message persists:

- a. Land as soon as possible.

Procedure Complete

#### • NOTE •

Selecting LEFT/RIGHT will control electric pump to ON. Pump latch is removed when cycling to LEFT/RIGHT tank selection, then back to AUTO.

To maintain fuel balance, monitor fuel quantity and alternate LEFT/RIGHT selection as needed.



## FUEL TEMP LOW Warning

### FUEL TEMP LOW

Fuel temperature below operating limit.

1. Fuel Temperature (L / R) ..... CHECK

◆ If in flight:

- a. Climb or descend to warmer conditions.

• NOTE •

Change altitude to increase OAT and increase fuel temperature.  
Due to temperature inversion, warmer temperatures may be at a  
higher altitude.

- If message extinguishes:

(1) Fuel Temperature ..... MONITOR  
Procedure Complete

- If message persists:

(1) Land as soon as practicable.  
Procedure Complete

◆ If on ground:

- a. Warm aircraft to increase fuel temperature.  
Procedure Complete

## FUEL VALVES BOTH FAIL Warning

### FUEL VALVES BOTH FAIL

Both fuel control valves have failed.

1. Fuel Quantity ..... CHECK
2. Airspeed ..... MAINTAIN FOR ADEQUATE ROLL CONTROL
3. Land as soon as possible.

Procedure Complete

#### • NOTE •

Simultaneously failed valves will cause a fuel imbalance which will increase with continued flight.

If both fuel control valves fail, they will remain in the last commanded position.

## GENERATOR 1 CURRENT Warning

### GENERATOR 1 CURRENT

Generator 1 current is too high.

1. GEN 1 Switch ..... OFF
2. GCU Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the GCU Reset Button.
3. ECS DISABLE ..... SELECT
4. GEN 1 Switch ..... ON

◆ If message extinguishes:

- a. ECS DISABLE ..... DESELECT

Procedure Complete

◆ If message persists:

- a. GEN 1 Switch ..... OFF
- b. BAT 1 Switch..... OFF
- c. Land as soon as practicable, avoiding IMC or night flight as able (reduced power redundancy).

Procedure Complete

#### • NOTE •

With Battery 1 and Generator 1 switched off, power will be removed from the entire main bus, including the hydraulic powerpack. The following equipment will also be inoperative:

- Landing gear control (to lower landing gear, perform [Alternate Gear Extension](#) checklist)
- ECS (such as AC Compressor and Fans)
- Redundant equipment (such as Fuel Valve 2 and AHRS 2 / ADC 2)
- Non-essential equipment (such as Electric Fuel Pump, Stick Shaker, Yaw SAS, Avionics Fan, Nose Landing Lights, and IFE USB)
- Optional equipment (such as Traffic/TCAS, Transponder2, Radar, EVS, 115VAC PWR, and Iridium WIFI Data)

## GENERATOR 1 OVERVOLTAGE Warning

### GENERATOR 1 OVERVOLTAGE

Generator 1 voltage is high.

1. GEN 1 Switch.....OFF
2. GCU Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the GCU Reset Button.
3. ECS DISABLE..... SELECT
4. GEN 1 Switch..... ON

◆ If message extinguishes:

- a. ECS DISABLE .....DESELECT

Procedure Complete

◆ If message persists:

- a. GEN 1 Switch .....OFF
- b. BAT 1 Switch.....OFF
- c. Land as soon as practicable, avoiding IMC or night flight as able (reduced power redundancy).

Procedure Complete

#### • NOTE •

With Battery 1 and Generator 1 switched off, power will be removed from the entire main bus, including the hydraulic powerpack. The following equipment will also be inoperative:

- Landing gear control (to lower landing gear, perform [Alternate Gear Extension](#) checklist)
- ECS (such as AC Compressor and Fans)
- Redundant equipment (such as Fuel Valve 2 and AHRS 2 / ADC 2)
- Non-essential equipment (such as Electric Fuel Pump, Stick Shaker, Yaw SAS, Avionics Fan, Nose Landing Lights, and IFE USB)
- Optional equipment (such as Traffic/TCAS, Transponder2, Radar, EVS, 115VAC PWR, and Iridium WIFI Data)

## Main Bus Circuit Breakers

| CIRCUIT BREAKER                              | CIRCUIT                               |
|----------------------------------------------|---------------------------------------|
| <b>PILOT CIRCUIT BREAKER PANEL (ROW B)</b>   |                                       |
| BLEED AIR FLOW CTRL (B2)                     | Bleed Air Flow Control Valve          |
| STICK SHAKER (B3)                            | Stick Shaker Motor                    |
| LANDING GEAR HYD (B4)                        | Landing Gear Hydraulics Power Control |
| <b>PILOT CIRCUIT BREAKER PANEL (ROW C)</b>   |                                       |
| PFD B<br>GTC 1B (C8)                         | GTC-1 Secondary, PFD Secondary        |
| MFD B<br>GTC 2B (C9)                         | MFD-Secondary, GTC-2 Secondary        |
| FUEL PUMP CTRL (C10)                         | Fuel Control Panel                    |
| IGNITER 1 (C11)                              | Igniter 1, Ignition Control Panel     |
| <b>COPILOT CIRCUIT BREAKER PANEL (ROW F)</b> |                                       |
| YAW SAS (F8)                                 | Yaw Servo                             |
| IFE CTRL (F7)                                | Convenience Power Relay               |
| IRIDIUM<br>WIFI DATA (F6)                    | Wi-Fi Datalink, Iridium Satellite     |
| AHRS 2<br>ADC 2 (F5)                         | AHRS 2, ADC 2                         |
| FUEL<br>VALVE 2 (F2)                         | Fuel Control Valve (FCV) 2            |
| AVIONICS FAN (F1)                            | Avionics Equipment Fan In Console     |

| <b>CIRCUIT BREAKER</b>                    | <b>CIRCUIT</b>                                          |
|-------------------------------------------|---------------------------------------------------------|
| <b>AFT CIRCUIT BREAKER PANEL (ROW K)</b>  |                                                         |
| IFE USB (K1)                              | In-Flight Entertainment                                 |
| FADEC PANEL (K2)                          | FADEC A Maintenance Panel,<br>FADEC B Maintenance Panel |
| TRAFFIC (K3)                              | TCAS-1                                                  |
| DME (K4)                                  | DME                                                     |
| FADEC B (K5)                              | Engine Control Electronics                              |
| XPONDER 2 (K6)                            | Transponder 2                                           |
| PAX LIGHTS<br>LIGHT CTRL (K7)             | Passenger task lights                                   |
| IN FLIGHT WIFI (K8)                       | In-Flight Internet (if installed)                       |
| <b>AFT CIRCUIT BREAKER PANEL (ROW L)</b>  |                                                         |
| 115 VAC PWR (L12)                         | DC-AC Inverter                                          |
| ECS BOOST FAN (L1)                        | ECS Booster Fan                                         |
| ECS GROUND FAN (L2)                       | ECS Ground Fan                                          |
| <b>NOSE CIRCUIT BREAKER PANEL (ROW M)</b> |                                                         |
| NOSE LAND LIGHT (M2)                      | Nose Landing Light                                      |
| RADAR (M3)                                | Weather Radar                                           |
| EVS (M4)                                  | Enhanced Vision System                                  |
| AC INLET DOOR (M6)                        | AC Door Actuator                                        |
| ROCKET WARMER (M7)                        | CAPS Rocket Warmer                                      |

## GENERATOR 2 CURRENT Warning

### GENERATOR 2 CURRENT

Generator 2 current is too high.

1. GEN 2 Switch ..... CYCLE
2. GEN 2 Amps.....CHECK WITHIN LIMITS

◆ If GEN 2 amps are within limits:

- a. No further action required.

Procedure Complete

◆ If GEN 2 amps are still not within limits:

- a. GEN 2 Switch ..... OFF

• NOTE •

GEN 1 will support GEN 2 electrical loads.

- b. Land as soon as practicable.

Procedure Complete

## GENERATOR 2 OVERVOLTAGE Warning

### GENERATOR 2 OVERVOLTAGE

Generator 2 voltage is high.

1. GEN 2 Switch ..... CYCLE

◆ If message persists:

- a. GEN 2 Switch ..... OFF

• NOTE •

GEN 1 will support GEN 2 electrical loads.

- b. Land as soon as practicable.

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

• NOTE •

Generator 2 voltage > 32 V.

## ITT Warning

### ITT

ITT limits exceeded.

◆ If in flight:

- a. Thrust Lever..... REDUCE
- b. ITT .....MONITOR

○ If message persists:

- (1) Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Shut down engine. Do not dispatch.

Procedure Complete



## LDG GEAR IS UP Warning

### LDG GEAR IS UP

Landing gear is not down and locked for landing.

◆ If in flight:

- a. Landing Gear ..... CHECK DOWN

○ If message persists:

- (1) Perform [Alternate Gear Extension](#) checklist.

Procedure Complete

◆ If on ground:

- a. Do not taxi.

- b. Landing Gear ..... CHECK DOWN

Procedure Complete

• NOTE •

This CAS message is inhibited at altitudes and speeds where the aircraft landing gear warning system (as designed) has determined that airspeed and/or altitudes are not within normal parameters to require the landing gear to be down.

The landing gear DOWN / LOCK / WOW sensors are powered through the EMER BUS and the LANDING GEAR IND CB.

## LDG GEAR SELECTED UP Warning

### LDG GEAR SELECTED UP

Ldg gear should be selected down while on ground.

◆ If on ground:

- a. Do not taxi.
- b. Landing Gear ..... SELECT DOWN

Procedure Complete

◆ If in flight:

- a. Disregard.

Procedure Complete

• NOTE •

This CAS message is displayed when the landing gear handle is in the UP position and the aircraft incorrectly indicates it is on the ground.

This CAS message may also indicate a possible WOW sensor failure.

## N1 Warning

### N1

N1 speed exceeded.

◆ If in flight:

- a. Thrust Lever..... REDUCE
- b. Monitor N1.

○ If message persists:

- (1) Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Shut down engine. Do not dispatch.

Procedure Complete

## N2 Warning

### N2

N2 speed exceeded.

◆ If in flight:

- a. Thrust Lever .....REDUCE
- b. Monitor N2.

○ If message persists:

- (1) Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Shut down engine. Do not dispatch.

Procedure Complete

## OIL CHIP DETECT Warning - On Ground

### OIL CHIP DETECT

Engine oil debris detected.

1. Shut down engine. Do not dispatch.

Procedure Complete

• NOTE •

Frequent oil chip detect messages during flight may cause the OIL CHIP DETECT Warning to occur on a subsequent flight. In that event, dispatch is not permitted unless maintenance is performed.

## OIL FILTER BYPASS Warning - On Ground

### OIL FILTER BYPASS

Oil filter diff press too high, bypass impending.

1. Shut down engine. Do not dispatch.

Procedure Complete

## OIL PRESSURE Warning

### OIL PRESSURE

Engine oil pressure exceedance.

◆ If in flight:

- a. Monitor oil system pressure and temperature.

○ If message persists:

- (1) Thrust Lever .....REDUCE (MINIMUM REQUIRED)
- (2) Avoid abrupt thrust lever movements.
- (3) Land as soon as possible.

Procedure Complete

◆ If on ground:

- a. Perform [Abnormal Engine Shutdown On Ground](#) checklist.

Procedure Complete

## OIL TEMPERATURE Warning

### OIL TEMPERATURE

Engine oil temperature exceedance.

1. Thrust Lever .....REDUCE (MINIMUM REQUIRED)
2. Monitor oil system pressure and temperature.

◆ If message persists:

- a. Land as soon as possible.

Procedure Complete

## STALL WARNING

### STALL WARNING

Stall warning active.

- |                      |        |
|----------------------|--------|
| 1. AOA .....         | REDUCE |
| 2. Thrust Lever..... | T/O    |

Procedure Complete

## STICK PUSHER Warning

### STICK PUSHER

Stick pusher active.

- |                      |        |
|----------------------|--------|
| 1. AOA .....         | REDUCE |
| 2. Thrust Lever..... | T/O    |

#### • CAUTION •

Do not attempt to overpower the stick pusher.

#### • NOTE •

If airspeed, altitude, power settings, and load factors are within normal limits, and there is no evidence of ice contamination, message should be considered a malfunction.

◆ If message is determined to be a malfunction:

- |                               |                |
|-------------------------------|----------------|
| a. Grip side stick firmly.    |                |
| b. AP DISC Button .....       | PRESS AND HOLD |
| c. Airplane Control.....      | MAINTAIN       |
| d. AP SERVOS CB (A3) .....    | PULL           |
| e. AP DISC Button .....       | RELEASE        |
| f. STICK SHAKER CB (B3) ..... | PULL           |

○ If in RVSM airspace:

- |                            |          |
|----------------------------|----------|
| (1) Airplane Control ..... | MAINTAIN |
| (2) Exit RVSM airspace.    |          |

(Continued on next page)

**(Continued)**

- g. Airspeed ..... MAINTAIN ABOVE 1.3  $V_s$
- h. Avoid stalls, low airspeed, and uncoordinated or abrupt control inputs.
- i. Land as soon as practicable.

Procedure Complete

**• NOTE •**

Autopilot, Stall Warning, and Stick Shaker will be unavailable.  
Green donut airspeed reference will be unavailable or unreliable.

Refer to the following procedures for  $V_{REF}$  speeds.

Low speed awareness indications on the airspeed tape should be considered unreliable and should not be referenced.

Refer to [Section 3A: Abnormal Procedures, "STICK PUSHER FAIL Caution"](#).

Refer to [Section 3A: Abnormal Procedures, "AOA FAIL Caution"](#).

Expect CAPS AUTOPILOT INOP CAS. Refer to [Section 3A: Abnormal Procedures, "CAPS AUTOPILOT INOP Advisory"](#).

| <b><math>V_{REF}</math> Speeds (KIAS)</b>                    |                |                |                |                |                |
|--------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|
| <b>FLAPS</b>                                                 | <b>4000 lb</b> | <b>4500 lb</b> | <b>5000 lb</b> | <b>5550 lb</b> | <b>6000 lb</b> |
| UP or UNKNOWN                                                | 89             | 95             | 100            | 104            | 109            |
| UP (Ice Contaminated Airframe/<br>STALL SPEED HIGH Advisory) | 115            | 122            | 128            | 135            | 140            |
| 50%                                                          | 82             | 87             | 91             | 96             | 100            |
| 50% (Ice Contaminated Airframe /STALL SPEED HIGH Advisory)   | 98             | 104            | 110            | 115            | 120            |
| 100%                                                         | 71             | 76             | 81             | 85             | 89             |

## TAKEOFF PITCH TRIM Warning - On Ground

### TAKEOFF PITCH TRIM

Pitch trim not in takeoff configuration.

1. Takeoff.....ABORT
2. Pitch Trim..... ADJUST TO TAKEOFF (GREEN BAND)  
Procedure Complete

## THR LEVER CALIBRATION Warning - On Ground

### THR LEVER CALIBRATION

FADEC is in TLA rigging mode.

1. Do not dispatch.  
Procedure Complete

## TT2 OVERHEAT Warning

### TT2 OVERHEAT

TT2 probes pending overheat.

1. ENG. IPS Switch..... OFF
2. Avoid/exit icing conditions.  
Procedure Complete

Intentionally Left Blank



# ***Section 3A: Abnormal Procedures***

## **Table of Contents**

|                                                |    |
|------------------------------------------------|----|
| Introduction.....                              | 5  |
| Crew Alert System (CAS) Messaging .....        | 5  |
| Cautions .....                                 | 5  |
| Advisories .....                               | 5  |
| Abnormal Procedures Guidance .....             | 5  |
| Circuit Breakers .....                         | 6  |
| Locating Circuit Breakers .....                | 6  |
| Procedure Division Symbols .....               | 7  |
| Landing Guidance .....                         | 7  |
| Land as soon as practicable.....               | 7  |
| Land as soon as possible.....                  | 8  |
| Abnormal Procedures.....                       | 9  |
| GIA 1 Failure .....                            | 9  |
| Loss of GPS Navigation .....                   | 11 |
| Dry Motoring Engine - Failed Engine Start..... | 12 |
| Flaps Inoperative Approach/Landing.....        | 12 |
| Landing With Failed Brakes .....               | 13 |
| Landing With Flat Tire .....                   | 14 |
| Alternate Gear Extension .....                 | 15 |
| Gear Indication(s) Not Down/Locked .....       | 16 |
| Communications Failure.....                    | 17 |
| Flight Through Volcanic Ash .....              | 17 |
| Abnormal Engine Shutdown On Ground .....       | 18 |
| Wing/Stab IPS Activated in RVSM Airspace ..... | 18 |
| AFCS Alerts.....                               | 19 |
| Autopilot Abnormal Disconnect .....            | 19 |
| Autopilot Failure .....                        | 19 |
| Pitch Axis Failure.....                        | 19 |
| Roll Axis Failure .....                        | 20 |
| Yaw Axis Failure.....                          | 20 |
| Autopilot Preflight Test Fail .....            | 20 |
| Elevator Mistrim Down .....                    | 21 |
| Elevator Mistrim Up .....                      | 21 |
| Aileron Mistrim Left/Right .....               | 22 |
| Abnormal CAS Procedures .....                  | 23 |
| AOA FAIL Caution.....                          | 23 |

---

|                                                    |    |
|----------------------------------------------------|----|
| AOA HEAT FAIL Caution .....                        | 25 |
| BLEED AIR Caution .....                            | 25 |
| BLEED LEAK Caution .....                           | 26 |
| BLEED OFF Caution .....                            | 27 |
| BLEED OPEN Caution .....                           | 27 |
| BLEED OVERPRESSURE Caution .....                   | 28 |
| CABIN LFE NOT SET Caution .....                    | 28 |
| CABIN PRESSURE CTRL FAIL Caution .....             | 29 |
| CAPS AUTOPILOT INOP Advisory .....                 | 29 |
| CAPS MISCOMPARE Advisory .....                     | 30 |
| DOOR OPEN Advisory - On Ground .....               | 30 |
| DOOR OPEN EMER EXIT Caution .....                  | 31 |
| ECS BLEED HOT Caution .....                        | 32 |
| ECS BLEED OVERHEAT Caution .....                   | 33 |
| ECS CONTROL FAIL Caution .....                     | 34 |
| ECS TEMP BACKUP Caution .....                      | 35 |
| EDM DISABLED Advisory .....                        | 35 |
| EMER BATTERY FAIL Caution .....                    | 36 |
| EMER BATTERY FAULT Caution .....                   | 37 |
| ENGINE START BAT TEMP Caution .....                | 37 |
| ENGINE START VOLTS Caution - On Ground .....       | 38 |
| ESP FAIL Advisory .....                            | 38 |
| ESP OFF Advisory .....                             | 38 |
| ESS BUS VOLTS Caution .....                        | 39 |
| EXTERNAL POWER CONNECTED Caution - On Ground ..... | 39 |
| FADEC NO DISPATCH Caution .....                    | 40 |
| FIRE DETECTOR FAIL Caution .....                   | 40 |
| FIRE EXT LOW L-R Caution .....                     | 41 |
| FIRE EXT LOW L Advisory .....                      | 41 |
| FIRE EXT LOW R Advisory .....                      | 41 |
| FLAPS AIRSPEED INHIBIT Caution .....               | 42 |
| FLAPS DISAGREE Caution .....                       | 42 |
| FLAPS FAIL Caution .....                           | 43 |
| FLAPS SELECTOR FAIL Caution .....                  | 43 |
| FLIGHT IDLE FAIL Caution .....                     | 43 |
| FUEL IMBALANCE Caution .....                       | 44 |
| FUEL LOW LEFT/RIGHT/TOTAL Caution .....            | 45 |
| FUEL PUMP FAIL Caution .....                       | 45 |
| FUEL PUMP ON Caution .....                         | 46 |
| FUEL PUMP ON Advisory .....                        | 47 |
| FUEL QTY MISCOMPARE Caution .....                  | 47 |
| FUEL SELECTOR FAIL Caution .....                   | 48 |

---

|                                                    |    |
|----------------------------------------------------|----|
| FUEL SHUTOFF Advisory .....                        | 48 |
| FUEL SHUTOFF FAIL Caution .....                    | 49 |
| FUEL TANK BALANCED Advisory .....                  | 49 |
| FUEL TEMP LOW Caution.....                         | 50 |
| FUEL VALVE #1 FAIL Caution.....                    | 51 |
| FUEL VALVE #2 FAIL Caution.....                    | 51 |
| FUEL VALVE AUTO FAIL Caution.....                  | 52 |
| GENERATOR 1 CURRENT Caution.....                   | 53 |
| GENERATOR 2 CURRENT Caution.....                   | 56 |
| GROUND IDLE FAIL Caution - On Ground .....         | 56 |
| HYDRAULIC OVERHEAT Caution .....                   | 57 |
| IPS BLEED OVERHEAT Caution.....                    | 60 |
| IPS ENG INLET OFF Caution .....                    | 60 |
| IPS ENG INLET ON Caution.....                      | 61 |
| IPS ENG INLET OVERHEAT Caution .....               | 61 |
| IPS ENG INLET UNDERPRESS Caution .....             | 62 |
| IPS ENG INLET UNDERPRESS Advisory - On Ground..... | 62 |
| IPS FLUID LOW Caution.....                         | 62 |
| IPS FLUID LOW Advisory - On Ground .....           | 63 |
| IPS WINDSHLD TIMER Caution .....                   | 63 |
| IPS WING/STAB FAIL Caution .....                   | 63 |
| IPS WING/STAB TEMP LOW Caution.....                | 64 |
| LDG GEAR FAIL Caution.....                         | 65 |
| LDG GEAR SELECTOR FAIL Caution .....               | 66 |
| MAIN BATTERY FAIL Caution.....                     | 66 |
| MAIN BATTERY FAULT Caution .....                   | 67 |
| MAIN BUS VOLTS Caution.....                        | 67 |
| MASK MICROPHONE Advisory.....                      | 68 |
| NO ADC MODES Caution.....                          | 68 |
| NO VERT MODES Caution.....                         | 68 |
| OIL FILTER BYPASS Caution .....                    | 69 |
| OXYGEN OFF Caution .....                           | 69 |
| OXYGEN ON Caution - On Ground .....                | 70 |
| OXYGEN ON Advisory - On Ground.....                | 70 |
| OXYGEN PASSENGER FAIL Caution.....                 | 71 |
| OXYGEN PRESSURE LOW Caution.....                   | 71 |
| PARK BRAKE Caution - On Ground .....               | 71 |
| PITCH MET ACTIVE Caution.....                      | 72 |
| PROBE HEAT FAIL L Caution .....                    | 72 |
| PROBE HEAT FAIL R Caution.....                     | 73 |
| PROBE HEAT OFF Caution.....                        | 74 |
| SFD ALT MISCOMPARE Caution.....                    | 74 |

---

|                                              |    |
|----------------------------------------------|----|
| SFD IAS MISCOMPARE Caution.....              | 75 |
| SFD NO-COMPARE Advisory .....                | 75 |
| SFD PITCH MISCOMPARE Caution.....            | 76 |
| SFD ROLL MISCOMPARE Caution .....            | 76 |
| STALL SPEED HIGH Advisory.....               | 77 |
| STALL WARNING FAIL Caution .....             | 77 |
| STARTER ENGAGED Caution .....                | 78 |
| STARTER ENGAGED Advisory .....               | 78 |
| STICK PUSHER FAIL Caution .....              | 79 |
| SURFACEWATCH FAIL Advisory .....             | 79 |
| TAKEOFF EMER BAT Caution .....               | 79 |
| TAKEOFF FLAPS Caution - On Ground .....      | 80 |
| TAKEOFF PITCH TRIM Advisory - On Ground..... | 80 |
| TAKEOFF THRUST Caution .....                 | 80 |
| TAKEOFF THRUST Advisory.....                 | 81 |
| TCV CONTROL FAIL Caution.....                | 82 |
| THROTTLE SERVO Caution.....                  | 83 |
| TT2 HEAT FAIL Caution.....                   | 83 |
| WHEEL SPEED FAULT Advisory .....             | 83 |
| WOW DISAGREE Advisory.....                   | 84 |
| WOW FAIL Caution .....                       | 84 |
| YAW SAS SERVO Caution .....                  | 85 |

## **Introduction**

This section provides procedures for handling abnormal system and/or flight conditions which, if followed, will maintain an acceptable level of airworthiness or reduce operational risk. The guidelines described in this section are to be used when an abnormal condition exists, and should be considered and applied as necessary.

• **WARNING** •

**If a Warning annunciation is illuminated in combination with any of the following Abnormal annunciations, the Warning annunciation takes precedence and shall be performed first.**

## **Crew Alert System (CAS) Messaging**

### ***Cautions***

Displayed in yellow against a black background, Caution CAS messages arise during situations that require immediate flight crew awareness and subsequent flight crew response.

- A flashing Caution CAS message with an accompanying aural alert requires more timely flight crew response.
- A flashing Caution CAS message with no accompanying aural alert requires flight crew response, dependent on workload.
- A non-flashing Caution CAS message with no accompanying aural alert requires attention, dependent on workload. It may also require performing maintenance or taking corrective action prior to next flight.

### ***Advisories***

Displayed in white against a black background, Advisory CAS messages arise during situations that require flight crew awareness and that may require subsequent flight crew response.

## **Abnormal Procedures Guidance**

Although this section provides procedures for handling most abnormal system and/or flight conditions that could arise in the aircraft, it is not a substitute for proper flight training, thorough knowledge of the airplane, and recognized piloting techniques and standards. A thorough study of the information in this AFM while on the ground will help you prepare for time-critical situations in the air.

Sound judgment as well as thorough knowledge of the aircraft, its characteristics, and the AFM procedures are essential in the handling of any abnormal system and/or flight condition. In addition to the outlined items in the Abnormal Procedures, the following steps are considered part of all abnormal situations:

- Maintain Aircraft Control
- Analyze the Situation
- Take Appropriate Action

## **Circuit Breakers**

Some procedures involve manipulating circuit breakers (CBs). The following criteria should be followed during “Circuit Breaker” steps:

- Intentional pulling of circuit breakers during flight, other than as required in specific procedures, is prohibited.
- When instructed to “SET”, the appropriate circuit breaker should be checked for normal condition. If the circuit breaker is not “SET”, it may be reset only once. If the circuit breaker opens again, do not reset.
- When instructed to “PULL”, the appropriate circuit breaker should only be pulled and not reset.
- When instructed to “CYCLE”, the appropriate circuit breaker should be pulled, delayed for several seconds, and reset only once. Allow sufficient cooling time for circuit breakers that are reset through a “CYCLE” procedure.

## **Locating Circuit Breakers**

Each circuit breaker panel row and column is labeled for easy reference. Each row is individually labeled A through C (on the Pilot CB panel) and D through F (on the Copilot CB panel). Each column is assigned a number. On the Aft CB panel, the columns are labeled K and L. Each row is assigned a number.

Circuit breakers are identified by a combination of the letter (A through F, K, or L) and the number. For instance, in the illustration below, the STICK SHAKER CB is located at B3 (‘B’ representing the row and ‘3’ representing the column).

**Figure 3A-1:** Example - Circuit Breaker Panel



SF50\_FM03A\_0048

## Procedure Division Symbols

For procedures requiring pilot decision, conditional steps are indented with a symbol to designate sub-sections within the procedure. On condition, the pilot makes a decision to identify the applicable sub-section. Following the initial decision, a further sub-division of the procedure may occur. In that event, one or more additional conditions guides the pilot through the remaining decisions. Once the applicable condition(s) are identified, the pilot follows the remaining steps until the indication “Procedure Complete” is reached.

The procedure symbol levels are:

- ◆ First Level
- Second Level
- Third Level

## Landing Guidance

### *Land as soon as practicable*

Extended flight is not recommended and the pilot should proceed to land the plane, but not necessarily at the first or closest landing location. The landing location and duration of flight remains at the discretion of the pilot in command, after considering all factors related to the safety of the flight. For example, landing may be deferred to an airfield which could provide

post-landing maintenance procedures or malfunction investigation. The choice of airfield may be determined by the pilot with consideration to weather, runway dimensions, surface conditions, and services available. However, in no case shall a suitable landing airfield be discounted in the name of convenience.

### ***Land as soon as possible***

Land without delay at the first site at which a safe landing can be made. Continued flight beyond an available airfield at which a safe landing can be made will result in increased risk and shall not be attempted. The nearest suitable landing site shall be determined by the pilot with consideration given to weather, surface dimensions and conditions, and pilot experience.



## **Abnormal Procedures**

### **GIA 1 Failure**

**• NOTE •**

GIA 1 interfaces with many systems; multiple CAS messages and synoptic X's are simultaneously presented in a GIA 1 failure.

Workload and stress is greatly reduced by recognizing this as a single manageable failure and not multiple or cascading system failures.

The most recognizable sign of GIA 1 failure is the combined loss of COM1/NAV1 and GPS1 coincident with multiple system CAS messages.

1. Aircraft Control .....MAINTAIN
2. Airspeed ..... MAINTAIN ABOVE 1.3 V<sub>S</sub>
3. Avoid stalls, low airspeed, and uncoordinated or abrupt control inputs.
4. Navigation.....SELECT GPS 2 OR NAV 2
5. Communication.....TUNE AND SELECT COM 2 AS ACTIVE
6. CAS Messages..... ASSESS AND ACKNOWLEDGE

**• NOTE •**

The following messages may result from loss of monitoring, not loss of function in related systems. AFM procedures associated with these conditions are not required in case of GIA 1 failure.

FUEL VALVES BOTH FAIL Warning

PROBE HEAT OFF Caution

IPS WING/STAB FAIL Caution

IPS ENGINE INLET OFF Caution

CABIN PRESSURE CTRL FAIL Caution

**(Continued on next page)**

**(Continued)**

**• NOTE •**

The following messages may be displayed, these are accurate if provided, but do not require pilot action unless specifically included in the action steps of the GIA 1 Failure checklist.

FLAPS ICE Warning

FADEC NO DISPATCH Caution

FLAPS FAIL Caution

LDG GEAR SELECTOR FAIL Caution

FUEL VALVE AUTO FAIL Caution

STICK PUSHER FAIL Caution

SFD NO COMPARE Advisory

STALL SPEED HIGH Advisory

7. Fuel Balance ..... MAINTAIN  
a. Fuel Quantity ..... CHECK  
b. Fuel Control Switch ..... FULLEST TANK

**• NOTE •**

To maintain fuel balance, monitor fuel quantity and alternate LEFT/RIGHT selection as needed.

8. Altitude ..... AS REQUIRED

**• NOTE •**

Although CABIN PRESSURE CTRL FAIL Caution is annunciated, only LFE functionality and depressurization on landing is affected.

**(Continued on next page)**

**(Continued)**

9. Flight into icing conditions .....AVOID

**• NOTE •**

System will provide several FAIL annunciations that are result of data loss, not system failure. With lack of active CAS monitor, avoiding ice conditions is best advised; if transit through ice is required, systems should be operated normally.

Approach and Landing using ice contaminated approach speeds at the current flap setting will be required if STALL SPEED HIGH advisory is posted.

FLAPS ICE message will be provided if flaps were at 100% at the time of GIA Failure.

Refer to [Section 5: Performance Data, "Landing Distance"](#) for appropriate Ice Contaminated landing distance chart.

10. Perform [Flaps Inoperative Approach/Landing](#) checklist.  
11. Perform [Alternate Gear Extension](#) checklist.  
12. CABIN PRESSURE DUMP Button (below 1500 AGL) ..... PRESS  
13. Land as soon as practicable.

Procedure Complete

## Loss of GPS Navigation

Amber GPS LOI text left of HSI and/or white system message on touchscreen controller.

**• NOTE •**

Stick pusher function will be inoperative.

1. Navigation.....VISUAL, OR SELECT VOR AND/OR DME  
2. Perform [STICK PUSHER FAIL Caution](#) checklist.

Procedure Complete

**• NOTE •**

Dispatch is not authorized with inoperative GPS, predicted degraded GPS (RAIM prediction), or known GPS outage (NOTAM) expected during the planned flight.

## Dry Motoring Engine - Failed Engine Start

### • NOTE •

Dry motoring is used to clear the engine of residual fuel following a failed engine start.

The engine must be visually inspected after dry motoring to ensure no fuel has pooled and the exhaust is dry before attempting another engine start.

1. Thrust Lever ..... IDLE
2. Engine Knob ..... OFF
3. BLEED Switch ..... FRESH
4. Engine Button ..... PRESS AND HOLD 15 - 30 SECONDS

Procedure Complete

### • NOTE •

Refer to [Section 2: Limitations, "Starter Generator Limits \(Start Mode\)"](#) for starter duty cycle limitations.

Battery damage may occur if motoring is prolonged.

High levels of bleed air extraction during engine start produces hotter ITT and slower start.

Engine Knob OFF inhibits Electric Fuel Pump.

## Flaps Inoperative Approach/Landing

1.  $V_{REF}$  ..... REFER TO GREEN DONUT
2. Landing Gear ..... DOWN AND LOCKED

Procedure Complete

### • NOTE •

Refer to [Section 5: Performance Data, "Flaps: UP"](#) for Flaps UP landing distance requirements.

## Landing With Failed Brakes

◆ If one brake is inoperative:

- a. Land on the side of runway corresponding to the inoperative brake.
- b. Maintain directional control using rudder and operational brake.
- c. Shut down engine as soon as practical.

Procedure Complete

◆ If both brakes are inoperative:

- a. Divert to the longest, widest runway with the most direct headwind.
- b. Land on downwind side of the runway.
- c. Use the rudder and aileron for directional control.
- d. Shut down engine as soon as practical.

Procedure Complete

• **NOTE** •

Rudder effectiveness will decrease with decreasing airspeed.

## Landing With Flat Tire

• WARNING •

**If a flat tire is suspected, do not retract landing gear since it may cause difficulties in extending the gear. Additionally, landings with the gear up are potentially more hazardous than with the gear down. The probability of aircraft damage is greater with gear up landings.**

**If a flat tire or tread separation occurs during takeoff and it is not possible to abort, land as soon as practicable. Tread separation could damage wing-mounted fuel system components. Continued flight could result in fuel imbalance or fuel exhaustion. A significant imbalance and/or exhaustion can result in loss of aircraft due to loss of control or loss of thrust leading to an off-airport landing.**

◆ If main gear tire is flat:

- a. Land on the side of the runway corresponding to the good tire.
- b. Maintain directional control with the brakes, rudder, and aileron.
- c. Do not taxi. Stop the airplane and perform a normal Engine Shut-down.

Procedure Complete

◆ If nose gear tire is flat:

- a. Land in the center of the runway.
- b. Hold the nosewheel off the ground as long as possible.
- c. Do not taxi. Stop the airplane and perform a normal Engine Shut-down.

Procedure Complete

## Alternate Gear Extension

1. Airspeed ..... BELOW 210 KIAS
2. LANDING GEAR HYD CB (B4)..... PULL
3. Landing Gear Handle .....DOWN
4. Alternate Gear Extension Handle..... PULL TO STOP, RESTOW
5. Verify 3 Green, DOWN / LOCKED indications.

◆ If landing gear is down and locked:

- a. No further action required.

Procedure Complete

◆ If landing gear is not down and locked:

- a. Alternate Gear Extension Handle ..... PULL TO STOP, RESTOW

• NOTE •

Pull the handle to the end of its available travel to ensure landing gear is fully unlocked.

- b. Reduce airspeed, or increase load factor (g) by increasing bank angle to assist in extending/locking the gear.
- c. Check landing gear indications. If landing gear indications are still not confirmed down, prepare for landing with possible gear issue.
- d. Perform visual verification of gear position by ATC (towered airport), if possible.

Procedure Complete

## Gear Indication(s) Not Down/Locked

◆ If in flight:

- a. Landing Gear .....CYCLE UP, THEN DOWN

○ If Landing Gear is down and locked:

- (1) Monitor landing gear indications.

Procedure Complete

○ If Landing Gear is not down and locked:

- (1) Perform **Alternate Gear Extension** checklist.

Procedure Complete

◆ If on ground:

- a. Do not taxi.

- b. Landing Gear .....CHECK DOWN

Procedure Complete

• NOTE •

Even though the gear assemblies are functioning properly, the root cause may be a failure of the gear sensors to indicate properly (Down/Up or Locked/Unlocked). If possible, visual verification by ATC (towered airport) can be used to confirm gear position.

Failure of the gear sensors to provide proper indication could cause the LDG GEAR FAIL Caution to display.



## Communications Failure

1. Switches, Controls ..... CHECK
2. Frequency.....CHANGE
3. Circuit Breakers.....SET
4. Headset ..... CHANGE OR SPEAKER
5. Attempt to use oxygen mask microphone or hand-held microphone and cabin speakers.

◆ If Precipitation-static is experienced during flight:

- a. WINDSHLD IPS Switch ..... ON

Procedure Complete

• **NOTE** •

If, after following the checklist procedure, communication is not restored, proceed with FAR/AIM lost communications procedures.

In the event of an audio panel power failure, the audio panel connects COM 1 to the pilot's headset.

If the oxygen mask microphone is used, emergency oxygen will be depleted. A hand-held microphone may optionally be furnished by the operator.

## Flight Through Volcanic Ash

• **CAUTION** •

Avoid flying through volcanic ash where possible.

1. Engine Parameters ..... MONITOR

◆ If engine parameters cannot be kept within limits:

- a. Land as soon as possible.

Procedure Complete

## Abnormal Engine Shutdown On Ground

◆ If normal engine shutdown is not successful:

- a. Thrust Lever..... IDLE
- b. ENGINE FIRE ACKNOWLEDGE Switch..... LIFT GUARD,  
TOGGLE

Procedure Complete

• **CAUTION** •

Do not press PUSH TO DISCHARGE buttons when the ENGINE FIRE ACKNOWLEDGE switch is on. Discharging the fire bottles will result in the bottles requiring servicing before subsequent flights.

• **NOTE** •

Activating the ENGINE FIRE ACKNOWLEDGE switch activates the fuel shutoff valve which will remove fuel to engine and arm the fire bottles.

Engine Knob OFF inhibits the Electric Fuel Pump.

## Wing/Stab IPS Activated in RVSM Airspace

◆ If activated by pilot:

- a. Advise ATC of non RVSM status.

Procedure Complete

◆ If activated uncommanded:

- a. Verify WING/STAB Switch OFF.
- b. WING/STAB IPS CB (A6)..... PULL

Procedure Complete

## **AFCS Alerts**

For more information on AFCS alerts, refer to SF50 Garmin Cockpit Reference Guide.

### **Autopilot Abnormal Disconnect**

Red AP flashing on PFD; continuous high-low aural tone.

1. AP DISC Button.....PRESS AND RELEASE
2. Airplane Control .....MAINTAIN

Procedure Complete

### **Autopilot Failure**

Red AFCS annunciator on PFD; red AP flashing on PFD; continuous high-low aural tone.

1. AP DISC Button.....PRESS AND RELEASE
2. Airplane Control .....MAINTAIN

◆ If in RVSM airspace:

- a. Exit RVSM airspace.

Procedure Complete

### **Pitch Axis Failure**

Red PTCH annunciator on PFD.

#### **• NOTE •**

If the pitch axis fails while autopilot is engaged, the autopilot will disengage as indicated by the red AP flashing on the PFD. The autopilot may be re-engaged; however, vertical Flight Director modes must be hand flown.

This condition will likely be accompanied by a “STICK PUSHER FAIL” CAS message.

1. Altitude.....MONITOR AND MAINTAIN AS POSSIBLE

Procedure Complete

## Roll Axis Failure

Red ROLL annunciator on PFD.

• NOTE •

If the roll axis fails while autopilot is engaged, the autopilot will disengage as indicated by the red AP flashing on the PFD. The autopilot may not be re-engaged.

ESP is inoperative.

1. Airplane Control ..... MAINTAIN  
Procedure Complete

## Yaw Axis Failure

Red YAW annunciator on PFD.

• NOTE •

The yaw damper will disconnect, but the autopilot may still be used for vertical and lateral guidance. Some lateral mistrim may become apparent, and manual rudder inputs may be required for turn coordination.

1. Continue flight.  
Procedure Complete

## Autopilot Preflight Test Fail

Red PFT annunciator on PFD.

1. Do not dispatch.  
Procedure Complete

• NOTE •

The autopilot, stick pusher, and ESP are inoperative.

## Elevator Mistrim Down

Yellow (down arrow) ELE annunciator on PFD.

• NOTE •

Be prepared for out of trim condition if AP is disengaged.

1. Manipulate elevator trim down to eliminate the mistrim condition.

◆ If condition persists:

- a. Disengage autopilot.
- b. Retrim aircraft as desired.

Procedure Complete

## Elevator Mistrim Up

Yellow (up arrow) ELE annunciator on PFD.

• NOTE •

Be prepared for out of trim condition if AP is disengaged.

1. Manipulate elevator trim up to eliminate the mistrim condition.

◆ If condition persists:

- a. Disengage autopilot.
- b. Retrim aircraft as desired.

Procedure Complete

## Aileron Mistrim Left/Right

Yellow (left/right arrow) AIL annunciator on PFD.

• NOTE •

Be prepared for out of trim condition if AP is disengaged.

1. Manipulate aileron trim as required to eliminate the mistrim condition.

• NOTE •

Momentary roll trim input in the direction of mistrim indication will typically correct the condition.

◆ If condition persists:

- a. Disengage autopilot.
- b. Retrim aircraft as desired.

Procedure Complete

Abnormal CAS Procedures

AOA FAIL Caution

AOA FAIL

AOA signal invalid.

- 1. Airspeed ..... REFER TO V<sub>REF</sub> SPEEDS TABLE
- 2. Avoid stalls, low airspeed, and uncoordinated or abrupt control inputs.
- 3. Land as soon as practicable.

Procedure Complete

| V <sub>REF</sub> Speeds (KIAS)                             |         |         |         |         |         |
|------------------------------------------------------------|---------|---------|---------|---------|---------|
| FLAPS                                                      | 4000 lb | 4500 lb | 5000 lb | 5550 lb | 6000 lb |
| UP or UNKNOWN                                              | 89      | 95      | 100     | 104     | 109     |
| UP (Ice Contaminated Airframe/ STALL SPEED HIGH Advisory)  | 115     | 122     | 128     | 135     | 140     |
| 50%                                                        | 82      | 87      | 91      | 96      | 100     |
| 50% (Ice Contaminated Airframe /STALL SPEED HIGH Advisory) | 98      | 104     | 110     | 115     | 120     |
| 100%                                                       | 71      | 76      | 81      | 85      | 89      |

(Continued on next page)

**(Continued)**

**• WARNING •**

**The aircraft is not stall protected. Stalls must be avoided when the AOA is inoperative. Excessive altitude loss may result if the aircraft is stalled. Departure from controlled flight or spin may occur during stall with uncoordinated aileron/rudder inputs.**

**Stall warning is not operative or reliable.**

**Stick pusher is not operative or reliable.**

**Expect STALL WARNING FAIL or STICK PUSHER FAIL CAS.**

**Stall speeds in turns or increased load factor are higher.**

**• NOTE •**

**Green donut airspeed reference will be unavailable or unreliable.**



## AOA HEAT FAIL Caution

### AOA HEAT FAIL

AOA heat failure.

1. PROBE HT RIGHT CB (E4) ..... CYCLE
2. PROBE HT LEFT CB (A8) ..... CYCLE
3. Avoid/exit icing conditions.

Procedure Complete

• NOTE •

In case of AOA heat failure during flight in icing conditions, AOA measurements may be erroneous.  
Refer to  $V_{REF}$  Speeds table.

## BLEED AIR Caution

### BLEED AIR

ECS bleed air system pressure low or failed.

1. BLEED Switch .....CYCLE FRESH, BLEED WITHIN < 1 SEC
2. Cabin Altitude ..... MONITOR

◆ If message extinguishes:

- a. Continue flight.

Procedure Complete

◆ If message persists:

- a. Land as soon as practicable.

Procedure Complete

## BLEED LEAK Caution

### BLEED LEAK

Bleed leak detected in tailcone.

1. Oxygen Mask ..... DON, 100%
2. BLEED Switch.....FRESH
3. MIC SELECT Switch ..... MASK MIC
4. Wait 15 seconds.

◆ If message extinguishes:

- a. Descend and land as soon as practicable.

• NOTE •

Cabin pressurization is no longer functional.

Procedure Complete

◆ If message persists:

- a. Perform Emergency Descent checklist as required.
- b. Thrust Lever..... REDUCE (MINIMUM REQUIRED)
- c. Land as soon as possible.

Procedure Complete

• NOTE •

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## BLEED OFF Caution

### BLEED OFF

Bleed air is switched to OFF/FRESH.

◆ If bleed is desired OFF:

- a. No further action required.

Procedure Complete

◆ If bleed is not desired OFF:

- a. BLEED Switch ..... CYCLE TO BLEED
- b. Verify message is extinguished.

○ If message persists:

- (1) Cabin Altitude ..... MONITOR
- (2) Monitor for CABIN ALTITUDE HIGH Warning.

Procedure Complete

## BLEED OPEN Caution

### BLEED OPEN

Shutoff solenoid failed to close.

1. BLEED Switch ..... CYCLE
2. BLEED AIR SHUTOFF CB (D7) ..... CYCLE
3. Land as soon as practicable.

Procedure Complete

## BLEED OVERPRESSURE Caution

### BLEED OVERPRESSURE

ECS bleed air system pressure high.

1. Thrust ..... REDUCE
2. BLEED Switch..... CYCLE FRESH, BLEED WITHIN < 1 SEC

◆ If message extinguishes:

- a. Continue flight.

Procedure Complete

◆ If message persists:

- a. Thrust .....MINIMIZE A/R
- b. Land as soon as practicable.

Procedure Complete

## CABIN LFE NOT SET Caution

### CABIN LFE NOT SET

Landing field elevation is not set.

1. Input destination airport in flight plan.  
OR
2. Manually set Landing Field Elevation on GTC.

Procedure Complete

## CABIN PRESSURE CTRL FAIL Caution

### CABIN PRESSURE CTRL FAIL

Detected failure of CPCS Controller.

1. Cabin Altitude ..... MONITOR
2. Monitor for **CABIN ALTITUDE HIGH Warning**.
3. Land as soon as practicable.

Procedure Complete

• NOTE •

If cabin pressure controller fails, CPCS will maintain current cabin altitude.

## CAPS AUTOPILOT INOP Advisory

### CAPS AUTOPILOT INOP

Minimize CAPS deployment speed.

◆ If in an emergency situation:

- a. Airspeed ..... MANUALLY REDUCE BELOW  
135 KIAS AND 145 KTAS BEFORE DEPLOYING PARACHUTE

Procedure Complete

## **CAPS MISCOMPARE Advisory**

### **CAPS MISCOMPARE**

CAPS service required.

• WARNING •

**CAPS may be unavailable.**

◆ If in flight:

- a. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Do not dispatch.

Procedure Complete

## **DOOR OPEN Advisory - On Ground**

### **DOOR OPEN**

One or more doors are open.

1. Advisory only. No further action required.

Procedure Complete

## DOOR OPEN EMER EXIT Caution

### DOOR OPEN EMER EXIT

Emergency exit door is open.

• WARNING •

**Stay clear of window if malfunction is believed to have occurred. Ensure seat belts are tightened and secured in order to avoid extraction from aircraft in the event of rapid door opening/decompression. Do not attempt to latch or unlatch the exit door or manipulate the exit door or latch.**

◆ If in flight:

- a. Visually check window latch handle position.
- b. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Verify window is latched.

Procedure Complete

## ECS BLEED HOT Caution

### ECS BLEED HOT

Bleed supply temperature is too hot.

1. Cabin Temp Knob..... FULL COLD

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

- a. TEMP BACKUP ..... SELECT
- b. Verify message is extinguished.

Procedure Complete

• *NOTE* •

This CAS message is displayed when bleed supply temperature reaches 250 °F (121 °C).



## ECS BLEED OVERHEAT Caution

### ECS BLEED OVERHEAT

Bleed supply temperature is too hot.

1. Cabin Temp Knob ..... FULL COLD

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

- a. TEMP BACKUP ..... SELECT

- b. Verify message is extinguished.

○ If message persists:

- (1) Thrust Lever ..... REDUCE AS PRACTICAL

- (2) Verify message is extinguished.

□ If message persists or if TCV CONTROL FAIL is displayed:

- (a) Oxygen Mask ..... DON, 100%

- (b) BLEED Switch ..... FRESH

- (c) MIC SELECT Switch ..... MASK MIC

- (d) Cabin Altitude ..... MONITOR

- (e) Monitor for **CABIN ALTITUDE HIGH Warning**.

- (f) Land as soon as practicable.

Procedure Complete

#### • NOTE •

This CAS message is displayed when bleed supply temperature reaches 270 °F (132 °C).

Refer to [Section 2: Limitations, "Oxygen"](#) for oxygen requirements.

## ECS CONTROL FAIL Caution

### ECS CONTROL FAIL

ECS controller has failed, OR  
All cabin temperature sensors have failed, OR  
All ECS fans in cabin have failed, OR  
Bleed duct temperature sensor has failed.

1. TEMP BACKUP ..... SELECT
2. Cabin Temp Knob ..... ADJUST FOR CABIN COMFORT
3. Monitor for ECS BLEED OVERHEAT Caution.
4. Perform **ECS BLEED OVERHEAT Caution** checklist, if required.

Procedure Complete

## ECS TEMP BACKUP Caution

### ECS TEMP BACKUP

Environmental control temperature backup mode.

◆ If TEMP BACKUP mode engaged intentionally:

- a. Cabin Temp Knob ..... ADJUST FOR CABIN COMFORT
- b. Monitor for ECS BLEED HOT or ECS BLEED OVERHEAT cautions.

○ If ECS BLEED HOT is displayed:

- (1) Perform ECS BLEED HOT caution checklist.

Procedure Complete

○ If ECS BLEED OVERHEAT is displayed:

- (1) Perform ECS BLEED OVERHEAT caution checklist.

Procedure Complete

◆ If TEMP BACKUP mode engaged unintentionally:

- a. TEMP BACKUP ..... DESELECT
- b. ECS Controls ..... A/R

Procedure Complete

#### • NOTE •

When in Temp Backup mode, changes in thrust and aircraft speed affect incoming bleed air temperature.

Prior to increasing thrust or reducing aircraft speed, it is recommended that the temperature setting be lowered (especially during low altitude level off, landing, and go-around conditions).

## EDM DISABLED Advisory

### EDM DISABLED

Emergency Descent Mode is disabled.

1. Advisory only. No further action required.

Procedure Complete

## EMER BATTERY FAIL Caution

### EMER BATTERY FAIL

Emergency battery (Bat 2) service is required.

1. Exit IMC as soon as practicable.
2. Land as soon as practicable.

Procedure Complete

• CAUTION •

Emergency battery may be unavailable.

• NOTE •

Battery FAIL indicates an internal condition which reduces capability of the battery. Possible conditions include a faulty temperature sensor, disconnected heater, etc.

## EMER BATTERY FAULT Caution

### EMER BATTERY FAULT

Emergency battery (Bat 2) fault detected.

1. BAT 2 Switch ..... CYCLE

◆ If message extinguishes:

- a. Continue flight.
- b. Service aircraft as soon as practicable.

◆ If message persists:

- a. Exit IMC as soon as practicable.
- b. Land as soon as practicable.

2. Contact Cirrus for corrective action.

Procedure Complete

• CAUTION •

Emergency battery is unavailable.

• NOTE •

Battery FAULT indicates a condition causing battery to disable charging and/or discharging until corrected. Possible conditions include under-voltage, over-voltage, over-current, short-circuit, and over-temperature.

## ENGINE START BAT TEMP Caution

### ENGINE START BAT TEMP

Main battery (Bat 1) is below minimum start temp.

1. Wait until message extinguishes before attempting start.

Procedure Complete

• CAUTION •

If start is attempted prior to message extinguishing, the battery may be temporarily inoperative, or it may result in an engine hot start or other faults.

## ENGINE START VOLTS Caution - On Ground

### ENGINE START VOLTS

Main battery (Bat 1) insufficient charge for reliable engine start.

• NOTE •

Avoid prolonged battery-only ground operation beyond 20 minutes combined operation, which may result in insufficient capacity for engine start.

Attempted start with insufficient battery capacity may result in battery low voltage protection circuit activation, which will temporarily make main battery unavailable.

Condition will automatically recover after main battery is adequately recharged.

1. Battery service, preheat and/or charging is required.

Procedure Complete

## ESP FAIL Advisory

### ESP FAIL

ESP is inoperative.

1. Advisory only. No further action required.

Procedure Complete

## ESP OFF Advisory

### ESP OFF

ESP selected off.

1. Advisory only. No further action required.

Procedure Complete

## ESS BUS VOLTS Caution

### ESS BUS VOLTS

Essential power bus voltage is low, OR  
Loss of forward essential power bus voltage.

1. Perform [GENERATOR 1 CURRENT Caution](#) and [GENERATOR 2 CURRENT Caution](#) Checklists.

Procedure Complete

• **NOTE** •

Indicates GEN 1 and GEN 2 failure and is associated with low Main Bus and Essential Bus voltages, GEN 1 and GEN 2 current indications, BAT 1 and BAT 2 discharge, and GEN 1 and GEN 2 Cautions.

## EXTERNAL POWER CONNECTED Caution - On Ground

### EXTERNAL POWER CONNECTED

External power is connected.

1. Ensure external power is removed before moving airplane.

Procedure Complete

• **NOTE** •

Indicates external power is connected.

## FADEC NO DISPATCH Caution

### FADEC NO DISPATCH

Reference FADEC Fault Code Messages.

1. ENG. IPS Switch (OAT < 50 °F) ..... VERIFY ON
  - a. Allow sufficient time for ice to clear from inlet probes.
2. FADEC Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the FADEC Stale Fault Reset Button.

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

- a. Continuance of flight is permitted.

Procedure Complete

• NOTE •

Consider selecting a destination where the necessary FADEC maintenance can be performed. Failure to do so will prevent further dispatch.

## FIRE DETECTOR FAIL Caution

### FIRE DETECTOR FAIL

Engine fire detector loop low pressure.

◆ If in flight:

- a. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Do not dispatch.

Procedure Complete



## **FIRE EXT LOW L-R Caution**

### **FIRE EXT LOW L-R**

Pressure is low in both fire extinguishers.

◆ If in flight:

- a. Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Do not dispatch.

Procedure Complete

• WARNING •

**May result in limited/reduced fire suppressant capability.**

## **FIRE EXT LOW L Advisory**

### **FIRE EXT LOW L**

Pressure is low in the left fire extinguisher.

1. Continue flight.
2. Service aircraft as soon as practical.

Procedure Complete

## **FIRE EXT LOW R Advisory**

### **FIRE EXT LOW R**

Pressure is low in the right fire extinguisher.

1. Continue flight.
2. Service aircraft as soon as practical.

Procedure Complete

## FLAPS AIRSPEED INHIBIT Caution

### FLAPS AIRSPEED INHIBIT

Flaps motion inhibited.

1. Airspeed..... INCREASE OR REDUCE, AS REQUIRED  
OR

2. Flaps .....RETURN TO PREVIOUS POSITION

Procedure Complete

## FLAPS DISAGREE Caution

### FLAPS DISAGREE

Flaps are not within commanded position limits.

#### • WARNING •

**Flaps motion is inhibited when a flap position disagree condition exists.**

1. Flaps ..... CYCLE TO DESIRED POSITION

◆ If message extinguishes:

a. Continue flight.

Procedure Complete

◆ If message persists:

a. Flaps .....MONITOR POSITION

○ If flap position is unknown:

(1) Airspeed.....BELOW 150 KIAS

b. Perform [Flaps Inoperative Approach/Landing](#) checklist.

Procedure Complete

## FLAPS FAIL Caution

### FLAPS FAIL

Flaps system inoperative, OR  
Flaps position sensor miscompare, OR  
Flaps sensors failed, position unknown, OR  
Flaps have not reached the selected position.

1. Place flaps in last known or safe position.
2. FLAPS CB (C2) ..... CYCLE

◆ If message persists:

- a. Perform [Flaps Inoperative Approach/Landing](#) checklist.

Procedure Complete

## FLAPS SELECTOR FAIL Caution

### FLAPS SELECTOR FAIL

Cannot determine flaps selector position.

1. Perform [Flaps Inoperative Approach/Landing](#) checklist.

Procedure Complete

## FLIGHT IDLE FAIL Caution

### FLIGHT IDLE FAIL

Engine idle N2 is too low.

1. Maintain at least 70% N2.
2. Land as soon as practicable.

Procedure Complete

#### • CAUTION •

When  $N2 < 70\%$ , Engine IPS and Wing/Stab IPS will be degraded.

#### • NOTE •

If  $N2 < 70\%$ , expect increased engine/go-around thrust acceleration time.

## FUEL IMBALANCE Caution

### FUEL IMBALANCE

Fuel imbalance is  $\geq 15$  Gallons.

1. Fuel Control Switch ..... AUTO
2. Fuel Quantity/Fuel Remaining.....COMPARE
  - a. Refer to EIS for fuel quantity (based on level sensors).
3. Fuel Control Switch ..... FULLEST TANK
4. Airspeed..... MAINTAIN FOR ADEQUATE ROLL CONTROL

◆ If message extinguishes or imbalance decreases:

- a. Continue flight.
- b. Fuel Control Switch..... AUTO

Procedure Complete

◆ If message persists or imbalance increases:

- a. Land as soon as possible.

Procedure Complete

#### • NOTE •

To maintain fuel balance, monitor fuel quantity and alternate  
LEFT/RIGHT selection as needed.

## FUEL LOW LEFT/RIGHT/TOTAL Caution

### FUEL LOW LEFT

### FUEL LOW RIGHT

### FUEL LOW TOTAL

Left fuel quantity is low, OR  
Right fuel quantity is low, OR  
Total fuel quantity is low.

1. Fuel Quantity..... CHECK
2. Thrust Lever..... REDUCE (MINIMUM REQUIRED)
3. Land as soon as practicable.

Procedure Complete

• *NOTE* •

AUTO Fuel Control will continue to balance fuel consumption  
during FUEL LOW LEFT/RIGHT/TOTAL caution.

## FUEL PUMP FAIL Caution

### FUEL PUMP FAIL

Fuel pump has failed.

1. Avoid abrupt thrust lever movements.
2. Land as soon as practicable.

Procedure Complete

## FUEL PUMP ON Caution

### FUEL PUMP ON

Fuel pump is ON due to low fuel pressure.

1. Fuel Control Switch ..... SWITCH TO OTHER TANK
    - a. Refer to EIS for current tank selection (AUTO selection shown in white; LEFT/RIGHT selection shown in cyan).
  2. Fuel Control Switch ..... AUTO
- ◆ If message extinguishes and does not reappear:
- a. Continue flight.

Procedure Complete

- ◆ If message persists or reappears:
- a. Fuel Control Switch ..... FULLEST TANK
  - b. Fuel Quantity ..... MAINTAIN BALANCE
  - c. Land as soon as practicable.

Procedure Complete

#### • NOTE •

Message indicates pump has been activated and latched ON. Latch is removed by setting the Fuel Control Switch to LEFT or RIGHT, then to AUTO.

To maintain fuel balance, monitor fuel quantity and alternate LEFT/RIGHT selection as needed.

## FUEL PUMP ON Advisory

### FUEL PUMP ON

Fuel pump is ON for normal operations.

1. Advisory only. No further action required.

Procedure Complete

• *NOTE* •

Fuel pump has been activated due to normal operations, either due to pilot action (selecting L or R tank) or by the avionics due to entering Start Mode.

## FUEL QTY MISCOMPARE Caution

### FUEL QTY MISCOMPARE

Sensed and totalized fuel quantity disagreement.

1. Fuel Quantity/Fuel Remaining ..... COMPARE
  - a. Refer to EIS for fuel quantity (based on level sensors).

Procedure Complete

## FUEL SELECTOR FAIL Caution

### FUEL SELECTOR FAIL

Cannot determine fuel selector position.

1. Fuel Control Switch ..... CHECK LEFT/AUTO/RIGHT

◆ If control is inoperable in all switch positions:

- a. Land as soon as possible.

Procedure Complete

• CAUTION •

A failed Fuel Control Switch will cause a fuel imbalance which will increase with continued flight.

◆ If LEFT and RIGHT Switch positions are operable:

- a. Fuel Quantity ..... CHECK
- b. Fuel Control Switch ..... FULLEST TANK, LEFT OR RIGHT ONLY
- c. Fuel Quantity ..... MAINTAIN BALANCE
- d. Land as soon as possible.

Procedure Complete

• NOTE •

To maintain fuel balance, monitor fuel quantity, and alternate LEFT/RIGHT selection as needed.

## FUEL SHUTOFF Advisory

### FUEL SHUTOFF

Fuel shutoff valve is closed.

1. ENGINE FIRE ACKNOWLEDGE Switch ..... CLOSE GUARD

Procedure Complete

• NOTE •

The above procedure opens the fuel shutoff valve. Disregard if fuel shutoff has been intentionally activated.



## FUEL SHUTOFF FAIL Caution

### FUEL SHUTOFF FAIL

Fuel shutoff valve not in commanded position.

◆ If during normal flight and engine is running:

- a. Land as soon as practicable.

Procedure Complete

◆ If airborne, and engine has shut down:

- a. ENGINE FIRE ACKNOWLEDGE Switch..... LIFT GUARD, TOGGLE, CLOSE GUARD
- b. Perform [Air Start](#) Checklist.

Procedure Complete

◆ If during Engine Fire In Flight Checklist:

- a. ENGINE FIRE ACKNOWLEDGE Switch..... CLOSE GUARD, LIFT GUARD, TOGGLE
- b. Continue with Engine Fire In Flight Checklist.

Procedure Complete

## FUEL TANK BALANCED Advisory

### FUEL TANK BALANCED

Fuel tank quantity balance has been detected.

1. Fuel Control Switch .....AUTO

Procedure Complete

## FUEL TEMP LOW Caution

### FUEL TEMP LOW

Fuel temperature is low, increase OAT.

1. Fuel Temperature (L / R) .....CHECK

◆ If in flight:

- a. Climb or descend to warmer conditions.

• NOTE •

Change altitude to increase OAT and increase fuel temperature.

Due to temperature inversion, warmer temperatures may be at a higher altitude and cooler temperatures may be at a lower altitude.

- b. Fuel Temperature.....MONITOR

○ If message extinguishes:

- (1) No further action required.

Procedure Complete

○ If message persists:

- (1) Land as soon as practicable.

Procedure Complete

◆ If on ground:

- a. Warm aircraft to increase fuel temperature.

Procedure Complete

## FUEL VALVE #1 FAIL Caution

### FUEL VALVE #1 FAIL

Fuel Control Valve #1 has failed.

1. Fuel Quantity ..... CHECK
2. Fuel Control Switch ..... FULLEST TANK
3. Fuel Quantity ..... MAINTAIN BALANCE
4. Land as soon as practicable.

Procedure Complete

• CAUTION •

Avoid AUTO selection, which will draw fuel through Fuel Valve #1 (which has failed).

• NOTE •

To maintain fuel balance, monitor fuel quantity and alternate LEFT/RIGHT selection as needed.

## FUEL VALVE #2 FAIL Caution

### FUEL VALVE #2 FAIL

Fuel Control Valve #2 has failed.

1. Fuel Quantity ..... CHECK
2. Fuel Control Switch ..... AUTO
3. Land as soon as practicable.

Procedure Complete

• NOTE •

The Fuel Control Switch should remain in AUTO position which draws fuel through Fuel Valve #1 which is operating normally. See EIS Display for selected valve position (AUTO).

Avoid using LEFT or RIGHT selection which activates the electric fuel pump and draws fuel through Fuel Valve #2 which has failed.

## FUEL VALVE AUTO FAIL Caution

### FUEL VALVE AUTO FAIL

Auto fuel tank selection failure, use Manual mode.

1. Fuel Quantity ..... CHECK
2. Fuel Control Switch ..... FULLEST TANK
3. Fuel Quantity ..... MAINTAIN BALANCE
4. Land as soon as practicable.

Procedure Complete

• **NOTE** •

To maintain fuel balance, monitor fuel quantity and alternate  
LEFT/RIGHT selection as needed.

## GENERATOR 1 CURRENT Caution

### GENERATOR 1 CURRENT

Generator 1 current is low.

1. GEN 1 Switch ..... OFF
2. GCU Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the GCU Reset Button.
3. GEN 1 Switch ..... ON

◆ If GEN 1 resets:

- a. Continue flight.

Procedure Complete

◆ If GEN 1 does not reset (low GEN 1 Current and Main Bus voltage):

- a. GEN 1 Switch ..... OFF

○ If flight conditions permit:

- (1) BAT 1 Switch ..... OFF
- (2) Land as soon as practicable, avoiding IMC or night flight as able (reduced power redundancy).

Procedure Complete

#### • NOTE •

With Battery 1 and Generator 1 switched off, power will be removed from the entire main bus, including the hydraulic powerpack. The following equipment will also be inoperative:

- Landing gear control (to lower landing gear, perform [Alternate Gear Extension](#) checklist)
- ECS (such as AC Compressor and Fans)
- Redundant equipment (such as Fuel Valve 2 and AHRS 2 / ADC 2)
- Non-essential equipment (such as Electric Fuel Pump, Stick Shaker, Yaw SAS, Avionics Fan, Nose Landing Lights, and IFE USB)
- Optional equipment (such as Traffic/TCAS, Transponder2, Radar, EVS, 115VAC PWR, and Iridium WIFI Data)

## **Main Bus Circuit Breakers**

| <b>CIRCUIT BREAKER</b>                        | <b>CIRCUIT</b>                        |
|-----------------------------------------------|---------------------------------------|
| <b>PILOT CIRCUIT BREAKER PANEL (ROW B)</b>    |                                       |
| BLEED AIR FLOW CTRL (B2)                      | Bleed Air Flow Control Valve          |
| STICK SHAKER (B3)                             | Stick Shaker Motor                    |
| LANDING GEAR HYD (B4)                         | Landing Gear Hydraulics Power Control |
| <b>PILOT CIRCUIT BREAKER PANEL (ROW C)</b>    |                                       |
| PFD B<br>GTC 1B (C8)                          | GTC-1 Secondary, PFD Secondary        |
| MFD B<br>GTC 2B (C9)                          | MFD-Secondary, GTC-2 Secondary        |
| FUEL PUMP<br>CTRL (C10)                       | Fuel Control Panel                    |
| IGNITER 1 (C11)                               | Igniter 1, Ignition Control Panel     |
| <b>COPILLOT CIRCUIT BREAKER PANEL (ROW F)</b> |                                       |
| YAW SAS (F8)                                  | Yaw Servo                             |
| IFE CTRL (F7)                                 | Convenience Power Relay               |
| IRIDIUM<br>WIFI DATA (F6)                     | Wi-Fi Datalink, Iridium Satellite     |
| AHRS 2<br>ADC 2 (F5)                          | AHRS 2, ADC 2                         |
| FUEL<br>VALVE 2 (F2)                          | Fuel Control Valve (FCV) 2            |
| AVIONICS FAN (F1)                             | Avionics Equipment Fan In Console     |

| <b>CIRCUIT BREAKER</b>                    | <b>CIRCUIT</b>                                       |
|-------------------------------------------|------------------------------------------------------|
| <b>AFT CIRCUIT BREAKER PANEL (ROW K)</b>  |                                                      |
| IFE USB (K1)                              | In-Flight Entertainment                              |
| FADEC PANEL (K2)                          | FADEC A Maintenance Panel, FADEC B Maintenance Panel |
| TRAFFIC (K3)                              | TCAS-1                                               |
| DME (K4)                                  | DME                                                  |
| FADEC B (K5)                              | Engine Control Electronics                           |
| XPONDER 2 (K6)                            | Transponder 2                                        |
| PAX LIGHTS<br>LIGHT CTRL (K7)             | Passenger task lights                                |
| IN FLIGHT WIFI (K8)                       | In-Flight Internet (if installed)                    |
| <b>AFT CIRCUIT BREAKER PANEL (ROW L)</b>  |                                                      |
| 115 VAC PWR (L12)                         | DC-AC Inverter                                       |
| ECS BOOST FAN (L1)                        | ECS Booster Fan                                      |
| ECS GROUND FAN (L2)                       | ECS Ground Fan                                       |
| <b>NOSE CIRCUIT BREAKER PANEL (ROW M)</b> |                                                      |
| NOSE LAND LIGHT (M2)                      | Nose Landing Light                                   |
| RADAR (M3)                                | Weather Radar                                        |
| EVS (M4)                                  | Enhanced Vision System                               |
| AC INLET DOOR (M6)                        | AC Door Actuator                                     |
| ROCKET WARMER (M7)                        | CAPS Rocket Warmer                                   |

## GENERATOR 2 CURRENT Caution

### GENERATOR 2 CURRENT

Generator 2 current is low.

1. GEN 2 Switch..... CYCLE
2. GEN 2 CB (C3).....CHECK, SET

◆ If GEN 2 resets:

- a. Continue flight.

Procedure Complete

◆ If GEN 2 does not reset (low GEN 2 current):

- a. GEN 2 Switch .....OFF

• NOTE •

GEN 1 will support GEN 2 electrical loads.

- b. Land as soon as practicable, avoiding IMC or night flight as able (reduced power redundancy).

Procedure Complete

## GROUND IDLE FAIL Caution - On Ground

### GROUND IDLE FAIL

Ground idle is unavailable.

1. Do not dispatch.

Procedure Complete

• CAUTION •

With increased idle speed, idle thrust will be greater than normal.

To prevent overheating of brakes, exercise proper taxiing techniques.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

• NOTE •

Idle speed is high for ground operation.



## HYDRAULIC OVERHEAT Caution

### HYDRAULIC OVERHEAT

Landing gear hydraulic power pack has overheated.

◆ If HYDRAULIC OVERHEAT Caution CAS is accompanied by LDG GEAR FAIL Caution CAS:

- a. Refer to [LDG GEAR FAIL Caution](#) checklist.

Procedure Complete

◆ If HYDRAULIC OVERHEAT Caution CAS is not accompanied by LDG GEAR FAIL Caution CAS:

- a. LANDING GEAR HYD CB (B4) ..... PULL
- b. Wait 2-3 minutes for HYDRAULIC OVERHEAT CAS message to extinguish.

○ If HYDRAULIC OVERHEAT CAS message extinguishes:

- (1) Perform [Alternate Gear Extension](#) checklist, as required.

Procedure Complete

○ If HYDRAULIC OVERHEAT CAS message persists, power pack may still be operating continuously:

- (1) BAT 1 Switch ..... OFF
- (2) GEN 1 Switch ..... OFF
- (3) Perform [Alternate Gear Extension](#) checklist, as required.
- (4) Land as soon as practicable.

Procedure Complete

#### • NOTE •

FUEL VALVE #2 FAIL Caution CAS will be periodically displayed during normal tank AUTO selection since the secondary valve is not powered by emergency power.

With Battery 1 and Generator 1 switched off, power will be removed from the entire main bus, including the hydraulic powerpack. Landing gear control will also be inoperative. To lower landing gear, perform [Alternate Gear Extension](#) checklist.

## **Main Bus Circuit Breakers**

| <b>CIRCUIT BREAKER</b>                       | <b>CIRCUIT</b>                        |
|----------------------------------------------|---------------------------------------|
| <b>PILOT CIRCUIT BREAKER PANEL (ROW B)</b>   |                                       |
| BLEED AIR FLOW CTRL (B2)                     | Bleed Air Flow Control Valve          |
| STICK SHAKER (B3)                            | Stick Shaker Motor                    |
| LANDING GEAR HYD (B4)                        | Landing Gear Hydraulics Power Control |
| <b>PILOT CIRCUIT BREAKER PANEL (ROW C)</b>   |                                       |
| PFD B<br>GTC 1B (C8)                         | GTC-1 Secondary, PFD Secondary        |
| MFD B<br>GTC 2B (C9)                         | MFD-Secondary, GTC-2 Secondary        |
| FUEL PUMP<br>CTRL (C10)                      | Fuel Control Panel                    |
| IGNITER 1 (C11)                              | Igniter 1, Ignition Control Panel     |
| <b>COPILOT CIRCUIT BREAKER PANEL (ROW F)</b> |                                       |
| YAW SAS (F8)                                 | Yaw Servo                             |
| IFE CTRL (F7)                                | Convenience Power Relay               |
| IRIDIUM<br>WIFI DATA (F6)                    | Wi-Fi Datalink, Iridium Satellite     |
| AHRS 2<br>ADC 2 (F5)                         | AHRS 2, ADC 2                         |
| FUEL<br>VALVE 2 (F2)                         | Fuel Control Valve (FCV) 2            |
| AVIONICS FAN (F1)                            | Avionics Equipment Fan In Console     |

| <b>CIRCUIT BREAKER</b>                    | <b>CIRCUIT</b>                                       |
|-------------------------------------------|------------------------------------------------------|
| <b>AFT CIRCUIT BREAKER PANEL (ROW K)</b>  |                                                      |
| IFE USB (K1)                              | In-Flight Entertainment                              |
| FADEC PANEL (K2)                          | FADEC A Maintenance Panel, FADEC B Maintenance Panel |
| TRAFFIC (K3)                              | TCAS-1                                               |
| DME (K4)                                  | DME                                                  |
| FADEC B (K5)                              | Engine Control Electronics                           |
| XPONDER 2 (K6)                            | Transponder 2                                        |
| PAX LIGHTS<br>LIGHT CTRL (K7)             | Passenger task lights                                |
| IN FLIGHT WIFI (K8)                       | In-Flight Internet (if installed)                    |
| <b>AFT CIRCUIT BREAKER PANEL (ROW L)</b>  |                                                      |
| 115 VAC PWR (L12)                         | DC-AC Inverter                                       |
| ECS BOOST FAN (L1)                        | ECS Booster Fan                                      |
| ECS GROUND FAN (L2)                       | ECS Ground Fan                                       |
| <b>NOSE CIRCUIT BREAKER PANEL (ROW M)</b> |                                                      |
| NOSE LAND LIGHT (M2)                      | Nose Landing Light                                   |
| RADAR (M3)                                | Weather Radar                                        |
| EVS (M4)                                  | Enhanced Vision System                               |
| AC INLET DOOR (M6)                        | AC Door Actuator                                     |
| ROCKET WARMER (M7)                        | CAPS Rocket Warmer                                   |

## IPS BLEED OVERHEAT Caution

### IPS BLEED OVERHEAT

Ice protection bleed cooling failure.

1. WING/STAB IPS Switch.....OFF
2. Thrust Lever..... REDUCE
3. Avoid/exit icing conditions.

Procedure Complete

## IPS ENG INLET OFF Caution

### IPS ENG INLET OFF

Engine IPS should be on.

1. ENGINE IPS Switch.....ON
2. Verify message is extinguished.

Procedure Complete

#### • NOTE •

This CAS message is displayed when the WING/STAB IPS Switch is selected ON without the ENGINE IPS Switch being selected ON.

## IPS ENG INLET ON Caution

### IPS ENG INLET ON

Engine IPS should be off, OR  
Shutoff solenoid failed to close.

◆ If Engine IPS should be off:

- a. ENGINE IPS Switch..... OFF

Procedure Complete

◆ If shutoff solenoid failed to close:

- a. ENGINE IPS Switch..... CYCLE
- b. ENGINE IPS Switch..... VERIFY OFF
- c. ENGINE IPS 1 CB (A6)..... VERIFY SET
- d. ENGINE IPS 2 CB (D9)..... VERIFY SET

Procedure Complete

• CAUTION •

Operation above temperature limitations may result in IPS ENG  
INLET OVERHEAT Caution.

## IPS ENG INLET OVERHEAT Caution

### IPS ENG INLET OVERHEAT

Inlet temp higher than required for anti-ice, OR  
Inlet temp higher than safe levels.

1. ENGINE IPS Switch..... CYCLE

◆ If message persists:

- a. ENGINE IPS Switch..... OFF
- b. Avoid/exit icing conditions.
- c. Reduce thrust as necessary to extinguish message.

Procedure Complete

## IPS ENG INLET UNDERPRESS Caution

### IPS ENG INLET UNDERPRESS

Engine IPS bleed flow low.

1. ENGINE IPS Switch..... CYCLE OFF, ON WITHIN < 1 SEC

◆ If message extinguishes:

- a. Continue flight.

Procedure Complete

◆ If message persists:

- a. Avoid/exit icing conditions.

Procedure Complete

## IPS ENG INLET UNDERPRESS Advisory - On Ground

### IPS ENG INLET UNDERPRESS

Engine IPS bleed flow low with engine at idle.

◆ If in icing conditions:

- a. Increase engine thrust to extinguish message.

Procedure Complete

## IPS FLUID LOW Caution

### IPS FLUID LOW

Less than 15 minutes of IPS fluid remaining.

1. Avoid/exit icing conditions.

Procedure Complete

#### • NOTE •

This CAS message is displayed when IPS fluid quantity  $\leq 0.2$  U.S. gallon (0.8 L).

## IPS FLUID LOW Advisory - On Ground

### IPS FLUID LOW

IPS fluid below minimum dispatch quantity.

1. Fill IPS fluid as desired.

Procedure Complete

• *NOTE* •

This CAS message is displayed when IPS fluid quantity  $\leq$  1 U.S. gallon (4 L).

## IPS WINDSHLD TIMER Caution

### IPS WINDSHLD TIMER

Loss of WS/Rad pump automatic timer control.

1. WINDSHLD IPS MAX Switch .....A/R
2. Avoid/exit icing conditions.

Procedure Complete

## IPS WING/STAB FAIL Caution

### IPS WING/STAB FAIL

Wing/Stabilizer IPS low system pressure, OR  
Wing/Stabilizer IPS zonal pressure failure, OR  
Wing/Stabilizer IPS controller failure, OR  
Low system pressure, Wing/Stab IPS inoperative.

1. Avoid/exit icing conditions.

Procedure Complete

## IPS WING/STAB TEMP LOW Caution

### IPS WING/STAB TEMP LOW

Min temp for boot operation is -40 °F.

◆ If activated by pilot:

- a. WING/STAB Switch .....OFF  
Procedure Complete

◆ If activation uncommanded:

- a. WING/STAB IPS CB (A6)..... PULL  
Procedure Complete



## LDG GEAR FAIL Caution

### LDG GEAR FAIL

Landing gear didn't reach the selected position.

1. Landing Gear Indications ..... VERIFY POSITION/LOCK
2. Landing Gear ..... CYCLE

#### • NOTE •

Even though the gear assemblies are functioning properly, the root cause may be a failure of the gear sensors to indicate properly (Down/Up or Locked/Unlocked).

Reversing the landing gear handle to the opposite position clears the inhibit of the hydraulic power pack enable (limited to 30 seconds of continuous operation).

#### ◆ If message extinguishes:

- a. Monitor landing gear indications.

Procedure Complete

#### ◆ If message persists:

- a. Monitor for HYDRAULIC OVERHEAT Caution.
- b. Perform visual verification of gear position by ATC (towered airport), if possible.
- c. Perform [Alternate Gear Extension](#) checklist.

Procedure Complete

## LDG GEAR SELECTOR FAIL Caution

### LDG GEAR SELECTOR FAIL

Cannot determine landing gear selector position.

1. Landing Gear ..... CYCLE

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists and landing gear is down:

- a. Do not cycle landing gear.
- b. Land as soon as practicable.

Procedure Complete

◆ If message persists and landing gear is up:

- a. Select gear down as required.

Procedure Complete

◆ If message persists and landing gear does not extend:

- a. Perform [Alternate Gear Extension](#).

Procedure Complete

## MAIN BATTERY FAIL Caution

### MAIN BATTERY FAIL

Main battery (Bat 1) service is required.

1. Exit IMC as soon as practicable.
2. Land as soon as practicable.

Procedure Complete

• CAUTION •

Air restart may not be possible.

• NOTE •

Battery FAIL indicates an internal condition which reduces capability of the battery. Possible conditions include a faulty temperature sensor, disconnected heater, etc.

## MAIN BATTERY FAULT Caution

### MAIN BATTERY FAULT

Main battery (Bat 1) fault detected.

1. BAT 1 Switch ..... CYCLE

◆ If message extinguishes:

- a. Continue flight.
- b. Service aircraft as soon as practicable.

◆ If message persists:

- a. Exit IMC as soon as practicable.
- b. Land as soon as practicable.

2. Contact Cirrus for corrective action.

Procedure Complete

• CAUTION •

Air restart is not possible.

• NOTE •

Battery FAULT indicates a condition causing battery to disable charging and/or discharging until corrected. Possible conditions include under-voltage, over-voltage, over-current, short-circuit, and over-temperature.

## MAIN BUS VOLTS Caution

### MAIN BUS VOLTS

Loss of forward main power bus voltage, OR  
Main power bus voltage is low.

1. Perform [GENERATOR 1 CURRENT Caution](#) Checklist.

Procedure Complete

• NOTE •

Indicates GEN 1 failure and is associated with low Main Bus voltage, GEN 1 current indications, BAT 1 discharge, and GENERATOR 1 CURRENT Caution.

## **MASK MICROPHONE Advisory**

### **MASK MICROPHONE**

A crew mask microphone is switched on.

◆ If not wearing oxygen mask:

- a. Turn MIC selection switch to normal headset position.

Procedure Complete

## **NO ADC MODES Caution**

### **NO ADC MODES**

Autopilot air data modes are not available.

• NOTE •

AP may only be engaged in pitch modes.

Procedure Complete

## **NO VERT MODES Caution**

### **NO VERT MODES**

Autopilot vertical modes are not available.

• NOTE •

AP may only be engaged in lateral modes.

Procedure Complete

## OIL FILTER BYPASS Caution

### OIL FILTER BYPASS

Oil filter diff press too high, bypass impending.

1. Avoid abrupt thrust lever movements.
2. Thrust Lever.....REDUCE (MINIMUM REQUIRED)
3. Land as soon as practicable.

Procedure Complete

## OXYGEN OFF Caution

### OXYGEN OFF

Oxygen has not been turned on.

1. OXYGEN MASTER Switch..... VERIFY ON
2. OXYGEN CREW CB (E7) .....VERIFY SET
3. OXYGEN CREW CB (E7) ..... CYCLE

◆ If message persists:

- a. Altitude ..... DESCEND (BELOW 10,000 FT), IF PRACTICAL

Procedure Complete

## OXYGEN ON Caution - On Ground

### OXYGEN ON

Oxygen is on.

1. OXYGEN MASTER Switch ..... VERIFY OFF

◆ If message persists:

- a. OXYGEN CREW CB (E7) ..... VERIFY SET
- b. OXYGEN CREW CB (E7) ..... CYCLE
- c. OXYGEN MASTER Switch ..... CYCLE

○ If message persists:

- (1) Service oxygen bottle as required.

Procedure Complete

• *NOTE* •

This CAS message is displayed during shutdown procedures.

## OXYGEN ON Advisory - On Ground

### OXYGEN ON

Oxygen is on.

1. Advisory only. No further action required.

Procedure Complete

• *NOTE* •

This CAS message is displayed while taxiing after landing.

## OXYGEN PASSENGER FAIL Caution

### OXYGEN PASSENGER FAIL

Passenger oxygen valve has failed closed.

• CAUTION •

Complete any emergency descent procedure prior to completing this checklist.

1. OXYGEN PAX CB (B1)..... VERIFY SET
2. OXYGEN PAX CB (B1)..... CYCLE
3. OXYGEN MASTER Switch..... VERIFY ON

◆ If message persists:

- a. Altitude .....DESCEND (BELOW 15,000 FT)

Procedure Complete

## OXYGEN PRESSURE LOW Caution

### OXYGEN PRESSURE LOW

Oxygen pressure below regulator control (250 PSI), OR  
Oxygen pressure low (500 PSI).

1. Altitude..... DESCEND (BELOW 10,000 FT), IF PRACTICAL
2. OXYGEN PAX CB (B1)..... PULL

Procedure Complete

• CAUTION •

If OXYGEN PAX CB is pulled, supplemental oxygen to the passengers will be unavailable.

## PARK BRAKE Caution - On Ground

### PARK BRAKE

Parking brake is set.

1. Parking Brake .....A/R

Procedure Complete

## PITCH MET ACTIVE Caution

### PITCH MET ACTIVE

Pitch manual electric trim is active.

1. Prepare for abnormal pitch movement upon AP disconnect.
2. Perform [Pitch Trim Runaway / Failure](#) Checklist.

Procedure Complete

## PROBE HEAT FAIL L Caution

### PROBE HEAT FAIL L

Left probe heat failure.

1. PROBE HT LEFT CB (A8) ..... CYCLE

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

○ If in flight:

(1) ADC 2 ..... SELECTED ACTIVE

☐ If ADC 2 is unavailable or if PROBE HEAT FAIL R Caution is also present, and icing encountered:

(a) Perform [Loss of Reliable Airspeed Indication](#) Checklist.

(b) Exit icing conditions.

Procedure Complete

○ If on ground:

(1) Do not dispatch.

Procedure Complete



## PROBE HEAT FAIL R Caution

### PROBE HEAT FAIL R

Right probe heat failure.

1. PROBE HT RIGHT CB (E4)..... CYCLE

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists:

○ If in flight:

(1) ADC 1 ..... SELECTED ACTIVE

□ If ADC 1 is unavailable or if PROBE HEAT FAIL L Caution is also present, and icing encountered:

(a) Perform [Loss of Reliable Airspeed Indication](#) Checklist.

(b) Exit icing conditions.

Procedure Complete

○ If on ground:

(1) Do not dispatch.

Procedure Complete

## PROBE HEAT OFF Caution

### PROBE HEAT OFF

Probe heat is required.

1. Probe Heat Switch..... ON

◆ If message extinguishes:

- a. No further action required.

Procedure Complete

◆ If message persists, and icing encountered:

- a. Airspeed ..... EXPECT NO RELIABLE INDICATION
- b. Perform [Loss of Reliable Airspeed Indication](#) checklist.
- c. Exit icing conditions.

Procedure Complete

## SFD ALT MISCOMPARE Caution

### SFD ALT MISCOMPARE

SFD altitude miscompare.

1. DISPLAY BACKUP Button ..... SELECT
2. Altitude ..... CROSS-CHECK SFD WITH PFD
3. Altitude ..... CROSS-CHECK PFD ADC #1 WITH ADC #2

#### • NOTE •

Select PFD mode on GTC #2 or GTC #3 and select SENSORS menu to select PFD air data source.

4. Select correct sensor source, if required.
5. Pull erroneous ADC circuit breaker, if required.

◆ If in RVSM airspace:

- a. Exit RVSM airspace.
6. Exit IMC as soon as practical.
  7. Land as soon as practicable.

Procedure Complete

## SFD IAS MISCOMPARE Caution

### SFD IAS MISCOMPARE

SFD airspeed miscompare.

1. DISPLAY BACKUP Button .....SELECT

2. Airspeed ..... CROSS-CHECK SFD WITH PFD

3. Airspeed ..... CROSS-CHECK PFD ADC #1 WITH ADC #2

• NOTE •

Select PFD mode on GTC #2 or GTC #3 and select SENSORS  
menu to select PFD air data source.

4. Select correct sensor source, if required.
5. Pull erroneous ADC circuit breaker, if required.
6. Exit IMC as soon as practical.
7. Land as soon as practicable.

Procedure Complete

## SFD NO-COMPARE Advisory

### SFD NO-COMPARE

SFD comparison data missing.

1. Exit IMC.
2. Land as soon as practicable.

Procedure Complete

## SFD PITCH MISCOMPARE Caution

### SFD PITCH MISCOMPARE

SFD pitch miscompare.

1. DISPLAY BACKUP Button..... SELECT

2. Attitude .....CROSS-CHECK SFD WITH PFD

3. Attitude.....CROSS-CHECK PFD AHRS #1 WITH AHRS #2

• NOTE •

Select PFD mode on GTC #2 or GTC #3 and select SENSORS menu to select PFD attitude source.

4. Select correct sensor source, if required.

5. Pull erroneous AHRS circuit breaker, if required.

6. Exit IMC as soon as practical.

7. Land as soon as practicable.

Procedure Complete

## SFD ROLL MISCOMPARE Caution

### SFD ROLL MISCOMPARE

SFD roll miscompare.

1. DISPLAY BACKUP Button..... SELECT

2. Attitude .....CROSS-CHECK SFD WITH PFD

3. Attitude.....CROSS-CHECK PFD AHRS #1 WITH AHRS #2

• NOTE •

Select PFD mode on GTC #2 or GTC #3 and select SENSORS menu to select PFD attitude source.

4. Select correct sensor source, if required.

5. Pull erroneous AHRS circuit breaker, if required.

6. Exit IMC as soon as practical.

7. Land as soon as practicable.

Procedure Complete

## STALL SPEED HIGH Advisory

### STALL SPEED HIGH

Stall protection system adjusted for icing.

- ◆ If airframe is no longer ice contaminated:
    - a. IPS Stall Offset Reset via MFD Controller:
      - (1) From Home, touch Aircraft Systems.
      - (2) Touch the IPS Stall Offset Reset Button.
- Procedure Complete

## STALL WARNING FAIL Caution

### STALL WARNING FAIL

Stall warning failure.

1. Airspeed .....MAINTAIN ABOVE 1.3  $V_s$
2. Avoid stalls, low airspeed, and uncoordinated or abrupt control inputs.
3. Land as soon as practicable.

Procedure Complete

• WARNING •

**The aircraft may not be stall protected. Stalls must be avoided when the stall warning is inoperative. Excessive altitude loss may result if the aircraft is stalled.**

**Departure from controlled flight or spin may occur during stall with uncoordinated aileron/rudder inputs.**

**Stall warning is not operative or reliable.**

**Stick pusher may not be operative or reliable.**

**Expect STICK PUSHER FAIL CAS.**

**Stall speeds in turns or increased load factor are higher.**

• NOTE •

Green donut airspeed reference will be unavailable or unreliable.

## STARTER ENGAGED Caution

### STARTER ENGAGED

Starter will not disengage (start relay stuck).

1. Start ..... ABORT
2. GCU Reset via MFD Controller:
  - a. From Home, touch Aircraft Systems.
  - b. Touch the GCU Reset Button.
3. BAT 1 Switch .....OFF
4. GEN 1 Switch.....OFF

◆ If in flight:

- a. Land as soon as practicable.

◆ If on ground:

- a. Perform [Shutdown](#) Checklist.

Procedure Complete

## STARTER ENGAGED Advisory

### STARTER ENGAGED

Engine start in progress.

#### • NOTE •

Receiving a STARTER ENGAGED CAS Advisory is normal  
during an engine start.

Procedure Complete

## STICK PUSHER FAIL Caution

### STICK PUSHER FAIL

Stick pusher failure.

1. Airspeed .....MAINTAIN ABOVE 1.3  $V_s$
2. Avoid stalls, low airspeed, and uncoordinated or abrupt control inputs.
3. Land as soon as practicable.

Procedure Complete

• WARNING •

**The aircraft is not stall protected. Stalls must be avoided when the stick pusher is inoperative. Excessive altitude loss may result if the aircraft is stalled.**

**Departure from controlled flight or spin may occur during stall with uncoordinated aileron/rudder inputs.**

**Stick pusher is not operative or reliable.**

**Stall speeds in turns or increased load factor are higher.**

## SURFACEWATCH FAIL Advisory

### SURFACEWATCH FAIL

SurfaceWatch failed.

• NOTE •

SurfaceWatch is unavailable.

Procedure Complete

## TAKEOFF EMER BAT Caution

### TAKEOFF EMER BAT

Emergency battery (Bat 2) is not charged.

Emergency battery (Bat 2) is cold.

1. Delay takeoff until message extinguishes.

Procedure Complete

## TAKEOFF FLAPS Caution - On Ground

### TAKEOFF FLAPS

Flaps not in takeoff configuration.

1. Takeoff ..... ABORT
2. Flaps ..... SELECT 50%

Procedure Complete

## TAKEOFF PITCH TRIM Advisory - On Ground

### TAKEOFF PITCH TRIM

Pitch trim not in takeoff configuration.

1. Pitch Trim Position ..... WITHIN GREEN ARC

Procedure Complete

## TAKEOFF THRUST Caution

### TAKEOFF THRUST

5 minute takeoff thrust limitation exceeded.

1. Thrust Lever ..... MCT

Procedure Complete

#### • CAUTION •

Engine operating costs are predicated on a maximum of 2 minutes of takeoff thrust on an average flight. Exceeding 5 minutes of takeoff thrust may disproportionately increase engine operating costs. Time at takeoff thrust is limited to 5 minutes maximum, beginning when the thrust lever reaches the T/O thrust detent.



## TAKEOFF THRUST Advisory

### TAKEOFF THRUST

Engine is in takeoff thrust.

1. Thrust Lever..... MCT

Procedure Complete

• CAUTION •

Engine operating costs are predicated on a maximum of 2 minutes of takeoff thrust on an average flight. Exceeding 5 minutes of takeoff thrust may disproportionately increase engine operating costs. Time at takeoff thrust is limited to 5 minutes maximum, beginning when the thrust lever reaches the T/O thrust detent.

• NOTE •

This CAS message indicates the engine has been producing takeoff thrust for more than 2 minutes.

## TCV CONTROL FAIL Caution

### TCV CONTROL FAIL

TCV motor controller has failed.

1. Cabin Temperature.....MONITOR

• NOTE •

TCV will typically fail open resulting in colder than normal cabin temperatures.

In the unlikely event that the TCV does not open upon failure, duct air temperature may be higher than desired and DUCT OVERHEAT Caution may occur.

- ◆ If message persists and cabin temperature is warmer than acceptable:

- Altitude.....DESCEND (BELOW 10,000 FT), IF PRACTICAL
- BLEED Switch .....FRESH
- CABIN PRESSURE DUMP Button.....PRESS

• NOTE •

Autopilot Emergency Descent Mode (EDM) is disabled when the cabin dump switch is activated.

- Land as soon as practicable.

Procedure Complete

- ◆ If message persists and cabin temperature is colder than acceptable:

- Altitude.....DESCEND TO WARMER OUTSIDE AIR TEMPERATURE, IF PRACTICAL
- Land as soon as practicable.

Procedure Complete

• NOTE •

A TCV CONTROL FAIL Caution may be followed by an ECS BLEED HOT Caution or an ECS BLEED OVERHEAT Caution.

## THROTTLE SERVO Caution

### THROTTLE SERVO

Throttle servo has failed.

• NOTE •

Autothrottle functions, including thrust lock, will be unavailable.  
CAPS activation will require manual retard of thrust lever to IDLE.

Procedure Complete

## TT2 HEAT FAIL Caution

### TT2 HEAT FAIL

TT2 probe heater failure.

1. ENG. IPS Switch..... CYCLE
2. ENGINE IPS 1 CB (A6) ..... VERIFY SET
3. ENGINE IPS 2 CB (D9) ..... VERIFY SET

◆ If message persists:

- a. Avoid/exit icing conditions.

Procedure Complete

## WHEEL SPEED FAULT Advisory

### WHEEL SPEED FAULT

Wheel speed sensor fault detected.

• NOTE •

Safe Return Emergency Autoland function will be degraded.  
Wheel speed sensor maintenance is required.

1. Advisory only. No further action required.

Procedure Complete

## WOW DISAGREE Advisory

### WOW DISAGREE

WOW indicating opposite on ground/in air states.

1. Continue flight.

Procedure Complete

• **NOTE** •

When WOW disagrees, avionics determines ground/in-air state based on airspeed.

Expect GROUND IDLE FAIL Caution after landing.

## WOW FAIL Caution

### WOW FAIL

WOW is indicating on ground state.

1. Maintain at least 70% N2.
2. Land as soon as practicable.

Procedure Complete

• **WARNING** •

**If above 10,000 ft, expect CABIN ALTITUDE Warning and aircraft to depressurize.**

• **CAUTION** •

Minimum engine RPM will default to Ground Idle in all configurations. Expect FLIGHT IDLE FAIL Caution with thrust at IDLE thrust in flight.

• **NOTE** •

Bleed air ground cooling fan will run.

The following indications should be ignored:

OIL CHIP DETECT Warning (false indication possible if in air)

OXYGEN ON Advisory

TAKEOFF PITCH TRIM Advisory

## YAW SAS SERVO Caution

### YAW SAS SERVO

Yaw SAS servo has failed.

◆ If in flight:

- a. Limit flaps to 0% or 50% in landing configuration.
- b. Perform [Flaps Inoperative Approach/Landing](#) checklist.
- c. Avoid any unnecessary maneuvers during flight. Limit maneuvers only to those necessary to safely land the aircraft.

Procedure Complete

◆ If on ground:

- a. Do not dispatch.

Procedure Complete

Intentionally Left Blank

## ***Section 4: Normal Procedures***

### **Table of Contents**

|                           |    |
|---------------------------|----|
| Introduction.....         | 3  |
| Normal Procedures .....   | 3  |
| Preflight Inspection..... | 3  |
| Before Engine Start ..... | 11 |
| Engine Start.....         | 11 |
| Before Taxi .....         | 13 |
| Before Takeoff.....       | 14 |
| Normal Takeoff.....       | 15 |
| Climb .....               | 16 |
| Cruise .....              | 16 |
| Descent .....             | 16 |
| Approach .....            | 17 |
| Before Landing .....      | 18 |
| Landing.....              | 19 |
| Go-Around.....            | 20 |
| After Landing .....       | 20 |
| Shutdown .....            | 21 |
| Icing Conditions.....     | 22 |
| Preflight Inspection..... | 22 |
| Before Takeoff.....       | 23 |
| IPS Activation .....      | 23 |

Intentionally Left Blank



## **Introduction**

This section provides amplified procedures for normal operation of the Vision SF50 aircraft.

• **NOTE** •

Refer to [Section 9: Log of Supplements](#) for optional equipment Normal Procedures.

## **Normal Procedures**

### **Preflight Inspection**

• **WARNING** •

**Before carrying out preflight inspections, ensure that all required maintenance has been accomplished. Review your flight plan and compute weight and balance and performance requirements. Throughout the walk-around: check all hinges, hinge pins, and bolts for security; check skin for damage, condition, and evidence of delamination; check all control surfaces for proper movement and excessive free play; check area around liquid reservoirs and lines for evidence of leaking.**

**Ensure there is no equipment or unwanted material near the engine exhaust and inlet.**

**Remove all securing devices. This includes AOA cover, engine exhaust plugs or cover, probe covers, static port covers, gust locks, and any aftermarket equipment such as exterior gust locks, canopy and wing covers, etc.**

**In cold weather, remove all frost (polished or not), ice, snow, or slush from fuselage, wing, stabilizers, control surfaces, and engine inlet.**

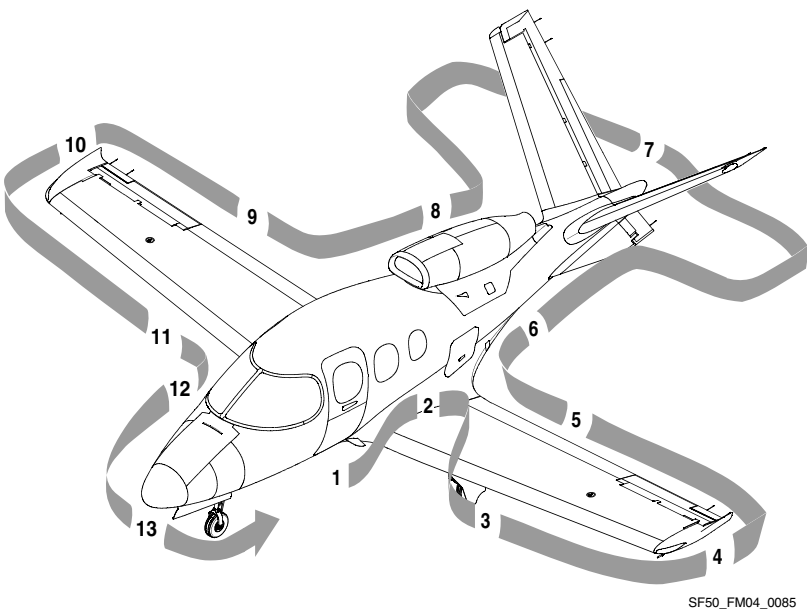
**Ensure that control surfaces are free of internal ice or debris. Check that wheels are free of snow and ice accumulation.**

**Failure to comply may result in significant aircraft damage, loss of aircraft, and/or loss of life.**

**(Continued on next page)**

**(Continued)**

**Figure 4-1: Recommended Walk-Around Sequence**



**1. Cabin**

- a. Gust Lock ..... REMOVED
- b. Cabin Door .. CHECK HINGE, LATCH AND SEAL CONDITION AND OPERATION
- c. Egress Window..... SECURED AND PIN REMOVED
- d. Seats, Interior Furnishing ..... SECURED AND LOCKED
  - (1) Crew Seats ..... LATCHED (FLIGHT RANGE POSITION)
  - (2) Passenger Seats .....GRAB TOP, ROCK FORWARD AND AFT
  - (3) Seat Levers, Latches.....VISUALLY INSPECT
  - (4) Latch Pins (Child Seats).....VISUALLY INSPECT
  - (5) Console, Relief Station (if installed) ..SECURED AND LOCKED
- e. Passenger Oxygen Masks ..... STOWED
- f. Required Documents..... CHECK
- g. Fuel Control Switch ..... AUTO

**(Continued on next page)**

**(Continued)**

- h. Alt Gear Extension Handle Cover Tab..... VISUALLY INSPECT
- i. ELT ..... ARMED
- j. Passenger Oxygen Switch .....AUTO
- k. Fire Extinguisher ..... CHARGED AND STOWED
- l. Circuit Breaker Panels ..... CHECK AND SET
- m. Landing Gear .....DOWN
- n. Bolster Switches ..... OFF
- o. Engine Knob..... OFF
- p. BAT 2 Switch..... ON
  - (1) Verify the PFD, GTC 1, and GTC 2 power up.
  - (2) Verify the MFD and GTC 3 do not power up.
- q. BAT 1 Switch ..... ON
  - (1) Verify ENGINE START BAT TEMP CAS message not present.

**• NOTE •**

If ENGINE START BAT TEMP CAS message is displayed,  
perform ENGINE START BAT TEMP checklist.

- (2) Verify the MFD and GTC 3 power up.

**• NOTE •**

During the initialization process, expect the FADEC System  
annunciators to cycle On & Off and between channels.

- r. Oxygen Pressure ..... CHECK
- s. OXYGEN MASTER Switch ..... ON
- t. Oxygen Mask(s) ..... CHECK

**• NOTE •**

It is recommended to verify oxygen flow using the press to test  
feature on the mask on at least a monthly basis.

- u. OXYGEN MIC SELECT ..... CHECK AND SET
- v. PROBE HEAT ..... ON AND CHECK NO CAS THEN OFF
- w. Lights .....ON AND CHECK OPERATION THEN OFF
- x. Autopilot..... ON
- y. AP DISC Button ..... PRESS

**(Continued on next page)**

**(Continued)**

- z. AP and YD PFD Annunciations.....CONFIRM
- aa. Aural AP Disconnect Tone.....CONFIRM
- ab. Fuel Control Switch.....LEFT
  - (1) Verify cyan L MAN selection box is displayed on the fuel section of the EIS.
  - (2) Verify no FUEL VALVE CAS messages are displayed.
- ac. Fuel Control Switch.....RIGHT
  - (1) Verify cyan MAN R selection box is displayed on the fuel section of the EIS.
  - (2) Verify no FUEL VALVE CAS messages are displayed.
- ad. Fuel Control Switch.....AUTO
  - (1) Verify white L or R selection box is displayed in the fuel section of the EIS.
- ae. BAT 1 and BAT 2 Switches .....OFF
- af. Test brakes by pressing brake pedals.

**• WARNING •**

**Do not fly aircraft if brake pedals on either side feel soft or spongy.**

**• NOTE •**

Complete the [Icing Conditions](#) Preflight Inspection Checklist if flight in icing conditions is expected or possible.

- 2. Engine Preflight
  - a. Engine Inlet.....CLEAR, COVER REMOVED, INSPECT
  - b. Engine Fan, Duct .....INSPECT, CLEAR
  - c. Engine Inlet Probes.....INSPECT
- 3. Left Forward Wing
  - a. LH Wing Root Vortex Generator.....CHECK
  - b. LH Fuel Collector Drain .....SAMPLE
  - c. LH Wing Deicer and Leading Edge.....CONDITION
  - d. LH MLG .....CONDITION
  - e. LH MLG Door.....CONDITION, SECURITY
  - f. LH Fuel Tank Drain .....SAMPLE
  - g. LH Fuel Cap.....CHECK FUEL LEVEL AND SECURE

**(Continued)**

**(Continued on next page)**

4. Left Wing Tip
  - a. LH Wing Tip, Strobe/Nav Lights ... CONDITION AND SECURITY
  - b. Pitot Probe ..... CLEAR, COVER REMOVED
5. Left Wing Trailing Edge and Left Main Gear
  - a. LH Aileron, Roll Trim Tab ..... CONDITION, MOTION
  - b. T-Strip ..... CHECK
  - c. Static Wicks ..... CHECK
  - d. LH Fuel Vent ..... CLEAR
  - e. LH Flap ..... CONDITION
  - f. LH MLG Brakes, Wheel and Tire ..... CONDITION
    - (1) Brake Pads .... VISUALLY INSPECT FRICTIONAL MATERIAL
      - (a) Center channel indicates thickness is greater than wear limit.
      - (b) Edges are free from voids or large sections of material missing.
    - (2) Brake Discs ..... VISUALLY INSPECT SURFACE CONDITION
      - (a) Surface appears smooth and uniform.
      - (b) Surface is free from irregularities such as large grooves, bumps, or deposits from brake pads.
    - (3) Brake Lines ..... VISUALLY INSPECT
      - (a) Flexible lines for chafe and distress.
      - (b) Lines for leaking.

**• NOTE •**

Refer to [Section 8: Handling and Servicing, "Brake Overheat Inspection"](#).

**(Continued on next page)**

**(Continued)**

6. Left Fuselage
  - a. Baggage Circuit Breaker Panel..... CHECK AND SET
  - b. Baggage..... CHECK, SECURE
  - c. Baggage Door .....SECURE
  - d. External Power Service Door .....SECURE
  - e. Static Ports ..... CONDITION, CLEAR
  - f. Nacelle ..... INSPECT
  - g. Pylon Panels..... INSPECT
  - h. Pylon NACA Inlet..... INSPECT
  - i. Pylon Louver..... CLEAR
  - j. Antennae .....CONDITION
  - k. OAT Probes .....CONDITION
  - l. Ram Scoop Inlet ..... CLEAR
  - m. Cabin Pressurization Static Port ..... CLEAR
7. Empennage
  - a. Engine Nozzle..... INSPECT
  - b. Second LP Turbine ..... INSPECT

**• NOTE •**

When inspecting the LP Turbine, look for Foreign Object Damage (FOD), overall damage, cracks, deformation, and evidence of pooled liquid (water, fuel, oil, etc.). If evidence of an oil leak is present, isolate the cause and correct.

A flashlight may be necessary to accomplish this step.

- c. Stabilizer Deicers.....CONDITION
- d. Ruddervators, Pitch Servo Tabs.....CONDITION, MOTION
- e. Ventral Fins .....CONDITION
- f. Yaw SAS .....CONDITION, MOTION
- g. Static Wicks ..... CHECK
- h. Tail Lights ..... CONDITION, SECURITY

**(Continued on next page)**

**(Continued)**

8. Right Fuselage
  - a. Engine Preflight ..... COMPLETE
    - (1) Pylon NACA Inlet..... CLEAR
    - (2) Engine Oil Level ..... CHECK
    - (3) Oil Cap.....CONDITION, SECURITY
  - b. Right Fuselage
    - (1) Static Ports ..... CONDITION, CLEAR
    - (2) Overboard Drains ..... CLEAR, NO LEAKS
    - (3) Cabin Pressurization Static Port ..... CLEAR
9. Right Wing Trailing Edge and Right Main Gear
  - a. RH MLG Brakes, Wheel and Tire ..... CONDITION
    - (1) Brake Pads..... VISUALLY INSPECT FRICTIONAL MATERIAL
      - (a) Center channel indicates thickness is greater than wear limit.
      - (b) Edges are free from voids or large sections of material missing.
    - (2) Brake Discs..... VISUALLY INSPECT SURFACE CONDITION
      - (a) Surface appears smooth and uniform.
      - (b) Surface is free from irregularities such as large grooves, bumps, or deposits from brake pads.
    - (3) Brake Lines..... VISUALLY INSPECT
      - (a) Flexible lines for chafe and distress.
      - (b) Lines for leaking.

**• NOTE •**

Refer to [Section 8: Handling and Servicing, "Brake Overheat Inspection"](#).

- b. RH Flap ..... CONDITION
- c. Static Wicks ..... CHECK
- d. RH Fuel Vent..... CLEAR
- e. RH Aileron, Roll Servo Tab.....CONDITION, MOVEMENT
- f. T-Strip ..... CHECK

**(Continued on next page)**

**(Continued)**

**10. Right Wing Tip**

- a. RH Wing Tip, Strobe/Nav Lights ..... CONDITION, SECURITY
- b. Pitot Probe ..... CLEAR, COVER REMOVED

**11. Right Forward Wing**

- a. RH Fuel Cap ..... CHECK FUEL LEVEL AND SECURE
- b. RH Fuel Tank Drain ..... SAMPLE
- c. RH MLG Door ..... CONDITION, SECURITY
- d. RH MLG ..... CONDITION
- e. RH Wing Deicer and Leading Edge ..... CONDITION
- f. RH Fuel Collector Drain ..... SAMPLE
- g. RH Wing Root Vortex Generator ..... CHECK

**12. Nose, Right Side**

- a. Egress Window ..... CONDITION, SECURITY (HANDLE FLUSH)
- b. Air Conditioning Air Inlet ..... CLEAR
- c. Windshield Spray Bar ..... CLEAR
- d. AOA Vane ..... CLEAR, COVER REMOVED,  
CONDITION, MOTION

**13. Nose**

- a. Lightning Diverters (if equipped) ..... CONDITION
- b. Radome Spray Nozzle ..... CONDITION
- c. EVS Camera ..... FREE & CLEAR
- d. NLG ..... CONDITION
- e. NLG Door ..... CONDITION, SECURITY
- f. NLG Wheel, Tire ..... CONDITION

**• CAUTION •**

NLG Wheel must be pointing forward.



## Before Engine Start

### • NOTE •

Extended use of Battery 1 without charging power will deplete the battery and may preclude battery engine start. Extended battery only ground operations (to obtain clearances, radio communications, entering and saving flight plans, etc.) may be accomplished utilizing Battery 2 to preserve starting battery capacity.

Do not change position of flaps until all avionics have completed initialization.

1. Preflight Inspection ..... COMPLETE
2. Door ..... LATCHED
3. Parking Brake ..... A/R
4. BAT 1 and BAT 2 Switches ..... ON
5. GEN 1 and GEN 2 Switches ..... ON
6. External GPU ..... AS NEEDED SET 28 VDC,  
800 A (MINIMUM), 1000 A (MAXIMUM)
7. Verify active FADEC channel is displayed below % thrust indication.

## Engine Start

### • CAUTION •

Point nose of the aircraft into the wind. If this is not possible (and if the speed of the tailwind component exceeds 10 knots), do not operate engine N1 speed above 60% until the aircraft rolling speed is greater than the tailwind component. Running the engine with a tailwind can cause the engine to surge or result in high ITT.

If engine maintenance has been performed, air in fuel lines may cause a hot start. Ensure that proper purging procedures have been accomplished prior to attempting a start. Be prepared to abort start.

If oil pressure is outside the limits, shut down engine.

### • NOTE •

Refer to [Section 2: Limitations, "Engine Start"](#) for engine start limitations.

**(Continued on next page)**

**(Continued)**

1. Strobe Lights ..... ON
2. CAS Messages ..... CHECK
3. Thrust Lever ..... IDLE
4. Brakes..... HOLD
5. Engine Knob ..... RUN
6. Fuel Pump ..... CHECK
  - a. Verify FUEL PUMP ON Advisory CAS is displayed.
  - b. Audibly confirm pump is ON.
7. Verify ENGINE START BAT TEMP or ENGINE START VOLTS CAS message is not displayed.
8. Engine Button.....PRESS
9. Engine Instruments.....MONITOR

**• CAUTION •**

At 25% N2, ensure positive N1 indication.

Be prepared to abort start. If any of the following occur, abort start:

- ITT exceeds limits (red line).
- No N1 indication by 25% N2.
- No evidence of lightoff within 8 seconds.
- Low oil pressure after start.

- 10.EIS .....CHECK ALL NORMAL
- 11.Engine IPS..... A/R
- 12.ECS Control Panel ..... SET
- 13.External GPU ..... DISCONNECT
- 14.Oxygen Master Switch..... VERIFY ON
- 15.Avionics Initialization ..... ALL INITS COMPLETE
- 16.CAPS Pin ..... REMOVE AND STOW
- 17.Passengers..... BRIEF
- 18.Seat Belts..... SECURED

**• NOTE •**

The BLEED switch should not be set to BLEED until BAT 1 and BAT 2 have been ON for 2 minutes or more.

- 19.BLEED Switch..... BLEED

## Before Taxi

### • WARNING •

**When operating aircraft on the ground in snow or icing conditions, N2 should be increased to 70% for 30 seconds every 10 minutes to ensure the inlet remains clear of ice. Failure to comply may result in ice accretions that may cause engine damage and possible loss of available thrust. This may lead to loss of aircraft and/or cause significant injury or loss of life.**

1. COM and NAV/GPS .....SET
2. ATIS/Clearance .....OBTAIN
3. Altimeter ..... SET/CHECK
4. Transponder .....SET
5. Heading/Initial ALT .....SET
6. Flight Controls ..... FREE & CORRECT

### • CAUTION •

If TT2 HEAT FAIL Caution occurs with Engine IPS ON, dispatch is not allowed.

### • NOTE •

Turn Engine IPS ON if OAT  $\leq 41^{\circ}\text{F}$  ( $5^{\circ}\text{C}$ ) with visible moisture or visibility of less than 1 mile (1.6 km).

7. Trim .....SET
8. Flaps ..... 50%
9. Electric Roll Trim.....SET
10. TO/GA Button ..... PRESS
11. Pitch Attitude .....  $10^{\circ}$
12. CAS Messages ..... CHECK
13. Seats must be in normal upright position.
14. Parking Brake ..... RELEASE
15. Brakes ..... CHECK

### • NOTE •

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## Before Takeoff

### • WARNING •

**When operating aircraft on the ground in snow or icing conditions, N2 should be increased to 70% for 30 seconds every 10 minutes to ensure the inlet remains clear of ice. Failure to comply may result in ice accretions that may cause engine damage and possible loss of available thrust. This may lead to loss of aircraft and/or cause significant injury or loss of life.**

1. Departure Briefing ..... COMPLETE
2. Landing Lights ..... A/R

### • CAUTION •

Above 41 °F (5 °C), probe heat may only be ON for 5 minutes (continuous) during ground operations. Extended use in warmer temperatures may damage the composite structure adjacent to the pitot probe.

3. PROBE HEAT Switch ..... ON
4. ENGINE IPS Switch ..... A/R
5. WING/STAB IPS Switch ..... OFF
6. CAS Messages ..... CHECK
7. Trim ..... SET
8. Flaps ..... 50%
9. Verify TAKEOFF EMER BAT CAS message is not displayed.
10. Multi-Function Windows ..... CONFIGURE

### • WARNING •

**Do not takeoff until oil temperature > 50 °F (10 °C). Failure to comply may cause possible engine damage.**

**Takeoff is prohibited with any frost (polished or not), ice, snow, or slush adhering to the wings, stabilizers, control surfaces, engine inlet, or fuselage forward of the engine inlet. A tactile inspection in accordance with Section 2, Limitations, must be accomplished no more than 5 minutes prior to takeoff when operating in ground icing conditions.**

### • CAUTION •

Flight operations with FADEC CH A or CH B inoperable are prohibited. Engine reliability and certification is predicated on having FADEC A and FADEC B available before dispatch. Attempting to operate with only one FADEC may result in loss of thrust or thrust control.

**(Continued on next page)**

**(Continued)**

**• NOTE •**

Prior to takeoff, engine should run at low thrust to stabilize engine temperature before Takeoff thrust is selected.

Following engine start, a warm engine should run at idle thrust for at least 2 minutes prior to takeoff.

If OAT < - 4 °F (-20 °C), run the engine at < 80% N2 until oil temperature has been above 81 °F (27 °C) for at least 5 minutes.

**Normal Takeoff**

1. Brakes .....HOLD
2. Thrust Lever..... T/O
3. Thrust Setting (N1% Indicator).....ALIGNED w/ BUG (T)
4. Engine Instruments .....CHECK NORMAL
5. Brakes ..... RELEASE
6.  $V_R$ ..... 85 KIAS
7. Pitch Attitude ..... 10°
8. Landing Gear ..... UP w/ POSITIVE RATE OF CLIMB
9. Flaps ..... UP AT 115 KIAS, CLEAR OF OBSTACLES
10. Airspeed .....PER CLIMB TABLES

**• NOTE •**

Rolling takeoffs are permitted and will increase takeoff distance.

## Climb

### • CAUTION •

Thrust set to Takeoff thrust is limited to 5 minutes maximum. If practical, reduce to MCT within 2 minutes of selecting Takeoff thrust. Engine life and maintenance costs are contingent upon observing the 2-minute limit of Takeoff thrust for every 72 minutes of flight.

### • NOTE •

Due to engine deck angle limitations, the pilot should avoid operation at deck angles of 20° or greater for more than 30 seconds.

1. Thrust Lever ..... MCT
2. Engine Parameters ..... MONITOR
3. Cabin Pressure ..... MONITOR
4. Ice Protection ..... A/R
5. Altimeter Transition Altitude ..... SET 29.92 INCHES / 1013 MB

## Cruise

1. Thrust Lever ..... A/R
2. Ice Protection ..... A/R

◆ If in RVSM Airspace:

○ If Wing/Stab IPS ON:

(1) Contact ATC and notify of non-RVSM status.

3. EIS ..... MONITOR

## Descent

1. Thrust Lever ..... A/R
2. Landing Field Elevation ..... CHECK
3. Ice Protection ..... A/R
4. Altimeter Transition Altitude ..... LOCAL SETTING
5. Seats and Seat Belts ..... SECURE
6. Brake Pressure ..... CHECK

## Approach

1. Approach Briefing ..... COMPLETE
2. All seats must be in normal upright position.
3. Ice Protection .....A/R
4. Lights .....A/R
5. Flaps ..... 50%
6. Airspeed ..... 140 KIAS

| <b>V<sub>REF</sub> Speeds (KIAS)</b>                          |                |                |                |                |                |
|---------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|
| <b>FLAPS</b>                                                  | <b>4000 lb</b> | <b>4500 lb</b> | <b>5000 lb</b> | <b>5500 lb</b> | <b>6000 lb</b> |
| UP or UNKNOWN                                                 | 89             | 95             | 100            | 104            | 109            |
| UP (Ice Contaminated Airframe/<br>STALL SPEED HIGH Advisory)  | 115            | 122            | 128            | 135            | 140            |
| 50%                                                           | 82             | 87             | 91             | 96             | 100            |
| 50% (Ice Contaminated Airframe /STALL<br>SPEED HIGH Advisory) | 98             | 104            | 110            | 115            | 120            |
| 100%                                                          | 71             | 76             | 81             | 85             | 89             |

## Before Landing

1. Landing Gear .....DOWN AND LOCKED
2. Flaps ..... A/R
3. Cabin Pressure Diff..... ~0 PSI
4. WINDSHLD IPS Switch .....OFF

• NOTE •

IPS fluid decreases flight visibility.

5. Autopilot .....OFF
6. Airspeed.....  $V_{REF}$  TO  $V_{REF}+10$
7. Brake Pressure .....CHECK



## Landing

1. Flaps ..... 100% (50%, IF AIRFRAME IS ICE CONTAMINATED)
  2. Airspeed ..  $V_{REF}$  (REFER TO LANDING CONSIDERATIONS TABLE)
  3. Thrust Lever ..... A/R
- After landing:
4. Brakes ..... A/R

**Table 1:** Landing Considerations

| Normal, Gusty, and Crosswind Landing Considerations |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal Approaches                                   | <ul style="list-style-type: none"> <li>• Maintain <math>V_{REF}+10</math> for all flap configurations (UP, 50%, and 100%).</li> <li>• AFM weight-adjusted <math>V_{REF}</math> or green donut on ASI tape are suitable for determining <math>V_{REF}</math> speed.</li> </ul>                                                                                                     |
| Gusty Approaches                                    | <ul style="list-style-type: none"> <li>• Maintain recommended approach speed plus half the gust factor.</li> </ul>                                                                                                                                                                                                                                                                |
| Normal Landings                                     | <ul style="list-style-type: none"> <li>• 100% flaps is recommended as it provides the shortest landing distance and provides the lowest landing energy for tire and brake wear.</li> <li>• 50% flap landings are required for approaches with ice contaminated airframe and are recommended when crosswinds are higher than 15 knots.</li> <li>• Do not prolong flare.</li> </ul> |
| Crosswind Landings                                  | <ul style="list-style-type: none"> <li>• Recommended technique is to maintain an additional 5 knots above normal approach and landing speeds to accommodate increased stall and stall warning speed when side slip is added.</li> <li>• Do not prolong flare.</li> </ul>                                                                                                          |

**• NOTE •**

Stall speed increases when aircraft is not coordinated. The aircraft provides stall warning at lower angle-of-attack (higher speed) when aircraft is held in slip or skid. This is relevant for crosswind landing cases. When transitioning from crab to side slip/wing low landing technique, expect an increase in stall speed by 3-5 knots for a full ball side slip.

## Go-Around

1. Thrust Lever..... T/O
2. TO/GA Button.....PRESS
3. Pitch Attitude..... 7°
4. Flaps ..... 50%
5. Landing Gear ..... UP w/ POSITIVE RATE OF CLIMB
6. Airspeed..... V<sub>x</sub>
7. Flaps .....UP AT 115 KIAS, CLEAR OF OBSTACLES
8. Climb Out..... RESUME NORMAL CLIMB OUT

## After Landing

1. Flaps ..... UP
2. WING/STAB IPS Switch.....OFF
3. ENGINE IPS Switch..... A/R

### • CAUTION •

Above 41 °F (5 °C), probe heat may only be ON for 5 minutes (continuous) during ground operations. Extended use in warmer temperatures may damage the composite structure adjacent to the pitot probe.

4. PROBE HEAT Switch.....OFF
5. External Lights..... A/R
6. Brakes..... CHECK

### • NOTE •

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## Shutdown

### • NOTE •

Allow engine to remain at ground idle for 2 minutes (minimum) prior to shutdown for hot section component temperatures to equalize.

1. Thrust Lever..... IDLE
2. ENGINE IPS Switch ..... OFF
3. BLEED Switch ..... FRESH
4. OXYGEN MASTER Switch..... OFF
5. Lights ..... OFF

### • CAUTION •

To avoid unnecessary FADEC System faults, do not remove aircraft power until  $N_2 < 24\%$ .

6. Engine Knob ..... OFF
7. Engine Button..... PRESS
8.  $N_2$ ..... BELOW 24%
9. CAPS Pin..... REPLACE
10. GEN 1 and GEN 2 Switches ..... OFF
11. BAT 1 and BAT 2 Switches ..... OFF
12. Parking Brake ..... A/R
13. Egress Door Pin..... REPLACE
14. Airplane..... SECURE

### • CAUTION •

Moving airplane with parking brake set may result in aircraft damage.

## **Icing Conditions**

### **Preflight Inspection**

**• WARNING •**

**In cold weather, remove all frost (polished or not), ice, snow, or slush from fuselage, wing, stabilizers, control surfaces, and engine inlet.**

**Ensure that control surfaces are free of internal ice or debris. Check that wheels are free of snow and ice accumulation.**

**Failure to comply may result in significant aircraft damage, loss of aircraft, and/or loss of life.**

**1. Cabin**

- a. Circuit Breakers..... SET
- b. BAT 1 and BAT 2 Switches ..... ON
- c. WINDSHLD IPS MAX Switch .....PRESS & HOLD UNTIL FLUID EXITS SPRAY BAR
- d. IPS Fluid ..... VERIFY QUANTITY

**• NOTE •**

IPS fluid minimum dispatch quantity is 1 U.S. gallon (4 L).

- e. ICE Light ..... ON

**2. Exterior**

- a. ICE Light ..... CHECK
- b. Nose Section ..... VERIFY EVIDENCE OF IPS FLUID

**• NOTE •**

Fluid will spray clear of the airframe.

- c. Engine Inlet..... INSPECT

**3. Cabin**

- a. PROBE HEAT Switch .....ON AND CHECK NO CAS THEN OFF
- b. ICE Light .....OFF
- c. BAT 1 and BAT 2 Switches.....OFF

## Before Takeoff

1. ENGINE IPS Switch .. ON AND CHECK OPERATION AND NO CAS THEN A/R
2. WING/STAB IPS Switch ... ON AND CHECK OPERATION AND NO CAS THEN OFF

• WARNING •

**Takeoff is prohibited with any frost (polished or not), ice, snow, or slush adhering to the wings, stabilizers, control surfaces, engine inlet, or fuselage forward of the engine inlet. A tactile inspection in accordance with Section 2, Limitations, must be accomplished no more than 5 minutes prior to takeoff when operating in ground icing conditions.**

## IPS Activation

***OAT ≤ 41 °F (5 °C) with visible moisture or visibility < 1 mile (1.6 km)***

In Flight:

1. PROBE HEAT Switch .....ON
2. ENGINE IPS Switch .....ON
3. WING/STAB IPS Switch ..... OFF
4. WINDSHLD IPS Switch ..... OFF

On Ground:

1. PROBE HEAT Switch .....ON
2. ENGINE IPS Switch .....ON

## ***At first sign of ice***

### **• WARNING •**

Ice contamination of the lifting surfaces may result in an inoperable or inadequate stall warning, which may cause an inadvertent stall. The resulting stall may lead to loss of control of the airplane. The

WING/STAB IPS switch must remain at the ON position until the entire wing is free of ice accretion and ice is no longer accreting on the aircraft. Stall Warning offset must remain ON until entire airframe is free of ice.

### **• NOTE •**

The left wing leading edge is the primary reference area for determining if ice is accumulating on the aircraft. Ice may be visible on other areas of the aircraft as well, such as the forward windshield.

1. PROBE HEAT Switch..... ON
2. ENGINE IPS Switch..... ON
3. WING/STAB IPS Switch..... ON

#### **◆ In RVSM Airspace:**

- a. Contact ATC and notify of non-RVSM status.

### **• NOTE •**

Verify STALL SPEED HIGH CAS Advisory is present.

4. WINDSHLD IPS Switch..... ON

### **• CAUTION •**

For every 10 minutes that the aircraft is operating on the ground during falling or blowing snow or any freezing moisture, a static engine run-up to 70% N2 for 30 seconds is required to ensure the engine intake, Tt2 probes, and Pt2 probes remain clear of ice.

Failure to comply may result in ice accretions that may cause engine damage and possible loss of available thrust.

### **• NOTE •**

Leave inlet IPS active for at least 2 minutes before turning it off again.

## ***Section 5: Performance Data***

### **Table of Contents**

|                                                                |    |
|----------------------------------------------------------------|----|
| Introduction.....                                              | 5  |
| Standard Charts .....                                          | 5  |
| Associated Conditions Affecting Performance .....              | 5  |
| Temperature Conversion.....                                    | 6  |
| OAT for International Standard Atmosphere (ISA) Condition..... | 7  |
| Pressure Conversion - Inches of Mercury to Millibars .....     | 8  |
| Fuel Quantity Conversion - U.S. Gallons to Liters .....        | 9  |
| Weight Conversion - Pounds to Kilograms.....                   | 10 |
| Distance Conversion: Feet to Meters .....                      | 11 |
| Length Conversion: Inches to Centimeters .....                 | 12 |
| Static Source Error Correction .....                           | 13 |
| Airspeed Correction.....                                       | 13 |
| Altitude Correction, Flaps: 0%.....                            | 15 |
| Altitude Correction, Flaps: 50%.....                           | 17 |
| Altitude Correction, Flaps: 100%.....                          | 18 |
| Stall Speeds .....                                             | 19 |
| Stall Speeds (Ice Contaminated/STALL SPEED HIGH) .....         | 20 |
| WING/STAB IPS: ON .....                                        | 20 |
| Wind Components.....                                           | 21 |
| Takeoff Power Setting (N1) .....                               | 22 |
| Takeoff Distance .....                                         | 22 |
| Takeoff Weight: 6000 lb (2722 kg) .....                        | 22 |
| Engine Anti-Ice: A/R .....                                     | 22 |
| Takeoff Weight: 5500 lb (2495 kg) .....                        | 24 |
| Engine Anti-Ice: A/R .....                                     | 24 |
| Takeoff Weight: 5000 lb (2268 kg) .....                        | 26 |
| Engine Anti-Ice: A/R .....                                     | 26 |
| Gear UP, Flaps 50% .....                                       | 28 |
| Takeoff Climb Gradient .....                                   | 28 |
| Takeoff Rate of Climb.....                                     | 34 |
| Enroute Obstacle Climb .....                                   | 40 |
| Engine Anti-Ice: Off .....                                     | 40 |
| Engine Anti-Ice: On .....                                      | 45 |
| Enroute Climb.....                                             | 49 |
| Climb Gradient .....                                           | 49 |
| Rate Of Climb .....                                            | 58 |

|                                                          |     |
|----------------------------------------------------------|-----|
| Enroute Ice Contaminated Climb.....                      | 67  |
| Climb Gradient.....                                      | 67  |
| Rate Of Climb (Ice Contaminated) .....                   | 75  |
| Time, Fuel, & Distance to Climb .....                    | 83  |
| ISA -10.....                                             | 83  |
| Aircraft Weight: 6000 lb (2722 kg) .....                 | 83  |
| Aircraft Weight: 5500 lb (2495 kg) .....                 | 84  |
| Aircraft Weight: 5000 lb (2268 kg) .....                 | 85  |
| Aircraft Weight: 4500 lb (2041 kg) .....                 | 86  |
| ISA.....                                                 | 87  |
| Aircraft Weight: 6000 lb (2722 kg) .....                 | 87  |
| Aircraft Weight: 5500 lb (2495 kg) .....                 | 88  |
| Aircraft Weight: 5000 lb (2268 kg) .....                 | 89  |
| Aircraft Weight: 4500 lb (2041 kg) .....                 | 90  |
| ISA +10.....                                             | 91  |
| Aircraft Weight: 6000 lb (2722 kg) .....                 | 91  |
| Aircraft Weight: 5500 lb (2495 kg) .....                 | 92  |
| Aircraft Weight: 5000 lb (2268 kg) .....                 | 93  |
| Aircraft Weight: 4500 lb (2041 kg) .....                 | 94  |
| Cruise Performance.....                                  | 95  |
| Descent Data.....                                        | 147 |
| Landing Distance.....                                    | 148 |
| Aircraft Weight: 5550 lb (2517 kg) .....                 | 149 |
| Flaps: 100%.....                                         | 149 |
| Flaps: 50%.....                                          | 151 |
| Flaps: 50% (Ice Contaminated/STALL SPEED HIGH) .....     | 153 |
| Flaps: UP .....                                          | 155 |
| Flaps: UP (Ice Contaminated/STALL SPEED HIGH) .....      | 155 |
| Aircraft Weight: 4500 lb (2041 kg) .....                 | 156 |
| Flaps: 100%.....                                         | 156 |
| Flaps: 50%.....                                          | 158 |
| Flaps: 50% (Ice Contaminated) .....                      | 160 |
| Landing Distance - Factored Data.....                    | 162 |
| Aircraft Weight: 5550 lb (2517 kg) .....                 | 162 |
| Flaps: 100%, Dry Runway .....                            | 162 |
| Flaps: 100%, Wet Runway .....                            | 164 |
| Flaps: 50%, Dry Runway .....                             | 166 |
| Flaps: 50%, Wet Runway .....                             | 168 |
| Flaps: 50% (Ice Cont/STALL SPEED HIGH), Dry Runway ..... | 170 |
| Flaps: 50% (Ice Cont/STALL SPEED HIGH), Wet Runway ..... | 172 |
| Aircraft Weight: 4500 lb (2041 kg) .....                 | 174 |
| Flaps: 100%, Dry Runway .....                            | 174 |



---

|                                                          |     |
|----------------------------------------------------------|-----|
| Flaps: 100%, Wet Runway .....                            | 176 |
| Flaps: 50%, Dry Runway .....                             | 178 |
| Flaps: 50%, Wet Runway .....                             | 180 |
| Flaps: 50% (Ice Cont/STALL SPEED HIGH), Dry Runway ..... | 182 |
| Flaps: 50% (Ice Cont/STALL SPEED HIGH), Wet Runway ..... | 184 |
| Landing Distance - Contaminated Runway .....             | 186 |
| Water Covered Runway .....                               | 186 |
| Snow Covered Runway .....                                | 187 |

Intentionally Left Blank

## **Introduction**

Performance data in this section are presented for operational planning so that you will know what performance to expect from the airplane under various ambient and field conditions. Performance data are presented for takeoff, climb, cruise (including range & endurance), and landing.

All data based on published normal procedures.

## **Standard Charts**

### **Associated Conditions Affecting Performance**

Computed performance data in this section are based upon data derived from actual flight testing with the airplane and engine in good condition and using average piloting techniques. Unless specifically noted in the “Conditions” notes presented with each table, ambient conditions are for a standard day. Flap position as well as thrust setting technique is similarly noted with each table.

The charts in this section provide data for ambient temperatures from -4 °F (-20 °C) to 104 °F (40 °C). If ambient temperature is below the chart value, use the lowest temperature shown to compute performance. This will result in more conservative performance calculations.

#### **• WARNING •**

**Use extreme caution at high density altitudes as performance degrades rapidly with increasing density altitude.**

For operations at altitudes below sea level, use Mean Sea Level (MSL) values.

RELATED LINKS:

[Section 1: General, "Meteorological Terminology".](#)

## Temperature Conversion

| Temp to Convert<br>°C or °F |     |     | Temp to Convert<br>°C or °F |     |     | Temp to Convert<br>°C or °F |     |     |
|-----------------------------|-----|-----|-----------------------------|-----|-----|-----------------------------|-----|-----|
| °C                          | < > | °F  | °C                          | < > | °F  | °C                          | < > | °F  |
| -50                         | -58 | -72 | -17                         | 2   | 36  | 17                          | 62  | 144 |
| -49                         | -56 | -69 | -16                         | 4   | 39  | 18                          | 64  | 147 |
| -48                         | -54 | -65 | -14                         | 6   | 43  | 19                          | 66  | 151 |
| -47                         | -52 | -62 | -13                         | 8   | 46  | 20                          | 68  | 154 |
| -46                         | -50 | -58 | -12                         | 10  | 50  | 21                          | 70  | 158 |
| -44                         | -48 | -54 | -11                         | 12  | 54  | 22                          | 72  | 162 |
| -43                         | -46 | -51 | -10                         | 14  | 57  | 23                          | 74  | 165 |
| -42                         | -44 | -47 | -9                          | 16  | 61  | 24                          | 76  | 169 |
| -41                         | -42 | -44 | -8                          | 18  | 64  | 26                          | 78  | 172 |
| -40                         | -40 | -40 | -7                          | 20  | 68  | 27                          | 80  | 176 |
| -39                         | -38 | -36 | -6                          | 22  | 72  | 28                          | 82  | 180 |
| -38                         | -36 | -33 | -4                          | 24  | 75  | 29                          | 84  | 183 |
| -37                         | -34 | -29 | -3                          | 26  | 79  | 30                          | 86  | 187 |
| -36                         | -32 | -26 | -2                          | 28  | 82  | 31                          | 88  | 190 |
| -34                         | -30 | -22 | -1                          | 30  | 86  | 32                          | 90  | 194 |
| -33                         | -28 | -18 | 0                           | 32  | 90  | 33                          | 92  | 198 |
| -32                         | -26 | -15 | 1                           | 34  | 93  | 34                          | 94  | 201 |
| -31                         | -24 | -11 | 2                           | 36  | 97  | 36                          | 96  | 205 |
| -30                         | -22 | -8  | 3                           | 38  | 100 | 37                          | 98  | 208 |
| -29                         | -20 | -4  | 4                           | 40  | 104 | 38                          | 100 | 212 |
| -28                         | -18 | 0   | 6                           | 42  | 108 | 39                          | 102 | 216 |
| -27                         | -16 | 3   | 7                           | 44  | 111 | 40                          | 104 | 219 |
| -26                         | -14 | 7   | 8                           | 46  | 115 | 41                          | 106 | 223 |
| -24                         | -12 | 10  | 9                           | 48  | 118 | 42                          | 108 | 226 |
| -23                         | -10 | 14  | 10                          | 50  | 122 | 43                          | 110 | 230 |
| -22                         | -8  | 18  | 11                          | 52  | 126 | 44                          | 112 | 234 |
| -21                         | -6  | 21  | 12                          | 54  | 129 | 46                          | 114 | 237 |
| -20                         | -4  | 25  | 13                          | 56  | 133 | 47                          | 116 | 241 |
| -19                         | -2  | 28  | 14                          | 58  | 136 | 48                          | 118 | 244 |
| -18                         | 0   | 32  | 16                          | 60  | 140 | 49                          | 120 | 248 |

## OAT for International Standard Atmosphere (ISA) Condition

| Press<br>Alt<br>Feet | ISA<br>-30 °C |     | ISA<br>-15 °C |     | ISA |     | ISA<br>+15 °C |     | ISA<br>+30 °C |     |
|----------------------|---------------|-----|---------------|-----|-----|-----|---------------|-----|---------------|-----|
|                      | °C            | °F  | °C            | °F  | °C  | °F  | °C            | °F  | °C            | °F  |
| SL                   | -15           | 5   | 0             | 32  | 15  | 59  | 30            | 86  | 45            | 113 |
| 1000                 | -17           | 1   | -2            | 28  | 13  | 55  | 28            | 82  | 43            | 109 |
| 2000                 | -19           | -2  | -4            | 25  | 11  | 52  | 26            | 79  | 41            | 106 |
| 3000                 | -21           | -6  | -6            | 21  | 9   | 48  | 24            | 75  | 39            | 102 |
| 4000                 | -23           | -9  | -8            | 18  | 7   | 45  | 22            | 72  | 37            | 99  |
| 5000                 | -25           | -13 | -10           | 14  | 5   | 41  | 20            | 68  | 35            | 95  |
| 6000                 | -27           | -17 | -12           | 10  | 3   | 37  | 18            | 64  | 33            | 91  |
| 7000                 | -29           | -20 | -14           | 7   | 1   | 34  | 16            | 61  | 31            | 88  |
| 8000                 | -31           | -24 | -16           | 3   | -1  | 30  | 14            | 57  | 29            | 84  |
| 9000                 | -33           | -27 | -18           | 0   | -3  | 27  | 12            | 54  | 27            | 81  |
| 10,000               | -35           | -31 | -20           | -4  | -5  | 23  | 10            | 50  | 25            | 77  |
| 11,000               | -37           | -35 | -22           | -8  | -7  | 19  | 8             | 46  | 23            | 73  |
| 12,000               | -39           | -38 | -24           | -11 | -9  | 16  | 6             | 43  | 21            | 70  |
| 13,000               | -41           | -42 | -26           | -15 | -11 | 12  | 4             | 39  | 19            | 66  |
| 14,000               | -43           | -45 | -28           | -18 | -13 | 9   | 2             | 36  | 17            | 63  |
| 15,000               | -45           | -49 | -30           | -22 | -15 | 5   | 0             | 32  | 15            | 59  |
| 16,000               | -47           | -53 | -32           | -26 | -17 | 1   | -2            | 28  | 13            | 55  |
| 17,000               | -49           | -56 | -34           | -29 | -19 | -2  | -4            | 25  | 11            | 52  |
| 18,000               | -51           | -60 | -36           | -33 | -21 | -6  | -6            | 21  | 9             | 48  |
| 19,000               | -53           | -63 | -38           | -36 | -23 | -9  | -8            | 18  | 7             | 45  |
| 20,000               | -55           | -67 | -40           | -40 | -25 | -13 | -10           | 14  | 5             | 41  |
| 21,000               | -57           | -71 | -42           | -44 | -27 | -17 | -12           | 10  | 3             | 37  |
| 22,000               | -59           | -74 | -44           | -47 | -29 | -20 | -14           | 7   | 1             | 34  |
| 23,000               | -61           | -78 | -46           | -51 | -31 | -24 | -16           | 3   | -1            | 30  |
| 24,000               | -63           | -81 | -48           | -54 | -33 | -27 | -18           | 0   | -3            | 27  |
| 25,000               | -65           | -85 | -50           | -58 | -35 | -31 | -20           | -4  | -5            | 23  |
| 26,000               | -67           | -89 | -52           | -62 | -37 | -35 | -22           | -8  | -7            | 19  |
| 27,000               | -69           | -92 | -54           | -65 | -39 | -38 | -24           | -11 | -9            | 16  |
| 28,000               | -71           | -96 | -56           | -69 | -41 | -42 | -26           | -15 | -11           | 12  |
| 29,000               | -72           | -98 | -57           | -71 | -42 | -44 | -27           | -17 | -12           | 10  |

| Press<br>Alt<br>Feet | ISA<br>-30 °C |      | ISA<br>-15 °C |     | ISA |     | ISA<br>+15 °C |     | ISA<br>+30 °C |    |
|----------------------|---------------|------|---------------|-----|-----|-----|---------------|-----|---------------|----|
|                      | °C            | °F   | °C            | °F  | °C  | °F  | °C            | °F  | °C            | °F |
| 30,000               | -74           | -102 | -59           | -75 | -44 | -48 | -29           | -21 | -14           | 6  |
| 31,000               | -76           | -106 | -61           | -79 | -46 | -52 | -31           | -25 | -16           | 2  |

Pressure Conversion - Inches of Mercury to Millibars

| Inches Of<br>Mercury | Millibars |
|----------------------|-----------|
| 28.0                 | 948       |
| 28.1                 | 951       |
| 28.2                 | 955       |
| 28.3                 | 958       |
| 28.4                 | 962       |
| 28.5                 | 965       |
| 28.6                 | 968       |
| 28.7                 | 972       |
| 28.8                 | 975       |
| 28.9                 | 979       |
| 29.0                 | 982       |
| 29.1                 | 985       |
| 29.2                 | 989       |
| 29.3                 | 992       |
| 29.4                 | 995       |
| 29.5                 | 999       |

| Inches Of<br>Mercury | Millibars |
|----------------------|-----------|
| 29.6                 | 1002      |
| 29.7                 | 1006      |
| 29.8                 | 1009      |
| 29.9                 | 1012      |
| 30.0                 | 1016      |
| 30.1                 | 1019      |
| 30.2                 | 1023      |
| 30.3                 | 1026      |
| 30.4                 | 1029      |
| 30.5                 | 1033      |
| 30.6                 | 1036      |
| 30.7                 | 1040      |
| 30.8                 | 1043      |
| 30.9                 | 1046      |
| 31.0                 | 1050      |

## Fuel Quantity Conversion - U.S. Gallons to Liters

• NOTE •

Fuel mass provided for reference assuming nominal 6.76 lb/gallon  
at 59 °F (15 °C).

| U.S. Gallons<br>(Liters) | Lb (Kg)        |
|--------------------------|----------------|
| 10 (37.9)                | 67.6 (30.7)    |
| 20 (75.7)                | 135.2 (61.3)   |
| 30 (113.6)               | 202.8 (92.0)   |
| 40 (151.4)               | 270.4 (122.7)  |
| 50 (189.3)               | 338.0 (153.3)  |
| 60 (227.1)               | 405.6 (184.0)  |
| 70 (265.0)               | 473.2 (214.6)  |
| 80 (302.8)               | 540.8 (245.3)  |
| 90 (340.7)               | 608.4 (276.0)  |
| 100 (378.5)              | 676.0 (306.6)  |
| 110 (416.4)              | 743.6 (337.3)  |
| 120 (454.2)              | 811.2 (368.0)  |
| 130 (492.1)              | 878.8 (398.6)  |
| 140 (530.0)              | 946.4 (429.3)  |
| 150 (567.8)              | 1014.0 (459.9) |

| U.S. Gallons<br>(Liters) | Lb (Kg)        |
|--------------------------|----------------|
| 160 (605.7)              | 1081.6 (490.6) |
| 170 (643.5)              | 1149.2 (521.3) |
| 180 (681.4)              | 1216.8 (551.9) |
| 190 (719.2)              | 1284.4 (582.6) |
| 200 (757.1)              | 1352.0 (613.3) |
| 210 (794.9)              | 1419.6 (643.9) |
| 220 (832.8)              | 1487.2 (674.6) |
| 230 (870.6)              | 1554.8 (705.2) |
| 240 (908.5)              | 1622.4 (735.9) |
| 250 (946.4)              | 1690.0 (766.6) |
| 260 (984.2)              | 1757.6 (797.2) |
| 270 (1022.1)             | 1825.2 (827.9) |
| 280 (1059.9)             | 1892.8 (858.6) |
| 290 (1097.8)             | 1960.4 (889.2) |
| 300 (1135.6)             | 2028.0 (919.9) |

## Weight Conversion - Pounds to Kilograms

| Pounds | Kilograms |
|--------|-----------|
| 6500   | 2948      |
| 6400   | 2903      |
| 6300   | 2858      |
| 6200   | 2812      |
| 6100   | 2767      |
| 6000   | 2722      |
| 5900   | 2676      |
| 5800   | 2631      |
| 5700   | 2585      |
| 5600   | 2540      |
| 5500   | 2495      |
| 5400   | 2449      |
| 5300   | 2404      |
| 5200   | 2359      |
| 5100   | 2313      |
| 5000   | 2268      |

| Pounds | Kilograms |
|--------|-----------|
| 4900   | 2223      |
| 4800   | 2177      |
| 4700   | 2132      |
| 4600   | 2087      |
| 4500   | 2041      |
| 4400   | 1996      |
| 4300   | 1950      |
| 4200   | 1905      |
| 4100   | 1860      |
| 4000   | 1814      |
| 3900   | 1769      |
| 3800   | 1724      |
| 3700   | 1678      |
| 3600   | 1633      |
| 3500   | 1588      |



**Distance Conversion: Feet to Meters**

| Feet | Meters |
|------|--------|
| 10   | 3      |
| 20   | 6      |
| 30   | 9      |
| 40   | 12     |
| 50   | 15     |
| 60   | 18     |
| 70   | 21     |
| 80   | 24     |
| 90   | 27     |
| 100  | 30     |
| 200  | 61     |
| 300  | 91     |
| 400  | 122    |
| 500  | 152    |

| Feet   | Meters |
|--------|--------|
| 600    | 183    |
| 700    | 213    |
| 800    | 244    |
| 900    | 274    |
| 1000   | 305    |
| 2000   | 610    |
| 3000   | 914    |
| 4000   | 1219   |
| 5000   | 1524   |
| 6000   | 1829   |
| 7000   | 2134   |
| 8000   | 2438   |
| 9000   | 2743   |
| 10,000 | 3048   |

**Length Conversion: Inches to Centimeters**

| Inches | Centimeters |
|--------|-------------|
| 1      | 2.54        |
| 2      | 5.08        |
| 3      | 7.62        |
| 4      | 10.16       |
| 5      | 12.70       |
| 6      | 15.24       |
| 7      | 17.78       |
| 8      | 20.32       |
| 9      | 22.86       |
| 10     | 25.40       |
| 11     | 27.94       |
| 12     | 30.48       |

| Inches | Centimeters |
|--------|-------------|
| 20     | 50.8        |
| 30     | 76.2        |
| 40     | 101.6       |
| 50     | 127         |
| 60     | 142.4       |
| 70     | 177.8       |
| 80     | 203.2       |
| 90     | 228.6       |
| 100    | 254         |
| 150    | 381         |
| 200    | 508         |
| 250    | 635         |

## Static Source Error Correction

### *Airspeed Correction*

**Conditions:**

- Thrust for level flight or MCT, whichever is less.
- For altitudes below FL180 use the subsequent table.
- For altitudes above FL280 indicated airspeed is equal to calibrated airspeed (zero static source error).
- For altitudes between FL180 and FL280 interpolate between the table below and zero static source error.

**• NOTE •**

Indicated airspeed values assume zero instrument error.

| KIAS | KCAS            |     |      |
|------|-----------------|-----|------|
|      | Flap Deflection |     |      |
|      | 0%              | 50% | 100% |
| 60   |                 |     | 62   |
| 70   |                 | 70  | 70   |
| 80   | 81              | 81  | 79   |
| 90   | 92              | 91  | 89   |
| 100  | 102             | 101 | 98   |
| 110  | 112             | 111 | 108  |
| 120  | 122             | 121 | 117  |
| 130  | 133             | 131 | 127  |
| 140  | 143             | 141 | 137  |
| 150  | 152             | 151 | 147  |
| 160  | 161             | 160 |      |
| 170  | 170             | 169 |      |
| 180  | 180             | 178 |      |
| 190  | 190             | 188 |      |
| 200  | 200             |     |      |
| 210  | 210             |     |      |
| 220  | 220             |     |      |
| 230  | 230             |     |      |
| 240  | 240             |     |      |
| 250  | 250             |     |      |

## ***Altitude Correction, Flaps: 0%***

### **Conditions:**

- Thrust for level flight or MCT, whichever is less.
- For altitudes below FL180 use the subsequent table.
- For altitudes above FL280 indicated airspeed is equal to calibrated airspeed (zero static source error).
- For altitudes between FL180 and FL280 interpolate between the table below and zero static source error.

### **• NOTE •**

Add correction to desired altitude to obtain indicated altitude to fly.

Indicated airspeed values assume zero instrument error.

| Indicated<br>Airspeed<br>(KIAS) | CORRECTION TO BE ADDED (ft) |      |        |        |        |        |        |
|---------------------------------|-----------------------------|------|--------|--------|--------|--------|--------|
|                                 | Indicated Altitude (ft MSL) |      |        |        |        |        |        |
|                                 | 0                           | 5000 | 10,000 | 15,000 | 20,000 | 25,000 | 28,000 |
| 80                              | 9                           | 10   | 12     | 14     | 17     | 20     | 22     |
| 90                              | 13                          | 15   | 17     | 20     | 24     | 28     | 32     |
| 100                             | 17                          | 19   | 23     | 27     | 31     | 37     | 42     |
| 110                             | 21                          | 24   | 29     | 33     | 40     | 47     | 52     |
| 120                             | 25                          | 30   | 35     | 41     | 48     | 57     | 63     |
| 130                             | 30                          | 35   | 41     | 48     | 56     | 67     | 75     |
| 140                             | 35                          | 40   | 47     | 55     | 65     | 78     | 86     |
| 150                             | 33                          | 39   | 45     | 53     | 62     | 74     | 82     |
| 160                             | 19                          | 23   | 26     | 31     | 37     | 43     | 48     |
| 170                             | 4                           | 5    | 5      | 6      | 7      | 9      | 10     |
| 180                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 190                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 200                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 210                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 220                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 230                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 240                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |
| 250                             | 0                           | 0    | 0      | 0      | 0      | 0      | 0      |

## ***Altitude Correction, Flaps: 50%***

### **Conditions:**

- Thrust for level flight or MCT, whichever is less.

### **• NOTE •**

Add correction to desired altitude to obtain indicated altitude to fly.

Indicated airspeed values assume zero instrument error.

| <b>Indicated<br/>Airspeed<br/>(KIAS)</b> | <b>CORRECTION TO BE ADDED (ft)</b> |             |               |
|------------------------------------------|------------------------------------|-------------|---------------|
|                                          | <b>Indicated Altitude (ft MSL)</b> |             |               |
|                                          | <b>0</b>                           | <b>5000</b> | <b>10,000</b> |
| <b>70</b>                                | 2                                  | 3           | 3             |
| <b>80</b>                                | 4                                  | 4           | 5             |
| <b>90</b>                                | 5                                  | 6           | 7             |
| <b>100</b>                               | 7                                  | 8           | 10            |
| <b>110</b>                               | 9                                  | 11          | 12            |
| <b>120</b>                               | 12                                 | 13          | 16            |
| <b>130</b>                               | 14                                 | 16          | 19            |
| <b>140</b>                               | 17                                 | 20          | 23            |
| <b>150</b>                               | 13                                 | 16          | 18            |
| <b>160</b>                               | -2                                 | -2          | -2            |
| <b>170</b>                               | -19                                | -22         | -25           |
| <b>180</b>                               | -29                                | -33         | -39           |
| <b>190</b>                               | -36                                | -42         | -49           |

**Altitude Correction, Flaps: 100%**

**Conditions:**

- Thrust for level flight or MCT, whichever is less.

**• NOTE •**

Add correction to desired altitude to obtain indicated altitude to fly.

Indicated airspeed values assume zero instrument error.

| Indicated<br>Airspeed<br>(KIAS) | CORRECTION TO BE ADDED (ft) |      |        |
|---------------------------------|-----------------------------|------|--------|
|                                 | Indicated Altitude (ft MSL) |      |        |
|                                 | 0                           | 5000 | 10,000 |
| 60                              | 8                           | 10   | 11     |
| 70                              | 2                           | 3    | 3      |
| 80                              | -4                          | -5   | -6     |
| 90                              | -10                         | -12  | -14    |
| 100                             | -16                         | -19  | -22    |
| 110                             | -22                         | -25  | -29    |
| 120                             | -27                         | -31  | -36    |
| 130                             | -31                         | -36  | -42    |
| 140                             | -35                         | -41  | -47    |
| 150                             | -45                         | -52  | -61    |



## Stall Speeds

| Bank Angle<br><br>Deg | STALL SPEEDS (KCAS) AT IDLE |     |      |
|-----------------------|-----------------------------|-----|------|
|                       | Flap Deflection             |     |      |
|                       | 0%                          | 50% | 100% |
| <b>6000 lb</b>        |                             |     |      |
| 0                     | 86                          | 77  | 67   |
| 15                    | 88                          | 78  | 68   |
| 30                    | 92                          | 83  | 72   |
| 45                    | 102                         | 92  | 80   |
| 60                    | 122                         | 109 | 95   |
| <b>5550 lb</b>        |                             |     |      |
| 0                     | 83                          | 74  | 64   |
| 15                    | 84                          | 75  | 66   |
| 30                    | 89                          | 80  | 69   |
| 45                    | 98                          | 88  | 77   |
| 60                    | 117                         | 105 | 91   |
| <b>5200 lb</b>        |                             |     |      |
| 0                     | 80                          | 72  | 62   |
| 15                    | 81                          | 73  | 63   |
| 30                    | 86                          | 77  | 67   |
| 45                    | 95                          | 85  | 74   |
| 60                    | 113                         | 101 | 88   |
| <b>4800 lb</b>        |                             |     |      |
| 0                     | 77                          | 69  | 60   |
| 15                    | 78                          | 70  | 61   |
| 30                    | 83                          | 74  | 64   |
| 45                    | 91                          | 82  | 71   |
| 60                    | 109                         | 97  | 85   |

## Stall Speeds (Ice Contaminated Airframe/STALL SPEED HIGH Advisory)

**WING/STAB IPS: ON**

• NOTE •

Actual stick pusher speed is dependent on physical ice accretion.

| Bank Angle<br><br>Deg | STALL SPEEDS (KCAS) AT IDLE |     |
|-----------------------|-----------------------------|-----|
|                       | Flap Deflection             |     |
|                       | 0%                          | 50% |
| <b>6000 lb</b>        |                             |     |
| 0                     | 108                         | 93  |
| 15                    | 110                         | 95  |
| 30                    | 116                         | 100 |
| 45                    | 128                         | 111 |
| <b>5550 lb</b>        |                             |     |
| 0                     | 104                         | 89  |
| 15                    | 106                         | 91  |
| 30                    | 112                         | 96  |
| 45                    | 124                         | 106 |
| <b>5200 lb</b>        |                             |     |
| 0                     | 101                         | 87  |
| 15                    | 102                         | 88  |
| 30                    | 108                         | 93  |
| 45                    | 120                         | 103 |
| <b>4800 lb</b>        |                             |     |
| 0                     | 97                          | 83  |
| 15                    | 98                          | 85  |
| 30                    | 104                         | 89  |
| 45                    | 115                         | 99  |

## Wind Components

Example:

Runway Heading ..... 10°

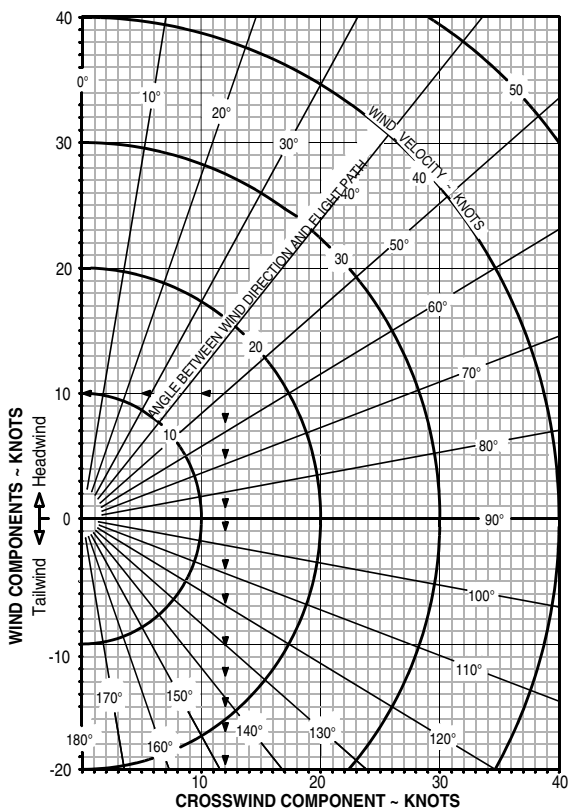
Wind Direction ..... 60°

Wind Velocity ..... 15 Knots

### • NOTE •

Maximum demonstrated crosswind is 16 knots for 100% flaps, and 18 knots for 50% flaps. Reduced flap settings enable higher crosswind limits.

**Figure 5-1: Wind Components**



SF50\_FM05\_0017

## Takeoff Power Setting (N1)

To achieve rated takeoff thrust setting, ensure that %N1 indications align with the cyan “T” takeoff thrust reference mark on the Engine Indication and Systems (EIS) %N1 indicator.

## Takeoff Distance

### Takeoff Weight: 6000 lb (2722 kg)

#### Engine Anti-Ice: A/R

Winds .....Calm  
Runway..... Dry, Level, Paved  
Thrust Lever ..... T/O  
Landing Gear ..... DOWN  
Flaps.....50%  
BLEED Switch .....ON

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 40% to the ground run and 35% to the total distance for every 10 knots of tailwind.

Downhill gradient: Subtract 2% from the ground run for every 1% of downhill gradient.

Uphill gradient: Add 14% to the ground run for every 1% of uphill gradient.

Unpaved runway: Add 21% to the total distance.

#### • NOTE •

The total distance is over a 50-foot (15-meter) obstacle.

Performance table below considers engine anti-ice off or on (maximum OAT for activation of Engine IPS is 10° C).

| <b>Takeoff Weight: 6000 lb (2722 kg)</b> |                          |                        |            |          |           |           |           |           |           |            |
|------------------------------------------|--------------------------|------------------------|------------|----------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>            | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |           |           |           |           | <b>ISA</b> |
|                                          |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                       | Gnd Roll                 | 1780                   | 1851       | 1924     | 1999      | 2231      | 2757      | 3418      | 4284      | 2036       |
|                                          | Total                    | 2789                   | 2900       | 3014     | 3133      | 3519      | 4415      | 5561      | 7093      | 3192       |
| 1000                                     | Gnd Roll                 | 1855                   | 1928       | 2005     | 2084      | 2386      | 2947      | 3652      | 4591      | 2106       |
|                                          | Total                    | 2907                   | 3022       | 3142     | 3268      | 3773      | 4732      | 5959      | 7626      | 3302       |
| 2000                                     | Gnd Roll                 | 1930                   | 2007       | 2086     | 2169      | 2586      | 3193      | 3970      | 5005      | 2175       |
|                                          | Total                    | 3025                   | 3146       | 3271     | 3402      | 4104      | 5147      | 6505      | 8350      | 3411       |
| 3000                                     | Gnd Roll                 | 2006                   | 2087       | 2170     | 2331      | 2806      | 3466      | 4323      | 5462      | 2316       |
|                                          | Total                    | 3145                   | 3272       | 3403     | 3666      | 4471      | 5609      | 7113      | 9156      | 3642       |
| 4000                                     | Gnd Roll                 | 2084                   | 2168       | 2255     | 2533      | 3048      | 3768      | 4711      | 5967      | 2452       |
|                                          | Total                    | 3268                   | 3400       | 3536     | 4000      | 4875      | 6122      | 7787      | 10,051    | 3864       |
| 5000                                     | Gnd Roll                 | 2165                   | 2252       | 2340     | 2737      | 3289      | 4069      | 5094      |           | 2542       |
|                                          | Total                    | 3395                   | 3532       | 3670     | 4336      | 5278      | 6634      | 8451      |           | 4009       |
| 6000                                     | Gnd Roll                 | 2242                   | 2332       | 2446     | 2983      | 3585      | 4440      | 5574      |           | 2613       |
|                                          | Total                    | 3515                   | 3657       | 3838     | 4747      | 5778      | 7271      | 9292      |           | 4121       |
| 7000                                     | Gnd Roll                 | 2329                   | 2439       | 2666     | 3291      | 3960      | 4905      | 6178      |           | 2737       |
|                                          | Total                    | 3651                   | 3827       | 4201     | 5263      | 6416      | 8078      | 10,361    |           | 4321       |
| 8000                                     | Gnd Roll                 | 2457                   | 2692       | 2942     | 3677      | 4429      | 5488      | 6932      |           | 2920       |
|                                          | Total                    | 3853                   | 4241       | 4654     | 5912      | 7216      | 9091      | 11,698    |           | 4619       |
| 9000                                     | Gnd Roll                 | 2735                   | 2996       | 3297     | 4141      | 4996      | 6197      |           |           | 3212       |
|                                          | Total                    | 4308                   | 4740       | 5242     | 6693      | 8187      | 10,325    |           |           | 5100       |
| 10,000                                   | Gnd Roll                 | 3046                   | 3336       | 3708     | 4661      | 5637      | 7005      |           |           | 3529       |
|                                          | Total                    | 4817                   | 5299       | 5926     | 7575      | 9289      | 11,741    |           |           | 5624       |

• NOTE •

Adjust pitch angle as needed to maintain  $V_X$  through 50 feet.

## Takeoff Weight: 5500 lb (2495 kg)

### Engine Anti-Ice: A/R

Winds .....Calm  
Runway..... Dry, Level, Paved  
Thrust Lever .....T/O  
Landing Gear ..... DOWN  
Flaps.....50%  
BLEED Switch .....ON

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 40% to the ground run and 36% to the total distance for every 10 knots of tailwind.

Downhill gradient: Subtract 1% from the ground run for every 1% of downhill gradient.

Uphill gradient: Add 13% to the ground run for every 1% of uphill gradient.

Unpaved runway: Add 21% to the total distance.

#### • NOTE •

The total distance is over a 50-foot (15-meter) obstacle.  
Performance table below considers engine anti-ice off or on  
(maximum OAT for activation of Engine IPS is 10° C).

| <b>Takeoff Weight: 5500 lb (2495 kg)</b> |                          |                        |            |          |           |           |           |           |           |            |
|------------------------------------------|--------------------------|------------------------|------------|----------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>            | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |           |           |           |           | <b>ISA</b> |
|                                          |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                       | Gnd Roll                 | 1632                   | 1697       | 1763     | 1833      | 2045      | 2528      | 3133      | 3927      | 1867       |
|                                          | Total                    | 2458                   | 2555       | 2656     | 2761      | 3099      | 3885      | 4887      | 6227      | 2812       |
| 1000                                     | Gnd Roll                 | 1700                   | 1768       | 1838     | 1911      | 2187      | 2702      | 3348      | 4209      | 1931       |
|                                          | Total                    | 2561                   | 2663       | 2769     | 2879      | 3322      | 4163      | 5236      | 6693      | 2910       |
| 2000                                     | Gnd Roll                 | 1769                   | 1840       | 1912     | 1989      | 2370      | 2927      | 3640      | 4588      | 1994       |
|                                          | Total                    | 2666                   | 2772       | 2882     | 2998      | 3613      | 4526      | 5714      | 7327      | 3006       |
| 3000                                     | Gnd Roll                 | 1839                   | 1913       | 1989     | 2137      | 2572      | 3177      | 3963      | 5006      | 2123       |
|                                          | Total                    | 2771                   | 2883       | 2998     | 3230      | 3935      | 4931      | 6247      | 8031      | 3208       |
| 4000                                     | Gnd Roll                 | 1911                   | 1988       | 2067     | 2322      | 2794      | 3454      | 4318      | 5469      | 2247       |
|                                          | Total                    | 2880                   | 2996       | 3116     | 3523      | 4290      | 5381      | 6836      | 8813      | 3404       |
| 5000                                     | Gnd Roll                 | 1985                   | 2064       | 2145     | 2508      | 3015      | 3730      | 4669      | 5919      | 2330       |
|                                          | Total                    | 2991                   | 3112       | 3234     | 3818      | 4643      | 5829      | 7417      | 9572      | 3531       |
| 6000                                     | Gnd Roll                 | 2055                   | 2138       | 2242     | 2735      | 3286      | 4070      | 5109      | 6487      | 2395       |
|                                          | Total                    | 3097                   | 3222       | 3381     | 4178      | 5081      | 6387      | 8153      | 10,540    | 3629       |
| 7000                                     | Gnd Roll                 | 2135                   | 2236       | 2444     | 3017      | 3630      | 4497      | 5663      |           | 2509       |
|                                          | Total                    | 3217                   | 3372       | 3700     | 4631      | 5641      | 7094      | 9088      |           | 3806       |
| 8000                                     | Gnd Roll                 | 2252                   | 2468       | 2697     | 3370      | 4060      | 5031      | 6355      |           | 2677       |
|                                          | Total                    | 3395                   | 3735       | 4098     | 5200      | 6342      | 7980      | 10,256    |           | 4068       |
| 9000                                     | Gnd Roll                 | 2507                   | 2747       | 3023     | 3796      | 4580      | 5681      | 7201      |           | 2944       |
|                                          | Total                    | 3794                   | 4174       | 4615     | 5885      | 7192      | 9060      | 11,690    |           | 4490       |
| 10,000                                   | Gnd Roll                 | 2792                   | 3058       | 3399     | 4273      | 5167      | 6422      | 8167      |           | 3235       |
|                                          | Total                    | 4242                   | 4665       | 5215     | 6658      | 8157      | 10,298    | 13,340    |           | 4950       |

• NOTE •

Adjust pitch angle as needed to maintain  $V_X$  through 50 feet.

## Takeoff Weight: 5000 lb (2268 kg)

### Engine Anti-Ice: A/R

Winds .....Calm  
Runway..... Dry, Level, Paved  
Thrust Lever .....T/O  
Landing Gear ..... DOWN  
Flaps.....50%  
BLEED Switch .....ON

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 40% to the ground run and 36% to the total distance for every 10 knots of tailwind.

Downhill gradient: Subtract 1% from the ground run for every 1% of downhill gradient.

Uphill gradient: Add 12% to the ground run for every 1% of uphill gradient.

Unpaved runway: Add 21% to the total distance.

#### • NOTE •

The total distance is over a 50-foot (15-meter) obstacle.  
Performance table below considers engine anti-ice off or on  
(maximum OAT for activation of Engine IPS is 10° C).



| <b>Takeoff Weight: 5000 lb (2268 kg)</b> |                          |                        |            |          |           |           |           |           |           |            |
|------------------------------------------|--------------------------|------------------------|------------|----------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>            | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |           |           |           |           | <b>ISA</b> |
|                                          |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                       | Gnd Roll                 | 1483                   | 1542       | 1603     | 1666      | 1859      | 2298      | 2848      | 3570      | 1697       |
|                                          | Total                    | 2147                   | 2232       | 2320     | 2412      | 2706      | 3388      | 4257      | 5417      | 2457       |
| 1000                                     | Gnd Roll                 | 1546                   | 1607       | 1671     | 1737      | 1988      | 2456      | 3044      | 3826      | 1755       |
|                                          | Total                    | 2237                   | 2326       | 2418     | 2515      | 2900      | 3629      | 4560      | 5821      | 2541       |
| 2000                                     | Gnd Roll                 | 1608                   | 1672       | 1739     | 1808      | 2155      | 2661      | 3309      | 4171      | 1813       |
|                                          | Total                    | 2328                   | 2421       | 2518     | 2618      | 3153      | 3945      | 4975      | 6370      | 2625       |
| 3000                                     | Gnd Roll                 | 1671                   | 1739       | 1808     | 1942      | 2338      | 2889      | 3602      | 4551      | 1930       |
|                                          | Total                    | 2420                   | 2518       | 2619     | 2820      | 3433      | 4297      | 5437      | 6980      | 2801       |
| 4000                                     | Gnd Roll                 | 1737                   | 1807       | 1879     | 2111      | 2540      | 3140      | 3926      | 4972      | 2043       |
|                                          | Total                    | 2515                   | 2617       | 2721     | 3076      | 3741      | 4688      | 5948      | 7657      | 2972       |
| 5000                                     | Gnd Roll                 | 1804                   | 1877       | 1950     | 2280      | 2741      | 3391      | 4245      | 5381      | 2118       |
|                                          | Total                    | 2613                   | 2718       | 2825     | 3332      | 4049      | 5077      | 6452      | 8315      | 3083       |
| 6000                                     | Gnd Roll                 | 1868                   | 1943       | 2038     | 2486      | 2987      | 3700      | 4645      | 5897      | 2177       |
|                                          | Total                    | 2705                   | 2814       | 2953     | 3645      | 4429      | 5561      | 7090      | 9153      | 3169       |
| 7000                                     | Gnd Roll                 | 1941                   | 2033       | 2222     | 2742      | 3300      | 4088      | 5148      | 6568      | 2281       |
|                                          | Total                    | 2810                   | 2945       | 3231     | 4039      | 4915      | 6174      | 7899      | 10,254    | 3322       |
| 8000                                     | Gnd Roll                 | 2048                   | 2243       | 2451     | 3064      | 3691      | 4574      | 5777      | 7406      | 2434       |
|                                          | Total                    | 2966                   | 3261       | 3577     | 4533      | 5523      | 6942      | 8910      | 11,629    | 3551       |
| 9000                                     | Gnd Roll                 | 2279                   | 2497       | 2748     | 3451      | 4163      | 5164      | 6546      | 8364      | 2677       |
|                                          | Total                    | 3313                   | 3643       | 4026     | 5129      | 6261      | 7878      | 10,152    | 13,204    | 3918       |
| 10,000                                   | Gnd Roll                 | 2538                   | 2780       | 3090     | 3885      | 4697      | 5838      | 7425      | 9411      | 2941       |
|                                          | Total                    | 3702                   | 4071       | 4548     | 5800      | 7099      | 8951      | 11,579    | 14,932    | 4318       |

• NOTE •

Adjust pitch angle as needed to maintain  $V_X$  through 50 feet.

**Gear UP, Flaps 50%**

**Takeoff Climb Gradient**

V<sub>x</sub> ..... 91 KIAS

Thrust Lever ..... T/O

Flaps..... 50%

Landing Gear ..... UP

| Engine Anti-Ice: A/R |             |                                         |      |      |      |
|----------------------|-------------|-----------------------------------------|------|------|------|
| Press<br>Alt<br>Feet | OAT<br>(°C) | Aircraft Weight (LB)                    |      |      |      |
|                      |             | 6000                                    | 5500 | 5000 | 4500 |
|                      |             | Climb Gradient (Feet per Nautical Mile) |      |      |      |
| 0                    | -40         | 1232                                    | 1421 | 1641 | 1901 |
|                      | -30         | 1231                                    | 1420 | 1639 | 1899 |
|                      | -20         | 1230                                    | 1419 | 1638 | 1898 |
|                      | -10         | 1239                                    | 1418 | 1637 | 1896 |
|                      | 0           | 1227                                    | 1415 | 1634 | 1893 |
|                      | 10          | 1223                                    | 1411 | 1629 | 1888 |
|                      | 20          | 1123                                    | 1302 | 1509 | 1754 |
|                      | 30          | 923                                     | 1084 | 1269 | 1488 |
|                      | 40          | 746                                     | 891  | 1057 | 1252 |
|                      | 50          | 584                                     | 714  | 862  | 1035 |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                                |             |             |             |
|-------------------------------|---------------------|------------------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>                    |             |             |             |
|                               |                     | <b>6000</b>                                    | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Climb Gradient (Feet per Nautical Mile)</b> |             |             |             |
| 1000                          | -40                 | 1232                                           | 1422        | 1642        | 1903        |
|                               | -30                 | 1233                                           | 1423        | 1643        | 1904        |
|                               | -20                 | 1231                                           | 1421        | 1640        | 1901        |
|                               | -10                 | 1231                                           | 1420        | 1640        | 1900        |
|                               | 0                   | 1228                                           | 1417        | 1637        | 1897        |
|                               | 10                  | 1225                                           | 1414        | 1633        | 1892        |
|                               | 20                  | 1094                                           | 1271        | 1475        | 1717        |
|                               | 30                  | 899                                            | 1058        | 1241        | 1457        |
|                               | 40                  | 725                                            | 868         | 1032        | 1225        |
|                               | 50                  | 563                                            | 692         | 839         | 1010        |
| 2000                          | -40                 | 1238                                           | 1428        | 1650        | 1912        |
|                               | -30                 | 1237                                           | 1427        | 1648        | 1910        |
|                               | -20                 | 1236                                           | 1426        | 1647        | 1909        |
|                               | -10                 | 1236                                           | 1426        | 1647        | 1909        |
|                               | 0                   | 1232                                           | 1422        | 1643        | 1904        |
|                               | 10                  | 1229                                           | 1419        | 1639        | 1899        |
|                               | 20                  | 1051                                           | 1224        | 1425        | 1662        |
|                               | 30                  | 862                                            | 1018        | 1198        | 1410        |
|                               | 40                  | 689                                            | 829         | 990         | 1179        |
|                               | 48                  | 561                                            | 689         | 836         | 1008        |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                                |             |             |             |
|-------------------------------|---------------------|------------------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>                    |             |             |             |
|                               |                     | <b>6000</b>                                    | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Climb Gradient (Feet per Nautical Mile)</b> |             |             |             |
| 3000                          | -40                 | 1243                                           | 1435        | 1657        | 1921        |
|                               | -30                 | 1243                                           | 1435        | 1657        | 1920        |
|                               | -20                 | 1242                                           | 1434        | 1656        | 1919        |
|                               | -10                 | 1241                                           | 1432        | 1654        | 1917        |
|                               | 0                   | 1237                                           | 1428        | 1649        | 1912        |
|                               | 10                  | 1191                                           | 1378        | 1595        | 1851        |
|                               | 20                  | 1008                                           | 1178        | 1374        | 1606        |
|                               | 30                  | 825                                            | 978         | 1154        | 1362        |
|                               | 40                  | 653                                            | 791         | 948         | 1133        |
|                               | 47                  | 543                                            | 670         | 816         | 985         |
| 4000                          | -40                 | 1252                                           | 1445        | 1669        | 1934        |
|                               | -30                 | 1251                                           | 1444        | 1667        | 1933        |
|                               | -20                 | 1251                                           | 1443        | 1667        | 1932        |
|                               | -10                 | 1248                                           | 1440        | 1663        | 1928        |
|                               | 0                   | 1246                                           | 1438        | 1661        | 1926        |
|                               | 10                  | 1142                                           | 1325        | 1537        | 1787        |
|                               | 20                  | 965                                            | 1132        | 1324        | 1551        |
|                               | 30                  | 786                                            | 936         | 1109        | 1311        |
|                               | 40                  | 618                                            | 753         | 907         | 1088        |
|                               | 45                  | 539                                            | 667         | 813         | 982         |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                                |             |             |             |
|-------------------------------|---------------------|------------------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>                    |             |             |             |
|                               |                     | <b>6000</b>                                    | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Climb Gradient (Feet per Nautical Mile)</b> |             |             |             |
| 5000                          | -40                 | 1052                                           | 1225        | 1426        | 1663        |
|                               | -30                 | 1052                                           | 1225        | 1426        | 1663        |
|                               | -20                 | 1050                                           | 1224        | 1424        | 1661        |
|                               | -10                 | 1049                                           | 1222        | 1422        | 1659        |
|                               | 0                   | 1046                                           | 1219        | 1420        | 1656        |
|                               | 10                  | 893                                            | 1052        | 1235        | 1451        |
|                               | 20                  | 724                                            | 868         | 1032        | 1225        |
|                               | 30                  | 553                                            | 681         | 827         | 997         |
|                               | 40                  | 391                                            | 504         | 632         | 781         |
|                               | 45                  | 317                                            | 423         | 543         | 682         |
| 6000                          | -40                 | 1062                                           | 1237        | 1440        | 1679        |
|                               | -30                 | 1063                                           | 1238        | 1440        | 1680        |
|                               | -20                 | 1061                                           | 1236        | 1438        | 1677        |
|                               | -10                 | 1060                                           | 1235        | 1437        | 1676        |
|                               | 0                   | 1057                                           | 1232        | 1434        | 1672        |
|                               | 10                  | 856                                            | 1012        | 1191        | 1403        |
|                               | 20                  | 692                                            | 833         | 995         | 1184        |
|                               | 30                  | 524                                            | 650         | 794         | 961         |
|                               | 40                  | 365                                            | 476         | 603         | 748         |
|                               | 44                  | 307                                            | 413         | 533         | 671         |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                                |             |             |             |
|-------------------------------|---------------------|------------------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>                    |             |             |             |
|                               |                     | <b>6000</b>                                    | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Climb Gradient (Feet per Nautical Mile)</b> |             |             |             |
| 7000                          | -40                 | 1074                                           | 1251        | 1455        | 1697        |
|                               | -30                 | 1073                                           | 1250        | 1454        | 1696        |
|                               | -20                 | 1072                                           | 1248        | 1452        | 1693        |
|                               | -10                 | 1070                                           | 1246        | 1450        | 1691        |
|                               | 0                   | 1010                                           | 1181        | 1378        | 1611        |
|                               | 10                  | 796                                            | 947         | 1121        | 1325        |
|                               | 20                  | 638                                            | 775         | 931         | 1114        |
|                               | 30                  | 475                                            | 597         | 736         | 897         |
|                               | 40                  | 321                                            | 428         | 550         | 691         |
|                               | 44                  | 263                                            | 366         | 482         | 615         |
| 8000                          | -40                 | 1061                                           | 1235        | 1437        | 1675        |
|                               | -30                 | 1059                                           | 1233        | 1435        | 1673        |
|                               | -20                 | 1058                                           | 1232        | 1433        | 1671        |
|                               | -10                 | 1008                                           | 1177        | 1373        | 1604        |
|                               | 0                   | 929                                            | 1091        | 1278        | 1499        |
|                               | 10                  | 722                                            | 865         | 1030        | 1223        |
|                               | 20                  | 581                                            | 716         | 871         | 1051        |
|                               | 30                  | 418                                            | 539         | 676         | 834         |
|                               | 40                  | 262                                            | 368         | 488         | 625         |
|                               | 43                  |                                                | 321         | 436         | 567         |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                                |             |             |             |
|-------------------------------|---------------------|------------------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>                    |             |             |             |
|                               |                     | <b>6000</b>                                    | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Climb Gradient (Feet per Nautical Mile)</b> |             |             |             |
| 9000                          | -40                 | 1071                                           | 1247        | 1451        | 1691        |
|                               | -30                 | 1071                                           | 1247        | 1451        | 1691        |
|                               | -20                 | 1021                                           | 1192        | 1390        | 1624        |
|                               | -10                 | 955                                            | 1120        | 1311        | 1536        |
|                               | 0                   | 868                                            | 1026        | 1207        | 1420        |
|                               | 10                  | 665                                            | 804         | 963         | 1148        |
|                               | 20                  | 527                                            | 658         | 808         | 981         |
|                               | 30                  | 370                                            | 487         | 620         | 772         |
|                               | 40                  |                                                | 321         | 437         | 569         |
|                               | 42                  |                                                | 290         | 403         | 531         |
| 10,000                        | -40                 | 1075                                           | 1251        | 1456        | 1697        |
|                               | -30                 | 1027                                           | 1199        | 1398        | 1633        |
|                               | -20                 | 968                                            | 1135        | 1327        | 1555        |
|                               | -10                 | 904                                            | 1065        | 1251        | 1470        |
|                               | 0                   | 809                                            | 962         | 1137        | 1343        |
|                               | 10                  | 612                                            | 746         | 900         | 1080        |
|                               | 20                  | 474                                            | 601         | 745         | 913         |
|                               | 30                  | 323                                            | 436         | 564         | 711         |
|                               | 40                  |                                                | 274         | 386         | 514         |
|                               | 41                  |                                                | 259         | 369         | 495         |

Takeoff Rate of Climb

V<sub>x</sub> ..... 91 KIAS

Thrust Lever ..... T/O

Flaps..... 50%

Landing Gear ..... UP

| Engine Anti-Ice: A/R |             |                                 |      |      |      |
|----------------------|-------------|---------------------------------|------|------|------|
| Press<br>Alt<br>Feet | OAT<br>(°C) | Aircraft Weight (LB)            |      |      |      |
|                      |             | 6000                            | 5500 | 5000 | 4500 |
|                      |             | Rate of Climb (Feet per Minute) |      |      |      |
| 0                    | -40         | 2099                            | 2421 | 2795 | 3238 |
|                      | -30         | 2055                            | 2371 | 2737 | 3170 |
|                      | -20         | 2012                            | 2321 | 2679 | 3104 |
|                      | -10         | 1972                            | 2275 | 2626 | 3043 |
|                      | 0           | 1933                            | 2229 | 2574 | 2982 |
|                      | 10          | 1892                            | 2183 | 2521 | 2921 |
|                      | 20          | 1707                            | 1979 | 2295 | 2667 |
|                      | 30          | 1381                            | 1621 | 1899 | 2225 |
|                      | 40          | 1097                            | 1310 | 1554 | 1841 |
|                      | 50          | 846                             | 1033 | 1248 | 1499 |



| <b>Engine Anti-Ice: A/R</b>   |                     |                                        |             |             |             |
|-------------------------------|---------------------|----------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>            |             |             |             |
|                               |                     | <b>6000</b>                            | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Rate of Climb (Feet per Minute)</b> |             |             |             |
| 1000                          | -40                 | 2126                                   | 2452        | 2832        | 3281        |
|                               | -30                 | 2081                                   | 2401        | 2772        | 3212        |
|                               | -20                 | 2037                                   | 2350        | 2714        | 3145        |
|                               | -10                 | 1997                                   | 2304        | 2660        | 3083        |
|                               | 0                   | 1956                                   | 2257        | 2607        | 3021        |
|                               | 10                  | 1915                                   | 2210        | 2553        | 2958        |
|                               | 20                  | 1682                                   | 1954        | 2268        | 2640        |
|                               | 30                  | 1359                                   | 1599        | 1877        | 2203        |
|                               | 40                  | 1078                                   | 1291        | 1535        | 1821        |
|                               | 50                  | 825                                    | 1013        | 1228        | 1478        |
| 2000                          | -40                 | 2158                                   | 2490        | 2876        | 3332        |
|                               | -30                 | 2110                                   | 2435        | 2813        | 3260        |
|                               | -20                 | 2068                                   | 2386        | 2756        | 3194        |
|                               | -10                 | 2027                                   | 2339        | 2701        | 3130        |
|                               | 0                   | 1985                                   | 2291        | 2646        | 3067        |
|                               | 10                  | 1943                                   | 2243        | 2591        | 3003        |
|                               | 20                  | 1634                                   | 1903        | 2215        | 2583        |
|                               | 30                  | 1317                                   | 1556        | 1831        | 2154        |
|                               | 40                  | 1036                                   | 1247        | 1489        | 1772        |
|                               | 48                  | 833                                    | 1024        | 1242        | 1497        |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                        |             |             |             |
|-------------------------------|---------------------|----------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>            |             |             |             |
|                               |                     | <b>6000</b>                            | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Rate of Climb (Feet per Minute)</b> |             |             |             |
| 3000                          | -40                 | 2193                                   | 2531        | 2923        | 3387        |
|                               | -30                 | 2145                                   | 2476        | 2859        | 3314        |
|                               | -20                 | 2101                                   | 2425        | 2800        | 3246        |
|                               | -10                 | 2058                                   | 2375        | 2744        | 3180        |
|                               | 0                   | 2015                                   | 2326        | 2687        | 3115        |
|                               | 10                  | 1906                                   | 2205        | 2551        | 2991        |
|                               | 20                  | 1585                                   | 1852        | 2161        | 2525        |
|                               | 30                  | 1274                                   | 1511        | 1784        | 2104        |
|                               | 40                  | 994                                    | 1203        | 1443        | 1724        |
|                               | 47                  | 816                                    | 1008        | 1227        | 1483        |
| 4000                          | -40                 | 2233                                   | 2577        | 2976        | 3449        |
|                               | -30                 | 2184                                   | 2521        | 2912        | 3375        |
|                               | -20                 | 2139                                   | 2469        | 2851        | 3305        |
|                               | -10                 | 2095                                   | 2418        | 2793        | 3237        |
|                               | 0                   | 2052                                   | 2368        | 2736        | 3171        |
|                               | 10                  | 1849                                   | 2145        | 2487        | 2893        |
|                               | 20                  | 1535                                   | 1799        | 2105        | 2466        |
|                               | 30                  | 1229                                   | 1464        | 1734        | 2051        |
|                               | 40                  | 950                                    | 1158        | 1395        | 1672        |
|                               | 45                  | 823                                    | 1018        | 1241        | 1500        |

| Engine Anti-Ice: A/R |             |                                 |      |      |      |
|----------------------|-------------|---------------------------------|------|------|------|
| Press<br>Alt<br>Feet | OAT<br>(°C) | Aircraft Weight (LB)            |      |      |      |
|                      |             | 6000                            | 5500 | 5000 | 4500 |
|                      |             | Rate of Climb (Feet per Minute) |      |      |      |
| 5000                 | -40         | 1998                            | 2211 | 2573 | 3001 |
|                      | -30         | 1857                            | 2164 | 2518 | 2937 |
|                      | -20         | 1818                            | 2119 | 2466 | 2876 |
|                      | -10         | 1780                            | 2074 | 2414 | 2816 |
|                      | 0           | 1743                            | 2031 | 2364 | 2758 |
|                      | 10          | 1461                            | 1721 | 2021 | 2374 |
|                      | 20          | 1165                            | 1396 | 1661 | 1972 |
|                      | 30          | 875                             | 1077 | 1308 | 1578 |
|                      | 40          | 609                             | 785  | 984  | 1216 |
|                      | 45          | 489                             | 653  | 839  | 1053 |
| 6000                 | -40         | 1939                            | 2258 | 2628 | 3064 |
|                      | -30         | 1898                            | 2211 | 2572 | 3000 |
|                      | -20         | 1858                            | 2165 | 2519 | 2938 |
|                      | -10         | 1820                            | 2120 | 2467 | 2877 |
|                      | 0           | 1783                            | 2077 | 2417 | 2819 |
|                      | 10          | 1417                            | 1675 | 1973 | 2323 |
|                      | 20          | 1126                            | 1355 | 1619 | 1927 |
|                      | 30          | 839                             | 1040 | 1270 | 1537 |
|                      | 40          | 575                             | 750  | 949  | 1179 |
|                      | 44          | 480                             | 646  | 833  | 1049 |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                        |             |             |             |
|-------------------------------|---------------------|----------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>            |             |             |             |
|                               |                     | <b>6000</b>                            | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Rate of Climb (Feet per Minute)</b> |             |             |             |
| 7000                          | -40                 | 1981                                   | 2308        | 2685        | 3131        |
|                               | -30                 | 1939                                   | 2259        | 2628        | 3064        |
|                               | -20                 | 1899                                   | 2211        | 2573        | 3000        |
|                               | -10                 | 1859                                   | 2165        | 2519        | 2938        |
|                               | 0                   | 1723                                   | 2014        | 2350        | 2747        |
|                               | 10                  | 1332                                   | 1585        | 1876        | 2218        |
|                               | 20                  | 1049                                   | 1274        | 1532        | 1833        |
|                               | 30                  | 769                                    | 966         | 1191        | 1452        |
|                               | 40                  | 511                                    | 682         | 876         | 1100        |
|                               | 44                  | 417                                    | 579         | 762         | 973         |
| 8000                          | -40                 | 1980                                   | 2305        | 2682        | 3127        |
|                               | -30                 | 1938                                   | 2256        | 2625        | 3060        |
|                               | -20                 | 1897                                   | 2209        | 2570        | 2996        |
|                               | -10                 | 1771                                   | 2069        | 2413        | 2819        |
|                               | 0                   | 1602                                   | 1882        | 2204        | 2585        |
|                               | 10                  | 1223                                   | 1466        | 1745        | 2071        |
|                               | 20                  | 968                                    | 1193        | 1451        | 1751        |
|                               | 30                  | 685                                    | 883         | 1107        | 1367        |
|                               | 40                  | 422                                    | 593         | 786         | 1008        |
|                               | 43                  |                                        | 514         | 698         | 910         |

| <b>Engine Anti-Ice: A/R</b>   |                     |                                        |             |             |             |
|-------------------------------|---------------------|----------------------------------------|-------------|-------------|-------------|
| <b>Press<br/>Alt<br/>Feet</b> | <b>OAT<br/>(°C)</b> | <b>Aircraft Weight (LB)</b>            |             |             |             |
|                               |                     | <b>6000</b>                            | <b>5500</b> | <b>5000</b> | <b>4500</b> |
|                               |                     | <b>Rate of Climb (Feet per Minute)</b> |             |             |             |
| 9000                          | -40                 | 2025                                   | 2357        | 2742        | 3196        |
|                               | -30                 | 1982                                   | 2307        | 2683        | 3128        |
|                               | -20                 | 1852                                   | 2163        | 2521        | 2945        |
|                               | -10                 | 1700                                   | 1993        | 2332        | 2732        |
|                               | 0                   | 1515                                   | 1790        | 2106        | 2478        |
|                               | 10                  | 1141                                   | 1379        | 1651        | 1970        |
|                               | 20                  | 888                                    | 1109        | 1360        | 1653        |
|                               | 30                  | 614                                    | 807         | 1027        | 1280        |
|                               | 40                  |                                        | 523         | 712         | 928         |
|                               | 42                  |                                        | 472         | 655         | 864         |
| 10,000                        | -40                 | 2055                                   | 2393        | 2784        | 3246        |
|                               | -30                 | 1923                                   | 2245        | 2618        | 3059        |
|                               | -20                 | 1777                                   | 2083        | 2437        | 2854        |
|                               | -10                 | 1628                                   | 1918        | 2252        | 2646        |
|                               | 0                   | 1430                                   | 1699        | 2009        | 2373        |
|                               | 10                  | 1061                                   | 1294        | 1561        | 1872        |
|                               | 20                  | 808                                    | 1025        | 1271        | 1557        |
|                               | 30                  | 542                                    | 732         | 946         | 1193        |
|                               | 40                  |                                        | 453         | 637         | 847         |
|                               | 41                  |                                        | 427         | 609         | 816         |

Enroute Obstacle Climb

Engine Anti-Ice: Off

Thrust Lever .....MCT  
Flaps..... UP  
Landing Gear ..... UP  
Anti-ice ..... OFF  
Airspeed ..... 120 KIAS

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 0                  | -54       | 2475                | 1057                | 2804                | 1197                | 3188                | 1362                | 3646                | 1557                |
|                    | -50       | 2452                | 1057                | 2778                | 1197                | 3158                | 1361                | 3613                | 1557                |
|                    | -40       | 2399                | 1057                | 2717                | 1197                | 3089                | 1361                | 3533                | 1557                |
|                    | -30       | 2348                | 1056                | 2659                | 1196                | 3023                | 1360                | 3458                | 1556                |
|                    | -20       | 2297                | 1055                | 2602                | 1195                | 2959                | 1358                | 3384                | 1554                |
|                    | -10       | 2251                | 1054                | 2550                | 1193                | 2899                | 1357                | 3316                | 1552                |
|                    | 0         | 2205                | 1051                | 2497                | 1191                | 2840                | 1354                | 3248                | 1549                |
|                    | 10        | 2077                | 1008                | 2356                | 1144                | 2682                | 1302                | 3072                | 1491                |
|                    | 20        | 1629                | 805                 | 1866                | 922                 | 2142                | 1058                | 2469                | 1220                |
|                    | 30        | 1247                | 627                 | 1448                | 727                 | 1680                | 844                 | 1955                | 982                 |
|                    | 40        | 926                 | 473                 | 1096                | 559                 | 1291                | 659                 | 1522                | 777                 |
|                    | 50        | 640                 | 332                 | 782                 | 406                 | 945                 | 490                 | 1135                | 589                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 2000               | -54       | 2484                | 1041                | 2815                | 1181                | 3203                | 1343                | 3666                | 1537                |
|                    | -50       | 2461                | 1041                | 2789                | 1180                | 3174                | 1343                | 3633                | 1537                |
|                    | -40       | 2407                | 1041                | 2728                | 1180                | 3104                | 1342                | 3552                | 1536                |
|                    | -30       | 2353                | 1039                | 2667                | 1178                | 3035                | 1341                | 3474                | 1534                |
|                    | -20       | 2304                | 1039                | 2612                | 1177                | 2972                | 1340                | 3402                | 1533                |
|                    | -10       | 2258                | 1037                | 2559                | 1176                | 2912                | 1338                | 3333                | 1531                |
|                    | 0         | 2210                | 1035                | 2506                | 1173                | 2851                | 1335                | 3264                | 1528                |
|                    | 10        | 1910                | 910                 | 2176                | 1037                | 2487                | 1185                | 2857                | 1362                |
|                    | 20        | 1482                | 719                 | 1708                | 828                 | 1970                | 955                 | 2281                | 1106                |
|                    | 30        | 1121                | 553                 | 1312                | 647                 | 1533                | 756                 | 1794                | 885                 |
|                    | 40        | 812                 | 407                 | 973                 | 488                 | 1159                | 581                 | 1377                | 690                 |
|                    | 45        | 666                 | 336                 | 813                 | 411                 | 982                 | 496                 | 1179                | 596                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 4000               | -54       | 2491                | 1021                | 2826                | 1158                | 3217                | 1319                | 3684                | 1510                |
|                    | -50       | 2468                | 1021                | 2800                | 1158                | 3187                | 1318                | 3650                | 1510                |
|                    | -40       | 2413                | 1020                | 2738                | 1157                | 3117                | 1318                | 3569                | 1509                |
|                    | -30       | 2360                | 1019                | 2677                | 1156                | 3048                | 1316                | 3490                | 1507                |
|                    | -20       | 2310                | 1018                | 2621                | 1154                | 2984                | 1314                | 3417                | 1505                |
|                    | -10       | 2262                | 1016                | 2566                | 1152                | 2922                | 1312                | 3346                | 1503                |
|                    | 0         | 2214                | 1013                | 2512                | 1149                | 2860                | 1308                | 3275                | 1498                |
|                    | 10        | 1772                | 826                 | 2028                | 945                 | 2326                | 1084                | 2680                | 1249                |
|                    | 20        | 1360                | 645                 | 1576                | 747                 | 1827                | 866                 | 2124                | 1007                |
|                    | 30        | 1015                | 489                 | 1198                | 578                 | 1410                | 680                 | 1659                | 800                 |
|                    | 40        | 712                 | 349                 | 867                 | 425                 | 1044                | 511                 | 1250                | 613                 |
| 6000               | -54       | 2401                | 962                 | 2730                | 1093                | 3114                | 1247                | 3572                | 1431                |
|                    | -50       | 2377                | 960                 | 2702                | 1092                | 3083                | 1246                | 3536                | 1429                |
|                    | -40       | 2324                | 960                 | 2643                | 1092                | 3015                | 1245                | 3458                | 1428                |
|                    | -30       | 2274                | 959                 | 2586                | 1091                | 2950                | 1244                | 3383                | 1427                |
|                    | -20       | 2226                | 958                 | 2531                | 1089                | 2887                | 1243                | 3312                | 1425                |
|                    | -10       | 2179                | 956                 | 2478                | 1087                | 2827                | 1240                | 3242                | 1423                |
|                    | 0         | 2024                | 905                 | 2307                | 1031                | 2637                | 1179                | 3029                | 1354                |
|                    | 10        | 1583                | 721                 | 1824                | 830                 | 2103                | 957                 | 2434                | 1108                |
|                    | 20        | 1194                | 553                 | 1397                | 647                 | 1632                | 756                 | 1910                | 884                 |
|                    | 30        | 868                 | 409                 | 1040                | 490                 | 1237                | 583                 | 1469                | 692                 |
|                    | 36        | 690                 | 328                 | 845                 | 402                 | 1022                | 486                 | 1228                | 584                 |



| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 8000               | -54       | 2239                | 876                 | 2556                | 1000                | 2925                | 1144                | 3364                | 1316                |
|                    | -50       | 2218                | 876                 | 2532                | 999                 | 2898                | 1144                | 3333                | 1315                |
|                    | -40       | 2168                | 875                 | 2475                | 999                 | 2833                | 1143                | 3258                | 1314                |
|                    | -30       | 2121                | 874                 | 2421                | 998                 | 2771                | 1142                | 3187                | 1313                |
|                    | -20       | 2077                | 873                 | 2370                | 997                 | 2713                | 1140                | 3120                | 1312                |
|                    | -10       | 2033                | 871                 | 2321                | 995                 | 2656                | 1138                | 3055                | 1309                |
|                    | 0         | 1777                | 776                 | 2039                | 890                 | 2344                | 1024                | 2706                | 1182                |
|                    | 10        | 1370                | 609                 | 1593                | 708                 | 1851                | 823                 | 2157                | 959                 |
|                    | 20        | 1010                | 457                 | 1198                | 542                 | 1416                | 640                 | 1671                | 756                 |
|                    | 30        | 702                 | 323                 | 861                 | 396                 | 1042                | 480                 | 1254                | 577                 |
|                    | 31        | 673                 | 310                 | 829                 | 382                 | 1008                | 464                 | 1216                | 560                 |
| 10,000             | -54       | 2081                | 795                 | 2386                | 911                 | 2740                | 1047                | 3162                | 1207                |
|                    | -50       | 2062                | 795                 | 2363                | 911                 | 2715                | 1046                | 3132                | 1207                |
|                    | -40       | 2015                | 794                 | 2310                | 910                 | 2653                | 1045                | 3061                | 1206                |
|                    | -30       | 1971                | 793                 | 2260                | 909                 | 2596                | 1044                | 2994                | 1205                |
|                    | -20       | 1929                | 792                 | 2212                | 908                 | 2541                | 1043                | 2931                | 1203                |
|                    | -10       | 1888                | 790                 | 2164                | 906                 | 2486                | 1040                | 2868                | 1200                |
|                    | 0         | 1549                | 660                 | 1793                | 764                 | 2075                | 885                 | 2410                | 1027                |
|                    | 10        | 1166                | 506                 | 1373                | 596                 | 1612                | 700                 | 1893                | 822                 |
|                    | 20        | 836                 | 369                 | 1011                | 447                 | 1211                | 535                 | 1446                | 639                 |
|                    | 27        | 628                 | 281                 | 783                 | 350                 | 959                 | 429                 | 1164                | 520                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 12,000             | -54       | 1932                | 720                 | 2225                | 830                 | 2567                | 957                 | 2971                | 1108                |
|                    | -50       | 1914                | 720                 | 2205                | 829                 | 2543                | 957                 | 2944                | 1107                |
|                    | -40       | 1871                | 719                 | 2155                | 829                 | 2486                | 956                 | 2877                | 1106                |
|                    | -30       | 1830                | 719                 | 2109                | 828                 | 2432                | 955                 | 2815                | 1105                |
|                    | -20       | 1792                | 718                 | 2064                | 827                 | 2381                | 954                 | 2756                | 1104                |
|                    | -10       | 1715                | 701                 | 1978                | 808                 | 2284                | 933                 | 2645                | 1081                |
|                    | 0         | 1321                | 550                 | 1546                | 644                 | 1806                | 752                 | 2113                | 879                 |
|                    | 10        | 967                 | 410                 | 1158                | 491                 | 1377                | 584                 | 1634                | 693                 |
|                    | 20        | 664                 | 286                 | 826                 | 356                 | 1010                | 435                 | 1224                | 528                 |
|                    | 22        | 605                 | 262                 | 761                 | 329                 | 938                 | 406                 | 1144                | 495                 |
| 14,000             | -54       | 1779                | 647                 | 2061                | 750                 | 2389                | 869                 | 2776                | 1010                |
|                    | -50       | 1762                | 647                 | 2041                | 749                 | 2366                | 868                 | 2749                | 1009                |
|                    | -40       | 1722                | 646                 | 1995                | 748                 | 2312                | 867                 | 2687                | 1008                |
|                    | -30       | 1684                | 645                 | 1951                | 748                 | 2261                | 866                 | 2628                | 1007                |
|                    | -20       | 1647                | 644                 | 1909                | 746                 | 2212                | 865                 | 2571                | 1005                |
|                    | -10       | 1473                | 587                 | 1717                | 684                 | 1999                | 797                 | 2331                | 929                 |
|                    | 0         | 1107                | 450                 | 1315                | 534                 | 1555                | 631                 | 1836                | 745                 |
|                    | 10        | 781                 | 323                 | 957                 | 396                 | 1158                | 479                 | 1393                | 576                 |
|                    | 17        | 586                 | 245                 | 743                 | 311                 | 921                 | 386                 | 1128                | 472                 |

Engine Anti-Ice: On

Thrust Lever ..... MCT  
Flaps..... UP  
Landing Gear..... UP  
Anti-ice ..... ON  
Airspeed ..... 120 KIAS

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 0                  | -54       | 2471                | 1056                | 2799                | 1196                | 3183                | 1360                | 3641                | 1555                |
|                    | -50       | 2448                | 1055                | 2773                | 1195                | 3154                | 1359                | 3607                | 1555                |
|                    | -40       | 2395                | 1055                | 2712                | 1195                | 3084                | 1359                | 3528                | 1554                |
|                    | -30       | 2344                | 1054                | 2655                | 1194                | 3019                | 1358                | 3453                | 1553                |
|                    | -20       | 2294                | 1053                | 2598                | 1193                | 2954                | 1356                | 3379                | 1551                |
|                    | -10       | 2247                | 1052                | 2546                | 1192                | 2895                | 1355                | 3311                | 1550                |
|                    | 0         | 2201                | 1050                | 2493                | 1189                | 2835                | 1352                | 3243                | 1547                |
|                    | 10        | 2045                | 993                 | 2321                | 1127                | 2644                | 1284                | 3029                | 1471                |
| 2000               | -54       | 2479                | 1040                | 2811                | 1179                | 3198                | 1341                | 3661                | 1535                |
|                    | -50       | 2456                | 1039                | 2785                | 1178                | 3169                | 1341                | 3627                | 1535                |
|                    | -40       | 2402                | 1039                | 2723                | 1178                | 3099                | 1340                | 3547                | 1534                |
|                    | -30       | 2349                | 1037                | 2663                | 1176                | 3030                | 1338                | 3468                | 1532                |
|                    | -20       | 2300                | 1037                | 2608                | 1175                | 2967                | 1337                | 3396                | 1531                |
|                    | -10       | 2254                | 1036                | 2555                | 1174                | 2907                | 1336                | 3328                | 1529                |
|                    | 0         | 2206                | 1033                | 2502                | 1171                | 2847                | 1333                | 3258                | 1525                |
|                    | 10        | 1878                | 895                 | 2142                | 1021                | 2449                | 1167                | 2815                | 1342                |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 4000               | -54       | 2486                | 1019                | 2821                | 1156                | 3212                | 1316                | 3678                | 1507                |
|                    | -50       | 2463                | 1019                | 2795                | 1156                | 3182                | 1316                | 3644                | 1507                |
|                    | -40       | 2409                | 1018                | 2733                | 1155                | 3112                | 1315                | 3563                | 1506                |
|                    | -30       | 2355                | 1017                | 2672                | 1153                | 3042                | 1313                | 3484                | 1504                |
|                    | -20       | 2306                | 1016                | 2616                | 1152                | 2978                | 1312                | 3411                | 1502                |
|                    | -10       | 2258                | 1014                | 2561                | 1150                | 2916                | 1310                | 3340                | 1500                |
|                    | 0         | 2207                | 1010                | 2504                | 1146                | 2851                | 1305                | 3265                | 1494                |
|                    | 10        | 1738                | 810                 | 1991                | 928                 | 2285                | 1065                | 2635                | 1227                |
| 6000               | -54       | 2396                | 960                 | 2725                | 1091                | 3109                | 1245                | 3566                | 1428                |
|                    | -50       | 2372                | 959                 | 2697                | 1090                | 3077                | 1243                | 3530                | 1426                |
|                    | -40       | 2320                | 958                 | 2638                | 1089                | 3009                | 1243                | 3452                | 1426                |
|                    | -30       | 2270                | 957                 | 2581                | 1089                | 2944                | 1242                | 3377                | 1424                |
|                    | -20       | 2222                | 956                 | 2526                | 1087                | 2882                | 1240                | 3306                | 1423                |
|                    | -10       | 2175                | 955                 | 2474                | 1086                | 2822                | 1238                | 3237                | 1421                |
|                    | 0         | 1990                | 890                 | 2269                | 1014                | 2595                | 1160                | 2983                | 1334                |
|                    | 10        | 1547                | 704                 | 1785                | 812                 | 2060                | 938                 | 2387                | 1087                |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 8000               | -54       | 2234                | 874                 | 2550                | 998                 | 2919                | 1142                | 3357                | 1313                |
|                    | -50       | 2213                | 874                 | 2527                | 997                 | 2892                | 1141                | 3326                | 1313                |
|                    | -40       | 2164                | 873                 | 2470                | 996                 | 2827                | 1140                | 3251                | 1312                |
|                    | -30       | 2117                | 872                 | 2416                | 995                 | 2765                | 1139                | 3181                | 1310                |
|                    | -20       | 2072                | 871                 | 2365                | 994                 | 2707                | 1138                | 3114                | 1309                |
|                    | -10       | 2028                | 869                 | 2315                | 992                 | 2650                | 1136                | 3048                | 1307                |
|                    | 0         | 1742                | 761                 | 2001                | 874                 | 2302                | 1005                | 2659                | 1161                |
|                    | 10        | 1332                | 592                 | 1551                | 690                 | 1806                | 803                 | 2106                | 936                 |
| 10,000             | -54       | 2076                | 793                 | 2380                | 909                 | 2734                | 1044                | 3155                | 1205                |
|                    | -50       | 2057                | 793                 | 2358                | 909                 | 2709                | 1044                | 3126                | 1205                |
|                    | -40       | 2010                | 792                 | 2305                | 908                 | 2647                | 1043                | 3055                | 1203                |
|                    | -30       | 1966                | 791                 | 2254                | 907                 | 2590                | 1042                | 2988                | 1202                |
|                    | -20       | 1924                | 790                 | 2206                | 906                 | 2534                | 1040                | 2924                | 1200                |
|                    | -10       | 1883                | 788                 | 2159                | 904                 | 2480                | 1038                | 2862                | 1198                |
|                    | 0         | 1512                | 645                 | 1753                | 747                 | 2031                | 866                 | 2361                | 1007                |
|                    | 10        | 1128                | 490                 | 1331                | 578                 | 1566                | 680                 | 1842                | 799                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight     |                     |                     |                     |                     |                     |                     |                     |
|--------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    |           | 6000 lb             |                     | 5500 lb             |                     | 5000 lb             |                     | 4500 lb             |                     |
|                    |           | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) | ROC<br>(FT/<br>MIN) | Grad<br>(FT/<br>NM) |
| 12,000             | -54       | 1927                | 718                 | 2220                | 828                 | 2561                | 955                 | 2964                | 1105                |
|                    | -50       | 1909                | 718                 | 2200                | 827                 | 2537                | 954                 | 2937                | 1105                |
|                    | -40       | 1866                | 717                 | 2150                | 827                 | 2480                | 953                 | 2871                | 1104                |
|                    | -30       | 1825                | 717                 | 2103                | 826                 | 2426                | 953                 | 2808                | 1103                |
|                    | -20       | 1787                | 716                 | 2059                | 825                 | 2375                | 951                 | 2749                | 1101                |
|                    | -10       | 1681                | 687                 | 1941                | 793                 | 2243                | 916                 | 2600                | 1062                |
|                    | 0         | 1283                | 534                 | 1505                | 626                 | 1761                | 733                 | 2062                | 858                 |
|                    | 10        | 928                 | 393                 | 1115                | 473                 | 1330                | 564                 | 1582                | 670                 |
| 14,000             | -54       | 1774                | 645                 | 2056                | 748                 | 2383                | 867                 | 2769                | 1007                |
|                    | -50       | 1757                | 645                 | 2036                | 747                 | 2360                | 866                 | 2743                | 1007                |
|                    | -40       | 1717                | 644                 | 1990                | 746                 | 2306                | 865                 | 2680                | 1005                |
|                    | -30       | 1679                | 643                 | 1946                | 745                 | 2255                | 864                 | 2621                | 1004                |
|                    | -20       | 1642                | 642                 | 1903                | 744                 | 2206                | 862                 | 2564                | 1002                |
|                    | -10       | 1441                | 574                 | 1681                | 670                 | 1959                | 781                 | 2287                | 912                 |
|                    | 0         | 1068                | 434                 | 1273                | 517                 | 1508                | 612                 | 1783                | 724                 |
|                    | 10        | 741                 | 306                 | 914                 | 378                 | 1111                | 459                 | 1340                | 554                 |

## Enroute Climb

### Climb Gradient

Thrust Lever ..... MCT  
 Flaps..... UP  
 Landing Gear..... UP  
 Anti-ice ..... OFF  
 Airspeed ..... Per subsequent Climb Gradient table

• NOTE •

Climb performance data provided results in approximately 95%  
best rate of climb.

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 0                  | -54       | 187             | 731                 | 186     | 820                 | 186     | 925                 | 185     | 1052                |
|                    | -50       | 187             | 730                 | 186     | 819                 | 186     | 925                 | 185     | 1051                |
|                    | -40       | 187             | 729                 | 186     | 818                 | 186     | 923                 | 185     | 1050                |
|                    | -30       | 187             | 729                 | 186     | 817                 | 185     | 927                 | 185     | 1054                |
|                    | -20       | 187             | 726                 | 186     | 819                 | 185     | 924                 | 185     | 1051                |
|                    | -10       | 186             | 728                 | 186     | 817                 | 185     | 922                 | 185     | 1049                |
|                    | 0         | 186             | 726                 | 186     | 814                 | 185     | 919                 | 184     | 1050                |
|                    | 10        | 183             | 691                 | 182     | 781                 | 181     | 883                 | 181     | 1005                |
|                    | 20        | 171             | 568                 | 170     | 648                 | 170     | 739                 | 169     | 852                 |
|                    | 30        | 160             | 455                 | 159     | 528                 | 157     | 613                 | 156     | 715                 |
|                    | 40        | 149             | 356                 | 147     | 424                 | 146     | 502                 | 145     | 595                 |
|                    | 50        | 138             | 262                 | 137     | 325                 | 136     | 397                 | 134     | 488                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 2000               | -54       | 187             | 719                 | 186     | 807                 | 186     | 911                 | 185     | 1037                |
|                    | -50       | 187             | 719                 | 186     | 807                 | 186     | 911                 | 185     | 1036                |
|                    | -40       | 187             | 718                 | 186     | 806                 | 186     | 910                 | 185     | 1040                |
|                    | -30       | 187             | 716                 | 186     | 808                 | 185     | 912                 | 185     | 1037                |
|                    | -20       | 187             | 715                 | 186     | 806                 | 185     | 910                 | 185     | 1035                |
|                    | -10       | 186             | 717                 | 186     | 805                 | 185     | 908                 | 185     | 1033                |
|                    | 0         | 186             | 714                 | 186     | 801                 | 185     | 909                 | 184     | 1034                |
|                    | 10        | 175             | 636                 | 174     | 722                 | 173     | 820                 | 172     | 943                 |
|                    | 20        | 163             | 519                 | 163     | 593                 | 161     | 684                 | 160     | 793                 |
|                    | 30        | 152             | 412                 | 151     | 483                 | 150     | 565                 | 148     | 667                 |
|                    | 40        | 142             | 314                 | 141     | 381                 | 140     | 457                 | 138     | 551                 |
|                    | 45        | 137             | 266                 | 136     | 330                 | 134     | 405                 | 133     | 495                 |



| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 4000               | -54       | 187             | 703                 | 186     | 793                 | 185     | 896                 | 185     | 1019                |
|                    | -50       | 187             | 703                 | 186     | 793                 | 185     | 895                 | 185     | 1019                |
|                    | -40       | 187             | 702                 | 186     | 792                 | 185     | 894                 | 185     | 1017                |
|                    | -30       | 186             | 703                 | 186     | 790                 | 185     | 892                 | 184     | 1019                |
|                    | -20       | 186             | 702                 | 186     | 788                 | 185     | 894                 | 184     | 1017                |
|                    | -10       | 186             | 700                 | 186     | 786                 | 185     | 892                 | 184     | 1015                |
|                    | 0         | 183             | 702                 | 182     | 789                 | 182     | 892                 | 181     | 1016                |
|                    | 10        | 171             | 582                 | 169     | 664                 | 168     | 761                 | 168     | 873                 |
|                    | 20        | 159             | 470                 | 158     | 544                 | 156     | 631                 | 155     | 736                 |
|                    | 30        | 148             | 369                 | 147     | 437                 | 145     | 519                 | 144     | 614                 |
|                    | 40        | 138             | 275                 | 137     | 338                 | 135     | 414                 | 134     | 504                 |
| 6000               | -54       | 191             | 660                 | 191     | 742                 | 190     | 842                 | 189     | 958                 |
|                    | -50       | 191             | 660                 | 191     | 741                 | 190     | 841                 | 189     | 958                 |
|                    | -40       | 191             | 658                 | 190     | 743                 | 190     | 839                 | 189     | 955                 |
|                    | -30       | 191             | 660                 | 190     | 741                 | 190     | 837                 | 189     | 953                 |
|                    | -20       | 191             | 658                 | 190     | 739                 | 189     | 839                 | 189     | 955                 |
|                    | -10       | 191             | 656                 | 190     | 740                 | 189     | 836                 | 189     | 952                 |
|                    | 0         | 181             | 624                 | 180     | 707                 | 179     | 802                 | 179     | 916                 |
|                    | 10        | 168             | 512                 | 167     | 588                 | 166     | 677                 | 165     | 780                 |
|                    | 20        | 155             | 407                 | 154     | 476                 | 153     | 560                 | 152     | 656                 |
|                    | 30        | 144             | 312                 | 143     | 376                 | 142     | 452                 | 140     | 544                 |
|                    | 36        | 138             | 256                 | 137     | 319                 | 135     | 392                 | 133     | 483                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 8000               | -54       | 186             | 608                 | 185     | 689                 | 185     | 780                 | 184     | 891                 |
|                    | -50       | 186             | 608                 | 186     | 685                 | 185     | 780                 | 184     | 890                 |
|                    | -40       | 186             | 607                 | 185     | 687                 | 185     | 778                 | 183     | 892                 |
|                    | -30       | 186             | 605                 | 185     | 685                 | 184     | 780                 | 183     | 890                 |
|                    | -20       | 186             | 607                 | 185     | 684                 | 184     | 778                 | 183     | 892                 |
|                    | -10       | 186             | 604                 | 185     | 684                 | 183     | 779                 | 183     | 889                 |
|                    | 0         | 172             | 549                 | 171     | 624                 | 170     | 716                 | 169     | 826                 |
|                    | 10        | 159             | 445                 | 158     | 516                 | 157     | 600                 | 156     | 700                 |
|                    | 20        | 147             | 346                 | 146     | 412                 | 145     | 491                 | 144     | 583                 |
|                    | 30        | 137             | 255                 | 136     | 315                 | 135     | 389                 | 133     | 476                 |
|                    | 31        | 137             | 244                 | 135     | 306                 | 134     | 379                 | 132     | 465                 |
| 10,000             | -54       | 181             | 557                 | 180     | 634                 | 179     | 724                 | 178     | 833                 |
|                    | -50       | 181             | 557                 | 180     | 634                 | 179     | 724                 | 178     | 833                 |
|                    | -40       | 181             | 558                 | 179     | 635                 | 179     | 722                 | 178     | 831                 |
|                    | -30       | 181             | 557                 | 179     | 634                 | 178     | 724                 | 178     | 829                 |
|                    | -20       | 181             | 555                 | 179     | 632                 | 178     | 722                 | 177     | 831                 |
|                    | -10       | 176             | 569                 | 175     | 644                 | 174     | 737                 | 174     | 845                 |
|                    | 0         | 163             | 480                 | 162     | 553                 | 161     | 640                 | 160     | 740                 |
|                    | 10        | 151             | 381                 | 150     | 448                 | 149     | 527                 | 148     | 622                 |
|                    | 20        | 141             | 287                 | 139     | 352                 | 138     | 425                 | 137     | 515                 |
|                    | 27        | 134             | 224                 | 133     | 283                 | 132     | 355                 | 131     | 436                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 12,000             | -54       | 176             | 514                 | 175     | 588                 | 174     | 675                 | 173     | 779                 |
|                    | -50       | 176             | 514                 | 175     | 588                 | 174     | 674                 | 173     | 778                 |
|                    | -40       | 176             | 513                 | 175     | 587                 | 174     | 673                 | 173     | 777                 |
|                    | -30       | 175             | 515                 | 174     | 588                 | 173     | 675                 | 173     | 776                 |
|                    | -20       | 175             | 514                 | 174     | 587                 | 173     | 674                 | 172     | 777                 |
|                    | -10       | 166             | 502                 | 165     | 577                 | 164     | 665                 | 163     | 771                 |
|                    | 0         | 155             | 408                 | 154     | 477                 | 153     | 558                 | 152     | 655                 |
|                    | 10        | 144             | 316                 | 143     | 380                 | 142     | 454                 | 141     | 543                 |
|                    | 20        | 135             | 229                 | 134     | 288                 | 133     | 357                 | 131     | 441                 |
|                    | 22        | 133             | 209                 | 132     | 270                 | 131     | 337                 | 130     | 418                 |
| 14,000             | -54       | 171             | 472                 | 170     | 542                 | 169     | 625                 | 167     | 725                 |
|                    | -50       | 171             | 471                 | 170     | 542                 | 169     | 625                 | 167     | 724                 |
|                    | -40       | 171             | 470                 | 170     | 541                 | 169     | 623                 | 167     | 722                 |
|                    | -30       | 171             | 469                 | 170     | 539                 | 169     | 622                 | 167     | 725                 |
|                    | -20       | 171             | 468                 | 169     | 540                 | 168     | 623                 | 167     | 722                 |
|                    | -10       | 158             | 434                 | 157     | 505                 | 155     | 587                 | 154     | 686                 |
|                    | 0         | 148             | 343                 | 147     | 408                 | 146     | 484                 | 145     | 574                 |
|                    | 10        | 139             | 253                 | 137     | 316                 | 136     | 386                 | 135     | 469                 |
|                    | 17        | 133             | 197                 | 132     | 253                 | 131     | 319                 | 129     | 400                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 16,000             | -54       | 164             | 438                 | 163     | 507                 | 162     | 589                 | 161     | 686                 |
|                    | -50       | 164             | 438                 | 163     | 507                 | 162     | 588                 | 161     | 685                 |
|                    | -40       | 164             | 437                 | 163     | 505                 | 162     | 586                 | 161     | 683                 |
|                    | -30       | 164             | 435                 | 163     | 504                 | 162     | 584                 | 161     | 681                 |
|                    | -20       | 164             | 434                 | 162     | 505                 | 161     | 586                 | 160     | 682                 |
|                    | -10       | 150             | 370                 | 149     | 437                 | 148     | 515                 | 147     | 608                 |
|                    | 0         | 142             | 280                 | 141     | 341                 | 139     | 415                 | 138     | 500                 |
|                    | 10        | 133             | 195                 | 132     | 251                 | 131     | 319                 | 130     | 397                 |
|                    | 14        | 130             | 162                 | 129     | 217                 | 128     | 281                 | 127     | 358                 |
| 18,000             | -54       | 162             | 402                 | 160     | 471                 | 159     | 549                 | 158     | 642                 |
|                    | -50       | 161             | 404                 | 160     | 470                 | 159     | 548                 | 158     | 641                 |
|                    | -40       | 161             | 402                 | 160     | 468                 | 159     | 546                 | 158     | 639                 |
|                    | -30       | 161             | 401                 | 160     | 467                 | 159     | 544                 | 157     | 640                 |
|                    | -20       | 156             | 397                 | 155     | 464                 | 153     | 545                 | 152     | 640                 |
|                    | -10       | 144             | 309                 | 142     | 374                 | 141     | 448                 | 140     | 540                 |
|                    | 0         | 136             | 225                 | 135     | 283                 | 133     | 353                 | 132     | 434                 |
|                    | 9         | 128             | 151                 | 127     | 207                 | 125     | 273                 | 124     | 352                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 20,000             | -54       | 157             | 371                 | 156     | 435                 | 154     | 513                 | 153     | 604                 |
|                    | -50       | 157             | 370                 | 156     | 434                 | 154     | 512                 | 153     | 603                 |
|                    | -40       | 157             | 369                 | 155     | 435                 | 154     | 511                 | 153     | 601                 |
|                    | -30       | 157             | 367                 | 155     | 434                 | 154     | 509                 | 153     | 599                 |
|                    | -20       | 149             | 333                 | 148     | 396                 | 147     | 470                 | 146     | 558                 |
|                    | -10       | 139             | 251                 | 138     | 311                 | 136     | 382                 | 135     | 468                 |
|                    | 0         | 130             | 172                 | 129     | 229                 | 127     | 296                 | 125     | 378                 |
|                    | 5         | 126             | 134                 | 125     | 189                 | 123     | 256                 | 121     | 334                 |
| 22,000             | -56       | 153             | 347                 | 152     | 407                 | 151     | 481                 | 150     | 569                 |
|                    | -50       | 153             | 343                 | 152     | 409                 | 151     | 482                 | 150     | 570                 |
|                    | -40       | 153             | 343                 | 152     | 406                 | 151     | 480                 | 150     | 567                 |
|                    | -30       | 153             | 341                 | 152     | 404                 | 151     | 477                 | 150     | 564                 |
|                    | -20       | 143             | 270                 | 142     | 329                 | 141     | 401                 | 140     | 483                 |
|                    | -10       | 133             | 194                 | 132     | 252                 | 131     | 318                 | 129     | 398                 |
|                    | 0         | 125             | 117                 | 124     | 172                 | 122     | 235                 | 121     | 308                 |
| 24,000             | -58       | 151             | 312                 | 150     | 372                 | 149     | 443                 | 148     | 527                 |
|                    | -50       | 151             | 311                 | 150     | 371                 | 149     | 441                 | 148     | 525                 |
|                    | -40       | 151             | 309                 | 150     | 369                 | 148     | 441                 | 147     | 525                 |
|                    | -30       | 144             | 317                 | 143     | 378                 | 142     | 452                 | 142     | 536                 |
|                    | -20       | 137             | 210                 | 136     | 266                 | 135     | 331                 | 133     | 411                 |
|                    | -10       | 129             | 141                 | 127     | 195                 | 126     | 257                 | 125     | 332                 |
|                    | -4        | 124             | 97                  | 123     | 149                 | 122     | 208                 | 121     | 278                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 26,000             | -59       | 148             | 270                 | 147     | 328                 | 146     | 397                 | 145     | 476                 |
|                    | -50       | 148             | 271                 | 147     | 328                 | 146     | 395                 | 145     | 474                 |
|                    | -40       | 148             | 269                 | 147     | 326                 | 145     | 395                 | 144     | 474                 |
|                    | -30       | 139             | 250                 | 138     | 311                 | 137     | 379                 | 136     | 461                 |
|                    | -20       | 131             | 155                 | 130     | 208                 | 129     | 272                 | 128     | 345                 |
|                    | -10       | 125             | 88                  | 124     | 139                 | 123     | 197                 | 121     | 266                 |
|                    | -8        | 123             | 75                  | 122     | 125                 | 121     | 182                 | 120     | 249                 |
| 28,000             | -61       | 145             | 228                 | 144     | 282                 | 142     | 348                 | 141     | 422                 |
|                    | -60       | 145             | 228                 | 144     | 282                 | 142     | 348                 | 141     | 423                 |
|                    | -50       | 144             | 228                 | 143     | 283                 | 142     | 345                 | 141     | 420                 |
|                    | -40       | 143             | 226                 | 142     | 280                 | 141     | 343                 | 140     | 418                 |
|                    | -30       | 135             | 185                 | 133     | 239                 | 132     | 304                 | 131     | 379                 |
|                    | -20       | 126             | 101                 | 125     | 151                 | 124     | 211                 | 123     | 280                 |
|                    | -13       | 122             | 57                  | 121     | 105                 | 120     | 160                 | 119     | 227                 |
| 30,000             | -63       | 142             | 185                 | 141     | 236                 | 139     | 298                 | 138     | 368                 |
|                    | -60       | 141             | 187                 | 141     | 237                 | 139     | 298                 | 138     | 368                 |
|                    | -50       | 141             | 187                 | 140     | 238                 | 138     | 300                 | 137     | 371                 |
|                    | -40       | 139             | 179                 | 138     | 230                 | 137     | 291                 | 136     | 361                 |
|                    | -30       | 130             | 127                 | 128     | 178                 | 127     | 239                 | 126     | 309                 |
|                    | -20       | 121             | 51                  | 120     | 99                  | 119     | 154                 | 118     | 219                 |
|                    | -18       | 120             | 38                  | 119     | 85                  | 118     | 139                 | 117     | 202                 |

| Press<br>Alt<br>FT | OAT<br>°C | Aircraft Weight |                     |         |                     |         |                     |         |                     |
|--------------------|-----------|-----------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|
|                    |           | 6000 lb         |                     | 5500 lb |                     | 5000 lb |                     | 4500 lb |                     |
|                    |           | KIAS            | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) | KIAS    | Grad<br>(FT/<br>NM) |
| 31,000             | -64       | 140             | 163                 | 139     | 213                 | 138     | 270                 | 136     | 340                 |
|                    | -60       | 140             | 164                 | 139     | 213                 | 138     | 270                 | 136     | 340                 |
|                    | -50       | 140             | 164                 | 139     | 214                 | 138     | 271                 | 136     | 340                 |
|                    | -40       | 137             | 156                 | 136     | 205                 | 135     | 265                 | 134     | 333                 |
|                    | -30       | 128             | 100                 | 127     | 149                 | 125     | 208                 | 124     | 275                 |
|                    | -20       | 119             | 28                  | 118     | 74                  | 117     | 128                 | 116     | 190                 |

Rate Of Climb

Thrust Lever .....MCT  
Flaps..... UP  
Landing Gear ..... UP  
Anti-ice ..... OFF  
Airspeed .....Per Rate Of Climb table below

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 0                      | -54         | 187                  | 2605                | 186  | 2914                | 186  | 3279                | 185  | 3719                |
|                        | -50         | 187                  | 2580                | 186  | 2887                | 186  | 3248                | 185  | 3684                |
|                        | -40         | 187                  | 2521                | 186  | 2820                | 186  | 3173                | 185  | 3599                |
|                        | -30         | 187                  | 2465                | 186  | 2758                | 185  | 3110                | 185  | 3527                |
|                        | -20         | 187                  | 2409                | 186  | 2701                | 185  | 3039                | 185  | 3447                |
|                        | -10         | 186                  | 2363                | 186  | 2644                | 185  | 2975                | 185  | 3374                |
|                        | 0           | 186                  | 2311                | 186  | 2586                | 185  | 2910                | 184  | 3307                |
|                        | 10          | 183                  | 2120                | 182  | 2382                | 181  | 2686                | 181  | 3051                |
|                        | 20          | 171                  | 1609                | 170  | 1826                | 170  | 2076                | 169  | 2380                |
|                        | 30          | 160                  | 1192                | 159  | 1372                | 157  | 1584                | 156  | 1835                |
|                        | 40          | 149                  | 858                 | 147  | 1014                | 146  | 1192                | 145  | 1402                |
|                        | 50          | 138                  | 578                 | 137  | 711                 | 136  | 862                 | 134  | 1044                |



| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 2000                   | -54         | 187                  | 2616                | 186  | 2927                | 186  | 3296                | 185  | 3739                |
|                        | -50         | 187                  | 2591                | 186  | 2899                | 186  | 3264                | 185  | 3703                |
|                        | -40         | 187                  | 2531                | 186  | 2833                | 186  | 3190                | 185  | 3625                |
|                        | -30         | 187                  | 2472                | 186  | 2772                | 185  | 3121                | 185  | 3540                |
|                        | -20         | 187                  | 2419                | 186  | 2713                | 185  | 3054                | 185  | 3465                |
|                        | -10         | 186                  | 2372                | 186  | 2655                | 185  | 2989                | 185  | 3391                |
|                        | 0           | 186                  | 2319                | 186  | 2596                | 185  | 2928                | 184  | 3322                |
|                        | 10          | 175                  | 1909                | 174  | 2156                | 173  | 2440                | 172  | 2788                |
|                        | 20          | 163                  | 1437                | 163  | 1639                | 161  | 1878                | 160  | 2165                |
|                        | 30          | 152                  | 1055                | 151  | 1227                | 150  | 1428                | 148  | 1669                |
|                        | 40          | 142                  | 742                 | 141  | 891                 | 140  | 1061                | 138  | 1265                |
|                        | 45          | 137                  | 601                 | 136  | 739                 | 134  | 898                 | 133  | 1085                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 4000                   | -54         | 187                  | 2616                | 186  | 2935                | 185  | 3306                | 185  | 3751                |
|                        | -50         | 187                  | 2590                | 186  | 2907                | 185  | 3273                | 185  | 3714                |
|                        | -40         | 187                  | 2531                | 186  | 2840                | 185  | 3198                | 185  | 3629                |
|                        | -30         | 186                  | 2476                | 186  | 2773                | 185  | 3123                | 184  | 3551                |
|                        | -20         | 186                  | 2422                | 186  | 2713                | 185  | 3061                | 184  | 3474                |
|                        | -10         | 186                  | 2369                | 186  | 2653                | 185  | 2994                | 184  | 3398                |
|                        | 0           | 183                  | 2290                | 182  | 2566                | 182  | 2893                | 181  | 3285                |
|                        | 10          | 171                  | 1747                | 169  | 1981                | 168  | 2256                | 168  | 2580                |
|                        | 20          | 159                  | 1301                | 158  | 1496                | 156  | 1725                | 155  | 1997                |
|                        | 30          | 148                  | 941                 | 147  | 1107                | 145  | 1302                | 144  | 1530                |
|                        | 40          | 138                  | 644                 | 137  | 786                 | 135  | 952                 | 134  | 1147                |
| 6000                   | -54         | 191                  | 2568                | 191  | 2877                | 190  | 3248                | 189  | 3688                |
|                        | -50         | 191                  | 2543                | 191  | 2849                | 190  | 3217                | 189  | 3652                |
|                        | -40         | 191                  | 2481                | 190  | 2785                | 190  | 3138                | 189  | 3562                |
|                        | -30         | 191                  | 2430                | 190  | 2722                | 190  | 3066                | 189  | 3481                |
|                        | -20         | 191                  | 2375                | 190  | 2660                | 189  | 3003                | 189  | 3409                |
|                        | -10         | 191                  | 2321                | 190  | 2606                | 189  | 2936                | 189  | 3333                |
|                        | 0           | 181                  | 2059                | 180  | 2321                | 179  | 2625                | 179  | 2989                |
|                        | 10          | 168                  | 1550                | 167  | 1768                | 166  | 2024                | 165  | 2324                |
|                        | 20          | 155                  | 1132                | 154  | 1314                | 153  | 1530                | 152  | 1783                |
|                        | 30          | 144                  | 797                 | 143  | 951                 | 142  | 1134                | 140  | 1349                |
|                        | 36          | 138                  | 620                 | 137  | 764                 | 135  | 929                 | 133  | 1127                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 8000                   | -54         | 186                  | 2358                | 185  | 2656                | 185  | 3001                | 184  | 3415                |
|                        | -50         | 186                  | 2336                | 186  | 2625                | 185  | 2972                | 184  | 3383                |
|                        | -40         | 186                  | 2280                | 185  | 2568                | 185  | 2902                | 183  | 3309                |
|                        | -30         | 186                  | 2228                | 185  | 2509                | 184  | 2841                | 183  | 3233                |
|                        | -20         | 186                  | 2182                | 185  | 2453                | 184  | 2776                | 183  | 3165                |
|                        | -10         | 186                  | 2132                | 185  | 2401                | 183  | 2718                | 183  | 3094                |
|                        | 0           | 172                  | 1768                | 171  | 2003                | 170  | 2285                | 169  | 2621                |
|                        | 10          | 159                  | 1311                | 158  | 1512                | 157  | 1746                | 156  | 2026                |
|                        | 20          | 147                  | 938                 | 146  | 1108                | 145  | 1308                | 144  | 1542                |
|                        | 30          | 137                  | 633                 | 136  | 778                 | 135  | 949                 | 133  | 1149                |
|                        | 31          | 137                  | 603                 | 135  | 748                 | 134  | 916                 | 132  | 1113                |
| 10,000                 | -54         | 181                  | 2153                | 180  | 2436                | 179  | 2768                | 178  | 3166                |
|                        | -50         | 181                  | 2132                | 180  | 2412                | 179  | 2741                | 178  | 3135                |
|                        | -40         | 181                  | 2085                | 179  | 2359                | 179  | 2675                | 178  | 3060                |
|                        | -30         | 181                  | 2038                | 179  | 2305                | 178  | 2619                | 178  | 2990                |
|                        | -20         | 181                  | 1991                | 179  | 2252                | 178  | 2559                | 177  | 2928                |
|                        | -10         | 176                  | 1949                | 175  | 2203                | 174  | 2506                | 174  | 2863                |
|                        | 0           | 163                  | 1507                | 162  | 1728                | 161  | 1987                | 160  | 2290                |
|                        | 10          | 151                  | 1096                | 150  | 1282                | 149  | 1499                | 148  | 1755                |
|                        | 20          | 141                  | 762                 | 139  | 924                 | 138  | 1108                | 137  | 1329                |
|                        | 27          | 134                  | 561                 | 133  | 704                 | 132  | 871                 | 131  | 1064                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 12,000                 | -54         | 176                  | 1979                | 175  | 2249                | 174  | 2566                | 173  | 2946                |
|                        | -50         | 176                  | 1960                | 175  | 2228                | 174  | 2542                | 173  | 2917                |
|                        | -40         | 176                  | 1915                | 175  | 2176                | 174  | 2482                | 173  | 2849                |
|                        | -30         | 175                  | 1876                | 174  | 2131                | 173  | 2431                | 173  | 2785                |
|                        | -20         | 175                  | 1834                | 174  | 2083                | 173  | 2377                | 172  | 2728                |
|                        | -10         | 166                  | 1671                | 165  | 1909                | 164  | 2188                | 163  | 2520                |
|                        | 0           | 155                  | 1259                | 154  | 1462                | 153  | 1698                | 152  | 1979                |
|                        | 10          | 144                  | 893                 | 143  | 1066                | 142  | 1266                | 141  | 1502                |
|                        | 20          | 135                  | 595                 | 134  | 744                 | 133  | 915                 | 131  | 1119                |
|                        | 22          | 133                  | 538                 | 132  | 684                 | 131  | 850                 | 130  | 1044                |
| 14,000                 | -54         | 171                  | 1808                | 170  | 2066                | 169  | 2367                | 167  | 2728                |
|                        | -50         | 171                  | 1790                | 170  | 2045                | 169  | 2344                | 167  | 2701                |
|                        | -40         | 171                  | 1747                | 170  | 1996                | 169  | 2287                | 167  | 2636                |
|                        | -30         | 171                  | 1707                | 170  | 1950                | 169  | 2235                | 167  | 2581                |
|                        | -20         | 171                  | 1667                | 169  | 1909                | 168  | 2188                | 167  | 2521                |
|                        | -10         | 158                  | 1415                | 157  | 1634                | 155  | 1890                | 154  | 2195                |
|                        | 0           | 148                  | 1036                | 147  | 1223                | 146  | 1441                | 145  | 1699                |
|                        | 10          | 139                  | 706                 | 137  | 870                 | 136  | 1056                | 135  | 1274                |
|                        | 17          | 133                  | 520                 | 132  | 665                 | 131  | 832                 | 129  | 1031                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 16,000                 | -54         | 164                  | 1661                | 163  | 1911                | 162  | 2203                | 161  | 2552                |
|                        | -50         | 164                  | 1644                | 163  | 1892                | 162  | 2181                | 161  | 2527                |
|                        | -40         | 164                  | 1604                | 163  | 1845                | 162  | 2128                | 161  | 2465                |
|                        | -30         | 164                  | 1565                | 163  | 1801                | 162  | 2077                | 161  | 2406                |
|                        | -20         | 164                  | 1529                | 162  | 1763                | 161  | 2033                | 160  | 2354                |
|                        | -10         | 150                  | 1182                | 149  | 1386                | 148  | 1623                | 147  | 1904                |
|                        | 0           | 142                  | 832                 | 141  | 1006                | 139  | 1211                | 138  | 1449                |
|                        | 10          | 133                  | 534                 | 132  | 684                 | 131  | 860                 | 130  | 1061                |
|                        | 14          | 130                  | 433                 | 129  | 575                 | 128  | 737                 | 127  | 930                 |
| 18,000                 | -54         | 162                  | 1538                | 160  | 1784                | 159  | 2068                | 158  | 2405                |
|                        | -50         | 161                  | 1526                | 160  | 1766                | 159  | 2047                | 158  | 2380                |
|                        | -40         | 161                  | 1487                | 160  | 1722                | 159  | 1995                | 158  | 2320                |
|                        | -30         | 161                  | 1451                | 160  | 1680                | 159  | 1947                | 157  | 2269                |
|                        | -20         | 156                  | 1373                | 155  | 1595                | 153  | 1858                | 152  | 2166                |
|                        | -10         | 144                  | 973                 | 142  | 1166                | 141  | 1387                | 140  | 1652                |
|                        | 0           | 136                  | 656                 | 135  | 820                 | 133  | 1012                | 132  | 1233                |
|                        | 9           | 128                  | 410                 | 127  | 557                 | 125  | 725                 | 124  | 922                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 20,000                 | -54         | 157                  | 1419                | 156  | 1655                | 154  | 1934                | 153  | 2260                |
|                        | -50         | 157                  | 1404                | 156  | 1638                | 154  | 1914                | 153  | 2237                |
|                        | -40         | 157                  | 1369                | 155  | 1601                | 154  | 1866                | 153  | 2181                |
|                        | -30         | 157                  | 1335                | 155  | 1562                | 154  | 1821                | 153  | 2128                |
|                        | -20         | 149                  | 1134                | 148  | 1341                | 147  | 1579                | 146  | 1862                |
|                        | -10         | 139                  | 784                 | 138  | 962                 | 136  | 1170                | 135  | 1416                |
|                        | 0           | 130                  | 494                 | 129  | 651                 | 127  | 831                 | 125  | 1046                |
|                        | 5           | 126                  | 369                 | 125  | 517                 | 123  | 688                 | 121  | 887                 |
| 22,000                 | -56         | 153                  | 1334                | 152  | 1563                | 151  | 1833                | 150  | 2154                |
|                        | -50         | 153                  | 1308                | 152  | 1541                | 151  | 1808                | 150  | 2124                |
|                        | -40         | 153                  | 1275                | 152  | 1499                | 151  | 1758                | 150  | 2066                |
|                        | -30         | 153                  | 1242                | 152  | 1460                | 151  | 1713                | 150  | 2013                |
|                        | -20         | 143                  | 905                 | 142  | 1097                | 141  | 1321                | 140  | 1582                |
|                        | -10         | 133                  | 597                 | 132  | 767                 | 131  | 959                 | 129  | 1187                |
|                        | 0           | 125                  | 333                 | 124  | 482                 | 122  | 652                 | 121  | 847                 |
| 24,000                 | -58         | 151                  | 1221                | 150  | 1448                | 149  | 1712                | 148  | 2023                |
|                        | -50         | 151                  | 1195                | 150  | 1417                | 149  | 1675                | 148  | 1979                |
|                        | -40         | 151                  | 1162                | 150  | 1378                | 148  | 1634                | 147  | 1931                |
|                        | -30         | 144                  | 1118                | 143  | 1329                | 142  | 1577                | 142  | 1865                |
|                        | -20         | 137                  | 692                 | 136  | 870                 | 135  | 1074                | 133  | 1318                |
|                        | -10         | 129                  | 430                 | 127  | 588                 | 126  | 768                 | 125  | 981                 |
|                        | -4          | 124                  | 282                 | 123  | 429                 | 122  | 595                 | 121  | 788                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 26,000                 | -59         | 148                  | 1074                | 147  | 1292                | 146  | 1548                | 145  | 1847                |
|                        | -50         | 148                  | 1049                | 147  | 1263                | 146  | 1509                | 145  | 1800                |
|                        | -40         | 148                  | 1020                | 147  | 1228                | 145  | 1472                | 144  | 1755                |
|                        | -30         | 139                  | 877                 | 138  | 1079                | 137  | 1309                | 136  | 1579                |
|                        | -20         | 131                  | 503                 | 130  | 670                 | 129  | 865                 | 128  | 1089                |
|                        | -10         | 125                  | 268                 | 124  | 417                 | 123  | 586                 | 121  | 784                 |
|                        | -8          | 123                  | 224                 | 122  | 369                 | 121  | 534                 | 120  | 725                 |
| 28,000                 | -61         | 145                  | 913                 | 144  | 1122                | 142  | 1367                | 141  | 1651                |
|                        | -60         | 145                  | 911                 | 144  | 1119                | 142  | 1364                | 141  | 1647                |
|                        | -50         | 144                  | 888                 | 143  | 1091                | 142  | 1325                | 141  | 1602                |
|                        | -40         | 143                  | 855                 | 142  | 1052                | 141  | 1280                | 140  | 1548                |
|                        | -30         | 135                  | 643                 | 133  | 826                 | 132  | 1040                | 131  | 1286                |
|                        | -20         | 126                  | 325                 | 125  | 483                 | 124  | 665                 | 123  | 874                 |
|                        | -13         | 122                  | 174                 | 121  | 318                 | 120  | 483                 | 119  | 676                 |
| 30,000                 | -63         | 142                  | 751                 | 141  | 950                 | 139  | 1183                | 138  | 1452                |
|                        | -60         | 141                  | 750                 | 141  | 945                 | 139  | 1176                | 138  | 1443                |
|                        | -50         | 141                  | 731                 | 140  | 924                 | 138  | 1151                | 137  | 1413                |
|                        | -40         | 139                  | 675                 | 138  | 861                 | 137  | 1079                | 136  | 1331                |
|                        | -30         | 130                  | 439                 | 128  | 610                 | 127  | 809                 | 126  | 1036                |
|                        | -20         | 121                  | 162                 | 120  | 312                 | 119  | 482                 | 118  | 678                 |
|                        | -18         | 120                  | 118                 | 119  | 263                 | 118  | 428                 | 117  | 619                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) | KIAS | ROC<br>(FT/<br>MIN) |
| 31,000                 | -64         | 140                  | 663                 | 139  | 857                 | 138  | 1080                | 136  | 1344                |
|                        | -60         | 140                  | 659                 | 139  | 851                 | 138  | 1071                | 136  | 1333                |
|                        | -50         | 140                  | 646                 | 139  | 834                 | 138  | 1049                | 136  | 1305                |
|                        | -40         | 137                  | 589                 | 136  | 771                 | 135  | 982                 | 134  | 1225                |
|                        | -30         | 128                  | 345                 | 127  | 511                 | 125  | 702                 | 124  | 921                 |
|                        | -20         | 119                  | 87                  | 118  | 232                 | 117  | 398                 | 116  | 588                 |



## **Enroute Ice Contaminated Climb**

### **Climb Gradient**

Thrust Lever ..... MCT  
 Flaps..... UP  
 Landing Gear..... UP  
 Airspeed ..... Per Climb Gradient table below

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 0                      | -54         | 140                  | 713                 | 139  | 817                 | 139  | 931                 | 138  | 1076                |
|                        | -50         | 140                  | 716                 | 139  | 816                 | 139  | 931                 | 138  | 1079                |
|                        | -40         | 140                  | 715                 | 139  | 816                 | 139  | 931                 | 138  | 1079                |
|                        | -30         | 140                  | 715                 | 139  | 816                 | 139  | 930                 | 138  | 1078                |
|                        | -20         | 140                  | 714                 | 139  | 814                 | 139  | 929                 | 138  | 1073                |
|                        | -10         | 140                  | 713                 | 139  | 813                 | 139  | 927                 | 138  | 1075                |
|                        | 0           | 140                  | 711                 | 139  | 811                 | 138  | 931                 | 138  | 1072                |
|                        | 10          | 139                  | 656                 | 137  | 759                 | 137  | 872                 | 137  | 1003                |
| 2000                   | -54         | 140                  | 700                 | 139  | 799                 | 138  | 918                 | 138  | 1058                |
|                        | -50         | 140                  | 699                 | 139  | 798                 | 138  | 918                 | 138  | 1057                |
|                        | -40         | 140                  | 699                 | 139  | 798                 | 138  | 917                 | 138  | 1057                |
|                        | -30         | 140                  | 698                 | 139  | 797                 | 138  | 916                 | 138  | 1055                |
|                        | -20         | 140                  | 697                 | 139  | 796                 | 138  | 915                 | 138  | 1054                |
|                        | -10         | 140                  | 696                 | 139  | 795                 | 138  | 913                 | 138  | 1052                |
|                        | 0           | 140                  | 694                 | 139  | 792                 | 138  | 911                 | 138  | 1049                |
|                        | 10          | 138                  | 562                 | 135  | 662                 | 135  | 763                 | 134  | 890                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 4000                   | -54         | 140                  | 680                 | 139  | 777                 | 138  | 895                 | 138  | 1031                |
|                        | -50         | 140                  | 680                 | 139  | 777                 | 138  | 894                 | 138  | 1031                |
|                        | -40         | 139                  | 682                 | 139  | 777                 | 138  | 894                 | 137  | 1034                |
|                        | -30         | 139                  | 680                 | 139  | 775                 | 138  | 892                 | 137  | 1032                |
|                        | -20         | 139                  | 680                 | 139  | 774                 | 138  | 894                 | 138  | 1027                |
|                        | -10         | 139                  | 678                 | 139  | 772                 | 138  | 892                 | 137  | 1029                |
|                        | 0           | 138                  | 674                 | 137  | 774                 | 137  | 888                 | 136  | 1024                |
|                        | 10          | 138                  | 479                 | 133  | 577                 | 133  | 673                 | 132  | 787                 |
| 6000                   | -54         | 141                  | 628                 | 140  | 723                 | 139  | 831                 | 138  | 963                 |
|                        | -50         | 140                  | 630                 | 140  | 723                 | 139  | 831                 | 138  | 963                 |
|                        | -40         | 140                  | 629                 | 140  | 721                 | 139  | 829                 | 138  | 961                 |
|                        | -30         | 140                  | 629                 | 140  | 721                 | 139  | 828                 | 138  | 960                 |
|                        | -20         | 140                  | 628                 | 140  | 719                 | 139  | 827                 | 138  | 958                 |
|                        | -10         | 140                  | 626                 | 140  | 718                 | 139  | 825                 | 138  | 956                 |
|                        | 0           | 138                  | 565                 | 137  | 655                 | 136  | 760                 | 135  | 882                 |
|                        | 10          | 138                  | 381                 | 133  | 471                 | 132  | 558                 | 132  | 657                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 8000                   | -54         | 138                  | 565                 | 138  | 650                 | 137  | 754                 | 136  | 876                 |
|                        | -50         | 138                  | 565                 | 138  | 650                 | 137  | 754                 | 136  | 875                 |
|                        | -40         | 138                  | 563                 | 138  | 649                 | 136  | 756                 | 136  | 874                 |
|                        | -30         | 138                  | 564                 | 138  | 649                 | 136  | 755                 | 136  | 873                 |
|                        | -20         | 138                  | 563                 | 138  | 648                 | 136  | 754                 | 136  | 872                 |
|                        | -10         | 138                  | 562                 | 137  | 648                 | 136  | 752                 | 136  | 870                 |
|                        | 0           | 137                  | 450                 | 134  | 540                 | 133  | 631                 | 132  | 743                 |
|                        | 10          | 137                  | 282                 | 132  | 364                 | 130  | 444                 | 129  | 536                 |
| 10,000                 | -54         | 137                  | 496                 | 136  | 581                 | 135  | 679                 | 134  | 795                 |
|                        | -50         | 137                  | 496                 | 136  | 581                 | 135  | 678                 | 134  | 795                 |
|                        | -40         | 137                  | 495                 | 136  | 580                 | 135  | 678                 | 134  | 794                 |
|                        | -30         | 137                  | 494                 | 136  | 580                 | 135  | 677                 | 134  | 793                 |
|                        | -20         | 137                  | 494                 | 136  | 579                 | 135  | 676                 | 134  | 792                 |
|                        | -10         | 137                  | 492                 | 136  | 577                 | 135  | 674                 | 134  | 789                 |
|                        | 0           | 137                  | 346                 | 132  | 434                 | 130  | 519                 | 130  | 614                 |
|                        | 10          | 137                  | 191                 | 131  | 267                 | 128  | 342                 | 127  | 422                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 12,000                 | -54         | 137                  | 423                 | 134  | 510                 | 133  | 599                 | 132  | 707                 |
|                        | -50         | 137                  | 423                 | 134  | 510                 | 133  | 599                 | 132  | 707                 |
|                        | -40         | 137                  | 423                 | 134  | 510                 | 133  | 601                 | 132  | 706                 |
|                        | -30         | 137                  | 423                 | 134  | 510                 | 133  | 600                 | 132  | 706                 |
|                        | -20         | 137                  | 422                 | 135  | 505                 | 133  | 597                 | 132  | 705                 |
|                        | -10         | 137                  | 381                 | 131  | 476                 | 130  | 567                 | 128  | 675                 |
|                        | 0           | 137                  | 237                 | 131  | 316                 | 128  | 397                 | 127  | 482                 |
|                        | 10          | 137                  | 97                  | 131  | 164                 | 125  | 235                 | 124  | 304                 |
| 14,000                 | -54         | 138                  | 353                 | 132  | 440                 | 131  | 524                 | 130  | 622                 |
|                        | -50         | 138                  | 352                 | 132  | 440                 | 131  | 524                 | 130  | 622                 |
|                        | -40         | 138                  | 352                 | 132  | 439                 | 131  | 524                 | 130  | 621                 |
|                        | -30         | 138                  | 352                 | 132  | 439                 | 131  | 523                 | 130  | 620                 |
|                        | -20         | 138                  | 351                 | 132  | 438                 | 131  | 522                 | 130  | 619                 |
|                        | -10         | 138                  | 273                 | 132  | 358                 | 128  | 444                 | 126  | 539                 |
|                        | 0           | 138                  | 138                 | 132  | 209                 | 125  | 287                 | 124  | 362                 |
|                        | 10          | 138                  | 12                  | 132  | 72                  | 125  | 135                 | 122  | 198                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 16,000                 | -54         | 138                  | 290                 | 132  | 374                 | 130  | 454                 | 129  | 549                 |
|                        | -50         | 138                  | 290                 | 132  | 374                 | 129  | 457                 | 129  | 549                 |
|                        | -40         | 138                  | 290                 | 132  | 374                 | 129  | 457                 | 129  | 548                 |
|                        | -30         | 138                  | 289                 | 132  | 373                 | 129  | 456                 | 129  | 546                 |
|                        | -20         | 138                  | 288                 | 132  | 372                 | 129  | 454                 | 129  | 545                 |
|                        | -10         | 138                  | 173                 | 132  | 248                 | 126  | 329                 | 125  | 411                 |
|                        | 0           | 138                  | 48                  | 132  | 110                 | 126  | 178                 | 122  | 247                 |
|                        | 10          | 138                  | -67                 | 132  | -16                 | 126  | 38                  | 120  | 94                  |
| 18,000                 | -54         | 137                  | 236                 | 132  | 312                 | 128  | 392                 | 128  | 476                 |
|                        | -50         | 137                  | 236                 | 132  | 312                 | 128  | 392                 | 128  | 476                 |
|                        | -40         | 137                  | 235                 | 132  | 311                 | 128  | 391                 | 128  | 475                 |
|                        | -30         | 137                  | 235                 | 132  | 310                 | 128  | 390                 | 128  | 474                 |
|                        | -20         | 137                  | 219                 | 132  | 294                 | 127  | 376                 | 127  | 459                 |
|                        | -10         | 137                  | 82                  | 132  | 146                 | 125  | 219                 | 123  | 290                 |
|                        | 0           | 137                  | -32                 | 132  | 20                  | 125  | 79                  | 121  | 137                 |
|                        | 9           | 137                  | -127                | 132  | -84                 | 125  | -37                 | 119  | 9                   |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 20,000                 | -54         | 137                  | 182                 | 132  | 253                 | 127  | 329                 | 126  | 408                 |
|                        | -50         | 137                  | 182                 | 132  | 252                 | 127  | 328                 | 126  | 407                 |
|                        | -40         | 137                  | 181                 | 132  | 252                 | 127  | 327                 | 126  | 406                 |
|                        | -30         | 137                  | 181                 | 132  | 251                 | 127  | 326                 | 126  | 405                 |
|                        | -20         | 137                  | 120                 | 132  | 186                 | 126  | 258                 | 125  | 330                 |
|                        | -10         | 137                  | 0                   | 132  | 55                  | 125  | 117                 | 122  | 177                 |
|                        | 0           | 137                  | -105                | 132  | -61                 | 125  | -13                 | 119  | 35                  |
|                        | 5           | 137                  | -153                | 132  | -113                | 125  | -70                 | 119  | -30                 |
| 22,000                 | -56         | 138                  | 138                 | 131  | 204                 | 127  | 274                 | 126  | 345                 |
|                        | -50         | 138                  | 137                 | 131  | 203                 | 126  | 274                 | 126  | 344                 |
|                        | -40         | 138                  | 135                 | 131  | 202                 | 126  | 272                 | 126  | 342                 |
|                        | -30         | 138                  | 134                 | 131  | 200                 | 126  | 270                 | 126  | 340                 |
|                        | -20         | 138                  | 29                  | 131  | 86                  | 125  | 146                 | 122  | 209                 |
|                        | -10         | 138                  | -79                 | 131  | -31                 | 125  | 19                  | 120  | 70                  |
|                        | 0           | 138                  | -175                | 131  | -137                | 125  | -99                 | 119  | -62                 |
| 24,000                 | -58         | 138                  | 88                  | 132  | 149                 | 126  | 212                 | 125  | 278                 |
|                        | -50         | 138                  | 87                  | 132  | 148                 | 126  | 212                 | 125  | 277                 |
|                        | -40         | 138                  | 87                  | 132  | 147                 | 126  | 210                 | 125  | 275                 |
|                        | -30         | 138                  | 75                  | 132  | 139                 | 126  | 204                 | 125  | 268                 |
|                        | -20         | 138                  | -58                 | 132  | -10                 | 125  | 41                  | 121  | 92                  |
|                        | -10         | 138                  | -149                | 132  | -109                | 125  | -68                 | 119  | -28                 |
|                        | -4          | 138                  | -202                | 132  | -167                | 125  | -133                | 119  | -101                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 26,000                 | -59         | 137                  | 36                  | 132  | 88                  | 126  | 146                 | 125  | 203                 |
|                        | -50         | 137                  | 35                  | 132  | 87                  | 126  | 145                 | 125  | 203                 |
|                        | -40         | 137                  | 34                  | 132  | 87                  | 126  | 144                 | 124  | 202                 |
|                        | -30         | 137                  | -10                 | 132  | 43                  | 126  | 100                 | 122  | 159                 |
|                        | -20         | 137                  | -129                | 132  | -89                 | 126  | -47                 | 120  | -5                  |
|                        | -10         | 137                  | -209                | 132  | -177                | 126  | -144                | 119  | -115                |
|                        | -8          | 137                  | -224                | 132  | -194                | 126  | -163                | 119  | -136                |
| 28,000                 | -61         | 138                  | -19                 | 131  | 29                  | 125  | 79                  | 124  | 129                 |
|                        | -60         | 138                  | -18                 | 131  | 29                  | 125  | 79                  | 124  | 130                 |
|                        | -50         | 138                  | -19                 | 131  | 29                  | 125  | 79                  | 124  | 129                 |
|                        | -40         | 138                  | -22                 | 131  | 25                  | 125  | 74                  | 124  | 124                 |
|                        | -30         | 138                  | -89                 | 131  | -45                 | 125  | 1                   | 121  | 47                  |
|                        | -20         | 138                  | -191                | 131  | -157                | 125  | -124                | 120  | -92                 |
|                        | -13         | 138                  | -241                | 131  | -212                | 125  | -185                | 119  | -163                |
| 30,000                 | -63         | 137                  | -68                 | 132  | -28                 | 125  | 14                  | 124  | 57                  |
|                        | -60         | 137                  | -67                 | 132  | -28                 | 125  | 14                  | 124  | 57                  |
|                        | -50         | 137                  | -67                 | 132  | -27                 | 125  | 16                  | 123  | 59                  |
|                        | -40         | 137                  | -79                 | 132  | -39                 | 125  | 2                   | 122  | 45                  |
|                        | -30         | 137                  | -155                | 132  | -120                | 125  | -83                 | 120  | -48                 |
|                        | -20         | 137                  | -243                | 132  | -217                | 125  | -191                | 119  | -170                |
|                        | -18         | 137                  | -257                | 132  | -233                | 125  | -209                | 119  | -190                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) | KIAS | Grad<br>(FT/<br>NM) |
| 31,000                 | -64         | 137                  | -95                 | 132  | -58                 | 126  | -20                 | 123  | 19                  |
|                        | -60         | 137                  | -94                 | 132  | -58                 | 126  | -19                 | 123  | 20                  |
|                        | -50         | 137                  | -92                 | 132  | -55                 | 126  | -17                 | 123  | 22                  |
|                        | -40         | 137                  | -105                | 132  | -69                 | 126  | -31                 | 122  | 7                   |
|                        | -30         | 137                  | -184                | 132  | -153                | 126  | -120                | 120  | -90                 |
|                        | -20         | 137                  | -266                | 132  | -243                | 126  | -221                | 119  | -204                |



## Rate Of Climb (Ice Contaminated)

Thrust Lever ..... MCT  
Flaps..... UP  
Landing Gear..... UP  
Airspeed ..... Per Rate Of Climb table below

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 0                      | -54         | 140                  | 1943                | 139  | 2208                | 139  | 2519                | 138  | 2889                |
|                        | -50         | 140                  | 1925                | 139  | 2188                | 139  | 2495                | 138  | 2862                |
|                        | -40         | 140                  | 1883                | 139  | 2140                | 139  | 2440                | 138  | 2799                |
|                        | -30         | 140                  | 1842                | 139  | 2094                | 139  | 2388                | 138  | 2739                |
|                        | -20         | 140                  | 1802                | 139  | 2049                | 139  | 2337                | 138  | 2680                |
|                        | -10         | 140                  | 1765                | 139  | 2006                | 139  | 2288                | 138  | 2625                |
|                        | 0           | 140                  | 1727                | 139  | 1964                | 138  | 2240                | 138  | 2569                |
|                        | 10          | 139                  | 1560                | 137  | 1781                | 137  | 2038                | 137  | 2345                |
| 2000                   | -54         | 140                  | 1942                | 139  | 2209                | 138  | 2521                | 138  | 2894                |
|                        | -50         | 140                  | 1924                | 139  | 2188                | 138  | 2498                | 138  | 2867                |
|                        | -40         | 140                  | 1881                | 139  | 2140                | 138  | 2443                | 138  | 2804                |
|                        | -30         | 140                  | 1838                | 139  | 2092                | 138  | 2387                | 138  | 2740                |
|                        | -20         | 140                  | 1800                | 139  | 2048                | 138  | 2338                | 138  | 2683                |
|                        | -10         | 140                  | 1763                | 139  | 2006                | 138  | 2289                | 138  | 2628                |
|                        | 0           | 140                  | 1724                | 139  | 1962                | 138  | 2240                | 138  | 2571                |
|                        | 10          | 138                  | 1353                | 135  | 1559                | 135  | 1797                | 134  | 2080                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 4000                   | -54         | 140                  | 1932                | 139  | 2200                | 138  | 2514                | 138  | 2888                |
|                        | -50         | 140                  | 1914                | 139  | 2180                | 138  | 2490                | 138  | 2861                |
|                        | -40         | 139                  | 1871                | 139  | 2131                | 138  | 2435                | 137  | 2797                |
|                        | -30         | 139                  | 1828                | 139  | 2082                | 138  | 2379                | 137  | 2733                |
|                        | -20         | 139                  | 1790                | 139  | 2038                | 138  | 2329                | 138  | 2676                |
|                        | -10         | 139                  | 1751                | 139  | 1995                | 138  | 2279                | 137  | 2619                |
|                        | 0           | 138                  | 1695                | 137  | 1933                | 137  | 2211                | 136  | 2541                |
|                        | 10          | 138                  | 1180                | 133  | 1376                | 133  | 1599                | 132  | 1863                |
| 6000                   | -54         | 141                  | 1839                | 140  | 2101                | 139  | 2406                | 138  | 2769                |
|                        | -50         | 140                  | 1822                | 140  | 2081                | 139  | 2384                | 138  | 2743                |
|                        | -40         | 140                  | 1779                | 140  | 2032                | 139  | 2328                | 138  | 2679                |
|                        | -30         | 140                  | 1741                | 140  | 1988                | 139  | 2277                | 138  | 2621                |
|                        | -20         | 140                  | 1703                | 140  | 1945                | 139  | 2228                | 138  | 2564                |
|                        | -10         | 140                  | 1667                | 140  | 1904                | 139  | 2180                | 138  | 2510                |
|                        | 0           | 138                  | 1451                | 137  | 1668                | 136  | 1922                | 135  | 2223                |
|                        | 10          | 138                  | 962                 | 133  | 1144                | 132  | 1346                | 132  | 1584                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 8000                   | -54         | 138                  | 1659                | 138  | 1908                | 137  | 2197                | 136  | 2541                |
|                        | -50         | 138                  | 1643                | 138  | 1890                | 137  | 2177                | 136  | 2517                |
|                        | -40         | 138                  | 1606                | 138  | 1847                | 136  | 2128                | 136  | 2461                |
|                        | -30         | 138                  | 1571                | 138  | 1807                | 136  | 2081                | 136  | 2407                |
|                        | -20         | 138                  | 1538                | 138  | 1768                | 136  | 2036                | 136  | 2355                |
|                        | -10         | 138                  | 1504                | 137  | 1730                | 136  | 1992                | 136  | 2304                |
|                        | 0           | 137                  | 1178                | 134  | 1378                | 133  | 1606                | 132  | 1876                |
|                        | 10          | 137                  | 726                 | 132  | 899                 | 130  | 1081                | 129  | 1293                |
| 10,000                 | -54         | 137                  | 1484                | 136  | 1721                | 135  | 1995                | 134  | 2321                |
|                        | -50         | 137                  | 1470                | 136  | 1705                | 135  | 1977                | 134  | 2299                |
|                        | -40         | 137                  | 1437                | 136  | 1666                | 135  | 1932                | 134  | 2247                |
|                        | -30         | 137                  | 1406                | 136  | 1630                | 135  | 1890                | 134  | 2198                |
|                        | -20         | 137                  | 1375                | 136  | 1594                | 135  | 1849                | 134  | 2150                |
|                        | -10         | 137                  | 1344                | 136  | 1559                | 135  | 1808                | 134  | 2103                |
|                        | 0           | 137                  | 927                 | 132  | 1117                | 130  | 1323                | 130  | 1565                |
|                        | 10          | 137                  | 503                 | 131  | 672                 | 128  | 839                 | 127  | 1027                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 12,000                 | -54         | 137                  | 1299                | 134  | 1526                | 133  | 1785                | 132  | 2091                |
|                        | -50         | 137                  | 1287                | 134  | 1512                | 133  | 1769                | 132  | 2072                |
|                        | -40         | 137                  | 1258                | 134  | 1478                | 133  | 1729                | 132  | 2025                |
|                        | -30         | 137                  | 1232                | 134  | 1447                | 133  | 1692                | 132  | 1981                |
|                        | -20         | 137                  | 1206                | 135  | 1416                | 133  | 1655                | 132  | 1939                |
|                        | -10         | 137                  | 1067                | 131  | 1275                | 130  | 1502                | 128  | 1768                |
|                        | 0           | 137                  | 650                 | 131  | 832                 | 128  | 1017                | 127  | 1229                |
|                        | 10          | 137                  | 262                 | 131  | 423                 | 125  | 580                 | 124  | 744                 |
| 14,000                 | -54         | 138                  | 1108                | 132  | 1329                | 131  | 1572                | 130  | 1858                |
|                        | -50         | 138                  | 1098                | 132  | 1316                | 131  | 1557                | 130  | 1840                |
|                        | -40         | 138                  | 1073                | 132  | 1286                | 131  | 1521                | 130  | 1798                |
|                        | -30         | 138                  | 1049                | 132  | 1258                | 131  | 1488                | 130  | 1759                |
|                        | -20         | 138                  | 1026                | 132  | 1230                | 131  | 1455                | 130  | 1720                |
|                        | -10         | 138                  | 783                 | 132  | 982                 | 128  | 1187                | 126  | 1423                |
|                        | 0           | 138                  | 389                 | 132  | 564                 | 125  | 738                 | 124  | 924                 |
|                        | 10          | 138                  | 34                  | 132  | 190                 | 125  | 341                 | 122  | 488                 |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 16,000                 | -54         | 138                  | 936                 | 132  | 1155                | 130  | 1385                | 129  | 1655                |
|                        | -50         | 138                  | 927                 | 132  | 1143                | 129  | 1372                | 129  | 1639                |
|                        | -40         | 138                  | 905                 | 132  | 1117                | 129  | 1340                | 129  | 1601                |
|                        | -30         | 138                  | 884                 | 132  | 1091                | 129  | 1309                | 129  | 1564                |
|                        | -20         | 138                  | 863                 | 132  | 1066                | 129  | 1280                | 129  | 1529                |
|                        | -10         | 138                  | 508                 | 132  | 698                 | 126  | 889                 | 125  | 1096                |
|                        | 0           | 138                  | 138                 | 132  | 304                 | 126  | 469                 | 122  | 633                 |
|                        | 10          | 138                  | -190                | 132  | -44                 | 126  | 98                  | 120  | 232                 |
| 18,000                 | -54         | 137                  | 779                 | 132  | 988                 | 128  | 1208                | 128  | 1462                |
|                        | -50         | 137                  | 771                 | 132  | 979                 | 128  | 1196                | 128  | 1447                |
|                        | -40         | 137                  | 752                 | 132  | 955                 | 128  | 1167                | 128  | 1412                |
|                        | -30         | 137                  | 734                 | 132  | 932                 | 128  | 1140                | 128  | 1380                |
|                        | -20         | 137                  | 671                 | 132  | 867                 | 127  | 1071                | 127  | 1301                |
|                        | -10         | 137                  | 248                 | 132  | 423                 | 125  | 603                 | 123  | 784                 |
|                        | 0           | 137                  | -94                 | 132  | 57                  | 125  | 213                 | 121  | 357                 |
|                        | 9           | 137                  | -369                | 132  | -235                | 125  | -98                 | 119  | 22                  |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 20,000                 | -54         | 137                  | 616                 | 132  | 820                 | 127  | 1031                | 126  | 1269                |
|                        | -50         | 137                  | 610                 | 132  | 812                 | 127  | 1021                | 126  | 1256                |
|                        | -40         | 137                  | 594                 | 132  | 792                 | 127  | 995                 | 126  | 1225                |
|                        | -30         | 137                  | 580                 | 132  | 773                 | 127  | 972                 | 126  | 1196                |
|                        | -20         | 137                  | 379                 | 132  | 561                 | 126  | 746                 | 125  | 944                 |
|                        | -10         | 137                  | 1                   | 132  | 163                 | 125  | 328                 | 122  | 486                 |
|                        | 0           | 137                  | -319                | 132  | -178                | 125  | -35                 | 119  | 93                  |
|                        | 5           | 137                  | -459                | 132  | -327                | 125  | -193                | 119  | -78                 |
| 22,000                 | -56         | 138                  | 480                 | 131  | 681                 | 127  | 883                 | 126  | 1109                |
|                        | -50         | 138                  | 470                 | 131  | 669                 | 126  | 869                 | 126  | 1091                |
|                        | -40         | 138                  | 455                 | 131  | 649                 | 126  | 844                 | 126  | 1060                |
|                        | -30         | 138                  | 442                 | 131  | 631                 | 126  | 820                 | 126  | 1031                |
|                        | -20         | 138                  | 93                  | 131  | 264                 | 125  | 431                 | 122  | 601                 |
|                        | -10         | 138                  | -251                | 131  | -95                 | 125  | 54                  | 120  | 193                 |
|                        | 0           | 138                  | -544                | 131  | -408                | 125  | -280                | 119  | -168                |
| 24,000                 | -58         | 138                  | 316                 | 132  | 511                 | 126  | 703                 | 125  | 912                 |
|                        | -50         | 138                  | 308                 | 132  | 500                 | 126  | 688                 | 125  | 893                 |
|                        | -40         | 138                  | 299                 | 132  | 485                 | 126  | 668                 | 125  | 868                 |
|                        | -30         | 138                  | 255                 | 132  | 451                 | 126  | 635                 | 125  | 828                 |
|                        | -20         | 138                  | -192                | 132  | -31                 | 125  | 124                 | 121  | 270                 |
|                        | -10         | 138                  | -486                | 132  | -339                | 125  | -202                | 119  | -78                 |
|                        | -4          | 138                  | -650                | 132  | -515                | 125  | -390                | 119  | -281                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 26,000                 | -59         | 137                  | 131                 | 132  | 312                 | 126  | 493                 | 125  | 683                 |
|                        | -50         | 137                  | 127                 | 132  | 303                 | 126  | 480                 | 125  | 666                 |
|                        | -40         | 137                  | 121                 | 132  | 294                 | 126  | 467                 | 124  | 648                 |
|                        | -30         | 137                  | -33                 | 132  | 141                 | 126  | 317                 | 122  | 488                 |
|                        | -20         | 137                  | -436                | 132  | -291                | 126  | -146                | 120  | -16                 |
|                        | -10         | 137                  | -694                | 132  | -566                | 126  | -440                | 119  | -332                |
|                        | -8          | 137                  | -744                | 132  | -619                | 126  | -497                | 119  | -392                |
| 28,000                 | -61         | 138                  | -71                 | 131  | 106                 | 125  | 274                 | 124  | 445                 |
|                        | -60         | 138                  | -70                 | 131  | 106                 | 125  | 274                 | 124  | 445                 |
|                        | -50         | 138                  | -69                 | 131  | 103                 | 125  | 267                 | 124  | 434                 |
|                        | -40         | 138                  | -82                 | 131  | 86                  | 125  | 247                 | 124  | 408                 |
|                        | -30         | 138                  | -317                | 131  | -153                | 125  | 4                   | 121  | 147                 |
|                        | -20         | 138                  | -666                | 131  | -524                | 125  | -394                | 120  | -281                |
|                        | -13         | 138                  | -829                | 131  | -699                | 125  | -582                | 119  | -486                |
| 30,000                 | -63         | 137                  | -267                | 132  | -106                | 125  | 50                  | 124  | 201                 |
|                        | -60         | 137                  | -263                | 132  | -103                | 125  | 52                  | 124  | 202                 |
|                        | -50         | 137                  | -256                | 132  | -99                 | 125  | 55                  | 123  | 203                 |
|                        | -40         | 137                  | -293                | 132  | -141                | 125  | 8                   | 122  | 148                 |
|                        | -30         | 137                  | -566                | 132  | -420                | 125  | -278                | 120  | -153                |
|                        | -20         | 137                  | -868                | 132  | -745                | 125  | -627                | 119  | -529                |
|                        | -18         | 137                  | -915                | 132  | -796                | 125  | -682                | 119  | -589                |

| Press<br>Alt<br>(Feet) | OAT<br>(°C) | Aircraft Weight (lb) |                     |      |                     |      |                     |      |                     |
|------------------------|-------------|----------------------|---------------------|------|---------------------|------|---------------------|------|---------------------|
|                        |             | 6000                 |                     | 5500 |                     | 5000 |                     | 4500 |                     |
|                        |             | KIAS                 | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) | KIAS | ROC<br>(FT/M<br>IN) |
| 31,000                 | -64         | 137                  | -379                | 132  | -223                | 126  | -73                 | 123  | 68                  |
|                        | -60         | 137                  | -372                | 132  | -218                | 126  | -70                 | 123  | 70                  |
|                        | -50         | 137                  | -356                | 132  | -206                | 126  | -62                 | 123  | 75                  |
|                        | -40         | 137                  | -398                | 132  | -251                | 126  | -108                | 122  | 24                  |
|                        | -30         | 137                  | -681                | 132  | -542                | 126  | -407                | 120  | -292                |
|                        | -20         | 137                  | -966                | 132  | -848                | 126  | -735                | 119  | -644                |



## Time, Fuel, & Distance to Climb

Thrust Lever ..... MCT  
Landing Gear ..... UP  
Flaps ..... UP  
Anti-ice ..... OFF  
Airspeed ..... Per subsequent tables

### ISA -10

**Aircraft Weight: 6000 lb (2722 kg)**

| ISA -10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 5           | 186  | 0             | 0                  | 0            | 0                |
| 2000             | 1           | 186  | 1             | 2                  | 13           | 2                |
| 4000             | -3          | 186  | 2             | 4                  | 26           | 5                |
| 6000             | -7          | 191  | 3             | 6                  | 39           | 7                |
| 8000             | -11         | 186  | 3             | 8                  | 52           | 10               |
| 10,000           | -15         | 180  | 4             | 9                  | 65           | 13               |
| 12,000           | -19         | 175  | 6             | 11                 | 79           | 16               |
| 14,000           | -23         | 171  | 7             | 13                 | 93           | 19               |
| 16,000           | -27         | 164  | 8             | 15                 | 107          | 23               |
| 18,000           | -31         | 161  | 9             | 17                 | 121          | 27               |
| 20,000           | -35         | 157  | 11            | 19                 | 136          | 32               |
| 22,000           | -39         | 153  | 12            | 21                 | 150          | 37               |
| 24,000           | -43         | 151  | 14            | 23                 | 166          | 42               |
| 26,000           | -47         | 148  | 16            | 26                 | 181          | 48               |
| 28,000           | -50         | 144  | 18            | 28                 | 198          | 55               |
| 30,000           | -54         | 141  | 20            | 30                 | 217          | 64               |
| 31,000           | -56         | 140  | 22            | 32                 | 227          | 69               |

***Aircraft Weight: 5500 lb (2495 kg)***

| ISA -10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 5           | 186  | 0             | 0                  | 0            | 0                |
| 2000             | 1           | 186  | 1             | 2                  | 12           | 2                |
| 4000             | -3          | 185  | 2             | 3                  | 23           | 4                |
| 6000             | -7          | 190  | 2             | 5                  | 35           | 6                |
| 8000             | -11         | 185  | 3             | 7                  | 46           | 9                |
| 10,000           | -15         | 179  | 4             | 8                  | 58           | 11               |
| 12,000           | -19         | 174  | 5             | 10                 | 70           | 14               |
| 14,000           | -23         | 170  | 6             | 12                 | 82           | 17               |
| 16,000           | -27         | 162  | 7             | 14                 | 94           | 21               |
| 18,000           | -31         | 160  | 8             | 15                 | 107          | 24               |
| 20,000           | -35         | 155  | 9             | 17                 | 119          | 28               |
| 22,000           | -39         | 152  | 11            | 19                 | 132          | 32               |
| 24,000           | -43         | 150  | 12            | 21                 | 145          | 37               |
| 26,000           | -47         | 147  | 14            | 22                 | 158          | 42               |
| 28,000           | -50         | 143  | 15            | 24                 | 172          | 48               |
| 30,000           | -54         | 140  | 17            | 26                 | 187          | 55               |
| 31,000           | -56         | 139  | 18            | 27                 | 195          | 59               |

**Aircraft Weight: 5000 lb (2268 kg)**

| ISA -10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 5           | 185  | 0             | 0                  | 0            | 0                |
| 2000             | 1           | 185  | 1             | 2                  | 10           | 2                |
| 4000             | -3          | 185  | 1             | 3                  | 21           | 4                |
| 6000             | -7          | 189  | 2             | 5                  | 31           | 6                |
| 8000             | -11         | 183  | 3             | 6                  | 41           | 8                |
| 10,000           | -15         | 178  | 4             | 7                  | 51           | 10               |
| 12,000           | -19         | 173  | 4             | 9                  | 62           | 12               |
| 14,000           | -23         | 168  | 5             | 10                 | 72           | 15               |
| 16,000           | -27         | 161  | 6             | 12                 | 83           | 18               |
| 18,000           | -31         | 159  | 7             | 13                 | 94           | 21               |
| 20,000           | -35         | 154  | 8             | 15                 | 105          | 25               |
| 22,000           | -39         | 151  | 9             | 16                 | 115          | 28               |
| 24,000           | -43         | 149  | 10            | 18                 | 126          | 32               |
| 26,000           | -47         | 146  | 12            | 19                 | 137          | 36               |
| 28,000           | -50         | 142  | 13            | 21                 | 149          | 41               |
| 30,000           | -54         | 139  | 15            | 23                 | 161          | 47               |
| 31,000           | -56         | 138  | 16            | 23                 | 168          | 50               |

***Aircraft Weight: 4500 lb (2041 kg)***

| ISA -10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 5           | 184  | 0             | 0                  | 0            | 0                |
| 2000             | 1           | 184  | 1             | 1                  | 9            | 2                |
| 4000             | -3          | 184  | 1             | 3                  | 18           | 3                |
| 6000             | -7          | 189  | 2             | 4                  | 27           | 5                |
| 8000             | -11         | 183  | 2             | 5                  | 36           | 7                |
| 10,000           | -15         | 177  | 3             | 7                  | 45           | 9                |
| 12,000           | -19         | 172  | 4             | 8                  | 54           | 11               |
| 14,000           | -23         | 167  | 5             | 9                  | 64           | 13               |
| 16,000           | -27         | 161  | 5             | 10                 | 73           | 16               |
| 18,000           | -31         | 158  | 6             | 12                 | 82           | 19               |
| 20,000           | -35         | 153  | 7             | 13                 | 91           | 21               |
| 22,000           | -39         | 150  | 8             | 14                 | 101          | 24               |
| 24,000           | -43         | 148  | 9             | 16                 | 110          | 28               |
| 26,000           | -47         | 145  | 10            | 17                 | 119          | 31               |
| 28,000           | -50         | 141  | 11            | 18                 | 129          | 36               |
| 30,000           | -54         | 138  | 13            | 19                 | 139          | 40               |
| 31,000           | -56         | 136  | 13            | 20                 | 144          | 43               |

## ISA

**Aircraft Weight: 6000 lb (2722 kg)**

| ISA              |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 15          | 177  | 0             | 0                  | 0            | 0                |
| 2000             | 11          | 173  | 1             | 2                  | 15           | 3                |
| 4000             | 7           | 174  | 2             | 4                  | 30           | 6                |
| 6000             | 3           | 175  | 3             | 6                  | 44           | 8                |
| 8000             | -1          | 173  | 4             | 9                  | 58           | 11               |
| 10,000           | -5          | 169  | 5             | 11                 | 72           | 14               |
| 12,000           | -9          | 165  | 7             | 13                 | 87           | 18               |
| 14,000           | -13         | 159  | 8             | 15                 | 102          | 22               |
| 16,000           | -17         | 159  | 9             | 17                 | 117          | 25               |
| 18,000           | -21         | 157  | 11            | 19                 | 132          | 30               |
| 20,000           | -25         | 152  | 12            | 21                 | 147          | 34               |
| 22,000           | -29         | 153  | 14            | 23                 | 162          | 39               |
| 24,000           | -33         | 151  | 15            | 25                 | 178          | 45               |
| 26,000           | -37         | 147  | 17            | 28                 | 195          | 51               |
| 28,000           | -40         | 144  | 19            | 30                 | 213          | 58               |
| 30,000           | -44         | 140  | 22            | 33                 | 233          | 67               |
| 31,000           | -46         | 139  | 23            | 34                 | 244          | 73               |

***Aircraft Weight: 5500 lb (2495 kg)***

| ISA              |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 15          | 177  | 0             | 0                  | 0            | 0                |
| 2000             | 11          | 172  | 1             | 2                  | 13           | 2                |
| 4000             | 7           | 173  | 2             | 4                  | 26           | 5                |
| 6000             | 3           | 175  | 3             | 6                  | 39           | 7                |
| 8000             | -1          | 173  | 4             | 8                  | 51           | 10               |
| 10,000           | -5          | 168  | 5             | 9                  | 64           | 13               |
| 12,000           | -9          | 164  | 6             | 11                 | 77           | 16               |
| 14,000           | -13         | 158  | 7             | 13                 | 90           | 19               |
| 16,000           | -17         | 158  | 8             | 15                 | 103          | 22               |
| 18,000           | -21         | 156  | 9             | 17                 | 116          | 26               |
| 20,000           | -25         | 152  | 11            | 19                 | 129          | 30               |
| 22,000           | -29         | 152  | 12            | 20                 | 142          | 34               |
| 24,000           | -33         | 150  | 13            | 22                 | 155          | 39               |
| 26,000           | -37         | 146  | 15            | 24                 | 169          | 44               |
| 28,000           | -40         | 142  | 17            | 26                 | 184          | 50               |
| 30,000           | -44         | 139  | 19            | 28                 | 200          | 58               |
| 31,000           | -46         | 138  | 20            | 29                 | 208          | 62               |

**Aircraft Weight: 5000 lb (2268 kg)**

| ISA              |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 15          | 176  | 0             | 0                  | 0            | 0                |
| 2000             | 11          | 171  | 1             | 2                  | 12           | 2                |
| 4000             | 7           | 172  | 2             | 3                  | 23           | 4                |
| 6000             | 3           | 174  | 3             | 5                  | 34           | 7                |
| 8000             | -1          | 172  | 3             | 7                  | 45           | 9                |
| 10,000           | -5          | 166  | 4             | 8                  | 56           | 11               |
| 12,000           | -9          | 163  | 5             | 10                 | 68           | 14               |
| 14,000           | -13         | 157  | 6             | 11                 | 79           | 17               |
| 16,000           | -17         | 157  | 7             | 13                 | 90           | 20               |
| 18,000           | -21         | 155  | 8             | 15                 | 101          | 23               |
| 20,000           | -25         | 150  | 9             | 16                 | 112          | 26               |
| 22,000           | -29         | 151  | 10            | 18                 | 124          | 30               |
| 24,000           | -33         | 149  | 12            | 19                 | 135          | 34               |
| 26,000           | -37         | 145  | 13            | 21                 | 147          | 38               |
| 28,000           | -40         | 141  | 14            | 23                 | 159          | 43               |
| 30,000           | -44         | 138  | 16            | 24                 | 172          | 49               |
| 31,000           | -46         | 136  | 17            | 25                 | 179          | 53               |

***Aircraft Weight: 4500 lb (2041 kg)***

| ISA              |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 15          | 175  | 0             | 0                  | 0            | 0                |
| 2000             | 11          | 171  | 1             | 2                  | 10           | 2                |
| 4000             | 7           | 172  | 1             | 3                  | 20           | 4                |
| 6000             | 3           | 173  | 2             | 4                  | 30           | 6                |
| 8000             | -1          | 171  | 3             | 6                  | 40           | 8                |
| 10,000           | -5          | 165  | 4             | 7                  | 49           | 10               |
| 12,000           | -9          | 162  | 4             | 9                  | 59           | 12               |
| 14,000           | -13         | 156  | 5             | 10                 | 69           | 15               |
| 16,000           | -17         | 156  | 6             | 11                 | 78           | 17               |
| 18,000           | -21         | 153  | 7             | 13                 | 88           | 20               |
| 20,000           | -25         | 150  | 8             | 14                 | 98           | 23               |
| 22,000           | -29         | 150  | 9             | 15                 | 107          | 26               |
| 24,000           | -33         | 147  | 10            | 17                 | 117          | 29               |
| 26,000           | -37         | 144  | 11            | 18                 | 127          | 33               |
| 28,000           | -40         | 140  | 12            | 19                 | 137          | 37               |
| 30,000           | -44         | 137  | 14            | 21                 | 147          | 42               |
| 31,000           | -46         | 135  | 14            | 22                 | 153          | 45               |



## ISA +10

**Aircraft Weight: 6000 lb (2722 kg)**

| ISA +10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 25          | 165  | 0             | 0                  | 0            | 0                |
| 2000             | 21          | 162  | 1             | 3                  | 18           | 4                |
| 4000             | 17          | 162  | 3             | 5                  | 35           | 7                |
| 6000             | 13          | 164  | 4             | 8                  | 52           | 11               |
| 8000             | 9           | 160  | 6             | 10                 | 69           | 14               |
| 10,000           | 5           | 156  | 7             | 13                 | 86           | 19               |
| 12,000           | 1           | 153  | 9             | 15                 | 103          | 23               |
| 14,000           | -3          | 150  | 11            | 18                 | 121          | 28               |
| 16,000           | -7          | 147  | 12            | 20                 | 139          | 33               |
| 18,000           | -11         | 144  | 14            | 23                 | 157          | 38               |
| 20,000           | -15         | 144  | 16            | 25                 | 176          | 44               |
| 22,000           | -19         | 141  | 19            | 28                 | 195          | 51               |
| 24,000           | -23         | 139  | 21            | 31                 | 215          | 59               |
| 26,000           | -27         | 135  | 24            | 34                 | 236          | 67               |
| 28,000           | -30         | 135  | 27            | 37                 | 258          | 76               |
| 30,000           | -34         | 130  | 30            | 40                 | 281          | 87               |
| 31,000           | -36         | 131  | 32            | 42                 | 293          | 93               |

***Aircraft Weight: 5500 lb (2495 kg)***

| ISA +10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 25          | 164  | 0             | 0                  | 0            | 0                |
| 2000             | 21          | 161  | 2             | 2                  | 16           | 3                |
| 4000             | 17          | 161  | 3             | 5                  | 31           | 6                |
| 6000             | 13          | 163  | 4             | 7                  | 45           | 9                |
| 8000             | 9           | 159  | 5             | 9                  | 60           | 13               |
| 10,000           | 5           | 155  | 6             | 11                 | 75           | 16               |
| 12,000           | 1           | 152  | 8             | 13                 | 90           | 20               |
| 14,000           | -3          | 149  | 9             | 15                 | 105          | 24               |
| 16,000           | -7          | 146  | 11            | 18                 | 120          | 29               |
| 18,000           | -11         | 143  | 12            | 20                 | 136          | 33               |
| 20,000           | -15         | 143  | 14            | 22                 | 151          | 38               |
| 22,000           | -19         | 140  | 16            | 24                 | 167          | 44               |
| 24,000           | -23         | 137  | 18            | 26                 | 183          | 50               |
| 26,000           | -27         | 134  | 20            | 29                 | 200          | 57               |
| 28,000           | -30         | 133  | 22            | 31                 | 217          | 64               |
| 30,000           | -34         | 128  | 25            | 33                 | 235          | 72               |
| 31,000           | -36         | 131  | 26            | 35                 | 244          | 77               |

**Aircraft Weight: 5000 lb (2268 kg)**

| ISA +10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 25          | 163  | 0             | 0                  | 0            | 0                |
| 2000             | 21          | 160  | 1             | 2                  | 14           | 3                |
| 4000             | 17          | 160  | 2             | 4                  | 27           | 5                |
| 6000             | 13          | 162  | 3             | 6                  | 40           | 8                |
| 8000             | 9           | 158  | 4             | 8                  | 52           | 11               |
| 10,000           | 5           | 153  | 5             | 10                 | 65           | 14               |
| 12,000           | 1           | 151  | 7             | 11                 | 78           | 17               |
| 14,000           | -3          | 148  | 8             | 13                 | 91           | 21               |
| 16,000           | -7          | 145  | 9             | 15                 | 104          | 25               |
| 18,000           | -11         | 142  | 11            | 17                 | 117          | 29               |
| 20,000           | -15         | 141  | 12            | 19                 | 130          | 33               |
| 22,000           | -19         | 139  | 14            | 21                 | 143          | 38               |
| 24,000           | -23         | 136  | 15            | 23                 | 157          | 43               |
| 26,000           | -27         | 133  | 17            | 24                 | 170          | 48               |
| 28,000           | -30         | 132  | 19            | 26                 | 184          | 54               |
| 30,000           | -34         | 127  | 21            | 28                 | 198          | 61               |
| 31,000           | -36         | 130  | 22            | 29                 | 205          | 64               |

***Aircraft Weight: 4500 lb (2041 kg)***

| ISA +10          |             |      |               |                    |              |                  |
|------------------|-------------|------|---------------|--------------------|--------------|------------------|
| Altitude<br>(ft) | OAT<br>(°C) | KIAS | Time<br>(min) | Fuel<br>(U.S. gal) | Fuel<br>(lb) | Distance<br>(nm) |
| 0                | 25          | 162  | 0             | 0                  | 0            | 0                |
| 2000             | 21          | 159  | 1             | 2                  | 12           | 2                |
| 4000             | 17          | 159  | 2             | 4                  | 23           | 5                |
| 6000             | 13          | 161  | 3             | 5                  | 34           | 7                |
| 8000             | 9           | 157  | 4             | 7                  | 45           | 10               |
| 10,000           | 5           | 152  | 5             | 8                  | 56           | 12               |
| 12,000           | 1           | 149  | 6             | 10                 | 67           | 15               |
| 14,000           | -3          | 147  | 7             | 12                 | 78           | 18               |
| 16,000           | -7          | 144  | 8             | 13                 | 89           | 21               |
| 18,000           | -11         | 141  | 9             | 15                 | 100          | 25               |
| 20,000           | -15         | 139  | 10            | 16                 | 111          | 28               |
| 22,000           | -19         | 138  | 12            | 18                 | 122          | 32               |
| 24,000           | -23         | 134  | 13            | 19                 | 134          | 36               |
| 26,000           | -27         | 132  | 14            | 21                 | 145          | 41               |
| 28,000           | -30         | 131  | 16            | 22                 | 156          | 45               |
| 30,000           | -34         | 125  | 17            | 24                 | 167          | 51               |
| 31,000           | -36         | 130  | 18            | 25                 | 173          | 54               |

**Cruise Performance**

- \*\*\* V<sub>MO</sub>/M<sub>MO</sub> Limited
- \*\* MCT
- \* Max Range Cruise (No Wind)

| Alt<br>FT | Aircraft<br>Weight<br>LB | Δ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|-------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 5000      | 6000                     | 20          | 25        | 89.6**  | 94                  | 242              | 25.6                                       |
|           |                          |             |           | 84.4    | 80                  | 218              | 27.3                                       |
|           |                          |             |           | 79.1    | 68                  | 194              | 28.7                                       |
|           |                          |             |           | 74.1*   | 58                  | 171              | 29.4                                       |
|           |                          | 10          | 15        | 92.2**  | 105                 | 257              | 24.5                                       |
|           |                          |             |           | 85.8    | 85                  | 227              | 26.9                                       |
|           |                          |             |           | 79.2    | 68                  | 197              | 28.9                                       |
|           |                          |             |           | 72.8*   | 56                  | 168              | 29.9                                       |
|           |                          | 0           | 5         | 93.8*** | 113                 | 268              | 23.6                                       |
|           |                          |             |           | 86.4    | 88                  | 233              | 26.5                                       |
|           |                          |             |           | 78.7    | 68                  | 198              | 29.2                                       |
|           |                          |             |           | 71.2*   | 54                  | 163              | 30.4                                       |
|           |                          | -10         | -5        | 92.1*** | 110                 | 264              | 24                                         |
|           |                          |             |           | 85      | 85                  | 230              | 26.9                                       |
|           |                          |             |           | 77.6    | 66                  | 196              | 29.6                                       |
|           |                          |             |           | 70.3*   | 52                  | 162              | 30.9                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 5000      | 5500                     | 20                 | 25        | 89.6**  | 94                  | 243              | 25.8                                       |
|           |                          |                    |           | 83.6    | 78                  | 216              | 27.8                                       |
|           |                          |                    |           | 77.3    | 64                  | 190              | 29.6                                       |
|           |                          |                    |           | 71.6*   | 54                  | 163              | 30.4                                       |
|           |                          | 10                 | 15        | 92.1**  | 105                 | 258              | 24.6                                       |
|           |                          |                    |           | 85.2    | 83                  | 226              | 27.3                                       |
|           |                          |                    |           | 77.7    | 65                  | 194              | 29.7                                       |
|           |                          |                    |           | 70.7*   | 52                  | 162              | 30.9                                       |
|           |                          | 0                  | 5         | 93.6*** | 113                 | 268              | 23.8                                       |
|           |                          |                    |           | 85.8    | 86                  | 232              | 26.9                                       |
|           |                          |                    |           | 77.5    | 65                  | 196              | 29.9                                       |
|           |                          |                    |           | 69.4*   | 51                  | 159              | 31.5                                       |
|           |                          | -10                | -5        | 91.9*** | 109                 | 264              | 24.2                                       |
|           |                          |                    |           | 84.3    | 83                  | 228              | 27.3                                       |
|           |                          |                    |           | 76.3    | 64                  | 193              | 30.3                                       |
|           |                          |                    |           | 68.4*   | 49                  | 158              | 32                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 5000      | 5000                     | 20                 | 25        | 89.5**  | 94                  | 244              | 25.9                                       |
|           |                          |                    |           | 83.1    | 76                  | 216              | 28.3                                       |
|           |                          |                    |           | 76.2    | 62                  | 188              | 30.4                                       |
|           |                          |                    |           | 69.8*   | 51                  | 160              | 31.4                                       |
|           |                          | 10                 | 15        | 92.1**  | 105                 | 259              | 24.7                                       |
|           |                          |                    |           | 84.8    | 82                  | 226              | 27.6                                       |
|           |                          |                    |           | 76.8    | 64                  | 193              | 30.3                                       |
|           |                          |                    |           | 69.2*   | 50                  | 160              | 31.9                                       |
|           |                          | 0                  | 5         | 93.4*** | 112                 | 268              | 23.9                                       |
|           |                          |                    |           | 85.1    | 84                  | 230              | 27.4                                       |
|           |                          |                    |           | 76      | 62                  | 192              | 30.7                                       |
|           |                          |                    |           | 67.2*   | 47                  | 154              | 32.5                                       |
|           |                          | -10                | -5        | 91.7*** | 108                 | 264              | 24.3                                       |
|           |                          |                    |           | 83.5    | 81                  | 226              | 27.9                                       |
|           |                          |                    |           | 74.4    | 60                  | 188              | 31.3                                       |
|           |                          |                    |           | 65.7*   | 45                  | 150              | 33                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 5000      | 4500                     | 20                 | 25        | 89.5**  | 94                  | 246              | 26                                         |
|           |                          |                    |           | 82.7    | 75                  | 216              | 28.6                                       |
|           |                          |                    |           | 75.2    | 60                  | 187              | 31.1                                       |
|           |                          |                    |           | 68.1*   | 49                  | 158              | 32.4                                       |
|           |                          | 10                 | 15        | 92.1**  | 105                 | 260              | 24.8                                       |
|           |                          |                    |           | 83.7    | 79                  | 223              | 28.2                                       |
|           |                          |                    |           | 74.3    | 59                  | 185              | 31.5                                       |
|           |                          |                    |           | 65.4*   | 45                  | 148              | 32.9                                       |
|           |                          | 0                  | 5         | 93.2*** | 111                 | 268              | 24.1                                       |
|           |                          |                    |           | 84.6    | 83                  | 229              | 27.7                                       |
|           |                          |                    |           | 74.9    | 60                  | 190              | 31.4                                       |
|           |                          |                    |           | 65.4*   | 45                  | 151              | 33.5                                       |
|           |                          | -10                | -5        | 91.5*** | 108                 | 264              | 24.5                                       |
|           |                          |                    |           | 83.3    | 80                  | 226              | 28.1                                       |
|           |                          |                    |           | 74      | 59                  | 188              | 31.8                                       |
|           |                          |                    |           | 64.8*   | 44                  | 151              | 34.1                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 10,000    | 6000                     | 20                 | 15        | 91.8**  | 87                  | 252              | 29                                         |
|           |                          |                    |           | 87.2    | 75                  | 230              | 30.7                                       |
|           |                          |                    |           | 82.8    | 65                  | 207              | 32                                         |
|           |                          |                    |           | 78.4*   | 56                  | 185              | 32.7                                       |
|           |                          | 10                 | 5         | 94.5**  | 98                  | 269              | 27.6                                       |
|           |                          |                    |           | 88.6    | 80                  | 240              | 30.1                                       |
|           |                          |                    |           | 82.6    | 65                  | 210              | 32.2                                       |
|           |                          |                    |           | 76.9*   | 54                  | 181              | 33.3                                       |
|           |                          | 0                  | -5        | 97.2*** | 110                 | 287              | 26                                         |
|           |                          |                    |           | 90.1    | 86                  | 251              | 29.3                                       |
|           |                          |                    |           | 82.8    | 67                  | 215              | 32.2                                       |
|           |                          |                    |           | 75.7*   | 53                  | 179              | 33.8                                       |
|           |                          | -10                | -15       | 95.7*** | 108                 | 283              | 26.3                                       |
|           |                          |                    |           | 88.6    | 83                  | 247              | 29.7                                       |
|           |                          |                    |           | 81.3    | 64                  | 211              | 32.8                                       |
|           |                          |                    |           | 74.2*   | 51                  | 175              | 34.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 10,000    | 5500                     | 20                 | 15        | 91.7**  | 87                  | 254              | 29.2                                       |
|           |                          |                    |           | 86.4    | 73                  | 228              | 31.3                                       |
|           |                          |                    |           | 81      | 61                  | 202              | 33.1                                       |
|           |                          |                    |           | 75.8*   | 52                  | 177              | 33.9                                       |
|           |                          | 10                 | 5         | 94.4**  | 98                  | 271              | 27.7                                       |
|           |                          |                    |           | 87.8    | 78                  | 238              | 30.7                                       |
|           |                          |                    |           | 81      | 62                  | 206              | 33.2                                       |
|           |                          |                    |           | 74.3*   | 50                  | 173              | 34.5                                       |
|           |                          | 0                  | -5        | 97.2*** | 110                 | 288              | 26.1                                       |
|           |                          |                    |           | 89.3    | 83                  | 249              | 29.8                                       |
|           |                          |                    |           | 81.1    | 63                  | 209              | 33.3                                       |
|           |                          |                    |           | 73.0*   | 48                  | 170              | 35.1                                       |
|           |                          | -10                | -15       | 95.5*** | 107                 | 283              | 26.5                                       |
|           |                          |                    |           | 87.7    | 80                  | 244              | 30.3                                       |
|           |                          |                    |           | 79.5    | 61                  | 205              | 33.9                                       |
|           |                          |                    |           | 71.5*   | 47                  | 166              | 35.7                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 10,000    | 5000                     | 20                 | 15        | 91.7**  | 87                  | 255              | 29.4                                       |
|           |                          |                    |           | 85.5    | 71                  | 226              | 31.9                                       |
|           |                          |                    |           | 79.1    | 58                  | 197              | 34.2                                       |
|           |                          |                    |           | 73.0*   | 48                  | 169              | 35.2                                       |
|           |                          | 10                 | 5         | 94.4**  | 98                  | 272              | 27.8                                       |
|           |                          |                    |           | 87      | 76                  | 236              | 31.3                                       |
|           |                          |                    |           | 79.2    | 58                  | 200              | 34.4                                       |
|           |                          |                    |           | 71.5*   | 46                  | 165              | 35.8                                       |
|           |                          | 0                  | -5        | 97.2*** | 110                 | 289              | 26.2                                       |
|           |                          |                    |           | 88.6    | 81                  | 247              | 30.3                                       |
|           |                          |                    |           | 79.5    | 60                  | 204              | 34.3                                       |
|           |                          |                    |           | 70.3*   | 44                  | 162              | 36.5                                       |
|           |                          | -10                | -15       | 95.3*** | 106                 | 283              | 26.7                                       |
|           |                          |                    |           | 86.8    | 78                  | 242              | 30.9                                       |
|           |                          |                    |           | 77.9    | 57                  | 200              | 34.9                                       |
|           |                          |                    |           | 68.9*   | 43                  | 159              | 37.1                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 10,000    | 4500                     | 20                 | 15        | 91.7**  | 87                  | 257              | 29.5                                       |
|           |                          |                    |           | 84.5    | 69                  | 224              | 32.7                                       |
|           |                          |                    |           | 77      | 54                  | 192              | 35.4                                       |
|           |                          |                    |           | 69.9*   | 43                  | 159              | 36.6                                       |
|           |                          | 10                 | 5         | 94.4**  | 98                  | 273              | 28                                         |
|           |                          |                    |           | 86.3    | 74                  | 235              | 31.7                                       |
|           |                          |                    |           | 77.7    | 56                  | 197              | 35.4                                       |
|           |                          |                    |           | 69.2*   | 43                  | 159              | 37.3                                       |
|           |                          | 0                  | -5        | 97.2*** | 110                 | 290              | 26.3                                       |
|           |                          |                    |           | 88.2    | 80                  | 246              | 30.7                                       |
|           |                          |                    |           | 78.5    | 58                  | 202              | 35.2                                       |
|           |                          |                    |           | 68.4*   | 42                  | 159              | 38                                         |
|           |                          | -10                | -15       | 95.1*** | 106                 | 283              | 26.8                                       |
|           |                          |                    |           | 86.2    | 77                  | 240              | 31.4                                       |
|           |                          |                    |           | 76.7    | 55                  | 197              | 35.9                                       |
|           |                          |                    |           | 66.9*   | 40                  | 155              | 38.6                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 15,000    | 6000                     | 20                 | 5         | 93.0** | 78                  | 258              | 33.2                                       |
|           |                          |                    |           | 89.3   | 68                  | 238              | 34.8                                       |
|           |                          |                    |           | 85.7   | 60                  | 217              | 35.9                                       |
|           |                          |                    |           | 82.3*  | 54                  | 197              | 36.5                                       |
|           |                          | 10                 | -5        | 95.8** | 88                  | 277              | 31.6                                       |
|           |                          |                    |           | 90.6   | 73                  | 249              | 34.1                                       |
|           |                          |                    |           | 85.5   | 61                  | 221              | 36.2                                       |
|           |                          |                    |           | 80.7*  | 52                  | 193              | 37.1                                       |
|           |                          | 0                  | -15       | 98.1** | 99                  | 294              | 29.8                                       |
|           |                          |                    |           | 91.7   | 78                  | 259              | 33.3                                       |
|           |                          |                    |           | 85.2   | 62                  | 224              | 36.3                                       |
|           |                          |                    |           | 79.2*  | 50                  | 189              | 37.7                                       |
|           |                          | -10                | -25       | 98.6** | 105                 | 302              | 28.7                                       |
|           |                          |                    |           | 91.9   | 81                  | 264              | 32.8                                       |
|           |                          |                    |           | 84.8   | 62                  | 226              | 36.4                                       |
|           |                          |                    |           | 78.0*  | 49                  | 188              | 38.3                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 15,000    | 5500                     | 20                 | 5         | 93.0**  | 78                  | 260              | 33.5                                       |
|           |                          |                    |           | 88.6    | 67                  | 237              | 35.5                                       |
|           |                          |                    |           | 84.1    | 57                  | 214              | 37.1                                       |
|           |                          |                    |           | 80.0*   | 50                  | 190              | 37.8                                       |
|           |                          | 10                 | -5        | 95.8**  | 88                  | 278              | 31.8                                       |
|           |                          |                    |           | 89.8    | 71                  | 247              | 34.8                                       |
|           |                          |                    |           | 83.9    | 58                  | 217              | 37.4                                       |
|           |                          |                    |           | 78.4*   | 48                  | 186              | 38.5                                       |
|           |                          | 0                  | -15       | 98.0**  | 99                  | 296              | 29.9                                       |
|           |                          |                    |           | 91      | 76                  | 258              | 33.9                                       |
|           |                          |                    |           | 83.8    | 59                  | 220              | 37.4                                       |
|           |                          |                    |           | 77.0*   | 47                  | 183              | 39.1                                       |
|           |                          | -10                | -25       | 98.6*** | 105                 | 304              | 28.8                                       |
|           |                          |                    |           | 91.2    | 78                  | 262              | 33.4                                       |
|           |                          |                    |           | 83.2    | 59                  | 221              | 37.6                                       |
|           |                          |                    |           | 75.6*   | 45                  | 180              | 39.8                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 15,000    | 5000                     | 20                 | 5         | 93.0**  | 78                  | 262              | 33.7                                       |
|           |                          |                    |           | 87.8    | 65                  | 236              | 36.3                                       |
|           |                          |                    |           | 82.5    | 55                  | 210              | 38.4                                       |
|           |                          |                    |           | 77.4*   | 46                  | 183              | 39.4                                       |
|           |                          | 10                 | -5        | 95.7**  | 88                  | 280              | 31.9                                       |
|           |                          |                    |           | 89.2    | 70                  | 247              | 35.5                                       |
|           |                          |                    |           | 82.5    | 55                  | 213              | 38.5                                       |
|           |                          |                    |           | 76.1*   | 45                  | 180              | 40.1                                       |
|           |                          | 0                  | -15       | 98.0**  | 99                  | 297              | 30                                         |
|           |                          |                    |           | 90.4    | 75                  | 257              | 34.4                                       |
|           |                          |                    |           | 82.4    | 56                  | 217              | 38.5                                       |
|           |                          |                    |           | 74.7*   | 43                  | 177              | 40.7                                       |
|           |                          | -10                | -25       | 98.6*** | 106                 | 305              | 28.9                                       |
|           |                          |                    |           | 90.5    | 77                  | 261              | 34                                         |
|           |                          |                    |           | 81.6    | 56                  | 216              | 38.8                                       |
|           |                          |                    |           | 73.0*   | 42                  | 172              | 41.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 15,000    | 4500                     | 20                 | 5         | 93.0**  | 78                  | 264              | 34                                         |
|           |                          |                    |           | 86.8    | 63                  | 234              | 37.1                                       |
|           |                          |                    |           | 80.6    | 51                  | 204              | 39.8                                       |
|           |                          |                    |           | 74.4*   | 42                  | 174              | 41                                         |
|           |                          | 10                 | -5        | 95.7**  | 88                  | 281              | 32.1                                       |
|           |                          |                    |           | 88.3    | 67                  | 244              | 36.2                                       |
|           |                          |                    |           | 80.6    | 52                  | 207              | 39.9                                       |
|           |                          |                    |           | 73.0*   | 41                  | 171              | 41.7                                       |
|           |                          | 0                  | -15       | 98.0**  | 99                  | 298              | 30.1                                       |
|           |                          |                    |           | 89.6    | 72                  | 254              | 35.1                                       |
|           |                          |                    |           | 80.7    | 53                  | 211              | 39.9                                       |
|           |                          |                    |           | 71.7*   | 39                  | 168              | 42.4                                       |
|           |                          | -10                | -25       | 98.6*** | 106                 | 306              | 28.9                                       |
|           |                          |                    |           | 89.7    | 75                  | 258              | 34.6                                       |
|           |                          |                    |           | 80      | 53                  | 211              | 40.1                                       |
|           |                          |                    |           | 70.1*   | 38                  | 164              | 43.1                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 17,000    | 6000                     | 20                 | 1         | 93.5** | 74                  | 260              | 35.1                                       |
|           |                          |                    |           | 90.3   | 66                  | 241              | 36.5                                       |
|           |                          |                    |           | 87     | 59                  | 223              | 37.6                                       |
|           |                          |                    |           | 84.0*  | 54                  | 204              | 38                                         |
|           |                          | 10                 | -9        | 96.1** | 83                  | 279              | 33.4                                       |
|           |                          |                    |           | 91.5   | 71                  | 253              | 35.8                                       |
|           |                          |                    |           | 87.1   | 60                  | 228              | 37.7                                       |
|           |                          |                    |           | 82.8*  | 52                  | 203              | 38.7                                       |
|           |                          | 0                  | -19       | 98.5** | 95                  | 298              | 31.3                                       |
|           |                          |                    |           | 92.7   | 76                  | 264              | 34.8                                       |
|           |                          |                    |           | 86.7   | 61                  | 231              | 37.9                                       |
|           |                          |                    |           | 81.1*  | 50                  | 198              | 39.3                                       |
|           |                          | -10                | -29       | 98.5** | 99                  | 302              | 30.6                                       |
|           |                          |                    |           | 92.2   | 76                  | 266              | 34.7                                       |
|           |                          |                    |           | 85.5   | 60                  | 229              | 38.2                                       |
|           |                          |                    |           | 79.3*  | 48                  | 193              | 39.9                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 17,000    | 5500                     | 20                 | 1         | 93.4** | 74                  | 262              | 35.4                                       |
|           |                          |                    |           | 89.4   | 64                  | 240              | 37.4                                       |
|           |                          |                    |           | 85.3   | 56                  | 218              | 38.9                                       |
|           |                          |                    |           | 81.6*  | 50                  | 196              | 39.5                                       |
|           |                          | 10                 | -9        | 96.1** | 83                  | 281              | 33.7                                       |
|           |                          |                    |           | 90.6   | 69                  | 251              | 36.6                                       |
|           |                          |                    |           | 85.2   | 57                  | 222              | 39.1                                       |
|           |                          |                    |           | 80.2*  | 48                  | 193              | 40.2                                       |
|           |                          | 0                  | -19       | 98.5** | 95                  | 299              | 31.4                                       |
|           |                          |                    |           | 91.9   | 74                  | 262              | 35.6                                       |
|           |                          |                    |           | 85     | 57                  | 225              | 39.2                                       |
|           |                          |                    |           | 78.5*  | 46                  | 188              | 40.9                                       |
|           |                          | -10                | -29       | 98.6** | 99                  | 304              | 30.8                                       |
|           |                          |                    |           | 91.7   | 75                  | 265              | 35.3                                       |
|           |                          |                    |           | 84.4   | 58                  | 226              | 39.3                                       |
|           |                          |                    |           | 77.5*  | 45                  | 188              | 41.5                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 17,000    | 5000                     | 20                 | 1         | 93.4** | 74                  | 265              | 35.7                                       |
|           |                          |                    |           | 88.5   | 62                  | 239              | 38.3                                       |
|           |                          |                    |           | 83.5   | 53                  | 213              | 40.3                                       |
|           |                          |                    |           | 78.8*  | 45                  | 187              | 41.2                                       |
|           |                          | 10                 | -9        | 96.1** | 83                  | 283              | 33.8                                       |
|           |                          |                    |           | 89.8   | 67                  | 250              | 37.4                                       |
|           |                          |                    |           | 83.5   | 54                  | 217              | 40.5                                       |
|           |                          |                    |           | 77.5*  | 44                  | 184              | 41.9                                       |
|           |                          | 0                  | -19       | 98.4** | 95                  | 301              | 31.6                                       |
|           |                          |                    |           | 91.2   | 72                  | 261              | 36.2                                       |
|           |                          |                    |           | 83.5   | 55                  | 221              | 40.4                                       |
|           |                          |                    |           | 76.2*  | 43                  | 181              | 42.5                                       |
|           |                          | -10                | -29       | 98.6** | 99                  | 305              | 30.8                                       |
|           |                          |                    |           | 90.7   | 73                  | 262              | 36.1                                       |
|           |                          |                    |           | 82.4   | 54                  | 219              | 40.8                                       |
|           |                          |                    |           | 74.5*  | 41                  | 177              | 43.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 17,000    | 4500                     | 20                 | 1         | 93.4** | 74                  | 267              | 36                                         |
|           |                          |                    |           | 87.8   | 61                  | 238              | 39.1                                       |
|           |                          |                    |           | 82     | 50                  | 210              | 41.7                                       |
|           |                          |                    |           | 76.4*  | 42                  | 181              | 42.9                                       |
|           |                          | 10                 | -9        | 96.1** | 84                  | 284              | 34                                         |
|           |                          |                    |           | 89.1   | 65                  | 248              | 38.2                                       |
|           |                          |                    |           | 81.9   | 51                  | 213              | 41.8                                       |
|           |                          |                    |           | 74.9*  | 41                  | 177              | 43.7                                       |
|           |                          | 0                  | -19       | 98.4** | 95                  | 302              | 31.7                                       |
|           |                          |                    |           | 90.5   | 70                  | 259              | 36.9                                       |
|           |                          |                    |           | 82     | 52                  | 217              | 41.7                                       |
|           |                          |                    |           | 73.5*  | 39                  | 174              | 44.4                                       |
|           |                          | -10                | -29       | 98.6** | 99                  | 306              | 30.9                                       |
|           |                          |                    |           | 90.1   | 71                  | 261              | 36.7                                       |
|           |                          |                    |           | 81     | 51                  | 215              | 42.1                                       |
|           |                          |                    |           | 72.0*  | 38                  | 170              | 45.1                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 19,000    | 6000                     | 20                 | -3        | 94.0** | 71                  | 261              | 37                                         |
|           |                          |                    |           | 91     | 64                  | 244              | 38.4                                       |
|           |                          |                    |           | 88.2   | 58                  | 227              | 39.3                                       |
|           |                          |                    |           | 85.6*  | 53                  | 210              | 39.7                                       |
|           |                          | 10                 | -13       | 96.5** | 79                  | 281              | 35.4                                       |
|           |                          |                    |           | 92.1   | 68                  | 256              | 37.7                                       |
|           |                          |                    |           | 88     | 58                  | 231              | 39.6                                       |
|           |                          |                    |           | 84.0*  | 51                  | 206              | 40.4                                       |
|           |                          | 0                  | -23       | 99.4** | 93                  | 303              | 32.7                                       |
|           |                          |                    |           | 93.6   | 74                  | 269              | 36.5                                       |
|           |                          |                    |           | 87.7   | 59                  | 235              | 39.7                                       |
|           |                          |                    |           | 82.3*  | 49                  | 201              | 41.1                                       |
|           |                          | -10                | -33       | 98.8** | 93                  | 303              | 32.6                                       |
|           |                          |                    |           | 92.6   | 73                  | 268              | 36.7                                       |
|           |                          |                    |           | 86.4   | 58                  | 233              | 40                                         |
|           |                          |                    |           | 80.8*  | 47                  | 198              | 41.7                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 19,000    | 5500                     | 20                 | -3        | 93.9** | 71                  | 264              | 37.4                                       |
|           |                          |                    |           | 90.2   | 62                  | 243              | 39.4                                       |
|           |                          |                    |           | 86.6   | 55                  | 222              | 40.7                                       |
|           |                          |                    |           | 83.2*  | 49                  | 202              | 41.3                                       |
|           |                          | 10                 | -13       | 96.5** | 79                  | 283              | 35.7                                       |
|           |                          |                    |           | 91.3   | 66                  | 255              | 38.5                                       |
|           |                          |                    |           | 86.3   | 55                  | 226              | 41                                         |
|           |                          |                    |           | 81.7*  | 47                  | 198              | 42                                         |
|           |                          | 0                  | -23       | 99.3** | 93                  | 305              | 32.9                                       |
|           |                          |                    |           | 93.1   | 72                  | 269              | 37.1                                       |
|           |                          |                    |           | 86.6   | 57                  | 233              | 40.8                                       |
|           |                          |                    |           | 80.5*  | 46                  | 197              | 42.7                                       |
|           |                          | -10                | -33       | 98.8** | 93                  | 305              | 32.7                                       |
|           |                          |                    |           | 91.9   | 71                  | 267              | 37.4                                       |
|           |                          |                    |           | 84.9   | 55                  | 228              | 41.5                                       |
|           |                          |                    |           | 78.4*  | 44                  | 190              | 43.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 19,000    | 5000                     | 20                 | -3        | 93.9** | 71                  | 267              | 37.8                                       |
|           |                          |                    |           | 89.5   | 60                  | 243              | 40.3                                       |
|           |                          |                    |           | 85     | 52                  | 219              | 42.2                                       |
|           |                          |                    |           | 80.8*  | 45                  | 195              | 43                                         |
|           |                          | 10                 | -13       | 96.5** | 79                  | 285              | 35.9                                       |
|           |                          |                    |           | 90.5   | 64                  | 253              | 39.4                                       |
|           |                          |                    |           | 84.6   | 52                  | 221              | 42.5                                       |
|           |                          |                    |           | 79.1*  | 43                  | 190              | 43.8                                       |
|           |                          | 0                  | -23       | 99.3** | 93                  | 306              | 33                                         |
|           |                          |                    |           | 92.3   | 70                  | 267              | 37.9                                       |
|           |                          |                    |           | 85     | 54                  | 227              | 42.2                                       |
|           |                          |                    |           | 77.9*  | 42                  | 188              | 44.5                                       |
|           |                          | -10                | -33       | 98.8** | 93                  | 307              | 32.8                                       |
|           |                          |                    |           | 91.2   | 69                  | 265              | 38.1                                       |
|           |                          |                    |           | 83.3   | 52                  | 223              | 42.9                                       |
|           |                          |                    |           | 76.0*  | 40                  | 182              | 45.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1    | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|--------|---------------------|------------------|--------------------------------------------|
| 19,000    | 4500                     | 20                 | -3        | 93.8** | 71                  | 269              | 38.1                                       |
|           |                          |                    |           | 88.4   | 58                  | 241              | 41.3                                       |
|           |                          |                    |           | 82.9   | 48                  | 212              | 43.9                                       |
|           |                          |                    |           | 77.7*  | 41                  | 184              | 44.9                                       |
|           |                          | 10                 | -13       | 96.4** | 79                  | 286              | 36.1                                       |
|           |                          |                    |           | 89.7   | 62                  | 251              | 40.3                                       |
|           |                          |                    |           | 82.8   | 49                  | 216              | 44                                         |
|           |                          |                    |           | 76.3*  | 40                  | 181              | 45.7                                       |
|           |                          | 0                  | -23       | 99.3** | 93                  | 308              | 33.2                                       |
|           |                          |                    |           | 91.4   | 68                  | 264              | 38.8                                       |
|           |                          |                    |           | 82.9   | 50                  | 220              | 43.9                                       |
|           |                          |                    |           | 74.7*  | 38                  | 177              | 46.5                                       |
|           |                          | -10                | -33       | 98.8** | 93                  | 308              | 32.9                                       |
|           |                          |                    |           | 90.5   | 68                  | 263              | 38.8                                       |
|           |                          |                    |           | 81.8   | 49                  | 219              | 44.3                                       |
|           |                          |                    |           | 73.4*  | 37                  | 174              | 47.3                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 21,000    | 6000                     | 20                 | -7        | 94.4**  | 67                  | 263              | 39.1                                       |
|           |                          |                    |           | 91.8    | 61                  | 246              | 40.3                                       |
|           |                          |                    |           | 89.3    | 56                  | 230              | 41.1                                       |
|           |                          |                    |           | 87.0*   | 52                  | 214              | 41.5                                       |
|           |                          | 10                 | -17       | 97.0**  | 76                  | 283              | 37.3                                       |
|           |                          |                    |           | 92.9    | 65                  | 259              | 39.6                                       |
|           |                          |                    |           | 88.9    | 57                  | 235              | 41.4                                       |
|           |                          |                    |           | 85.4*   | 50                  | 210              | 42.2                                       |
|           |                          | 0                  | -27       | 100.1** | 89                  | 306              | 34.6                                       |
|           |                          |                    |           | 94.4    | 71                  | 273              | 38.4                                       |
|           |                          |                    |           | 88.7    | 58                  | 239              | 41.5                                       |
|           |                          |                    |           | 83.7*   | 48                  | 206              | 42.9                                       |
|           |                          | -10                | -37       | 99.2**  | 88                  | 305              | 34.6                                       |
|           |                          |                    |           | 93      | 70                  | 270              | 38.7                                       |
|           |                          |                    |           | 87.2    | 56                  | 236              | 42                                         |
|           |                          |                    |           | 82.0*   | 46                  | 202              | 43.5                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 21,000    | 5500                     | 20                 | -7        | 94.4**  | 67                  | 266              | 39.6                                       |
|           |                          |                    |           | 91.1    | 60                  | 247              | 41.3                                       |
|           |                          |                    |           | 87.9    | 53                  | 228              | 42.6                                       |
|           |                          |                    |           | 84.9*   | 48                  | 209              | 43.1                                       |
|           |                          | 10                 | -17       | 97.0**  | 76                  | 286              | 37.6                                       |
|           |                          |                    |           | 92.3    | 64                  | 259              | 40.4                                       |
|           |                          |                    |           | 87.8    | 54                  | 233              | 42.8                                       |
|           |                          |                    |           | 83.6*   | 47                  | 206              | 43.9                                       |
|           |                          | 0                  | -27       | 100.0** | 88                  | 308              | 34.8                                       |
|           |                          |                    |           | 93.9    | 70                  | 273              | 39.1                                       |
|           |                          |                    |           | 87.7    | 56                  | 238              | 42.7                                       |
|           |                          |                    |           | 82.1*   | 46                  | 203              | 44.6                                       |
|           |                          | -10                | -37       | 99.2**  | 88                  | 307              | 34.8                                       |
|           |                          |                    |           | 92.5    | 68                  | 270              | 39.4                                       |
|           |                          |                    |           | 86      | 54                  | 233              | 43.3                                       |
|           |                          |                    |           | 80.1*   | 43                  | 197              | 45.3                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 21,000    | 5000                     | 20                 | -7        | 94.3**  | 67                  | 269              | 40.1                                       |
|           |                          |                    |           | 90.2    | 58                  | 246              | 42.4                                       |
|           |                          |                    |           | 86.1    | 50                  | 223              | 44.2                                       |
|           |                          |                    |           | 82.3*   | 44                  | 200              | 45                                         |
|           |                          | 10                 | -17       | 96.9**  | 76                  | 288              | 37.9                                       |
|           |                          |                    |           | 91.5    | 62                  | 258              | 41.4                                       |
|           |                          |                    |           | 86      | 51                  | 227              | 44.4                                       |
|           |                          |                    |           | 80.9*   | 43                  | 197              | 45.7                                       |
|           |                          | 0                  | -27       | 100.0** | 89                  | 310              | 35                                         |
|           |                          |                    |           | 93      | 68                  | 271              | 39.9                                       |
|           |                          |                    |           | 86      | 52                  | 232              | 44.3                                       |
|           |                          |                    |           | 79.3*   | 42                  | 193              | 46.5                                       |
|           |                          | -10                | -37       | 99.2**  | 88                  | 308              | 34.9                                       |
|           |                          |                    |           | 91.8    | 67                  | 269              | 40.2                                       |
|           |                          |                    |           | 84.6    | 51                  | 229              | 44.8                                       |
|           |                          |                    |           | 77.8*   | 40                  | 189              | 47.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 21,000    | 4500                     | 20                 | -7        | 94.3**  | 67                  | 272              | 40.4                                       |
|           |                          |                    |           | 89.3    | 56                  | 245              | 43.6                                       |
|           |                          |                    |           | 84.2    | 47                  | 218              | 46                                         |
|           |                          |                    |           | 79.5*   | 41                  | 191              | 47                                         |
|           |                          | 10                 | -17       | 96.9**  | 76                  | 290              | 38.1                                       |
|           |                          |                    |           | 90.5    | 60                  | 256              | 42.4                                       |
|           |                          |                    |           | 84.2    | 48                  | 222              | 46.1                                       |
|           |                          |                    |           | 78.1*   | 39                  | 188              | 47.8                                       |
|           |                          | 0                  | -27       | 100.0** | 89                  | 311              | 35.1                                       |
|           |                          |                    |           | 92.2    | 66                  | 269              | 40.8                                       |
|           |                          |                    |           | 84.2    | 49                  | 226              | 46                                         |
|           |                          |                    |           | 76.5*   | 38                  | 184              | 48.6                                       |
|           |                          | -10                | -37       | 99.2**  | 89                  | 310              | 35                                         |
|           |                          |                    |           | 91      | 65                  | 266              | 41                                         |
|           |                          |                    |           | 82.7    | 48                  | 223              | 46.6                                       |
|           |                          |                    |           | 74.9*   | 36                  | 179              | 49.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 23,000    | 6000                     | 20                 | -11       | 94.8**  | 64                  | 263              | 41.3                                       |
|           |                          |                    |           | 92.9    | 60                  | 251              | 42.1                                       |
|           |                          |                    |           | 91.1    | 56                  | 239              | 42.8                                       |
|           |                          |                    |           | 89.5*   | 53                  | 227              | 43.1                                       |
|           |                          | 10                 | -21       | 97.5**  | 73                  | 287              | 39.1                                       |
|           |                          |                    |           | 94.2    | 64                  | 265              | 41.4                                       |
|           |                          |                    |           | 90.7    | 56                  | 243              | 43                                         |
|           |                          |                    |           | 87.6*   | 50                  | 221              | 43.9                                       |
|           |                          | 0                  | -31       | 100.5** | 84                  | 308              | 36.7                                       |
|           |                          |                    |           | 94.8    | 68                  | 276              | 40.4                                       |
|           |                          |                    |           | 89.8    | 56                  | 244              | 43.4                                       |
|           |                          |                    |           | 85.2*   | 47                  | 212              | 44.7                                       |
|           |                          | -10                | -41       | 99.6**  | 83                  | 306              | 36.8                                       |
|           |                          |                    |           | 93.4    | 67                  | 273              | 40.7                                       |
|           |                          |                    |           | 88.2    | 54                  | 240              | 44                                         |
|           |                          |                    |           | 83.4*   | 46                  | 207              | 45.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 23,000    | 5500                     | 20                 | -11       | 94.7**  | 64                  | 267              | 41.9                                       |
|           |                          |                    |           | 91.7    | 57                  | 249              | 43.5                                       |
|           |                          |                    |           | 89      | 52                  | 231              | 44.6                                       |
|           |                          |                    |           | 86.3*   | 47                  | 213              | 45.1                                       |
|           |                          | 10                 | -21       | 97.5**  | 73                  | 290              | 39.5                                       |
|           |                          |                    |           | 93.1    | 62                  | 263              | 42.5                                       |
|           |                          |                    |           | 88.6    | 53                  | 236              | 44.8                                       |
|           |                          |                    |           | 84.7*   | 45                  | 209              | 45.8                                       |
|           |                          | 0                  | -31       | 100.5** | 84                  | 309              | 37                                         |
|           |                          |                    |           | 94.2    | 67                  | 275              | 41.3                                       |
|           |                          |                    |           | 88.3    | 53                  | 240              | 44.9                                       |
|           |                          |                    |           | 83.1*   | 44                  | 205              | 46.6                                       |
|           |                          | -10                | -41       | 99.6**  | 83                  | 308              | 37                                         |
|           |                          |                    |           | 92.9    | 66                  | 273              | 41.4                                       |
|           |                          |                    |           | 87.1    | 53                  | 238              | 45.3                                       |
|           |                          |                    |           | 81.7*   | 43                  | 203              | 47.3                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 23,000    | 5000                     | 20                 | -11       | 94.7**  | 64                  | 270              | 42.5                                       |
|           |                          |                    |           | 91      | 56                  | 249              | 44.6                                       |
|           |                          |                    |           | 87.5    | 49                  | 228              | 46.3                                       |
|           |                          |                    |           | 84.1*   | 44                  | 207              | 47                                         |
|           |                          | 10                 | -21       | 97.4**  | 73                  | 292              | 39.8                                       |
|           |                          |                    |           | 92.3    | 60                  | 262              | 43.5                                       |
|           |                          |                    |           | 87.2    | 50                  | 232              | 46.6                                       |
|           |                          |                    |           | 82.3*   | 42                  | 202              | 47.8                                       |
|           |                          | 0                  | -31       | 100.4** | 84                  | 312              | 37.3                                       |
|           |                          |                    |           | 93.6    | 65                  | 274              | 42.2                                       |
|           |                          |                    |           | 87      | 51                  | 237              | 46.5                                       |
|           |                          |                    |           | 80.9*   | 41                  | 199              | 48.6                                       |
|           |                          | -10                | -41       | 99.5**  | 83                  | 310              | 37.2                                       |
|           |                          |                    |           | 92.2    | 64                  | 271              | 42.4                                       |
|           |                          |                    |           | 85.4    | 49                  | 232              | 47.1                                       |
|           |                          |                    |           | 79.1*   | 39                  | 194              | 49.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 23,000    | 4500                     | 20                 | -11       | 94.7**  | 64                  | 273              | 42.9                                       |
|           |                          |                    |           | 90      | 54                  | 248              | 45.9                                       |
|           |                          |                    |           | 85.4    | 46                  | 222              | 48.3                                       |
|           |                          |                    |           | 81.1*   | 40                  | 197              | 49.2                                       |
|           |                          | 10                 | -21       | 97.4**  | 73                  | 294              | 40.1                                       |
|           |                          |                    |           | 91.5    | 59                  | 261              | 44.5                                       |
|           |                          |                    |           | 85.5    | 47                  | 227              | 48.4                                       |
|           |                          |                    |           | 79.7*   | 39                  | 194              | 50                                         |
|           |                          | 0                  | -31       | 100.4** | 84                  | 313              | 37.4                                       |
|           |                          |                    |           | 92.8    | 63                  | 272              | 43.1                                       |
|           |                          |                    |           | 85.2    | 48                  | 230              | 48.4                                       |
|           |                          |                    |           | 78.0*   | 37                  | 189              | 50.8                                       |
|           |                          | -10                | -41       | 99.5**  | 84                  | 312              | 37.3                                       |
|           |                          |                    |           | 91.7    | 62                  | 270              | 43.3                                       |
|           |                          |                    |           | 84.1    | 47                  | 229              | 48.7                                       |
|           |                          |                    |           | 76.8*   | 36                  | 188              | 51.6                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 25,000    | 6000                     | 20                 | -15       | 95.3**  | 61                  | 264              | 43.5                                       |
|           |                          |                    |           | 93.6    | 57                  | 252              | 44.2                                       |
|           |                          |                    |           | 92      | 54                  | 241              | 44.7                                       |
|           |                          |                    |           | 90.4*   | 51                  | 229              | 44.9                                       |
|           |                          | 10                 | -25       | 97.7**  | 69                  | 287              | 41.4                                       |
|           |                          |                    |           | 94.6    | 61                  | 266              | 43.7                                       |
|           |                          |                    |           | 91.4    | 54                  | 244              | 45.1                                       |
|           |                          |                    |           | 88.5*   | 49                  | 223              | 45.7                                       |
|           |                          | 0                  | -35       | 101.8** | 81                  | 313              | 38.6                                       |
|           |                          |                    |           | 95.8    | 67                  | 281              | 42                                         |
|           |                          |                    |           | 91.2    | 55                  | 250              | 45.2                                       |
|           |                          |                    |           | 86.8*   | 47                  | 219              | 46.5                                       |
|           |                          | -10                | -45       | 100.1** | 79                  | 308              | 39.1                                       |
|           |                          |                    |           | 94      | 65                  | 277              | 42.5                                       |
|           |                          |                    |           | 89.5    | 54                  | 246              | 45.8                                       |
|           |                          |                    |           | 85.1*   | 46                  | 215              | 47.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 25,000    | 5500                     | 20                 | -15       | 95.2**  | 61                  | 269              | 44.3                                       |
|           |                          |                    |           | 92.6    | 55                  | 252              | 45.6                                       |
|           |                          |                    |           | 90.1    | 51                  | 236              | 46.6                                       |
|           |                          |                    |           | 88.0*   | 47                  | 219              | 47                                         |
|           |                          | 10                 | -25       | 97.6**  | 69                  | 291              | 41.9                                       |
|           |                          |                    |           | 94.2    | 60                  | 268              | 44.6                                       |
|           |                          |                    |           | 90.5    | 53                  | 245              | 46.5                                       |
|           |                          |                    |           | 87.1*   | 46                  | 222              | 47.7                                       |
|           |                          | 0                  | -35       | 101.7** | 81                  | 315              | 38.9                                       |
|           |                          |                    |           | 95.1    | 65                  | 280              | 43                                         |
|           |                          |                    |           | 89.7    | 53                  | 246              | 46.9                                       |
|           |                          |                    |           | 84.7*   | 44                  | 212              | 48.5                                       |
|           |                          | -10                | -45       | 100.1** | 79                  | 310              | 39.3                                       |
|           |                          |                    |           | 93.3    | 63                  | 276              | 43.6                                       |
|           |                          |                    |           | 87.9    | 51                  | 241              | 47.5                                       |
|           |                          |                    |           | 82.9*   | 42                  | 207              | 49.3                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 25,000    | 5000                     | 20                 | -15       | 95.2**   | 61                  | 272              | 45                                         |
|           |                          |                    |           | 91.8     | 54                  | 252              | 46.9                                       |
|           |                          |                    |           | 88.6     | 48                  | 233              | 48.5                                       |
|           |                          |                    |           | 85.7*    | 43                  | 213              | 49.2                                       |
|           |                          | 10                 | -25       | 97.6**   | 70                  | 294              | 42.2                                       |
|           |                          |                    |           | 93       | 58                  | 265              | 45.9                                       |
|           |                          |                    |           | 88.3     | 49                  | 237              | 48.7                                       |
|           |                          |                    |           | 84.0*    | 42                  | 208              | 50                                         |
|           |                          | 0                  | -35       | 101.7**  | 81                  | 317              | 39.2                                       |
|           |                          |                    |           | 94.4     | 63                  | 279              | 44.1                                       |
|           |                          |                    |           | 88.1     | 50                  | 241              | 48.6                                       |
|           |                          |                    |           | 82.3*    | 40                  | 204              | 50.8                                       |
|           |                          | -10                | -45       | 100.1*** | 79                  | 312              | 39.6                                       |
|           |                          |                    |           | 92.7     | 62                  | 275              | 44.6                                       |
|           |                          |                    |           | 86.4     | 48                  | 237              | 49.4                                       |
|           |                          |                    |           | 80.6*    | 39                  | 199              | 51.6                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 25,000    | 4500                     | 20                 | -15       | 95.1**   | 61                  | 276              | 45.5                                       |
|           |                          |                    |           | 90.9     | 52                  | 252              | 48.3                                       |
|           |                          |                    |           | 86.9     | 45                  | 228              | 50.5                                       |
|           |                          |                    |           | 83.1*    | 40                  | 205              | 51.5                                       |
|           |                          | 10                 | -25       | 97.6**   | 70                  | 296              | 42.6                                       |
|           |                          |                    |           | 92.3     | 56                  | 264              | 47                                         |
|           |                          |                    |           | 86.7     | 46                  | 232              | 50.7                                       |
|           |                          |                    |           | 81.5*    | 38                  | 201              | 52.3                                       |
|           |                          | 0                  | -35       | 101.6*** | 81                  | 319              | 39.4                                       |
|           |                          |                    |           | 93.8     | 62                  | 278              | 45.1                                       |
|           |                          |                    |           | 86.7     | 47                  | 238              | 50.4                                       |
|           |                          |                    |           | 80.0*    | 37                  | 198              | 53.2                                       |
|           |                          | -10                | -45       | 99.6***  | 78                  | 312              | 40                                         |
|           |                          |                    |           | 91.9     | 59                  | 272              | 45.8                                       |
|           |                          |                    |           | 84.8     | 45                  | 232              | 51.3                                       |
|           |                          |                    |           | 78.2*    | 36                  | 192              | 54                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 27,000    | 6000                     | 20                 | -18       | 95.6**  | 57                  | 263              | 46                                         |
|           |                          |                    |           | 94.4    | 55                  | 254              | 46.4                                       |
|           |                          |                    |           | 93.2    | 52                  | 245              | 46.6                                       |
|           |                          |                    |           | 92.1*   | 50                  | 236              | 46.8                                       |
|           |                          | 10                 | -28       | 97.7**  | 65                  | 286              | 44                                         |
|           |                          |                    |           | 95.5    | 59                  | 270              | 45.5                                       |
|           |                          |                    |           | 93.4    | 54                  | 255              | 46.9                                       |
|           |                          |                    |           | 91.4*   | 50                  | 239              | 47.4                                       |
|           |                          | 0                  | -38       | 102.2** | 76                  | 313              | 41.2                                       |
|           |                          |                    |           | 96.5    | 64                  | 284              | 44.3                                       |
|           |                          |                    |           | 92.3    | 54                  | 255              | 47.2                                       |
|           |                          |                    |           | 88.5*   | 47                  | 226              | 48.4                                       |
|           |                          | -10                | -48       | 100.7** | 74                  | 308              | 41.6                                       |
|           |                          |                    |           | 94.7    | 62                  | 279              | 44.9                                       |
|           |                          |                    |           | 90.4    | 52                  | 250              | 48                                         |
|           |                          |                    |           | 86.6*   | 45                  | 221              | 49.1                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 27,000    | 5500                     | 20                 | -18       | 95.6**   | 57                  | 268              | 47                                         |
|           |                          |                    |           | 93.4     | 53                  | 254              | 48                                         |
|           |                          |                    |           | 91.4     | 49                  | 240              | 48.7                                       |
|           |                          |                    |           | 89.5*    | 46                  | 226              | 48.9                                       |
|           |                          | 10                 | -28       | 97.6**   | 65                  | 290              | 44.6                                       |
|           |                          |                    |           | 94.3     | 56                  | 267              | 47.3                                       |
|           |                          |                    |           | 91       | 50                  | 244              | 49.1                                       |
|           |                          |                    |           | 87.8*    | 44                  | 222              | 49.8                                       |
|           |                          | 0                  | -38       | 102.2*** | 76                  | 316              | 41.5                                       |
|           |                          |                    |           | 96       | 63                  | 286              | 45                                         |
|           |                          |                    |           | 91.5     | 52                  | 255              | 48.7                                       |
|           |                          |                    |           | 87.1*    | 45                  | 225              | 50.5                                       |
|           |                          | -10                | -48       | 100.3*** | 73                  | 310              | 42.1                                       |
|           |                          |                    |           | 93.9     | 61                  | 278              | 45.9                                       |
|           |                          |                    |           | 89.2     | 50                  | 247              | 49.7                                       |
|           |                          |                    |           | 84.8*    | 42                  | 216              | 51.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 27,000    | 5000                     | 20                 | -18       | 95.5**   | 57                  | 273              | 47.8                                       |
|           |                          |                    |           | 92.5     | 51                  | 255              | 49.5                                       |
|           |                          |                    |           | 89.7     | 46                  | 236              | 50.8                                       |
|           |                          |                    |           | 87.2*    | 42                  | 218              | 51.4                                       |
|           |                          | 10                 | -28       | 97.6**   | 65                  | 294              | 45.1                                       |
|           |                          |                    |           | 93.5     | 55                  | 267              | 48.6                                       |
|           |                          |                    |           | 89.2     | 47                  | 240              | 51.1                                       |
|           |                          |                    |           | 85.5*    | 41                  | 213              | 52.3                                       |
|           |                          | 0                  | -38       | 101.7*** | 75                  | 316              | 42.1                                       |
|           |                          |                    |           | 94.6     | 60                  | 280              | 46.7                                       |
|           |                          |                    |           | 89       | 48                  | 245              | 51.2                                       |
|           |                          |                    |           | 83.7*    | 39                  | 209              | 53.1                                       |
|           |                          | -10                | -48       | 99.6***  | 72                  | 310              | 42.8                                       |
|           |                          |                    |           | 92.7     | 58                  | 275              | 47.5                                       |
|           |                          |                    |           | 87.2     | 46                  | 240              | 51.9                                       |
|           |                          |                    |           | 82.0*    | 38                  | 205              | 54                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 27,000    | 4500                     | 20                 | -18       | 95.5**   | 57                  | 277              | 48.4                                       |
|           |                          |                    |           | 91.7     | 50                  | 255              | 50.9                                       |
|           |                          |                    |           | 88.1     | 44                  | 233              | 53                                         |
|           |                          |                    |           | 84.7*    | 39                  | 211              | 54                                         |
|           |                          | 10                 | -28       | 97.6**   | 65                  | 297              | 45.5                                       |
|           |                          |                    |           | 92.8     | 54                  | 267              | 49.8                                       |
|           |                          |                    |           | 87.8     | 45                  | 238              | 53.1                                       |
|           |                          |                    |           | 83.3*    | 38                  | 208              | 54.9                                       |
|           |                          | 0                  | -38       | 101.2*** | 74                  | 316              | 42.7                                       |
|           |                          |                    |           | 93.7     | 58                  | 278              | 48.1                                       |
|           |                          |                    |           | 87.3     | 45                  | 240              | 53.2                                       |
|           |                          |                    |           | 81.4*    | 36                  | 202              | 55.8                                       |
|           |                          | -10                | -48       | 99.1***  | 71                  | 310              | 43.3                                       |
|           |                          |                    |           | 91.8     | 56                  | 272              | 48.8                                       |
|           |                          |                    |           | 85.6     | 44                  | 235              | 54                                         |
|           |                          |                    |           | 79.7*    | 35                  | 198              | 56.7                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 28,000    | 6000                     | 20                 | -20       | 95.7**  | 55                  | 261              | 47.2                                       |
|           |                          |                    |           | 94.9    | 54                  | 254              | 47.5                                       |
|           |                          |                    |           | 94      | 52                  | 247              | 47.7                                       |
|           |                          |                    |           | 93.2*   | 50                  | 241              | 47.7                                       |
|           |                          | 10                 | -30       | 98.5**  | 64                  | 289              | 45                                         |
|           |                          |                    |           | 96      | 59                  | 273              | 46.5                                       |
|           |                          |                    |           | 94      | 54                  | 257              | 47.8                                       |
|           |                          |                    |           | 92.0*   | 50                  | 241              | 48.5                                       |
|           |                          | 0                  | -40       | 102.2** | 73                  | 311              | 42.7                                       |
|           |                          |                    |           | 97      | 63                  | 285              | 45.5                                       |
|           |                          |                    |           | 93.1    | 54                  | 259              | 48                                         |
|           |                          |                    |           | 89.9*   | 47                  | 234              | 49.4                                       |
|           |                          | -10                | -50       | 100.8** | 71                  | 307              | 43                                         |
|           |                          |                    |           | 95.1    | 61                  | 280              | 46.2                                       |
|           |                          |                    |           | 91      | 52                  | 253              | 48.9                                       |
|           |                          |                    |           | 87.6*   | 45                  | 226              | 50.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 28,000    | 5500                     | 20                 | -20       | 95.7**   | 55                  | 268              | 48.3                                       |
|           |                          |                    |           | 94.4     | 53                  | 259              | 49                                         |
|           |                          |                    |           | 93.1     | 50                  | 250              | 49.5                                       |
|           |                          |                    |           | 91.8*    | 48                  | 241              | 49.8                                       |
|           |                          | 10                 | -30       | 98.4**   | 64                  | 293              | 45.7                                       |
|           |                          |                    |           | 95.3     | 57                  | 273              | 47.9                                       |
|           |                          |                    |           | 92.7     | 51                  | 254              | 49.7                                       |
|           |                          |                    |           | 90.0*    | 46                  | 235              | 50.7                                       |
|           |                          | 0                  | -40       | 102.2*** | 73                  | 315              | 43.1                                       |
|           |                          |                    |           | 95.9     | 61                  | 284              | 46.7                                       |
|           |                          |                    |           | 91.4     | 50                  | 253              | 50.3                                       |
|           |                          |                    |           | 87.1*    | 43                  | 222              | 51.7                                       |
|           |                          | -10                | -50       | 100.3*** | 71                  | 308              | 43.7                                       |
|           |                          |                    |           | 94.1     | 59                  | 279              | 47.3                                       |
|           |                          |                    |           | 89.8     | 49                  | 249              | 50.9                                       |
|           |                          |                    |           | 85.7*    | 42                  | 220              | 52.5                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 28,000    | 5000                     | 20                 | -20       | 95.6**   | 55                  | 273              | 49.3                                       |
|           |                          |                    |           | 92.9     | 50                  | 256              | 50.8                                       |
|           |                          |                    |           | 90.3     | 46                  | 238              | 52                                         |
|           |                          |                    |           | 88.0*    | 42                  | 221              | 52.5                                       |
|           |                          | 10                 | -30       | 98.2**   | 64                  | 296              | 46.2                                       |
|           |                          |                    |           | 94       | 54                  | 269              | 49.8                                       |
|           |                          |                    |           | 89.9     | 46                  | 243              | 52.3                                       |
|           |                          |                    |           | 86.2*    | 40                  | 216              | 53.4                                       |
|           |                          | 0                  | -40       | 101.5*** | 72                  | 315              | 43.8                                       |
|           |                          |                    |           | 94.7     | 58                  | 281              | 48.2                                       |
|           |                          |                    |           | 89.5     | 47                  | 246              | 52.5                                       |
|           |                          |                    |           | 84.5*    | 39                  | 212              | 54.3                                       |
|           |                          | -10                | -50       | 99.4***  | 69                  | 308              | 44.5                                       |
|           |                          |                    |           | 92.7     | 56                  | 275              | 49                                         |
|           |                          |                    |           | 87.6     | 45                  | 241              | 53.3                                       |
|           |                          |                    |           | 82.7*    | 38                  | 208              | 55.1                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 28,000    | 4500                     | 20                 | -20       | 95.6**   | 55                  | 277              | 50                                         |
|           |                          |                    |           | 91.9     | 49                  | 255              | 52.5                                       |
|           |                          |                    |           | 88.3     | 43                  | 233              | 54.5                                       |
|           |                          |                    |           | 85.2*    | 38                  | 211              | 55.3                                       |
|           |                          | 10                 | -30       | 98.1**   | 64                  | 299              | 46.7                                       |
|           |                          |                    |           | 93.2     | 52                  | 268              | 51.2                                       |
|           |                          |                    |           | 88.1     | 44                  | 238              | 54.6                                       |
|           |                          |                    |           | 83.7*    | 37                  | 208              | 56.2                                       |
|           |                          | 0                  | -40       | 100.9*** | 71                  | 315              | 44.4                                       |
|           |                          |                    |           | 93.6     | 56                  | 278              | 49.8                                       |
|           |                          |                    |           | 87.5     | 44                  | 240              | 54.8                                       |
|           |                          |                    |           | 81.8*    | 35                  | 203              | 57.1                                       |
|           |                          | -10                | -50       | 98.8***  | 68                  | 308              | 45.1                                       |
|           |                          |                    |           | 91.9     | 54                  | 273              | 50.4                                       |
|           |                          |                    |           | 86.1     | 43                  | 238              | 55.4                                       |
|           |                          |                    |           | 80.7*    | 35                  | 203              | 58                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1     | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|---------|---------------------|------------------|--------------------------------------------|
| 29,000    | 6000                     | 20                 | -22       | 95.9**  | 54                  | 260              | 48.3                                       |
|           |                          |                    |           | 95.4    | 53                  | 256              | 48.5                                       |
|           |                          |                    |           | 94.9    | 52                  | 251              | 48.7                                       |
|           |                          |                    |           | 94.3*   | 51                  | 247              | 48.8                                       |
|           |                          | 10                 | -32       | 99.3**  | 63                  | 291              | 46.2                                       |
|           |                          |                    |           | 96.4    | 58                  | 274              | 47.6                                       |
|           |                          |                    |           | 94.3    | 52                  | 257              | 48.9                                       |
|           |                          |                    |           | 92.3*   | 48                  | 240              | 49.6                                       |
|           |                          | 0                  | -42       | 102.2** | 70                  | 309              | 44.3                                       |
|           |                          |                    |           | 97.1    | 60                  | 284              | 47                                         |
|           |                          |                    |           | 93.3    | 53                  | 259              | 49.2                                       |
|           |                          |                    |           | 90.4*   | 46                  | 234              | 50.5                                       |
|           |                          | -10                | -52       | 100.8** | 68                  | 305              | 44.6                                       |
|           |                          |                    |           | 95.4    | 59                  | 279              | 47.6                                       |
|           |                          |                    |           | 91.4    | 51                  | 254              | 49.9                                       |
|           |                          |                    |           | 88.4*   | 45                  | 229              | 51.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 29,000    | 5500                     | 20                 | -22       | 95.8**   | 54                  | 267              | 49.6                                       |
|           |                          |                    |           | 94.8     | 52                  | 259              | 50.3                                       |
|           |                          |                    |           | 93.7     | 50                  | 251              | 50.7                                       |
|           |                          |                    |           | 92.6*    | 48                  | 243              | 50.9                                       |
|           |                          | 10                 | -32       | 99.1**   | 63                  | 295              | 46.8                                       |
|           |                          |                    |           | 95.8     | 56                  | 276              | 49                                         |
|           |                          |                    |           | 93.2     | 50                  | 256              | 50.9                                       |
|           |                          |                    |           | 90.6*    | 46                  | 237              | 51.8                                       |
|           |                          | 0                  | -42       | 102.2*** | 70                  | 313              | 44.7                                       |
|           |                          |                    |           | 96.3     | 59                  | 285              | 48                                         |
|           |                          |                    |           | 92.1     | 50                  | 257              | 51.2                                       |
|           |                          |                    |           | 88.4*    | 43                  | 229              | 52.8                                       |
|           |                          | -10                | -52       | 100.3*** | 68                  | 307              | 45.2                                       |
|           |                          |                    |           | 94.2     | 57                  | 278              | 48.9                                       |
|           |                          |                    |           | 89.8     | 48                  | 249              | 52.3                                       |
|           |                          |                    |           | 86.0*    | 41                  | 220              | 53.6                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 29,000    | 5000                     | 20                 | -22       | 95.8**   | 54                  | 273              | 50.7                                       |
|           |                          |                    |           | 93.7     | 50                  | 259              | 52                                         |
|           |                          |                    |           | 91.6     | 46                  | 245              | 53                                         |
|           |                          |                    |           | 89.6*    | 43                  | 231              | 53.6                                       |
|           |                          | 10                 | -32       | 99.0**   | 63                  | 299              | 47.4                                       |
|           |                          |                    |           | 94.7     | 54                  | 273              | 50.8                                       |
|           |                          |                    |           | 90.8     | 46                  | 247              | 53.4                                       |
|           |                          |                    |           | 87.2*    | 40                  | 221              | 54.6                                       |
|           |                          | 0                  | -42       | 101.4*** | 69                  | 314              | 45.5                                       |
|           |                          |                    |           | 95       | 57                  | 282              | 49.5                                       |
|           |                          |                    |           | 90.3     | 47                  | 250              | 53.7                                       |
|           |                          |                    |           | 85.7*    | 39                  | 219              | 55.5                                       |
|           |                          | -10                | -52       | 99.3***  | 66                  | 307              | 46.2                                       |
|           |                          |                    |           | 92.8     | 54                  | 275              | 50.5                                       |
|           |                          |                    |           | 88.1     | 45                  | 243              | 54.7                                       |
|           |                          |                    |           | 83.6*    | 38                  | 212              | 56.4                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 29,000    | 4500                     | 20                 | -22       | 95.7**   | 54                  | 278              | 51.5                                       |
|           |                          |                    |           | 92.4     | 48                  | 257              | 53.9                                       |
|           |                          |                    |           | 89.1     | 42                  | 237              | 55.7                                       |
|           |                          |                    |           | 86.3*    | 38                  | 216              | 56.6                                       |
|           |                          | 10                 | -32       | 98.9**   | 63                  | 302              | 47.9                                       |
|           |                          |                    |           | 93.8     | 52                  | 272              | 52.4                                       |
|           |                          |                    |           | 89       | 43                  | 242              | 55.8                                       |
|           |                          |                    |           | 84.6*    | 37                  | 212              | 57.5                                       |
|           |                          | 0                  | -42       | 100.7*** | 68                  | 314              | 46.2                                       |
|           |                          |                    |           | 93.7     | 54                  | 278              | 51.3                                       |
|           |                          |                    |           | 88.1     | 43                  | 243              | 56.2                                       |
|           |                          |                    |           | 82.8*    | 35                  | 207              | 58.4                                       |
|           |                          | -10                | -52       | 98.6***  | 65                  | 307              | 46.9                                       |
|           |                          |                    |           | 91.7     | 52                  | 272              | 52.2                                       |
|           |                          |                    |           | 86.2     | 42                  | 237              | 57.1                                       |
|           |                          |                    |           | 81.1*    | 34                  | 203              | 59.3                                       |



| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 30,000    | 6000                     | 20                 | -24       | 96.1*/** | 52                  | 259              | 49.4                                       |
|           |                          | 10                 | -34       | 99.9**   | 62                  | 293              | 47.5                                       |
|           |                          |                    |           | 96.8     | 56                  | 274              | 48.8                                       |
|           |                          |                    |           | 94.4     | 51                  | 256              | 50                                         |
|           |                          |                    |           | 92.5*    | 47                  | 237              | 50.6                                       |
|           |                          | 0                  | -44       | 102.2**  | 67                  | 307              | 45.8                                       |
|           |                          |                    |           | 97.1     | 58                  | 282              | 48.5                                       |
|           |                          |                    |           | 93.2     | 51                  | 257              | 50.5                                       |
|           |                          |                    |           | 90.6*    | 45                  | 232              | 51.4                                       |
|           |                          | -10                | -54       | 100.8**  | 66                  | 303              | 46.1                                       |
|           |                          |                    |           | 95.4     | 56                  | 277              | 49.2                                       |
|           |                          |                    |           | 91.4     | 49                  | 252              | 51.3                                       |
|           |                          |                    |           | 88.6*    | 43                  | 227              | 52.2                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 30,000    | 5500                     | 20                 | -24       | 96.0**   | 53                  | 267              | 50.9                                       |
|           |                          |                    |           | 95       | 50                  | 259              | 51.5                                       |
|           |                          |                    |           | 93.9     | 48                  | 250              | 52                                         |
|           |                          |                    |           | 92.8*    | 46                  | 242              | 52.2                                       |
|           |                          | 10                 | -34       | 99.8**   | 62                  | 297              | 48.2                                       |
|           |                          |                    |           | 96.2     | 55                  | 278              | 50.1                                       |
|           |                          |                    |           | 93.7     | 50                  | 258              | 52                                         |
|           |                          |                    |           | 91.2*    | 45                  | 239              | 53                                         |
|           |                          | 0                  | -44       | 102.2**  | 67                  | 311              | 46.3                                       |
|           |                          |                    |           | 96.8     | 58                  | 286              | 49.4                                       |
|           |                          |                    |           | 92.7     | 50                  | 260              | 52.2                                       |
|           |                          |                    |           | 89.4*    | 43                  | 234              | 53.9                                       |
|           |                          | -10                | -54       | 100.3*** | 65                  | 305              | 46.9                                       |
|           |                          |                    |           | 94.5     | 55                  | 278              | 50.4                                       |
|           |                          |                    |           | 90.3     | 47                  | 251              | 53.3                                       |
|           |                          |                    |           | 87.0*    | 41                  | 224              | 54.8                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 30,000    | 5000                     | 20                 | -24       | 96.0**   | 53                  | 274              | 52.1                                       |
|           |                          |                    |           | 93.7     | 48                  | 258              | 53.6                                       |
|           |                          |                    |           | 91.5     | 44                  | 242              | 54.5                                       |
|           |                          |                    |           | 89.4*    | 41                  | 226              | 54.8                                       |
|           |                          | 10                 | -34       | 99.7**   | 62                  | 301              | 48.8                                       |
|           |                          |                    |           | 95       | 53                  | 275              | 52                                         |
|           |                          |                    |           | 91.4     | 45                  | 249              | 54.8                                       |
|           |                          |                    |           | 87.9*    | 40                  | 223              | 55.8                                       |
|           |                          | 0                  | -44       | 101.4*** | 66                  | 312              | 47.2                                       |
|           |                          |                    |           | 95.1     | 55                  | 282              | 51.1                                       |
|           |                          |                    |           | 90.7     | 46                  | 251              | 55.1                                       |
|           |                          |                    |           | 86.4*    | 39                  | 221              | 56.7                                       |
|           |                          | -10                | -54       | 99.3***  | 64                  | 305              | 47.9                                       |
|           |                          |                    |           | 93.1     | 53                  | 275              | 52.1                                       |
|           |                          |                    |           | 88.6     | 44                  | 245              | 56                                         |
|           |                          |                    |           | 84.4*    | 37                  | 215              | 57.6                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 30,000    | 4500                     | 20                 | -24       | 95.9**   | 53                  | 279              | 53                                         |
|           |                          |                    |           | 92.8     | 47                  | 258              | 55.4                                       |
|           |                          |                    |           | 89.7     | 42                  | 238              | 57.1                                       |
|           |                          |                    |           | 86.9*    | 38                  | 218              | 57.8                                       |
|           |                          | 10                 | -34       | 99.5**   | 62                  | 304              | 49.3                                       |
|           |                          |                    |           | 94.1     | 51                  | 274              | 53.7                                       |
|           |                          |                    |           | 89.4     | 42                  | 243              | 57.3                                       |
|           |                          |                    |           | 85.1*    | 36                  | 213              | 58.8                                       |
|           |                          | 0                  | -44       | 100.5*** | 65                  | 312              | 48.1                                       |
|           |                          |                    |           | 93.8     | 53                  | 279              | 52.9                                       |
|           |                          |                    |           | 88.6     | 42                  | 245              | 57.6                                       |
|           |                          |                    |           | 83.7*    | 35                  | 211              | 59.7                                       |
|           |                          | -10                | -54       | 98.4***  | 63                  | 305              | 48.8                                       |
|           |                          |                    |           | 91.7     | 50                  | 272              | 53.9                                       |
|           |                          |                    |           | 86.5     | 40                  | 238              | 58.7                                       |
|           |                          |                    |           | 81.6*    | 34                  | 204              | 60.7                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 31,000    | 6000                     | 20                 | -26       | 96.3*/** | 51                  | 258              | 50.4                                       |
|           |                          | 10                 | -36       | 100.6**  | 60                  | 294              | 48.8                                       |
|           |                          |                    |           | 97.5     | 55                  | 276              | 50                                         |
|           |                          |                    |           | 95       | 51                  | 258              | 51                                         |
|           |                          |                    |           | 93.3*    | 47                  | 241              | 51.4                                       |
|           |                          | 0                  | -46       | 102.2**  | 64                  | 305              | 47.5                                       |
|           |                          |                    |           | 97.5     | 56                  | 282              | 50                                         |
|           |                          |                    |           | 93.8     | 50                  | 259              | 51.5                                       |
|           |                          |                    |           | 91.4*    | 45                  | 236              | 52.3                                       |
|           |                          | -10                | -56       | 100.8**  | 63                  | 300              | 47.8                                       |
|           |                          |                    |           | 95.8     | 55                  | 277              | 50.7                                       |
|           |                          |                    |           | 92       | 49                  | 254              | 52.3                                       |
|           |                          |                    |           | 89.5*    | 43                  | 231              | 53.1                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 31,000    | 5500                     | 20                 | -26       | 96.3**   | 51                  | 267              | 52.2                                       |
|           |                          |                    |           | 95.3     | 49                  | 259              | 52.7                                       |
|           |                          |                    |           | 94.4     | 47                  | 251              | 53.2                                       |
|           |                          |                    |           | 93.5*    | 45                  | 243              | 53.3                                       |
|           |                          | 10                 | -36       | 100.4**  | 60                  | 299              | 49.6                                       |
|           |                          |                    |           | 96.7     | 54                  | 278              | 51.5                                       |
|           |                          |                    |           | 93.9     | 48                  | 258              | 53.3                                       |
|           |                          |                    |           | 91.5*    | 44                  | 237              | 54.2                                       |
|           |                          | 0                  | -46       | 102.2**  | 64                  | 309              | 48                                         |
|           |                          |                    |           | 96.7     | 55                  | 284              | 51.2                                       |
|           |                          |                    |           | 92.7     | 48                  | 258              | 53.7                                       |
|           |                          |                    |           | 89.7*    | 42                  | 233              | 55.1                                       |
|           |                          | -10                | -56       | 100.5*** | 63                  | 304              | 48.5                                       |
|           |                          |                    |           | 94.8     | 54                  | 278              | 52                                         |
|           |                          |                    |           | 90.7     | 46                  | 252              | 54.6                                       |
|           |                          |                    |           | 87.7*    | 40                  | 227              | 56                                         |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 31,000    | 5000                     | 20                 | -26       | 96.2**   | 51                  | 274              | 53.5                                       |
|           |                          |                    |           | 94.6     | 48                  | 262              | 54.7                                       |
|           |                          |                    |           | 93       | 45                  | 250              | 55.6                                       |
|           |                          |                    |           | 91.4*    | 42                  | 238              | 55.9                                       |
|           |                          | 10                 | -36       | 100.3**  | 60                  | 303              | 50.2                                       |
|           |                          |                    |           | 95.5     | 52                  | 277              | 53.2                                       |
|           |                          |                    |           | 92       | 45                  | 252              | 56.1                                       |
|           |                          |                    |           | 88.7*    | 40                  | 226              | 57                                         |
|           |                          | 0                  | -46       | 101.5*** | 63                  | 311              | 49                                         |
|           |                          |                    |           | 95.3     | 53                  | 282              | 52.8                                       |
|           |                          |                    |           | 90.9     | 45                  | 252              | 56.5                                       |
|           |                          |                    |           | 87.0*    | 38                  | 223              | 57.9                                       |
|           |                          | -10                | -56       | 99.3***  | 61                  | 304              | 49.7                                       |
|           |                          |                    |           | 93.3     | 51                  | 275              | 53.7                                       |
|           |                          |                    |           | 89       | 43                  | 247              | 57.4                                       |
|           |                          |                    |           | 85.2*    | 37                  | 218              | 58.8                                       |

| Alt<br>FT | Aircraft<br>Weight<br>LB | $\Delta$ ISA<br>°C | OAT<br>°C | %N1      | Fuel<br>Flow<br>GPH | Airspeed<br>KTAS | Specific<br>Range<br>Nm/<br>10 U.S.<br>Gal |
|-----------|--------------------------|--------------------|-----------|----------|---------------------|------------------|--------------------------------------------|
| 31,000    | 4500                     | 20                 | -26       | 96.1**   | 51                  | 280              | 54.5                                       |
|           |                          |                    |           | 93.3     | 46                  | 261              | 56.9                                       |
|           |                          |                    |           | 90.5     | 41                  | 242              | 58.4                                       |
|           |                          |                    |           | 87.9*    | 38                  | 223              | 59.1                                       |
|           |                          | 10                 | -36       | 100.2**  | 60                  | 307              | 50.8                                       |
|           |                          |                    |           | 94.6     | 50                  | 277              | 55                                         |
|           |                          |                    |           | 90.2     | 42                  | 247              | 58.7                                       |
|           |                          |                    |           | 86.0*    | 36                  | 217              | 60.2                                       |
|           |                          | 0                  | -46       | 100.4*** | 62                  | 311              | 50                                         |
|           |                          |                    |           | 93.9     | 51                  | 279              | 54.6                                       |
|           |                          |                    |           | 89.2     | 42                  | 247              | 59.1                                       |
|           |                          |                    |           | 84.5*    | 35                  | 215              | 61.1                                       |
|           |                          | -10                | -56       | 98.3***  | 60                  | 304              | 50.8                                       |
|           |                          |                    |           | 91.8     | 49                  | 272              | 55.6                                       |
|           |                          |                    |           | 87       | 40                  | 240              | 60.3                                       |
|           |                          |                    |           | 82.5*    | 33                  | 208              | 62.1                                       |



## **Descent Data**

Airspeed .....0.51 MACH/245 KIAS

Throttle ..... A/R

| Pressure<br>Altitude<br>(ft) | Flight Path Angle =<br>-2.8 |                     |              | Flight Path Angle =<br>-3.7 |                     |              | Flight Path Angle =<br>-4.6 |                     |              |
|------------------------------|-----------------------------|---------------------|--------------|-----------------------------|---------------------|--------------|-----------------------------|---------------------|--------------|
|                              | Time<br>(min:<br>sec)       | Fuel<br>(US<br>gal) | Dist<br>(NM) | Time<br>(min:<br>sec)       | Fuel<br>(US<br>gal) | Dist<br>(NM) | Time<br>(min:<br>sec)       | Fuel<br>(US<br>gal) | Dist<br>(NM) |
| 31,000                       | 21:39                       | 27                  | 106          | 16:22                       | 19                  | 80           | 13:09                       | 14                  | 65           |
| 30,000                       | 20:58                       | 26                  | 103          | 15:52                       | 18                  | 78           | 12:45                       | 14                  | 62           |
| 29,000                       | 20:18                       | 26                  | 99           | 15:21                       | 18                  | 75           | 12:20                       | 13                  | 60           |
| 28,000                       | 19:38                       | 25                  | 96           | 14:51                       | 18                  | 73           | 11:56                       | 13                  | 58           |
| 27,000                       | 18:58                       | 25                  | 93           | 14:21                       | 17                  | 70           | 11:32                       | 13                  | 56           |
| 26,000                       | 18:19                       | 24                  | 89           | 13:51                       | 17                  | 68           | 11:08                       | 13                  | 54           |
| 25,000                       | 17:39                       | 24                  | 86           | 13:21                       | 17                  | 65           | 10:43                       | 12                  | 52           |
| 24,000                       | 17:00                       | 23                  | 83           | 12:51                       | 16                  | 63           | 10:19                       | 12                  | 50           |
| 23,000                       | 16:20                       | 22                  | 79           | 12:21                       | 16                  | 60           | 9:56                        | 12                  | 48           |
| 22,000                       | 15:41                       | 22                  | 76           | 11:52                       | 15                  | 57           | 9:32                        | 11                  | 46           |
| 21,000                       | 15:02                       | 21                  | 73           | 11:22                       | 15                  | 55           | 9:08                        | 11                  | 44           |
| 20,000                       | 14:24                       | 20                  | 69           | 10:53                       | 14                  | 52           | 8:45                        | 11                  | 42           |
| 19,000                       | 13:45                       | 20                  | 66           | 10:24                       | 14                  | 50           | 8:21                        | 10                  | 40           |
| 17,000                       | 12:28                       | 18                  | 59           | 9:26                        | 13                  | 45           | 7:34                        | 9                   | 36           |
| 15,000                       | 11:09                       | 16                  | 52           | 8:26                        | 11                  | 40           | 6:47                        | 8                   | 32           |
| 10,000                       | 7:42                        | 11                  | 35           | 5:50                        | 8                   | 26           | 4:41                        | 6                   | 21           |
| 5000                         | 4:00                        | 6                   | 17           | 3:01                        | 4                   | 13           | 2:25                        | 3                   | 11           |
| 0                            | 0:00                        | 0                   | 0            | 0:00                        | 0                   | 0            | 0:00                        | 0                   | 0            |

## Landing Distance

| <b>V<sub>REF</sub> Speeds (KIAS)</b>                       |                |                |                |                |                 |
|------------------------------------------------------------|----------------|----------------|----------------|----------------|-----------------|
| <b>FLAPS</b>                                               | <b>4000 lb</b> | <b>4500 lb</b> | <b>5000 lb</b> | <b>5550 lb</b> | <b>6000 lb*</b> |
| UP or UNKNOWN                                              | 89             | 95             | 100            | 104            | 109             |
| UP (Ice Contaminated Airframe/STALL SPEED HIGH Advisory)   | 115            | 122            | 128            | 135            | 140             |
| 50%                                                        | 82             | 87             | 91             | 96             | 100             |
| 50% (Ice Contaminated Airframe /STALL SPEED HIGH Advisory) | 98             | 104            | 110            | 115            | 120             |
| 100%                                                       | 71             | 76             | 81             | 85             | 89              |

\* Maximum Landing Weight (MLW) is 5550 lb (2495 kg).

**• NOTE •**

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## Aircraft Weight: 5550 lb (2517 kg)

### Flaps: 100%

V<sub>REF</sub> ..... 85 KIAS

Winds ..... Calm

Runway ..... Dry, Level, Paved

Thrust Lever ..... IDLE

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 44% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

#### • NOTE •

The grayed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 100%</b> |                  |                 |      |      |      |      |    |      |
|--------------------------------------------------------|------------------|-----------------|------|------|------|------|----|------|
| Press<br>Alt<br>Feet                                   | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |    | ISA  |
|                                                        |                  | 0               | 10   | 20   | 30   | 40   | 50 |      |
| SL                                                     | Gnd Roll         | 1537            | 1593 | 1650 | 1706 | 1762 |    | 1622 |
|                                                        | Total            | 2430            | 2498 | 2566 | 2635 | 2704 |    | 2532 |
| 1000                                                   | Gnd Roll         | 1594            | 1652 | 1711 | 1769 | 1827 |    | 1670 |
|                                                        | Total            | 2498            | 2569 | 2641 | 2712 | 2784 |    | 2591 |
| 2000                                                   | Gnd Roll         | 1653            | 1714 | 1774 | 1835 | 1895 |    | 1720 |
|                                                        | Total            | 2570            | 2644 | 2719 | 2793 | 2868 |    | 2652 |
| 3000                                                   | Gnd Roll         | 1715            | 1778 | 1841 | 1903 |      |    | 1772 |
|                                                        | Total            | 2646            | 2723 | 2800 | 2878 |      |    | 2716 |
| 4000                                                   | Gnd Roll         | 1780            | 1845 | 1910 | 1975 |      |    | 1826 |
|                                                        | Total            | 2725            | 2806 | 2886 | 2967 |      |    | 2782 |
| 5000                                                   | Gnd Roll         | 1847            | 1915 | 1983 | 2050 |      |    | 1882 |
|                                                        | Total            | 2809            | 2892 | 2976 | 3060 |      |    | 2851 |
| 6000                                                   | Gnd Roll         | 1918            | 1988 | 2059 | 2129 |      |    | 1940 |
|                                                        | Total            | 2896            | 2983 | 3071 | 3159 |      |    | 2923 |
| 7000                                                   | Gnd Roll         | 1992            | 2065 | 2138 |      |      |    | 2000 |
|                                                        | Total            | 2988            | 3079 | 3170 |      |      |    | 2998 |
| 8000                                                   | Gnd Roll         | 2093            | 2170 | 2247 |      |      |    | 2087 |
|                                                        | Total            | 3120            | 3216 | 3312 |      |      |    | 3112 |
| 9000                                                   | Gnd Roll         | 2218            | 2300 | 2381 |      |      |    | 2195 |
|                                                        | Total            | 3286            | 3388 | 3490 |      |      |    | 3257 |
| 10,000                                                 | Gnd Roll         | 2350            | 2437 |      |      |      |    | 2309 |
|                                                        | Total            | 3462            | 3570 |      |      |      |    | 3410 |

• NOTE •

Unfactored data: the above published distances do not include factors that may be required by operational rules.

## Flaps: 50%

V<sub>REF</sub> ..... 96 KIAS

Winds ..... Calm

Runway ..... Dry, Level, Paved

Thrust Lever ..... IDLE

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 38% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

### • NOTE •

The grayed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50%</b> |                          |                        |           |           |           |           |           |            |
|-------------------------------------------------------|--------------------------|------------------------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>                         | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |           |           |           |           |           | <b>ISA</b> |
|                                                       |                          | <b>0</b>               | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                                    | Gnd Roll                 | 2028                   | 2103      | 2177      | 2251      | 2325      | 2400      | 2140       |
|                                                       | Total                    | 3061                   | 3154      | 3247      | 3340      | 3433      | 3527      | 3200       |
| 1000                                                  | Gnd Roll                 | 2103                   | 2180      | 2257      | 2334      | 2411      | 2488      | 2204       |
|                                                       | Total                    | 3155                   | 3251      | 3347      | 3444      | 3542      | 3639      | 3280       |
| 2000                                                  | Gnd Roll                 | 2182                   | 2261      | 2341      | 2421      | 2501      | 2581      | 2270       |
|                                                       | Total                    | 3252                   | 3353      | 3453      | 3554      | 3655      | 3757      | 3363       |
| 3000                                                  | Gnd Roll                 | 2263                   | 2346      | 2429      | 2512      | 2595      | 2677      | 2338       |
|                                                       | Total                    | 3355                   | 3459      | 3564      | 3669      | 3774      | 3880      | 3449       |
| 4000                                                  | Gnd Roll                 | 2349                   | 2435      | 2521      | 2607      | 2693      | 2778      | 2409       |
|                                                       | Total                    | 3462                   | 3571      | 3680      | 3789      | 3899      | 4009      | 3539       |
| 5000                                                  | Gnd Roll                 | 2438                   | 2527      | 2616      | 2706      | 2795      |           | 2483       |
|                                                       | Total                    | 3575                   | 3688      | 3802      | 3916      | 4030      |           | 3633       |
| 6000                                                  | Gnd Roll                 | 2531                   | 2624      | 2716      | 2809      | 2902      |           | 2560       |
|                                                       | Total                    | 3693                   | 3811      | 3930      | 4048      | 4167      |           | 3730       |
| 7000                                                  | Gnd Roll                 | 2629                   | 2725      | 2821      | 2917      | 3014      |           | 2640       |
|                                                       | Total                    | 3818                   | 3940      | 4064      | 4187      | 4311      |           | 3831       |
| 8000                                                  | Gnd Roll                 | 2762                   | 2864      | 2965      | 3066      | 3167      |           | 2754       |
|                                                       | Total                    | 3994                   | 4123      | 4253      | 4383      | 4514      |           | 3983       |
| 9000                                                  | Gnd Roll                 | 2927                   | 3034      | 3142      | 3249      | 3356      |           | 2897       |
|                                                       | Total                    | 4214                   | 4351      | 4490      | 4628      | 4767      |           | 4175       |
| 10,000                                                | Gnd Roll                 | 3102                   | 3215      | 3329      | 3442      |           |           | 3047       |
|                                                       | Total                    | 4447                   | 4593      | 4740      | 4887      |           |           | 4376       |

• NOTE •

Unfactored data: The above published distances do not include factors that may be required by operational rules.

**Flaps: 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory)**

V<sub>REF</sub> ..... 115 KIAS  
Winds ..... Calm  
Runway ..... Dry, Level, Paved  
Thrust Lever ..... IDLE

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 37% to the ground run and 33% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 6% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 7% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50% (Ice Contaminated)</b> |                          |                        |            |          |           |
|--------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                            | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                          |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                       | Gnd Roll                 | 2667                   | 2772       | 2878     | 2983      |
|                                                                          | Total                    | 4188                   | 4334       | 4481     | 4628      |
| 1000                                                                     | Gnd Roll                 | 2765                   | 2875       | 2984     | 3093      |
|                                                                          | Total                    | 4324                   | 4476       | 4629     | 4782      |
| 2000                                                                     | Gnd Roll                 | 2868                   | 2982       | 3095     | 3208      |
|                                                                          | Total                    | 4467                   | 4626       | 4784     | 4943      |
| 3000                                                                     | Gnd Roll                 | 2976                   | 3093       | 3211     | 3328      |
|                                                                          | Total                    | 4617                   | 4782       | 4947     | 5112      |
| 4000                                                                     | Gnd Roll                 | 3088                   | 3210       | 3332     | 3454      |
|                                                                          | Total                    | 4774                   | 4946       | 5117     | 5289      |
| 5000                                                                     | Gnd Roll                 | 3205                   | 3332       | 3458     | 3585      |
|                                                                          | Total                    | 4939                   | 5117       | 5296     | 5475      |
| 6000                                                                     | Gnd Roll                 | 3328                   | 3459       | 3591     | 3722      |
|                                                                          | Total                    | 5112                   | 5297       | 5483     | 5669      |
| 7000                                                                     | Gnd Roll                 | 3456                   | 3593       | 3729     | 3866      |
|                                                                          | Total                    | 5293                   | 5486       | 5679     | 5873      |
| 8000                                                                     | Gnd Roll                 | 3632                   | 3776       | 3919     | 4063      |
|                                                                          | Total                    | 5546                   | 5750       | 5954     | 6158      |
| 9000                                                                     | Gnd Roll                 | 3849                   | 4001       | 4153     | 4305      |
|                                                                          | Total                    | 5861                   | 6077       | 6294     | 6511      |
| 10,000                                                                   | Gnd Roll                 | 4078                   | 4239       | 4400     | 4561      |
|                                                                          | Total                    | 6195                   | 6424       | 6655     | 6885      |

• NOTE •

Unfactored data: The above published distances do not include factors that may be required by operational rules.



## **Flaps: UP**

$V_{REF}$  ..... 104 KIAS

• NOTE •

To calculate Flaps UP landing distance, add 38% to the total distance of Flaps 50%.

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## **Flaps: UP (Ice Contaminated Airframe/STALL SPEED HIGH Advisory)**

$V_{REF}$  ..... 136 KIAS

• NOTE •

To calculate Flaps UP landing distance, add 52% to the total distance of Flaps 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory).

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## Aircraft Weight: 4500 lb (2041 kg)

### Flaps: 100%

V<sub>REF</sub> ..... 76 KIAS

Winds ..... Calm

Runway ..... Dry, Level, Paved

Thrust Lever ..... IDLE

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 45% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

#### • NOTE •

The grayed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 100%</b> |                          |                        |           |           |           |           |           |            |
|--------------------------------------------------------|--------------------------|------------------------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>                          | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |           |           |           |           |           | <b>ISA</b> |
|                                                        |                          | <b>0</b>               | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                                     | Gnd Roll                 | 1246                   | 1292      | 1338      | 1383      | 1429      | 1474      | 1315       |
|                                                        | Total                    | 1755                   | 1811      | 1867      | 1924      | 1980      | 2037      | 1839       |
| 1000                                                   | Gnd Roll                 | 1292                   | 1340      | 1387      | 1434      | 1482      | 1529      | 1354       |
|                                                        | Total                    | 1812                   | 1870      | 1928      | 1987      | 2046      | 2105      | 1888       |
| 2000                                                   | Gnd Roll                 | 1340                   | 1389      | 1439      | 1488      | 1537      | 1586      | 1395       |
|                                                        | Total                    | 1871                   | 1932      | 1992      | 2053      | 2114      | 2175      | 1938       |
| 3000                                                   | Gnd Roll                 | 1391                   | 1442      | 1492      | 1543      | 1594      |           | 1437       |
|                                                        | Total                    | 1933                   | 1996      | 2059      | 2122      | 2186      |           | 1990       |
| 4000                                                   | Gnd Roll                 | 1443                   | 1496      | 1549      | 1602      | 1654      |           | 1480       |
|                                                        | Total                    | 1998                   | 2063      | 2129      | 2195      | 2261      |           | 2044       |
| 5000                                                   | Gnd Roll                 | 1498                   | 1553      | 1608      | 1662      | 1717      |           | 1526       |
|                                                        | Total                    | 2066                   | 2134      | 2203      | 2271      | 2340      |           | 2101       |
| 6000                                                   | Gnd Roll                 | 1555                   | 1612      | 1669      | 1726      | 1783      |           | 1573       |
|                                                        | Total                    | 2137                   | 2208      | 2280      | 2351      | 2422      |           | 2159       |
| 7000                                                   | Gnd Roll                 | 1615                   | 1674      | 1733      | 1793      | 1852      |           | 1622       |
|                                                        | Total                    | 2212                   | 2286      | 2360      | 2434      | 2509      |           | 2220       |
| 8000                                                   | Gnd Roll                 | 1697                   | 1760      | 1822      | 1884      |           |           | 1692       |
|                                                        | Total                    | 2317                   | 2395      | 2473      | 2551      |           |           | 2311       |
| 9000                                                   | Gnd Roll                 | 1799                   | 1865      | 1930      | 1996      |           |           | 1780       |
|                                                        | Total                    | 2448                   | 2531      | 2614      | 2697      |           |           | 2424       |
| 10,000                                                 | Gnd Roll                 | 1906                   | 1976      | 2045      | 2115      |           |           | 1872       |
|                                                        | Total                    | 2586                   | 2674      | 2762      | 2850      |           |           | 2544       |

• NOTE •

Unfactored data: The above published distances do not include factors that may be required by operational rules.

**Flaps: 50%**

V<sub>REF</sub> ..... 87 KIAS  
Winds ..... Calm  
Runway..... Dry, Level, Paved  
Thrust Lever .....IDLE

Headwind: Subtract 7% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 39% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50%</b> |                          |                        |           |           |           |           |           |            |
|-------------------------------------------------------|--------------------------|------------------------|-----------|-----------|-----------|-----------|-----------|------------|
| <b>Press<br/>Alt<br/>Feet</b>                         | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |           |           |           |           |           | <b>ISA</b> |
|                                                       |                          | <b>0</b>               | <b>10</b> | <b>20</b> | <b>30</b> | <b>40</b> | <b>50</b> |            |
| SL                                                    | Gnd Roll                 | 1645                   | 1705      | 1765      | 1825      | 1885      | 1946      | 1735       |
|                                                       | Total                    | 2226                   | 2299      | 2373      | 2448      | 2522      | 2596      | 2336       |
| 1000                                                  | Gnd Roll                 | 1705                   | 1768      | 1830      | 1893      | 1955      | 2018      | 1787       |
|                                                       | Total                    | 2300                   | 2377      | 2454      | 2531      | 2608      | 2685      | 2400       |
| 2000                                                  | Gnd Roll                 | 1769                   | 1834      | 1898      | 1963      | 2028      | 2093      | 1840       |
|                                                       | Total                    | 2378                   | 2458      | 2538      | 2618      | 2698      | 2778      | 2466       |
| 3000                                                  | Gnd Roll                 | 1835                   | 1902      | 1969      | 2037      | 2104      | 2171      | 1896       |
|                                                       | Total                    | 2460                   | 2542      | 2626      | 2709      | 2792      | 2876      | 2535       |
| 4000                                                  | Gnd Roll                 | 1904                   | 1974      | 2044      | 2113      | 2183      | 2253      | 1954       |
|                                                       | Total                    | 2545                   | 2631      | 2718      | 2804      | 2891      | 2978      | 2606       |
| 5000                                                  | Gnd Roll                 | 1977                   | 2049      | 2121      | 2194      | 2266      | 2338      | 2013       |
|                                                       | Total                    | 2634                   | 2724      | 2814      | 2904      | 2994      | 3084      | 2680       |
| 6000                                                  | Gnd Roll                 | 2052                   | 2127      | 2203      | 2278      | 2353      | 2428      | 2076       |
|                                                       | Total                    | 2728                   | 2822      | 2915      | 3009      | 3102      | 3196      | 2757       |
| 7000                                                  | Gnd Roll                 | 2131                   | 2209      | 2287      | 2365      | 2443      | 2522      | 2140       |
|                                                       | Total                    | 2826                   | 2924      | 3021      | 3118      | 3216      | 3313      | 2837       |
| 8000                                                  | Gnd Roll                 | 2240                   | 2322      | 2404      | 2486      | 2568      | 2650      | 2233       |
|                                                       | Total                    | 2963                   | 3066      | 3168      | 3270      | 3373      | 3476      | 2955       |
| 9000                                                  | Gnd Roll                 | 2373                   | 2460      | 2547      | 2634      | 2721      | 2808      | 2349       |
|                                                       | Total                    | 3133                   | 3242      | 3350      | 3459      | 3568      | 3677      | 3103       |
| 10,000                                                | Gnd Roll                 | 2515                   | 2607      | 2699      | 2791      | 2883      | 2975      | 2471       |
|                                                       | Total                    | 3313                   | 3428      | 3544      | 3659      | 3775      | 3891      | 3258       |

• NOTE •

Unfactored data: The above published distances do not include factors that may be required by operational rules.

### ***Flaps: 50% (Ice Contaminated)***

V<sub>REF</sub> ..... 104 KIAS  
Winds ..... Calm  
Runway..... Dry, Level, Paved  
Thrust Lever .....IDLE

**Headwind**: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

**Tailwind**: Add 37% to the ground run and 34% to the total distance for every 10 knots of tailwind.

**Uphill gradient**: Subtract 6% from the ground run for every 1% of uphill gradient.

**Downhill gradient**: Add 7% to the ground run for every 1% of downhill gradient.

**Unpaved runway**: Add 20% to the total distance.

**• NOTE •**

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50% (Ice Contaminated)</b> |                          |                        |            |          |           |
|--------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                            | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                          |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                       | Gnd Roll                 | 2162                   | 2248       | 2333     | 2419      |
|                                                                          | Total                    | 2998                   | 3108       | 3218     | 3329      |
| 1000                                                                     | Gnd Roll                 | 2242                   | 2331       | 2419     | 2508      |
|                                                                          | Total                    | 3101                   | 3215       | 3330     | 3444      |
| 2000                                                                     | Gnd Roll                 | 2326                   | 2417       | 2509     | 2601      |
|                                                                          | Total                    | 3209                   | 3327       | 3446     | 3565      |
| 3000                                                                     | Gnd Roll                 | 2413                   | 2508       | 2603     | 2699      |
|                                                                          | Total                    | 3321                   | 3445       | 3568     | 3692      |
| 4000                                                                     | Gnd Roll                 | 2504                   | 2603       | 2702     | 2800      |
|                                                                          | Total                    | 3439                   | 3567       | 3696     | 3824      |
| 5000                                                                     | Gnd Roll                 | 2599                   | 2702       | 2804     | 2907      |
|                                                                          | Total                    | 3562                   | 3696       | 3829     | 3963      |
| 6000                                                                     | Gnd Roll                 | 2698                   | 2805       | 2912     | 3018      |
|                                                                          | Total                    | 3691                   | 3830       | 3969     | 4108      |
| 7000                                                                     | Gnd Roll                 | 2802                   | 2913       | 3024     | 3134      |
|                                                                          | Total                    | 3827                   | 3971       | 4115     | 4260      |
| 8000                                                                     | Gnd Roll                 | 2945                   | 3061       | 3178     | 3294      |
|                                                                          | Total                    | 4014                   | 4166       | 4318     | 4470      |
| 9000                                                                     | Gnd Roll                 | 3121                   | 3244       | 3367     | 3491      |
|                                                                          | Total                    | 4247                   | 4408       | 4569     | 4730      |
| 10,000                                                                   | Gnd Roll                 | 3307                   | 3437       | 3568     | 3698      |
|                                                                          | Total                    | 4493                   | 4664       | 4835     | 5006      |

• NOTE •

Unfactored data: The above published distances do not include factors that may be required by operational rules.

## **Landing Distance - Factored Data**

The landing distance data in the subsequent tables includes a 1.67 factor for dry pavement and a 1.92 factor ( $1.67 \times 1.15$ ) for wet pavement.

### **Aircraft Weight: 5550 lb (2517 kg)**

#### ***Flaps: 100%, Dry Runway***

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 44% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

#### **• NOTE •**

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.



| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 100%; Runway: Dry</b> |                  |                 |      |      |      |      |    |      |
|---------------------------------------------------------------------|------------------|-----------------|------|------|------|------|----|------|
| Press<br>Alt<br>Feet                                                | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |    | ISA  |
|                                                                     |                  | 0               | 10   | 20   | 30   | 40   | 50 |      |
| SL                                                                  | Gnd Roll         | 2567            | 2660 | 2756 | 2849 | 2943 |    | 2709 |
|                                                                     | Total            | 4058            | 4172 | 4285 | 4400 | 4516 |    | 4228 |
| 1000                                                                | Gnd Roll         | 2662            | 2759 | 2857 | 2954 | 3051 |    | 2789 |
|                                                                     | Total            | 4172            | 4290 | 4410 | 4529 | 4649 |    | 4327 |
| 2000                                                                | Gnd Roll         | 2761            | 2862 | 2963 | 3064 | 3165 |    | 2872 |
|                                                                     | Total            | 4292            | 4415 | 4541 | 4664 | 4790 |    | 4429 |
| 3000                                                                | Gnd Roll         | 2864            | 2969 | 3074 | 3178 |      |    | 2959 |
|                                                                     | Total            | 4419            | 4547 | 4676 | 4806 |      |    | 4536 |
| 4000                                                                | Gnd Roll         | 2973            | 3081 | 3190 | 3298 |      |    | 3049 |
|                                                                     | Total            | 4551            | 4686 | 4820 | 4955 |      |    | 4646 |
| 5000                                                                | Gnd Roll         | 3084            | 3198 | 3312 | 3424 |      |    | 3143 |
|                                                                     | Total            | 4691            | 4830 | 4970 | 5110 |      |    | 4761 |
| 6000                                                                | Gnd Roll         | 3203            | 3320 | 3439 | 3555 |      |    | 3240 |
|                                                                     | Total            | 4836            | 4982 | 5129 | 5276 |      |    | 4881 |
| 7000                                                                | Gnd Roll         | 3327            | 3449 | 3570 |      |      |    | 3340 |
|                                                                     | Total            | 4990            | 5142 | 5294 |      |      |    | 5007 |
| 8000                                                                | Gnd Roll         | 3495            | 3624 | 3752 |      |      |    | 3485 |
|                                                                     | Total            | 5210            | 5371 | 5531 |      |      |    | 5197 |
| 9000                                                                | Gnd Roll         | 3704            | 3841 | 3976 |      |      |    | 3666 |
|                                                                     | Total            | 5488            | 5658 | 5828 |      |      |    | 5439 |
| 10,000                                                              | Gnd Roll         | 3925            | 4070 |      |      |      |    | 3856 |
|                                                                     | Total            | 5782            | 5962 |      |      |      |    | 5695 |

### ***Flaps: 100%, Wet Runway***

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 44% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

**• NOTE •**

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 100%; Runway: Wet</b> |                  |                 |      |      |      |      |    |      |
|---------------------------------------------------------------------|------------------|-----------------|------|------|------|------|----|------|
| Press<br>Alt<br>Feet                                                | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |    | ISA  |
|                                                                     |                  | 0               | 10   | 20   | 30   | 40   | 50 |      |
| SL                                                                  | Gnd Roll         | 2951            | 3059 | 3168 | 3276 | 3383 |    | 3114 |
|                                                                     | Total            | 4666            | 4796 | 4927 | 5059 | 5192 |    | 4861 |
| 1000                                                                | Gnd Roll         | 3060            | 3172 | 3285 | 3396 | 3508 |    | 3206 |
|                                                                     | Total            | 4796            | 4932 | 5071 | 5207 | 5345 |    | 4975 |
| 2000                                                                | Gnd Roll         | 3174            | 3291 | 3406 | 3523 | 3638 |    | 3302 |
|                                                                     | Total            | 4934            | 5076 | 5220 | 5363 | 5507 |    | 5092 |
| 3000                                                                | Gnd Roll         | 3293            | 3414 | 3535 | 3654 |      |    | 3402 |
|                                                                     | Total            | 5080            | 5228 | 5376 | 5526 |      |    | 5215 |
| 4000                                                                | Gnd Roll         | 3418            | 3542 | 3667 | 3792 |      |    | 3506 |
|                                                                     | Total            | 5232            | 5388 | 5541 | 5697 |      |    | 5341 |
| 5000                                                                | Gnd Roll         | 3546            | 3677 | 3807 | 3936 |      |    | 3613 |
|                                                                     | Total            | 5393            | 5553 | 5714 | 5875 |      |    | 5474 |
| 6000                                                                | Gnd Roll         | 3683            | 3817 | 3953 | 4088 |      |    | 3725 |
|                                                                     | Total            | 5560            | 5727 | 5896 | 6065 |      |    | 5612 |
| 7000                                                                | Gnd Roll         | 3825            | 3965 | 4105 |      |      |    | 3840 |
|                                                                     | Total            | 5737            | 5912 | 6086 |      |      |    | 5756 |
| 8000                                                                | Gnd Roll         | 4019            | 4166 | 4314 |      |      |    | 4007 |
|                                                                     | Total            | 5990            | 6175 | 6359 |      |      |    | 5975 |
| 9000                                                                | Gnd Roll         | 4259            | 4416 | 4572 |      |      |    | 4214 |
|                                                                     | Total            | 6309            | 6505 | 6701 |      |      |    | 6253 |
| 10,000                                                              | Gnd Roll         | 4512            | 4679 |      |      |      |    | 4433 |
|                                                                     | Total            | 6647            | 6854 |      |      |      |    | 6547 |

## ***Flaps: 50%, Dry Runway***

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 38% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

### **• NOTE •**

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50%; Runway: Dry</b> |                  |                 |      |      |      |      |      |      |
|--------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                               | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                    |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                 | Gnd Roll         | 3387            | 3512 | 3636 | 3759 | 3883 | 4008 | 3574 |
|                                                                    | Total            | 5112            | 5267 | 5422 | 5578 | 5733 | 5890 | 5344 |
| 1000                                                               | Gnd Roll         | 3512            | 3641 | 3769 | 3898 | 4026 | 4155 | 3681 |
|                                                                    | Total            | 5269            | 5429 | 5589 | 5751 | 5915 | 6077 | 5478 |
| 2000                                                               | Gnd Roll         | 3644            | 3776 | 3909 | 4043 | 4177 | 4310 | 3791 |
|                                                                    | Total            | 5431            | 5600 | 5767 | 5935 | 6104 | 6274 | 5616 |
| 3000                                                               | Gnd Roll         | 3779            | 3918 | 4056 | 4195 | 4334 | 4471 | 3904 |
|                                                                    | Total            | 5603            | 5777 | 5952 | 6127 | 6303 | 6480 | 5760 |
| 4000                                                               | Gnd Roll         | 3923            | 4066 | 4210 | 4354 | 4497 | 4639 | 4023 |
|                                                                    | Total            | 5782            | 5964 | 6146 | 6328 | 6511 | 6695 | 5910 |
| 5000                                                               | Gnd Roll         | 4071            | 4220 | 4369 | 4519 | 4668 |      | 4147 |
|                                                                    | Total            | 5970            | 6159 | 6349 | 6540 | 6730 |      | 6067 |
| 6000                                                               | Gnd Roll         | 4227            | 4382 | 4536 | 4691 | 4846 |      | 4275 |
|                                                                    | Total            | 6167            | 6364 | 6563 | 6760 | 6959 |      | 6229 |
| 7000                                                               | Gnd Roll         | 4390            | 4551 | 4711 | 4871 | 5033 |      | 4409 |
|                                                                    | Total            | 6376            | 6580 | 6787 | 6992 | 7199 |      | 6398 |
| 8000                                                               | Gnd Roll         | 4613            | 4783 | 4952 | 5120 | 5289 |      | 4599 |
|                                                                    | Total            | 6670            | 6885 | 7103 | 7320 | 7538 |      | 6652 |
| 9000                                                               | Gnd Roll         | 4888            | 5067 | 5247 | 5426 | 5605 |      | 4838 |
|                                                                    | Total            | 7037            | 7266 | 7498 | 7729 | 7961 |      | 6972 |
| 10,000                                                             | Gnd Roll         | 5180            | 5369 | 5559 | 5748 |      |      | 5088 |
|                                                                    | Total            | 7426            | 7670 | 7916 | 8161 |      |      | 7308 |

### ***Flaps: 50%, Wet Runway***

Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 38% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

• NOTE •

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50%; Runway: Wet</b> |                  |                 |      |      |      |      |      |      |
|--------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                               | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                    |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                 | Gnd Roll         | 3894            | 4038 | 4180 | 4322 | 4464 | 4608 | 4109 |
|                                                                    | Total            | 5877            | 6056 | 6234 | 6413 | 6591 | 6772 | 6144 |
| 1000                                                               | Gnd Roll         | 4038            | 4186 | 4333 | 4481 | 4629 | 4777 | 4232 |
|                                                                    | Total            | 6058            | 6242 | 6426 | 6612 | 6801 | 6987 | 6298 |
| 2000                                                               | Gnd Roll         | 4189            | 4341 | 4495 | 4648 | 4802 | 4956 | 4358 |
|                                                                    | Total            | 6244            | 6438 | 6630 | 6824 | 7018 | 7213 | 6457 |
| 3000                                                               | Gnd Roll         | 4345            | 4504 | 4664 | 4823 | 4982 | 5140 | 4489 |
|                                                                    | Total            | 6442            | 6641 | 6843 | 7044 | 7246 | 7450 | 6622 |
| 4000                                                               | Gnd Roll         | 4510            | 4675 | 4840 | 5005 | 5171 | 5334 | 4625 |
|                                                                    | Total            | 6647            | 6856 | 7066 | 7275 | 7486 | 7697 | 6795 |
| 5000                                                               | Gnd Roll         | 4681            | 4852 | 5023 | 5196 | 5366 |      | 4767 |
|                                                                    | Total            | 6864            | 7081 | 7300 | 7519 | 7738 |      | 6975 |
| 6000                                                               | Gnd Roll         | 4860            | 5038 | 5215 | 5393 | 5572 |      | 4915 |
|                                                                    | Total            | 7091            | 7317 | 7546 | 7772 | 8001 |      | 7162 |
| 7000                                                               | Gnd Roll         | 5048            | 5232 | 5416 | 5601 | 5787 |      | 5069 |
|                                                                    | Total            | 7331            | 7565 | 7803 | 8039 | 8277 |      | 7356 |
| 8000                                                               | Gnd Roll         | 5303            | 5499 | 5693 | 5887 | 6081 |      | 5288 |
|                                                                    | Total            | 7668            | 7916 | 8166 | 8415 | 8667 |      | 7647 |
| 9000                                                               | Gnd Roll         | 5620            | 5825 | 6033 | 6238 | 6444 |      | 5562 |
|                                                                    | Total            | 8091            | 8354 | 8621 | 8886 | 9153 |      | 8016 |
| 10,000                                                             | Gnd Roll         | 5956            | 6173 | 6392 | 6609 |      |      | 5850 |
|                                                                    | Total            | 8538            | 8819 | 9101 | 9383 |      |      | 8402 |

---

***Flaps: 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory), Dry Runway***

**|** Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

**|** Tailwind: Add 37% to the ground run and 33% to the total distance for every 10 knots of tailwind.

**|** Uphill gradient: Subtract 6% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 7% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).



| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50% (Ice Contaminated);<br/>Runway: Dry</b> |                          |                        |            |          |           |
|-------------------------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                                             | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                                           |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                                        | Gnd Roll                 | 4454                   | 4630       | 4806     | 4982      |
|                                                                                           | Total                    | 6994                   | 7238       | 7482     | 7728      |
| 1000                                                                                      | Gnd Roll                 | 4618                   | 4801       | 4983     | 5166      |
|                                                                                           | Total                    | 7222                   | 7476       | 7730     | 7985      |
| 2000                                                                                      | Gnd Roll                 | 4790                   | 4979       | 5168     | 5358      |
|                                                                                           | Total                    | 7461                   | 7725       | 7989     | 8255      |
| 3000                                                                                      | Gnd Roll                 | 4969                   | 5166       | 5362     | 5558      |
|                                                                                           | Total                    | 7711                   | 7986       | 8261     | 8537      |
| 4000                                                                                      | Gnd Roll                 | 5157                   | 5361       | 5564     | 5768      |
|                                                                                           | Total                    | 7973                   | 8259       | 8546     | 8833      |
| 5000                                                                                      | Gnd Roll                 | 5353                   | 5564       | 5776     | 5987      |
|                                                                                           | Total                    | 8248                   | 8546       | 8844     | 9143      |
| 6000                                                                                      | Gnd Roll                 | 5558                   | 5777       | 5997     | 6216      |
|                                                                                           | Total                    | 8536                   | 8846       | 9157     | 9468      |
| 7000                                                                                      | Gnd Roll                 | 5772                   | 6000       | 6228     | 6456      |
|                                                                                           | Total                    | 8839                   | 9161       | 9485     | 9809      |
| 8000                                                                                      | Gnd Roll                 | 6066                   | 6305       | 6545     | 6784      |
|                                                                                           | Total                    | 9262                   | 9602       | 9942     | 10,284    |
| 9000                                                                                      | Gnd Roll                 | 6428                   | 6681       | 6935     | 7189      |
|                                                                                           | Total                    | 9788                   | 10,149     | 10,511   | 10,874    |
| 10,000                                                                                    | Gnd Roll                 | 6810                   | 7079       | 7349     | 7618      |
|                                                                                           | Total                    | 10,345                 | 10,729     | 11,113   | 11,499    |

---

***Flaps: 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory), Wet Runway***

**|** Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

**|** Tailwind: Add 37% to the ground run and 33% to the total distance for every 10 knots of tailwind.

**|** Uphill gradient: Subtract 6% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 7% to the ground run for every 1% of downhill gradient.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 5550 lb (2517 kg); Flaps: 50% (Ice Contaminated);<br/>Runway: Wet</b> |                          |                        |            |          |           |
|-------------------------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                                             | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                                           |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                                        | Gnd Roll                 | 5121                   | 5323       | 5525     | 5727      |
|                                                                                           | Total                    | 8041                   | 8321       | 8603     | 8885      |
| 1000                                                                                      | Gnd Roll                 | 5310                   | 5519       | 5729     | 5939      |
|                                                                                           | Total                    | 8303                   | 8595       | 8887     | 9181      |
| 2000                                                                                      | Gnd Roll                 | 5507                   | 5725       | 5942     | 6160      |
|                                                                                           | Total                    | 8578                   | 8881       | 9185     | 9491      |
| 3000                                                                                      | Gnd Roll                 | 5713                   | 5939       | 6165     | 6390      |
|                                                                                           | Total                    | 8865                   | 9181       | 9498     | 9815      |
| 4000                                                                                      | Gnd Roll                 | 5929                   | 6163       | 6397     | 6631      |
|                                                                                           | Total                    | 9167                   | 9495       | 9825     | 10,155    |
| 5000                                                                                      | Gnd Roll                 | 6154                   | 6397       | 6640     | 6883      |
|                                                                                           | Total                    | 9483                   | 9825       | 10,168   | 10,512    |
| 6000                                                                                      | Gnd Roll                 | 6390                   | 6642       | 6894     | 7147      |
|                                                                                           | Total                    | 9814                   | 10,170     | 10,527   | 10,885    |
| 7000                                                                                      | Gnd Roll                 | 6636                   | 6898       | 7160     | 7422      |
|                                                                                           | Total                    | 10,162                 | 10,533     | 10,904   | 11,277    |
| 8000                                                                                      | Gnd Roll                 | 6974                   | 7249       | 7525     | 7800      |
|                                                                                           | Total                    | 10,648                 | 11,039     | 11,431   | 11,824    |
| 9000                                                                                      | Gnd Roll                 | 7390                   | 7682       | 7974     | 8266      |
|                                                                                           | Total                    | 11,254                 | 11,669     | 12,085   | 12,502    |
| 10,000                                                                                    | Gnd Roll                 | 7830                   | 8139       | 8449     | 8758      |
|                                                                                           | Total                    | 11,894                 | 12,335     | 12,777   | 13,220    |

## Aircraft Weight: 4500 lb (2041 kg)

### Flaps: 100%, Dry Runway

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 45% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

#### • NOTE •

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 100%; Runway: Dry</b> |                  |                 |      |      |      |      |      |      |
|---------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                                | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                     |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                  | Gnd Roll         | 2081            | 2158 | 2234 | 2310 | 2386 | 2462 | 2196 |
|                                                                     | Total            | 2931            | 3024 | 3118 | 3213 | 3307 | 3402 | 3071 |
| 1000                                                                | Gnd Roll         | 2158            | 2238 | 2316 | 2395 | 2475 | 2553 | 2261 |
|                                                                     | Total            | 3026            | 3123 | 3220 | 3318 | 3417 | 3515 | 3153 |
| 2000                                                                | Gnd Roll         | 2238            | 2320 | 2403 | 2485 | 2567 | 2649 | 2330 |
|                                                                     | Total            | 3125            | 3226 | 3327 | 3429 | 3530 | 3632 | 3236 |
| 3000                                                                | Gnd Roll         | 2323            | 2408 | 2492 | 2577 | 2662 |      | 2400 |
|                                                                     | Total            | 3228            | 3333 | 3439 | 3544 | 3651 |      | 3323 |
| 4000                                                                | Gnd Roll         | 2410            | 2498 | 2587 | 2675 | 2762 |      | 2472 |
|                                                                     | Total            | 3337            | 3445 | 3555 | 3666 | 3776 |      | 3413 |
| 5000                                                                | Gnd Roll         | 2502            | 2594 | 2685 | 2776 | 2867 |      | 2548 |
|                                                                     | Total            | 3450            | 3564 | 3679 | 3793 | 3908 |      | 3509 |
| 6000                                                                | Gnd Roll         | 2597            | 2692 | 2787 | 2882 | 2978 |      | 2627 |
|                                                                     | Total            | 3569            | 3687 | 3808 | 3926 | 4045 |      | 3606 |
| 7000                                                                | Gnd Roll         | 2697            | 2796 | 2894 | 2994 | 3093 |      | 2709 |
|                                                                     | Total            | 3694            | 3818 | 3941 | 4065 | 4190 |      | 3707 |
| 8000                                                                | Gnd Roll         | 2834            | 2939 | 3043 | 3146 |      |      | 2826 |
|                                                                     | Total            | 3869            | 4000 | 4130 | 4260 |      |      | 3859 |
| 9000                                                                | Gnd Roll         | 3004            | 3115 | 3223 | 3333 |      |      | 2973 |
|                                                                     | Total            | 4088            | 4227 | 4365 | 4504 |      |      | 4048 |
| 10,000                                                              | Gnd Roll         | 3183            | 3300 | 3415 | 3532 |      |      | 3126 |
|                                                                     | Total            | 4319            | 4466 | 4613 | 4760 |      |      | 4248 |

### ***Flaps: 100%, Wet Runway***

Headwind: Subtract 8% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 49% to the ground run and 45% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

• NOTE •

The greyed-out cells in the subsequent table signify that the aircraft is not capable of meeting the regulatory balked landing climb gradient for these conditions.

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 100%; Runway: Wet</b> |                  |                 |      |      |      |      |      |      |
|---------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                                | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                     |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                  | Gnd Roll         | 2392            | 2481 | 2569 | 2655 | 2744 | 2830 | 2525 |
|                                                                     | Total            | 3370            | 3477 | 3585 | 3694 | 3802 | 3911 | 3531 |
| 1000                                                                | Gnd Roll         | 2481            | 2573 | 2663 | 2753 | 2845 | 2936 | 2600 |
|                                                                     | Total            | 3479            | 3590 | 3702 | 3815 | 3928 | 4042 | 3625 |
| 2000                                                                | Gnd Roll         | 2573            | 2667 | 2763 | 2857 | 2951 | 3045 | 2678 |
|                                                                     | Total            | 3592            | 3709 | 3825 | 3942 | 4059 | 4176 | 3721 |
| 3000                                                                | Gnd Roll         | 2671            | 2769 | 2865 | 2963 | 3060 |      | 2759 |
|                                                                     | Total            | 3711            | 3832 | 3953 | 4074 | 4197 |      | 3821 |
| 4000                                                                | Gnd Roll         | 2771            | 2872 | 2974 | 3076 | 3176 |      | 2842 |
|                                                                     | Total            | 3836            | 3961 | 4088 | 4214 | 4341 |      | 3924 |
| 5000                                                                | Gnd Roll         | 2876            | 2982 | 3087 | 3191 | 3297 |      | 2930 |
|                                                                     | Total            | 3967            | 4097 | 4230 | 4360 | 4493 |      | 4034 |
| 6000                                                                | Gnd Roll         | 2986            | 3095 | 3204 | 3314 | 3423 |      | 3020 |
|                                                                     | Total            | 4103            | 4239 | 4378 | 4514 | 4650 |      | 4145 |
| 7000                                                                | Gnd Roll         | 3101            | 3214 | 3327 | 3443 | 3556 |      | 3114 |
|                                                                     | Total            | 4247            | 4389 | 4531 | 4673 | 4817 |      | 4262 |
| 8000                                                                | Gnd Roll         | 3258            | 3379 | 3498 | 3617 |      |      | 3249 |
|                                                                     | Total            | 4449            | 4598 | 4748 | 4898 |      |      | 4437 |
| 9000                                                                | Gnd Roll         | 3454            | 3581 | 3706 | 3832 |      |      | 3418 |
|                                                                     | Total            | 4700            | 4860 | 5019 | 5178 |      |      | 4654 |
| 10,000                                                              | Gnd Roll         | 3660            | 3794 | 3926 | 4061 |      |      | 3594 |
|                                                                     | Total            | 4965            | 5134 | 5303 | 5472 |      |      | 4884 |

## ***Flaps: 50%, Dry Runway***

Headwind: Subtract 7% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 39% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

### **• NOTE •**

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).



| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50%; Runway: Dry</b> |                  |                 |      |      |      |      |      |      |
|--------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                               | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                    |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                 | Gnd Roll         | 2747            | 2847 | 2948 | 3048 | 3148 | 3250 | 2897 |
|                                                                    | Total            | 3717            | 3839 | 3963 | 4088 | 4212 | 4335 | 3901 |
| 1000                                                               | Gnd Roll         | 2847            | 2953 | 3056 | 3161 | 3265 | 3370 | 2984 |
|                                                                    | Total            | 3841            | 3970 | 4098 | 4227 | 4355 | 4484 | 4008 |
| 2000                                                               | Gnd Roll         | 2954            | 3063 | 3170 | 3278 | 3387 | 3495 | 3073 |
|                                                                    | Total            | 3971            | 4105 | 4238 | 4372 | 4506 | 4639 | 4118 |
| 3000                                                               | Gnd Roll         | 3064            | 3176 | 3288 | 3402 | 3514 | 3626 | 3166 |
|                                                                    | Total            | 4108            | 4245 | 4385 | 4524 | 4663 | 4803 | 4233 |
| 4000                                                               | Gnd Roll         | 3180            | 3297 | 3413 | 3529 | 3646 | 3763 | 3263 |
|                                                                    | Total            | 4250            | 4394 | 4539 | 4683 | 4828 | 4973 | 4352 |
| 5000                                                               | Gnd Roll         | 3302            | 3422 | 3542 | 3664 | 3784 | 3904 | 3362 |
|                                                                    | Total            | 4399            | 4549 | 4699 | 4850 | 5000 | 5150 | 4476 |
| 6000                                                               | Gnd Roll         | 3427            | 3552 | 3679 | 3804 | 3930 | 4055 | 3467 |
|                                                                    | Total            | 4556            | 4713 | 4868 | 5025 | 5180 | 5337 | 4604 |
| 7000                                                               | Gnd Roll         | 3559            | 3689 | 3819 | 3950 | 4080 | 4212 | 3574 |
|                                                                    | Total            | 4719            | 4883 | 5045 | 5207 | 5371 | 5533 | 4738 |
| 8000                                                               | Gnd Roll         | 3741            | 3878 | 4015 | 4152 | 4289 | 4426 | 3729 |
|                                                                    | Total            | 4948            | 5120 | 5291 | 5461 | 5633 | 5805 | 4935 |
| 9000                                                               | Gnd Roll         | 3963            | 4108 | 4253 | 4399 | 4544 | 4689 | 3923 |
|                                                                    | Total            | 5232            | 5414 | 5595 | 5777 | 5959 | 6141 | 5182 |
| 10,000                                                             | Gnd Roll         | 4200            | 4354 | 4507 | 4661 | 4815 | 4968 | 4127 |
|                                                                    | Total            | 5533            | 5725 | 5918 | 6111 | 6304 | 6498 | 5441 |

## ***Flaps: 50%, Wet Runway***

Headwind: Subtract 7% from the ground run and 7% from the total distance for every 10 knots of headwind.

Tailwind: Add 42% to the ground run and 39% to the total distance for every 10 knots of tailwind.

Uphill gradient: Subtract 5% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 6% to the ground run for every 1% of downhill gradient.

### **• NOTE •**

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50%; Runway: Wet</b> |                  |                 |      |      |      |      |      |      |
|--------------------------------------------------------------------|------------------|-----------------|------|------|------|------|------|------|
| Press<br>Alt<br>Feet                                               | Distance<br>Feet | TEMPERATURE ~°C |      |      |      |      |      | ISA  |
|                                                                    |                  | 0               | 10   | 20   | 30   | 40   | 50   |      |
| SL                                                                 | Gnd Roll         | 3158            | 3274 | 3389 | 3504 | 3619 | 3736 | 3331 |
|                                                                    | Total            | 4274            | 4414 | 4556 | 4700 | 4842 | 4984 | 4485 |
| 1000                                                               | Gnd Roll         | 3274            | 3395 | 3514 | 3635 | 3754 | 3875 | 3431 |
|                                                                    | Total            | 4416            | 4564 | 4712 | 4860 | 5007 | 5155 | 4608 |
| 2000                                                               | Gnd Roll         | 3396            | 3521 | 3644 | 3769 | 3894 | 4019 | 3533 |
|                                                                    | Total            | 4566            | 4719 | 4873 | 5027 | 5180 | 5334 | 4735 |
| 3000                                                               | Gnd Roll         | 3523            | 3652 | 3780 | 3911 | 4040 | 4168 | 3640 |
|                                                                    | Total            | 4723            | 4881 | 5042 | 5201 | 5361 | 5522 | 4867 |
| 4000                                                               | Gnd Roll         | 3656            | 3790 | 3924 | 4057 | 4191 | 4326 | 3752 |
|                                                                    | Total            | 4886            | 5052 | 5219 | 5384 | 5551 | 5718 | 5004 |
| 5000                                                               | Gnd Roll         | 3796            | 3934 | 4072 | 4212 | 4351 | 4489 | 3865 |
|                                                                    | Total            | 5057            | 5230 | 5403 | 5576 | 5748 | 5921 | 5146 |
| 6000                                                               | Gnd Roll         | 3940            | 4084 | 4230 | 4374 | 4518 | 4662 | 3986 |
|                                                                    | Total            | 5238            | 5418 | 5597 | 5777 | 5956 | 6136 | 5293 |
| 7000                                                               | Gnd Roll         | 4092            | 4241 | 4391 | 4541 | 4691 | 4842 | 4109 |
|                                                                    | Total            | 5426            | 5614 | 5800 | 5987 | 6175 | 6361 | 5447 |
| 8000                                                               | Gnd Roll         | 4301            | 4458 | 4616 | 4773 | 4931 | 5088 | 4287 |
|                                                                    | Total            | 5689            | 5887 | 6083 | 6278 | 6476 | 6674 | 5674 |
| 9000                                                               | Gnd Roll         | 4556            | 4723 | 4890 | 5057 | 5224 | 5391 | 4510 |
|                                                                    | Total            | 6015            | 6225 | 6432 | 6641 | 6851 | 7060 | 5958 |
| 10,000                                                             | Gnd Roll         | 4829            | 5005 | 5182 | 5359 | 5535 | 5712 | 4744 |
|                                                                    | Total            | 6361            | 6582 | 6804 | 7025 | 7248 | 7471 | 6255 |

---

***Flaps: 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory), Dry Runway***

**|** Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

**|** Tailwind: Add 37% to the ground run and 34% to the total distance for every 10 knots of tailwind.

**|** Uphill gradient: Subtract 6% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 7% to the ground run for every 1% of downhill gradient.

Unpaved runway: Add 20% to the total distance.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50% (Ice Contaminated);<br/>Runway: Dry</b> |                          |                        |            |          |           |
|-------------------------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                                             | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                                           |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                                        | Gnd Roll                 | 3611                   | 3754       | 3896     | 4039      |
|                                                                                           | Total                    | 5007                   | 5191       | 5375     | 5559      |
| 1000                                                                                      | Gnd Roll                 | 3745                   | 3892       | 4040     | 4188      |
|                                                                                           | Total                    | 5179                   | 5370       | 5561     | 5752      |
| 2000                                                                                      | Gnd Roll                 | 3884                   | 4037       | 4191     | 4344      |
|                                                                                           | Total                    | 5358                   | 5557       | 5755     | 5954      |
| 3000                                                                                      | Gnd Roll                 | 4029                   | 4188       | 4348     | 4507      |
|                                                                                           | Total                    | 5546                   | 5752       | 5959     | 6165      |
| 4000                                                                                      | Gnd Roll                 | 4181                   | 4346       | 4512     | 4677      |
|                                                                                           | Total                    | 5743                   | 5957       | 6172     | 6386      |
| 5000                                                                                      | Gnd Roll                 | 4340                   | 4512       | 4683     | 4854      |
|                                                                                           | Total                    | 5949                   | 6171       | 6394     | 6618      |
| 6000                                                                                      | Gnd Roll                 | 4506                   | 4684       | 4862     | 5040      |
|                                                                                           | Total                    | 6165                   | 6396       | 6628     | 6860      |
| 7000                                                                                      | Gnd Roll                 | 4680                   | 4865       | 5050     | 5235      |
|                                                                                           | Total                    | 6390                   | 6631       | 6872     | 7113      |
| 8000                                                                                      | Gnd Roll                 | 4918                   | 5112       | 5307     | 5501      |
|                                                                                           | Total                    | 6704                   | 6957       | 7211     | 7465      |
| 9000                                                                                      | Gnd Roll                 | 5212                   | 5417       | 5623     | 5829      |
|                                                                                           | Total                    | 7092                   | 7361       | 7630     | 7900      |
| 10,000                                                                                    | Gnd Roll                 | 5522                   | 5740       | 5958     | 6176      |
|                                                                                           | Total                    | 7503                   | 7788       | 8074     | 8360      |

---

***Flaps: 50% (Ice Contaminated Airframe/STALL SPEED HIGH Advisory), Wet Runway***

**|** Headwind: Subtract 7% from the ground run and 6% from the total distance for every 10 knots of headwind.

**|** Tailwind: Add 37% to the ground run and 34% to the total distance for every 10 knots of tailwind.

**|** Uphill gradient: Subtract 6% from the ground run for every 1% of uphill gradient.

Downhill gradient: Add 7% to the ground run for every 1% of downhill gradient.

**• NOTE •**

Refer to [Section 2: Limitations, "Brakes"](#) for high-energy braking events.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

| <b>Aircraft Weight: 4500 lb (2041 kg); Flaps: 50% (Ice Contaminated);<br/>Runway: Wet</b> |                          |                        |            |          |           |
|-------------------------------------------------------------------------------------------|--------------------------|------------------------|------------|----------|-----------|
| <b>Press<br/>Alt<br/>Feet</b>                                                             | <b>Distance<br/>Feet</b> | <b>TEMPERATURE ~°C</b> |            |          |           |
|                                                                                           |                          | <b>-20</b>             | <b>-10</b> | <b>0</b> | <b>10</b> |
| SL                                                                                        | Gnd Roll                 | 4152                   | 4316       | 4480     | 4644      |
|                                                                                           | Total                    | 5757                   | 5968       | 6179     | 6391      |
| 1000                                                                                      | Gnd Roll                 | 4305                   | 4475       | 4645     | 4815      |
|                                                                                           | Total                    | 5954                   | 6173       | 6393     | 6613      |
| 2000                                                                                      | Gnd Roll                 | 4465                   | 4642       | 4818     | 4994      |
|                                                                                           | Total                    | 6161                   | 6388       | 6617     | 6845      |
| 3000                                                                                      | Gnd Roll                 | 4632                   | 4815       | 4998     | 5181      |
|                                                                                           | Total                    | 6377                   | 6613       | 6851     | 7088      |
| 4000                                                                                      | Gnd Roll                 | 4807                   | 4997       | 5187     | 5377      |
|                                                                                           | Total                    | 6603                   | 6849       | 7095     | 7342      |
| 5000                                                                                      | Gnd Roll                 | 4990                   | 5187       | 5384     | 5581      |
|                                                                                           | Total                    | 6840                   | 7095       | 7352     | 7608      |
| 6000                                                                                      | Gnd Roll                 | 5181                   | 5385       | 5590     | 5795      |
|                                                                                           | Total                    | 7087                   | 7353       | 7620     | 7887      |
| 7000                                                                                      | Gnd Roll                 | 5381                   | 5593       | 5806     | 6018      |
|                                                                                           | Total                    | 7347                   | 7624       | 7901     | 8178      |
| 8000                                                                                      | Gnd Roll                 | 5654                   | 5878       | 6101     | 6324      |
|                                                                                           | Total                    | 7708                   | 7999       | 8290     | 8582      |
| 9000                                                                                      | Gnd Roll                 | 5992                   | 6228       | 6465     | 6702      |
|                                                                                           | Total                    | 8154                   | 8463       | 8773     | 9082      |
| 10,000                                                                                    | Gnd Roll                 | 6349                   | 6599       | 6850     | 7101      |
|                                                                                           | Total                    | 8626                   | 8954       | 9283     | 9611      |

## **Landing Distance - Contaminated Runway**

### **Water or Slush Covered Runway**

| <b>Dry GND Roll Distance</b> | <b>Water/Slush Covered (inches)</b> |            |            |            |            |
|------------------------------|-------------------------------------|------------|------------|------------|------------|
|                              | <b>0.125</b>                        | <b>0.2</b> | <b>0.3</b> | <b>0.4</b> | <b>0.5</b> |
| 1200                         | 2190                                | 1870       | 1740       | 1630       | 1540       |
| 1400                         | 2550                                | 2180       | 2030       | 1910       | 1800       |
| 1600                         | 2920                                | 2500       | 2310       | 2180       | 2060       |
| 1800                         | 3280                                | 2810       | 2600       | 2450       | 2310       |
| 2000                         | 3650                                | 3120       | 2890       | 2720       | 2570       |
| 2200                         | 4010                                | 3430       | 3180       | 2990       | 2830       |
| 2400                         | 4370                                | 3740       | 3470       | 3260       | 3080       |
| 2600                         | 4740                                | 4050       | 3760       | 3540       | 3340       |
| 2800                         | 5100                                | 4360       | 4050       | 3810       | 3590       |
| 3000                         | 5470                                | 4670       | 4330       | 4080       | 3850       |
| 3200                         | 5830                                | 4990       | 4620       | 4350       | 4110       |
| 3400                         | 6200                                | 5300       | 4910       | 4620       | 4360       |
| 3600                         | 6560                                | 5610       | 5200       | 4890       | 4620       |
| 3800                         | 6920                                | 5920       | 5490       | 5170       | 4880       |
| 4000                         | 7290                                | 6230       | 5780       | 5440       | 5130       |
| 4200                         | 7650                                | 6540       | 6070       | 5710       | 5390       |
| 4400                         | 8020                                | 6850       | 6360       | 5980       | 5650       |
| 4600                         | 8380                                | 7160       | 6640       | 6250       | 5900       |
| 4800                         | 8740                                | 7480       | 6930       | 6520       | 6160       |
| 5000                         | 9110                                | 7790       | 7220       | 6800       | 6410       |

**• NOTE •**

This data is considered supplemental and is not FAA approved. Data is primarily for runways where greater than 0.1 inch (3.0 mm) of contaminant is observed (FICON 4 or worse). Pilots/operators should recognize that these values are considered minimums and actual distances may be greater when landing in these conditions. Pilots/operators should also recognize that the risk of hydroplaning will increase at higher altitudes and warmer temperatures due to higher speeds at touchdown.



## Snow Covered Runway

| Dry GND<br>Roll<br>Distance | Slush/Wet Snow Covered (inches) |      |      |      |      | Dry Snow<br>(inches) | Compact<br>Snow |
|-----------------------------|---------------------------------|------|------|------|------|----------------------|-----------------|
|                             | 0.125                           | 0.2  | 0.3  | 0.4  | 0.5  | 1.0                  |                 |
| 1200                        | 1950                            | 1890 | 1810 | 1740 | 1670 | 1600                 | 1900            |
| 1400                        | 2270                            | 2200 | 2110 | 2020 | 1950 | 1870                 | 2210            |
| 1600                        | 2600                            | 2510 | 2410 | 2310 | 2220 | 2130                 | 2530            |
| 1800                        | 2920                            | 2830 | 2710 | 2600 | 2500 | 2400                 | 2850            |
| 2000                        | 3250                            | 3140 | 3010 | 2890 | 2780 | 2660                 | 3160            |
| 2200                        | 3570                            | 3450 | 3310 | 3180 | 3050 | 2930                 | 3480            |
| 2400                        | 3900                            | 3770 | 3610 | 3470 | 3330 | 3200                 | 3790            |
| 2600                        | 4220                            | 4080 | 3910 | 3760 | 3610 | 3460                 | 4110            |
| 2800                        | 4540                            | 4390 | 4210 | 4040 | 3890 | 3730                 | 4420            |
| 3000                        | 4870                            | 4710 | 4510 | 4330 | 4160 | 3990                 | 4740            |
| 3200                        | 5190                            | 5020 | 4810 | 4620 | 4440 | 4260                 | 5060            |
| 3400                        | 5520                            | 5330 | 5110 | 4910 | 4720 | 4530                 | 5370            |
| 3600                        | 5840                            | 5650 | 5420 | 5200 | 5000 | 4790                 | 5690            |
| 3800                        | 6160                            | 5960 | 5720 | 5490 | 5270 | 5060                 | 6000            |
| 4000                        | 6490                            | 6270 | 6020 | 5780 | 5550 | 5320                 | 6320            |
| 4200                        | 6810                            | 6590 | 6320 | 6060 | 5830 | 5590                 | 6630            |
| 4400                        | 7140                            | 6900 | 6620 | 6350 | 6100 | 5860                 | 6950            |
| 4600                        | 7460                            | 7210 | 6920 | 6640 | 6380 | 6120                 | 7270            |
| 4800                        | 7790                            | 7530 | 7220 | 6930 | 6660 | 6390                 | 7580            |
| 5000                        | 8110                            | 7840 | 7520 | 7220 | 6940 | 6650                 | 7900            |

• NOTE •

This data is considered supplemental and is not FAA approved.

Data is primarily for runways where greater than 0.1 inch (3.0 mm) of contaminant is observed (FICON 4 or worse).

Pilots/operators should recognize that these values are considered minimums and actual distances may be greater when landing in these conditions. Pilots/operators should also recognize that the risk of hydroplaning will increase at higher altitudes and warmer temperatures due to higher speeds at touchdown.

Intentionally Left Blank

## ***Section 6: Weight and Balance***

### **Table of Contents**

|                                                  |    |
|--------------------------------------------------|----|
| Introduction.....                                | 3  |
| Weight and Balance Record.....                   | 3  |
| Weight and Balance Data.....                     | 3  |
| Loading Instructions .....                       | 3  |
| Weight and Balance Determination for Flight..... | 5  |
| Weight and Center of Gravity Limit .....         | 5  |
| Weight/Fuselage Station (FS) Chart .....         | 6  |
| Weight and Balance Loading Form .....            | 7  |
| Weight & Balance Record .....                    | 10 |
| Equipment List.....                              | 11 |

Intentionally Left Blank

## **Introduction**

This section describes the procedure for establishing the basic empty weight and center of gravity of the airplane. Sample forms are provided for reference. A comprehensive list of all equipment available for this airplane is included at the back of this section.

It is the responsibility of the pilot to ensure that the airplane is loaded properly and that all changes to the basic empty weight and center of gravity are recorded.

**• NOTE •**

Refer to AMM Chapter 8: Leveling & Weighing for instructions.

## **Weight and Balance Record**

### **Weight and Balance Data**

Refer to “As-Delivered” Weight and Balance Data.

### **Loading Instructions**

It is the responsibility of the pilot to ensure that the airplane is properly loaded and operated within the prescribed weight and center of gravity limits. The following information enables the pilot to calculate the total weight and center of gravity for loading. The calculated Center of Gravity is then compared to the [Weight and Center of Gravity Limit](#) chart for a determination of proper loading.

Airplane loading determinations are calculated using the Weight & Balance Loading Form.

1. Basic Empty Weight – Enter the current Basic Empty Weight and Moment from the Weight & Balance Record.
2. Front Row Seat (Seats 1 and 2) Occupants – Enter the total weight and moment/1000 for the front row seat occupants from the Loading Data.
3. Middle Row Seats (Seats 3 and 5) – Enter total weight and moment/1000 for the middle row seats from the Loading Data.
4. Middle Row Outboard Seat (Seats 3 and 5) Occupants – Enter total weight and moment/1000 for the middle row seat occupants from the Loading Data.
5. Middle Row Inboard Seat (Seat 4) – Enter total weight and moment/1000 for the middle row seat from the Loading Data.

**• NOTE •**

Seat 4 may be located in any of three locations (forward, mid, and aft).

6. Middle Row Inboard Seat (Seat 4) Occupant – Enter total weight and moment/1000 for the middle row seat occupant from the Loading Data.

7. XC Seats (Seats 6 and 7) – Enter the total weight and moment/1000 for the XC seats, if installed, from the Loading Data.
8. XC Seat Occupants – Enter the total weight and moment/1000 for the XC Seat occupants from the Loading Data.
9. Lavatory – Enter the total weight and moment/1000 for the lavatory from the Loading Data.
10. Convenience Console – Enter the total weight and moment/1000 for the convenience console from the Loading Data.
11. Baggage and Cargo X-Tend Baggage – Enter weight and moment/1000 from the Loading Data. Assume the most adverse location.

• NOTE •

Load limit is 50 lb/sq. ft.

12. IPS Fluid – Enter the weight and moment/1000 of IPS Fluid loaded on the airplane from the Loading Data.
13. Zero Fuel Condition – If desired, subtotal the weights and moment/1000 from steps 1 through 9. This includes all useful load items, excluding fuel.
14. Fuel Loading – Enter the weight and moment/1000 of usable fuel loaded on the airplane from the Loading Data.
15. Ramp Condition (weight and moment of aircraft before taxi) – Subtotal the weight and moment/1000.
16. Fuel for start, taxi, and run-up – This value is pre-entered on the form. Normally, fuel used for start and taxi before takeoff is approximately 40 lb (6 U.S. gallons).
17. Takeoff Condition Weight – Subtract item 13 from item 12. Must be less than 6000 lb (2722 kg).

RELATED TABLE/FIGURE:

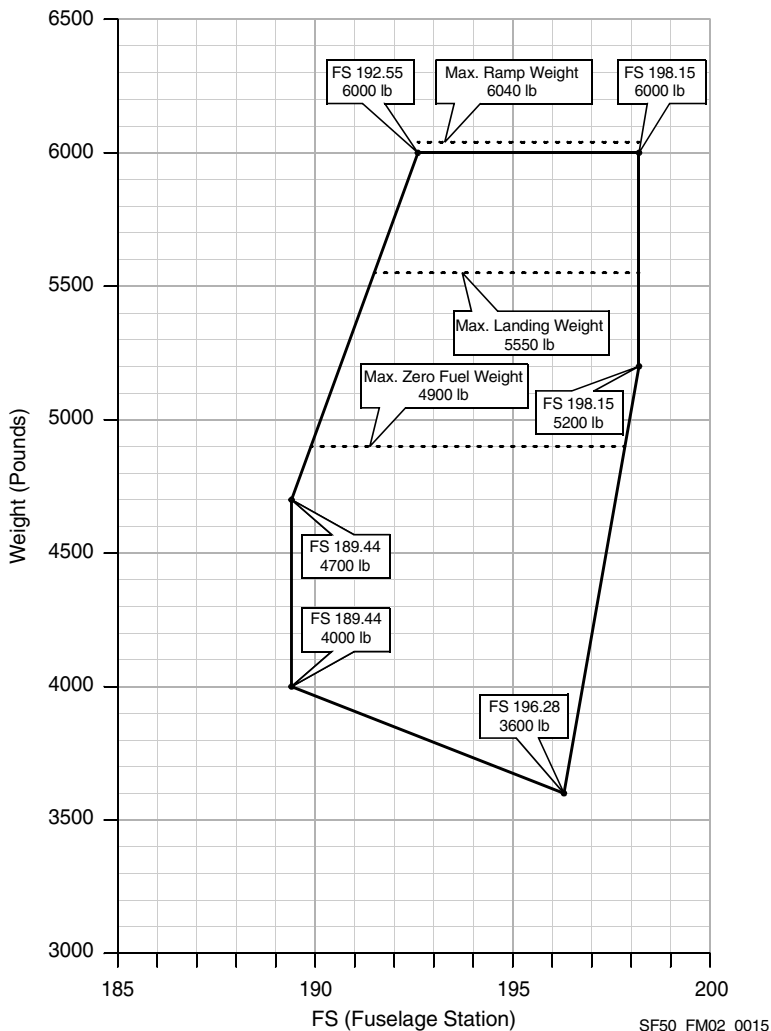
Refer to "Figure 6-1: Center of Gravity Envelope".

Refer to Section 6: Weight and Balance, "Weight/Fuselage Station (FS) Chart".

## Weight and Balance Determination for Flight

### Weight and Center of Gravity Limit

**Figure 6-1:** Center of Gravity Envelope



## Weight/Fuselage Station (FS) Chart

| Weight<br>(lb) | FS - Min.<br>(inches) | FS - Max.<br>(inches) | Weight<br>(lb) | FS - Min.<br>(inches) | FS - Max.<br>(inches) |
|----------------|-----------------------|-----------------------|----------------|-----------------------|-----------------------|
| 3600           | 196.28                | 196.28                | 4800           | 189.68                | 197.62                |
| 3650           | 195.43                | 196.34                | 4850           | 189.80                | 197.68                |
| 3700           | 194.57                | 196.40                | 4900           | 189.92                | 197.74                |
| 3750           | 193.72                | 196.46                | 4950           | 190.04                | 197.80                |
| 3800           | 192.86                | 196.51                | 5000           | 190.16                | 197.86                |
| 3850           | 192.01                | 196.57                | 5050           | 190.28                | 197.92                |
| 3900           | 191.15                | 196.63                | 5100           | 190.40                | 197.97                |
| 3950           | 190.30                | 196.69                | 5150           | 190.52                | 198.03                |
| 4000           | 189.44                | 196.75                | 5200           | 190.64                | 198.09                |
| 4050           | 189.44                | 196.81                | 5250           | 190.76                | 198.15                |
| 4100           | 189.44                | 196.86                | 5300           | 190.88                | 198.15                |
| 4150           | 189.44                | 196.92                | 5350           | 191.00                | 198.15                |
| 4200           | 189.44                | 196.98                | 5400           | 191.11                | 198.15                |
| 4250           | 189.44                | 197.04                | 5450           | 191.23                | 198.15                |
| 4300           | 189.44                | 197.10                | 5500           | 191.35                | 198.15                |
| 4350           | 189.44                | 197.16                | 5550           | 191.47                | 198.15                |
| 4400           | 189.44                | 197.22                | 5600           | 191.59                | 198.15                |
| 4450           | 189.44                | 197.27                | 5650           | 191.71                | 198.15                |
| 4500           | 189.44                | 197.33                | 5700           | 191.83                | 198.15                |
| 4550           | 189.44                | 197.39                | 5750           | 191.95                | 198.15                |
| 4600           | 189.44                | 197.45                | 5800           | 192.07                | 198.15                |
| 4650           | 189.44                | 197.51                | 5850           | 192.19                | 198.15                |
| 4700           | 189.44                | 197.57                | 5900           | 192.31                | 198.15                |
| 4750           | 189.56                | 197.62                | 5950           | 192.43                | 198.15                |
|                |                       |                       | 6000           | 192.55                | 198.15                |



Weight and Balance Loading Form

• NOTE •

The Takeoff Condition Weight must not exceed 6000 lb (2722 kg).  
The Takeoff Condition Moment must be within the Minimum  
Moment to Maximum Moment range at the Takeoff Condition  
Weight.

Serial Num:  
Date:  
Reg. Num:  
Initials:

RELATED TABLE/FIGURE:  
[Refer to “Figure 6-1: Center of Gravity Envelope”.](#)

| Item | Description                                                                                                    | Weight<br>lb | FS Inches<br>[(Moment/1000)<br>*1000]/Weight | Moment/1000<br>(Weight*FS)/<br>1000 |
|------|----------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------|-------------------------------------|
| 1.   | Basic Empty<br>Weight<br>Includes unusable<br>fuel & full oil, and<br>Seats 1 & 2 in the<br>forward positions. |              |                                              |                                     |
| 2.   | Front Row Seat<br>Occupants (Seats 1<br>& 2)<br>Pilot & Passenger<br>(total)                                   |              | 132.9                                        |                                     |
| 3.   | Add Seats 3 and 5                                                                                              |              | 183.2                                        |                                     |
| 4.   | Middle Row<br>Outboard Seat<br>Occupants (Seats 3<br>& 5)                                                      |              | 178.4                                        |                                     |
| 5.   | Middle Row<br>Inboard Seat (Seat<br>4)                                                                         |              | 183.2 (Fwd)                                  |                                     |
|      |                                                                                                                |              | 197.2 (Mid)                                  |                                     |
|      |                                                                                                                |              | 211.2 (Aft)                                  |                                     |

| Item | Description                                                                    | Weight<br>lb | FS Inches<br>[(Moment/1000)<br>*1000]/Weight | Moment/1000<br>(Weight*FS)/<br>1000 |
|------|--------------------------------------------------------------------------------|--------------|----------------------------------------------|-------------------------------------|
| 6.   | Middle Row<br>Inboard Seat<br>Occupant (Seat 4)                                |              | 178.4 (Fwd)                                  |                                     |
|      |                                                                                |              | 192.4 (Mid)                                  |                                     |
|      |                                                                                |              | 206.4 (Aft)                                  |                                     |
| 7.   | XC Seats (Seats 6 & 7)                                                         |              | 217.9                                        |                                     |
| 8.   | XC Seat Occupants<br>(Seats 6 & 7)                                             |              | 209.6                                        |                                     |
| 9.   | Lavatory                                                                       |              | 209.5                                        |                                     |
| 10.  | Convenience<br>Console                                                         |              | 179.9                                        |                                     |
| 11.  | Baggage Area<br>(300 lb max.)                                                  |              | 247.0                                        |                                     |
|      | Cargo X-Tend<br>Baggage Area<br>(40 lb max.)                                   |              | 282.0 (Cargo<br>X-tend)                      |                                     |
|      | Note: 50 lb/sq. ft<br>load limit                                               |              |                                              |                                     |
| 12.  | IPS Fluid<br>3 US Gallons<br>@ 9.18 lb/gal.<br>Maximum                         |              | 249                                          |                                     |
| 13.  | Zero Fuel<br>Condition Weight<br>Subtotal item 1<br>thru 12<br>4900 lb maximum |              |                                              |                                     |
| 14.  | Fuel Loading<br>296 U.S. gallons @<br>6.76 lb/gal.<br>Maximum                  |              | 203                                          |                                     |

| Item | Description                                                                                                                                 | Weight<br>lb | FS Inches<br>[(Moment/1000)<br>*1000]/Weight | Moment/1000<br>(Weight*FS)/<br>1000 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------|-------------------------------------|
| 15.  | Ramp Condition<br>Weight<br>Subtotal items 13<br>and 14, must be<br>less than<br>6040 lb                                                    |              |                                              |                                     |
| 16.  | Fuel for start, taxi,<br>and run-up<br>Normally, fuel<br>used for start and<br>taxi before takeoff<br>is approx. 40 lb<br>(6 U.S. gallons). |              | 203                                          |                                     |
| 17.  | Takeoff Condition<br>Weight<br>Subtract item 16<br>from item 15, must<br>be less than 6000 lb                                               |              |                                              |                                     |

Use this form to maintain a continuous history of changes and modifications to airplane structure or equipment affecting weight and balance:

[illegible]

## **Equipment List**

This list will be determined after the final equipment has been installed in the aircraft.

Intentionally Left Blank

# ***Section 7: Systems Description***

## **Table of Contents**

### **• NOTE •**

Content for Section 7: Systems Description is located in the Pilot's Information Manual (PIM).

Intentionally Left Blank



# ***Section 8: Handling and Servicing***

## **Table of Contents**

|                                                      |    |
|------------------------------------------------------|----|
| Introduction.....                                    | 3  |
| Operator's Publications .....                        | 3  |
| Service Publications .....                           | 3  |
| Obtaining Publications.....                          | 4  |
| Airplane Records and Certificates.....               | 4  |
| Airworthiness Directives .....                       | 5  |
| Airplane Inspection Periods.....                     | 5  |
| Hour Meters.....                                     | 5  |
| Annual Inspection.....                               | 6  |
| 100-Hour Inspection .....                            | 6  |
| Cirrus Aircraft Progressive Inspection Program ..... | 6  |
| Ground Handling .....                                | 7  |
| Application of External Power .....                  | 7  |
| Towing.....                                          | 8  |
| Taxiing.....                                         | 9  |
| Parking .....                                        | 10 |
| Tie Down .....                                       | 11 |
| Leveling .....                                       | 13 |
| Jacking .....                                        | 13 |
| Ground Deicing .....                                 | 13 |
| Safe Return Autoland Runway Recovery.....            | 17 |
| Servicing.....                                       | 17 |
| Landing Gear Servicing.....                          | 17 |
| Brake Servicing .....                                | 18 |
| Hydraulic Power Pack Servicing .....                 | 19 |
| Tire Inflation.....                                  | 20 |
| Engine Oil Servicing .....                           | 20 |
| Fuel System Servicing.....                           | 22 |
| Ice Protection System (IPS) Servicing .....          | 27 |
| Battery Service .....                                | 28 |
| Oxygen System Servicing .....                        | 28 |
| Cleaning Exterior Surfaces.....                      | 29 |
| General Exterior Surfaces.....                       | 29 |
| Exterior Windshield and Windows.....                 | 35 |
| Engine Compartment .....                             | 36 |
| Heated Inlet.....                                    | 37 |

---

|                                                      |    |
|------------------------------------------------------|----|
| Landing Gear .....                                   | 38 |
| Deicers .....                                        | 38 |
| Cleaning Interior Surfaces .....                     | 44 |
| Interior Windshield and Windows .....                | 44 |
| Instrument Panel and Electronic Display Screens..... | 46 |
| Headliner and Trim Panels .....                      | 46 |
| Leather Upholstery and Seats .....                   | 47 |
| Carpets .....                                        | 47 |

## **Introduction**

This section provides general guidelines for handling, servicing, and maintaining your aircraft. In order to ensure continued safe and efficient operation of your airplane, keep in contact with your Authorized Cirrus Service Center to obtain the latest information pertaining to your aircraft.

## **Operator's Publications**

The FAA Approved AFM and Pilot's Information Manual (PIM) are provided at delivery. Additional or replacement copies may be obtained from Cirrus Aircraft.

## **Service Publications**

The following service publications are available for purchase from Cirrus Aircraft:

- Airplane Maintenance Manual (AMM) – Maintenance Manual divided into chapters as specified by GAMA and S1000D covering inspection, servicing, maintenance, troubleshooting, and repair of the airplane structure, systems, and wiring. A current copy of the AMM is provided at delivery.
- Wiring Manual (WM) - Manual covering maintenance, troubleshooting, testing, and repair of the airplane electrical wiring.
- Illustrated Parts Catalog (IPC) - Catalog prepared to aid operators and mechanics to identify and procure replacement airplane parts.
- CAPS Component Maintenance Manual (CMM) - Maintenance Manual with Illustrated Parts List prepared to enable an authorized Cirrus Aircraft CAPS technician to restore the system to a functional condition.
- Engine Operators and Maintenance Manual – Cirrus Aircraft provides a Williams Engine Operator's and Maintenance Manual at the time of delivery. Engine and engine accessory overhaul manuals can be obtained from the original equipment manufacturer.
- Avionics Component Operator and Maintenance Manuals -- Cirrus Aircraft provides all available operator's manuals at the time of delivery. Maintenance manuals, if available, may be obtained from the original equipment manufacturer.

Cirrus Aircraft publishes Service Bulletins, Service Advisories, and Service Information Letters. Copies can be obtained from Cirrus Aircraft at [www.cirrusaircraft.com](http://www.cirrusaircraft.com).

- Service Bulletins are of special importance. When a Service Bulletin affecting your plane is published, comply with it promptly.

- Service Advisories are used to notify you of optional Service Bulletins, supplier Service Bulletins or Service Information Letters affecting your airplane, and maintenance data or corrections not requiring a Service Bulletin. Pay careful attention to the Service Advisory information.

## Obtaining Publications

AFMs and aircraft service publications can be obtained from Cirrus Aircraft at [www.cirrusaircraft.com](http://www.cirrusaircraft.com), or the Cirrus Connection at [www.cirrusconnection.com](http://www.cirrusconnection.com).

## Airplane Records and Certificates

The FAA requires that certain data, certificates, and licenses be displayed or carried aboard the airplane at all times. Additionally, other documents must be made available upon request. The mnemonic acronym “ARROW” is often used to help remember the required documents.

RELATED TABLE/FIGURE:

Refer to “Table 1: Required Documents”.

### • NOTE •

Owners of aircraft not registered in the United States should check with the registering authority for additional requirements.

**Table 1:** Required Documents

| Required Documents |                                              | Note                                                                                          |
|--------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------|
| A                  | Airworthiness Certificate<br>FAA Form 8100-2 | Must be displayed at all times.                                                               |
| R                  | Registration Certificate<br>FAA Form 8050-3  | Must be in the aircraft for all operations.                                                   |
| R                  | Radio Station License<br>FCC Form 556        | Required only for flight operations outside the United States.                                |
| O                  | Operating Instructions                       | FAA Approved AFM and associated aircraft placards fulfill this requirement.                   |
| W                  | Weight & Balance Data                        | Included in FAA Approved AFM. Data must include current empty weight, CG, and equipment list. |

| Other Documents   | Note                                 |
|-------------------|--------------------------------------|
| Airplane Logbook  | Must be made available upon request. |
| Engine Logbook    | Must be made available upon request. |
| Pilot's Checklist | Available in cockpit at all times.   |

## **Airworthiness Directives**

The FAA publishes Airworthiness Directives (ADs) that apply to specific aircraft and aircraft appliances or accessories. ADs are mandatory changes and must be complied with within a time limit set forth in the AD. Operators should periodically check with Cirrus Service Centers or A&P mechanic to verify receipt of the latest issued AD for their airplane.

## **Airplane Inspection Periods**

### **• NOTE •**

14 CFR 1.1 defines time in service, with respect to maintenance time records, as “the time from the moment an aircraft leaves the surface of the earth until it touches it at the next point of landing.”

The inspection items specified in the Annual/100-Hour Inspection have been determined by the average aircraft use rate of the typical owner. Non-commercially operated aircraft that are flown significantly more than 100 hours per year should consider additional inspections commensurate with the hours flown.

100-Hour Inspection or enrollment in a Progressive Inspection Program should be considered in addition to the normally required Annual Inspection. The Annual Inspection interval may also be shortened to accommodate high utilization rate.

## **Hour Meters**

The #1 hour meter, labeled HOBBS, is located forward of the cabin door and to the left of the pilot seat. It starts the meter and begins recording when the pilot turns the Engine knob to RUN and momentarily presses the Engine button. BAT 1 and either GEN 1 or GEN 2 need to be ON. It stops tracking time when the pilot turns the Engine knob to OFF and momentarily presses the Engine button (which stops the engine).

The #2 hour meter, labeled FLIGHT, is located forward of the HOBBS Hour Meter. It begins recording when the wheels are off the ground (Weight-On-Wheels is False) and the ground fan is OFF. It stops recording when the wheels are on the ground (Weight-On-Wheels is True) and the ground fan is ON.

The FLIGHT Hour Meter should be used to track maintenance time intervals as it records time-in-service more accurately than the HOBBS Hour Meter.

## Annual Inspection

Unless enrolled in a Progressive Inspection Program, the U.S. Code of Federal Regulations requires all civil aircraft must undergo a thorough Annual Inspection every twelve calendar months. Annual Inspections are due on the last day of the twelfth month following the last Annual Inspection. For example: If an Annual Inspection was performed on 19 November 2015, the next Annual Inspection will be due 30 November 2016. Annual Inspections must be accomplished regardless of the number of hours flown the previous year and can only be performed by a licensed Airframe and Powerplant (A&P) mechanic holding an Inspection Authorization (IA). Annual inspections can only be performed by facilities approved by Cirrus Aircraft. The inspection is listed, in detail, in Chapter 5 of the Airplane Maintenance Manual.

## 100-Hour Inspection

If the airplane is used to carry persons or provide flight instruction for hire, the Code of Federal Regulations requires that the airplane undergo a 100-Hour Inspection every 100 hours of flight operation in addition to the Annual Inspection requirement. The scope of the 100-Hour Inspection is identical to the Annual Inspection, except that it can be accomplished by a licensed A&P mechanic. The 100-hour interval may be exceeded by not more than 10 flight hours in order to reach a place where the inspection can be accomplished. Any flight hours used to reach an inspection station must be deducted from the next 100-Hour Inspection interval. The inspection is listed, in detail, in Chapter 5 of the Airplane Maintenance Manual.

## Cirrus Aircraft Progressive Inspection Program

In lieu of the above requirements, an airplane may be inspected using a Progressive Inspection Program in accordance with 14 CFR Part 91.409(d).

The Cirrus Aircraft Progressive Inspection Program provides for the complete inspection of the airplane utilizing a five-phase cyclic inspection program.

**400 flight hours:** A total of eight inspections are accomplished over the course of 400 flight hours, with an inspection occurring every 50 flight hours.

**800 flight hours:** A total of sixteen inspections are accomplished over the course of 800 flight hours, with an inspection occurring every 50 flight hours.

The inspection items to be covered in the Progressive Inspection are very similar to the Annual Inspection items. The Progressive Inspection will accomplish a full Inspection of the airplane at 400 (or 800) flight hours or at 12 calendar months.

The inspections are listed, in detail, in Chapter 5 of the Airplane Maintenance Manual.

## **Ground Handling**

### **Application of External Power**

An external power receptacle, located on the LH wing fairing behind the baggage door, permits the use of an external power unit for cold weather starting and maintenance procedures.

#### ***To Apply External Power to Airplane***

---

##### **• CAUTION •**

In accordance with the manufacturer's recommendation, external power should not be used to start the airplane with a dead battery.

Lithium Ion batteries may be recharged while remaining in the airplane. Battery maintenance must be performed in accordance with the appropriate AMM procedures.

Do not leave external power connected to the aircraft for extended periods of time without monitoring. Prolonged charging time may damage BAT 1.

Do not leave external power connected to the aircraft when the aircraft power is OFF, even if the external power source is turned OFF.

##### **• NOTE •**

If external power is connected and BAT 1 is not switched ON, BAT 1 will still be charged by external power.

Refer to [Section 2: Limitations, "Engine Start \(General\)"](#) for external power unit start limitations.

1. Ensure external power unit is regulated to 28 VDC.
2. Verify BAT 1, BAT 2, GEN 1, and GEN 2 power switches are set to OFF.
3. Plug external power unit into the receptacle.
4. Set BAT 1 and BAT 2 switches to ON. 28 VDC from the external power unit will energize the main, essential, and emergency distribution buses. The airplane may now be started or electrical equipment operated.

##### **• CAUTION •**

Depleting battery power, or running it down to low levels, may cause issues when starting the engine. If maintenance on avionics systems is to be performed, it is recommended that external power be used.

• NOTE •

Avionics power is provided by BAT 1 and BAT 2.

## ***To Remove External Power from Airplane***

1. Pull external power unit plug.

## **Towing**

The airplane may be moved on the ground by the use of a nose wheel steering bar or by power equipment that will not damage or excessively strain the nose gear assembly. The steering bar is engaged by attaching it into the nose wheel axle.

• NOTE •

Do not use any control surfaces or stabilizers to move the airplane.  
If a tow bar is not available, use the wing roots as push points.

Use care when towing the airplane when the main gear is obstructed by mud or snow. Excessive force will damage the aircraft or landing gear.

Towing provisions are included in the nose wheel axle; 0.556 inch (1.412 cm) outer diameter.

## ***Recommended Towing Equipment***

| ITEM         | P/N or SPEC | DESCRIPTION   | SUPPLIER     |
|--------------|-------------|---------------|--------------|
| Tow Bar      | Any         | Tow airplane. | Any Supplier |
| Wheel Chocks | WCK1001     | Block wheels. | Cirrus       |



## ***To Tow Airplane***

**• NOTE •**

Be especially cognizant of hangar door clearances.

1. Refer to [Section 1: General, "Airplane Three View"](#) for turning radius clearances.
2. Attach tow bar/adaptor onto the nose wheel axle.

**• NOTE •**

If towing with tow vehicle, attach tow bar to tow vehicle.

3. Release parking brake.
4. Remove chocks.
5. Move airplane to desired location.
6. Set parking brake in accordance with [Parking](#) procedure in this section.

**• CAUTION •**

Ensure nose wheel is facing forward.

7. Install chocks.
8. Remove tow bar.

**• NOTE •**

If towing with tow vehicle, remove tow bar from tow vehicle.

9. Set parking brake.

---

## **Taxiing**

**• CAUTION •**

Only rated pilots and mechanics should taxi the aircraft or run the engine.

Verify that taxi area is clear before beginning taxi.

Do not operate the engine at high thrust when running up or taxiing over ground containing loose stones, gravel, or any loose material that may cause damage to aircraft.

Taxi with minimum thrust needed for forward movement. Excessive braking may result in overheated or damaged brakes.

Refer to [Section 10: Safety Information, "Taxiing, Steering, and Braking Practices"](#).

## ***To Taxi Airplane***

1. Remove chocks.
2. Start engine in accordance with Airplane Flight Manual procedures.
3. Release parking brake.
4. Advance thrust lever to initiate taxi. Immediately after initiating taxi, apply the brakes to determine their effectiveness. To ascertain steering effectiveness during taxi, use differential braking to make slight turns.

### **• CAUTION •**

Observe wing clearance when taxiing near buildings or other stationary objects. If possible, station an observer outside the airplane.

Avoid holes and ruts when taxiing over uneven ground.

5. Taxi airplane to desired location.
6. Shut down engine in accordance with Airplane Flight Manual procedures.
7. Set parking brake in accordance with [Parking](#) procedure in this section.
8. Install chocks.
9. In gusty or stormy weather, moor airplane.

---

## **Parking**

The airplane should be parked to protect the airplane from weather and to prevent it from becoming a hazard to other aircraft. The parking brake may release or exert excessive pressure because of heat buildup after heavy braking or during wide temperature swings. Therefore, if the airplane is to be left unattended or is to be left overnight, chock and tie down the airplane.

If the airplane is parked for 30 days or more, the use of an external power unit for starting is recommended.

If parked for longer periods (60 days or more), the battery may become discharged. The battery may be recharged using a constant potential charger.

## ***Recommended Parking Equipment***

| ITEM         | P/N or SPEC | DESCRIPTION                                             | SUPPLIER |
|--------------|-------------|---------------------------------------------------------|----------|
| Wheel Chocks | WCK1001     | Block wheels.                                           | Cirrus   |
| Tail Stand   | 39003-001   | Support tail (if required under heavy snow conditions). | Cirrus   |

## ***To Park Airplane***

1. Position airplane on level surface and headed into the wind.
2. Retract flaps.

### **• CAUTION •**

Use care if setting parking brake during cold weather when accumulated moisture may freeze brakes, or when brakes are overheated.

3. Set parking brake by first applying brake pressure using the toe brakes and then pulling the parking brake handle aft.
4. Install chocks.
5. In gusty or stormy weather, tie down airplane in accordance with [Tie Down](#) procedure in this section.
6. Install pitot heat, AOA vane, engine inlet covers, and exhaust plugs. Be sure to remove the covers before flight.
7. Ensure cabin and baggage doors are locked when the airplane is left unattended.

---

## **Tie Down**

The airplane should be moored for immovability, security, and protection. FAA Advisory Circular AC 20-35C, Tie-down Sense, contains additional information regarding preparation for severe weather, tie down, and related information.

## ***Recommended Tie Down Equipment***

| ITEM                | P/N or SPEC             | DESCRIPTION                                         | SUPPLIER              |
|---------------------|-------------------------|-----------------------------------------------------|-----------------------|
| Wheel Chocks        | WCK1001                 | Block wheels.                                       | Cirrus                |
| Inlet Cover         | JETENGCVR               | Prevent entry of moisture and/or foreign particles. | Bruce's Custom Covers |
| Pitot Tube Covers   | PITOT2014               | Prevent entry of moisture and/or foreign particles. | Cirrus                |
| Static Ground Cable | Steel Cable with Clamps | Ground airplane.                                    | Any Source            |
| Rope                | 3/8 Braided             | Tie down wing and tail.                             | Any Source            |

## ***To Tie Down Airplane***

1. Position airplane on level surface and headed into the wind.

### **• CAUTION •**

Do not set parking brake during cold weather, when accumulated moisture may freeze brakes, or when brakes are overheated.

2. Secure gust lock strap to immobilize the side sticks.
3. Retract flaps.
4. Set parking brake in accordance with [Parking](#) procedure in this section.
5. Install chocks.
6. Install wing tie-down rings.

### **• NOTE •**

The wing tie-down rings are included in the Flight and Operation Accessories (Flyaway kit).

7. Secure tie-down ropes to the wing tie-down rings and to the NLG tie-down ring at approximately 45°-angles to the ground.

• CAUTION •

To prevent eye bolt damage in heavy winds, anchor points for wing tie-downs should not be more than 20 feet (6 meters) apart.

Use bowline knots, square knots, or the midshipman's hitch (also known as a taut line hitch or half-hitch). Do not use plain slipknots.

When using rope or non-synthetic material, leave sufficient slack to avoid damage to the airplane should the ropes contract.

8. In high snow load conditions, install tail stand at aft tail mooring point (located on the underside of the tail).
9. Install pitot heat, AOA vane, engine inlet, and exhaust covers to prevent entry of foreign materials. Be sure to remove the covers before flight.

---

## Leveling

Refer to AMM Chapter 8: Leveling & Weighing, Weighing the Airplane procedures for instructions and illustration.

## Jacking

Refer to AMM Chapter 7: Lifting & Shoring, Jacking the Airplane procedures for list of required tools and for illustration.

## Ground Deicing

• WARNING •

**Only SAE Type 1 Deicing Fluid is approved for use on the SF50 aircraft. Type II, III, & IV are prohibited. Follow the fluid manufacturer's recommendation for limitations and heating instructions. Follow the FAA guidelines on recommended holdover times.**

The application of heated, diluted SAE Type 1 deicing fluid removes frost, snow and/or ice from the airplane, and provides limited anti-icing protection.

Heated solutions of SAE Type 1 deicing fluid are more effective than unheated fluid. The temperature of the fluid applied to the aircraft must not exceed 180 °F (82 °C). Cooled SAE Type 1 deicing fluid may also be used if allowed by the fluid manufacturer. Always check with the fluid manufacturer to determine the recommended heating procedure.

SAE Type 1 deicing fluids are used in a diluted state as appropriate for the ambient temperature conditions.

SAE Type 1 deicing fluids have very limited holdover times. Refer to the FAA guidelines for information on applicable holdover times.

A heated hangar is an excellent option to deice aircraft and should be utilized whenever possible. Care must be exercised to ensure all melted precipitation is removed from the aircraft to prevent refreezing when aircraft is moved into cooler air.

Preliminary removal of heavy accumulation should be accomplished prior by brooms or another similar method. Exercise caution around antennas, windows, flight controls, deice boots, vanes, probes, etc.

When removing ice, snow, slush, or frost from aircraft surfaces, take care to prevent it from entering and accumulating in auxiliary intakes or control surface hinge areas. Manually remove snow from wing and tail surfaces forward towards the leading edge and remove control surfaces back towards the trailing edge.

RELATED TABLE/FIGURE:

Refer to “Figure 8-1: Deice Zones”.

## **To Deice Airplane**

### **• NOTE •**

Before Type 1 deicing procedures begin, deice personnel should familiarize themselves with areas to be sprayed and areas to avoid with a direct spray of fluid.

Aircraft should be parked clear of obstacles.

Aircraft should have engine OFF and all electrical equipment OFF. Flaps should be set to 50%.

1. Close doors. Install covers and plugs as necessary.

### **• CAUTION •**

Avoid excessive pressure when applying fluid. Pressure should be adjusted to be able to traverse distance from nozzle to aircraft without damaging paint or structure.

The temperature of the fluid applied to the aircraft must not exceed 180 °F (82 °C).

Never spray deicing fluid or water on or into any air inlet or exhaust area, open tubes or ports (i.e., pitot tubes, static ports, total air temperature probe, AOA probes, vents, ducts, engine, etc.). Never spray landing gear components. Avoid direct spraying of static wicks.

• NOTE •

Fluid spray direction must be from nose to tail and top to bottom to prevent fluid accumulation in sheltered areas.

Avoid spraying heated fluids onto cabin and cockpit windows, baggage door, egress hatch, and cabin door.

Aircraft must be sprayed symmetrically. Left and right wings and left and right tail must be treated and not one side only.

It is recommended to start deicing in an area visible from the cockpit, to allow for visual inspection of deiced areas for ice accumulation.

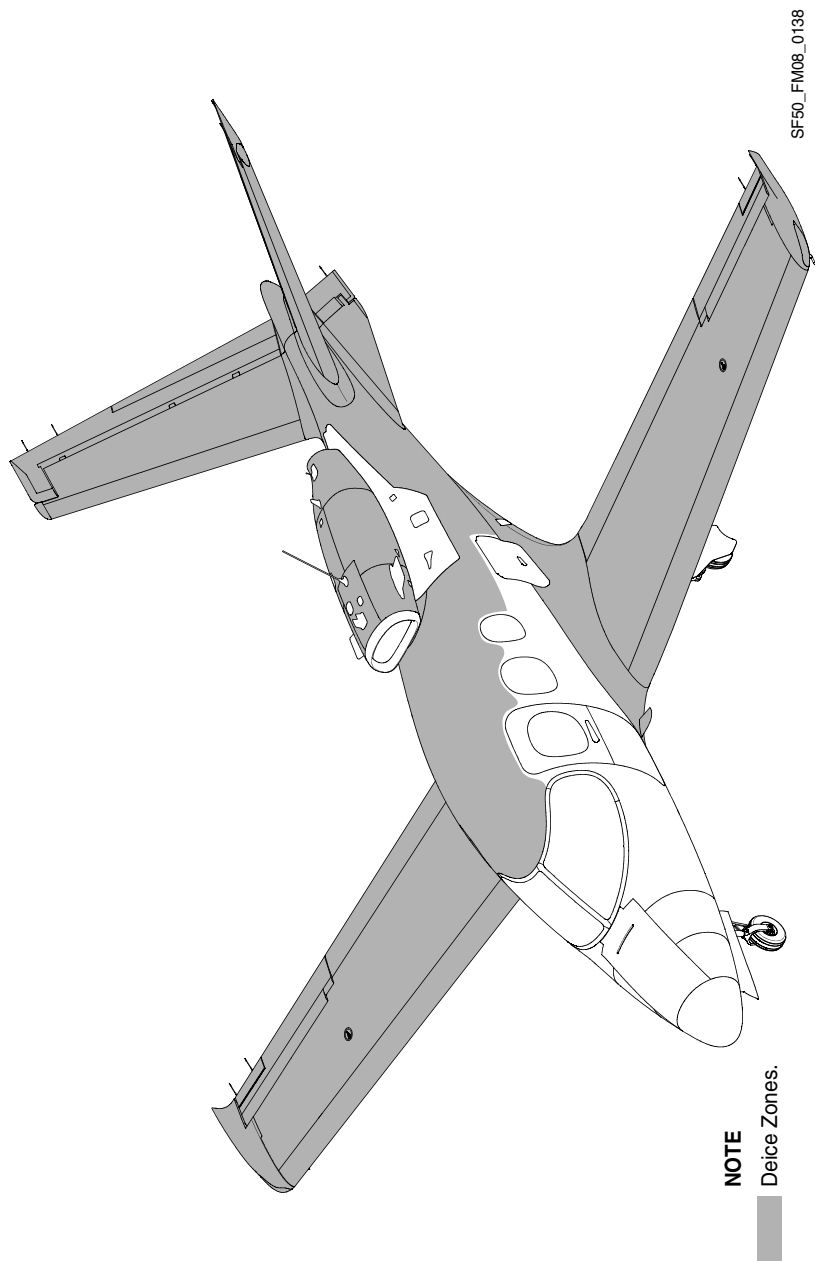
2. Spray wing and tail surfaces from tip to root and from leading edge to trailing edge.
3. Spray along fuselage centerline and then outboard, letting fluid cascade down and across windows.

• NOTE •

Use soft flannel cloth to minimize scratching when clearing windshield.

4. Ensure windshield is free of residual fluid to ensure adequate visibility prior to taxi.

**Figure 8-1: Deice Zones**





## Safe Return Autoland Runway Recovery

### • NOTE •

Brakes will remain set after emergency automatic landing and after power is removed from aircraft.

Perform the following steps to move the aircraft away from the runway after an emergency automatic landing. Brake reset must be accomplished any time the automatic braking system is activated as part of Safe Return Autoland. Brake reset is not intended for use during flight.

### ***Post-Autoland Aircraft Recovery***

1. Evacuate passengers from aircraft.
2. Place landing gear selector in DOWN position.
3. Shut down engine.
  - a. Place engine knob in OFF position.
  - b. Press engine button.
4. Set BAT 1 and BAT 2 switches in OFF positions.
5. Chock or secure airplane.
6. Press green brake reset button located on the right side of parking brake for 5 seconds.
7. Tow airplane away in accordance with [Towing](#) procedure in this section.
8. Perform Hard/Overweight Landing procedure in accordance with AMM Chapter 5-50: Unscheduled Maintenance Checks.

## **Servicing**

### **Landing Gear Servicing**

The nose landing gear is an air/oil oleo shock strut; the main landing gear is an air/oil oleo shock strut. The landing gear system is serviced with MIL-PRF-87257 hydraulic fluid and filled with compressed, dry nitrogen. The strut pressure should be checked every 50 hours and the struts should be serviced at the Annual/100-Hour inspection.

Landing gear servicing must be accomplished in accordance with AMM procedures.

## Brake Servicing

### Brake Replenishing

The brake system is filled with MIL-PRF-87257 hydraulic brake fluid. The fluid level should be checked at every oil change and at the annual/100-hour inspection, replenishing the system when necessary. The brake reservoir is located in the RH nose bay.

• NOTE •

If the entire system must be refilled, refer to AMM Chapter 12:  
Servicing, Brake Fluid Replenishing.

### To Replenish Brake Fluid

1. Install chocks.
2. Release parking brake.
3. Remove upper right nose access panel to gain access to hydraulic fluid reservoir.
4. Clean reservoir plug and area around plug before opening reservoir cap.
5. Remove plug and add MIL-PRF-87257 hydraulic fluid as necessary to fill reservoir.
6. Install plug, inspect area for leaks, and then install and secure upper right nose access panel.

---

### Brake Overheat Inspection

The brakes shall be inspected before every flight for general condition, evidence of overheating, and deterioration.

The aircraft shall not be operated with overheated, damaged, or leaking brakes. Conditions include, but are not limited to:

- Leaking brake fluid at the caliper. This can be observed by checking for evidence of fluid on the ground or deposited on the inside of the landing gear panels.

Overheated components, such as:

- Brake Pads: The profile of the edges is deformed and may include large voids or sections of missing material.
- Brake Discs: Surface is irregular and may include large grooves, bumps, or deposits from brake pads.

## **Brake Assembly Wear Limits**

### **To Inspect the Brake Assemblies**

• **NOTE** •

Refer to the manufacturer's approved Instructions For Continued Airworthiness for more information.

1. Wipe off any debris from brake caliper assembly that may obstruct inspection.
  2. Check brake pad assemblies and discs for deterioration and maximum permissible wear.
    - a Verify brake pad assembly frictional material minimum thickness is not less than 0.040 inch (1 mm).  
Proper pad frictional material thickness is indicated by the presence of a center channel in the frictional material of each pad.
    - b Verify brake disc minimum thickness is not less than 0.433 inch (11 mm).
    - c Verify brake disc / wheel interface maximum gap is not more than 0.020 inch (0.51 mm).
- 

## **Hydraulic Power Pack Servicing**

The hydraulic power pack system is filled with MIL-PRF-87257 hydraulic brake fluid. The fluid level should be checked at every oil change and at the annual/100-hour inspection, replenishing the system when necessary.

The hydraulic power pack reservoir is located in the LH nose bay.

• **NOTE** •

Refer to AMM Chapter 12: Replenishing and Chapter 32: Extension and Retraction for servicing procedures.

---

## ***To Replenish Hydraulic Power Pack Fluid***

1. Pull LANDING GEAR HYD circuit breaker.
  2. Remove upper left nose access panel to gain access to hydraulic power pack fluid reservoir.
  3. Clean Vent/Fill Cap Assembly, and area around it, before opening assembly.
  4. Remove Vent/Fill Cap Assembly and internal strainer.
  5. Add MIL-PRF-87257 hydraulic fluid as necessary to fill reservoir.
  6. Install assembly, inspect area for leaks, and then install and secure upper left nose access panel.
  7. Set LANDING GEAR HYD circuit breaker.
- 

## **Tire Inflation**

The main landing gear wheel assemblies use 18 × 5.5, 8-ply rating tubeless tires. The nose wheel assembly uses a 5.00 × 5, 10-ply rated tubeless tire. Always keep tires inflated to the rated pressure to obtain optimum performance and maximum service.

For maximum service from the tires, keep them inflated to the proper pressure. When checking tire pressure, examine the tires for wear, cuts, nicks, bruises and excessive wear.

All wheels and tires are balanced before original installation and the relationship of tire and wheel should be maintained upon reinstallation. In the installation of new components, it may be necessary to rebalance the wheels with the tires mounted. Unbalanced wheels or tires can cause extreme vibration in the landing gear.

## ***To Inflate Tires***

1. Remove valve stem cap and verify tire pressure with a dial-type tire pressure gauge.
  2. Inflate nose tire to 88±5 psi (607±34 kPa) and main wheel tires to 105±5 psi (724±34 kPa).
  3. Replace valve stem cap.
- 

## **Engine Oil Servicing**

It is recommended that the oil be changed every 24 months/600 operating hours (sooner under unfavorable operating conditions).

An oil filler cap is located at the 2 o'clock position on the right-hand side of the engine, next to the front mounting pad. The sight glass (located at the 3 o'clock position) is marked with two levels: FULL and ADD.

| ITEM                | DESCRIPTION                                                                                                   | VOLUME<br>(Quarts) |
|---------------------|---------------------------------------------------------------------------------------------------------------|--------------------|
| Oil Tank Total      | Represents volume at which filler port will overflow.                                                         | 3.0                |
| Oil Tank "Full Oil" | Represents volume required to reach FULL mark.                                                                | 2.2                |
| Oil Tank "Add Oil"  | Represents volume required to reach ADD mark.                                                                 | 1.6                |
| Oil Tank Usable     | Represents volume which the engine can consume between FULL level and the onset of oil pressure fluctuations. | 1.7                |

• **CAUTION** •

The engine should not be operated with observed oil level (sight glass indication) below the ADD mark.

• **NOTE** •

To correctly track engine oil consumption, do not add oil until observed level reaches the ADD mark on sight glass.

All engine oil level readings should be made at least 5 minutes after engine shutdown.

Use only oils conforming to Teledyne Continental Specification SAE J 1899 (Ashless Dispersant Lubrication Oil).

For a list of approved engine lubricants, refer to [Section 2: Limitations, "Oil"](#).

### ***To Check and Add Oil***

1. Open oil check door located on lower RH side of nacelle. Verify oil level via sight glass.
2. If oil level is at or below ADD mark on sight glass, open oil fill door on upper RH side of nacelle.
3. Top-up oil level and install filler cap.
4. Close and secure access panels.
5. In the engine logbook, record the amount and type of oil added.

## Fuel System Servicing

### Fuel Filtration Element

A 20-micron disposable fuel filter element is located on the engine FCU. In the event that the fuel filter becomes blocked, a filter bypass valve located adjacent to the filter will allow fuel to bypass the fuel filter for continued engine operation. A FUEL FILTER BYPASS warning is displayed prior to the bypass valve actuating.

Refer to AMM Chapter 28: Fuel, Fuel Screen/Element servicing procedures.

### Fuel Requirements

Approved Fuel Grades:

Jet A Specification Fuel .....ASTM D1655

Jet A-1 Specification Fuel.....ASTM D1655

JP-8 Specification Fuel ..... MIL-DTL-83133

### Filling Fuel Tanks

Observe all safety precautions required when handling aviation fuel. Fuel fillers are located on the upper surface of the wing. Each wing holds a maximum of 148 US gallons (560 L) (usable) with a total aircraft capacity of 296 US gallons (1120 L) (usable). When using less than full fuel capacity, the fuel should be distributed equally between each side.

If takeoff weight limitations for the next flight permit, the fuel tanks should be filled after each flight to minimize condensation.

#### • WARNING •

**During fueling, have a fire extinguisher available.**

**Ground fuel truck or cart to suitable earth ground.**

**Do not fill tank within 100 ft (30.5 m) of any energized electrical equipment capable of producing a spark.**

**Smoking or open flames are prohibited within 100 ft (30.5 m) of airplane or refuel vehicle.**

**Do not operate radios or electrical equipment during refuel operations.**

**Do not operate any electrical switches.**

#### • NOTE •

Recommended fueling best practice is to alternate between wings in 75-gallon (284-liter) increments when fueling from a single nozzle. Larger fuel imbalance may lead to fuel venting from low wing if aircraft “leans” towards a heavy wing as fuel reaches maximum tank capacity.

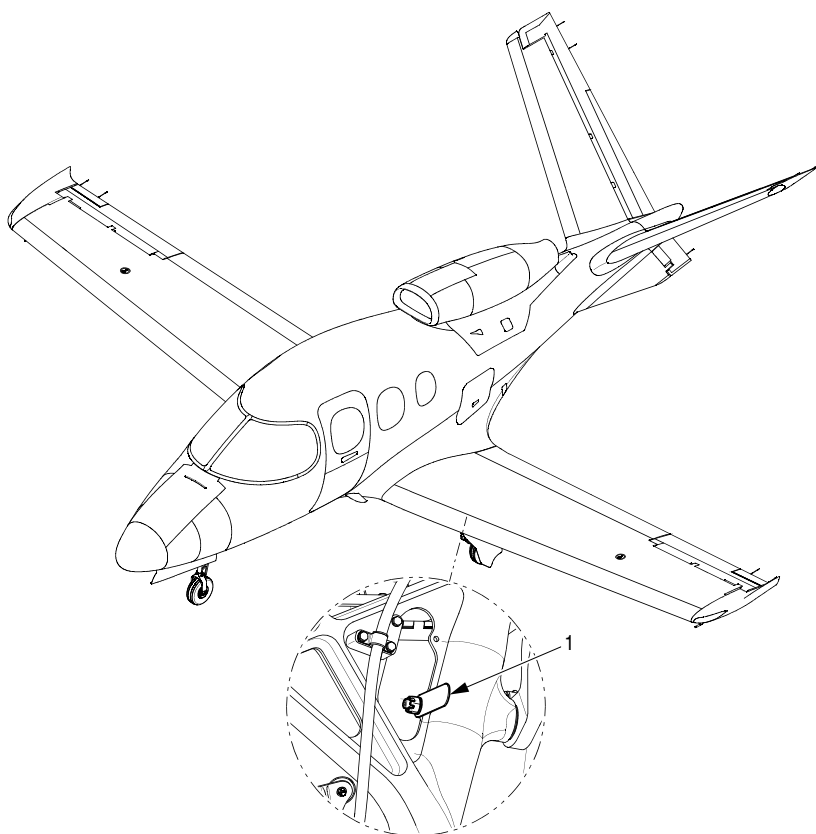
• NOTE •

Ground fuel nozzle and fuel truck to one of the MLG ground tabs (at Left or Right MLG Trunnion/Side-brace pin). All grounding points are connected to each other through the aircraft structure which also provides lightning protection.

RELATED TABLE/FIGURE:

[Refer to “Figure 8-2: Fuel Grounding Points”.](#)

**Figure 8-2:** Fuel Grounding Points



**LEGEND**

1. MLG Ground Tab

SF50\_FM08\_0018A



## **To Refuel Airplane**

1. Place fire extinguisher near fuel tank being filled.
2. Connect ground wire from refuel nozzle to airplane ground point, from airplane ground point to fuel truck or cart, and from fuel truck or cart to a suitable earth ground.
3. Place rubber protective cover over wing around fuel filler.

**• NOTE •**

Do not permit fuel nozzle to come in contact with bottom of fuel tanks.

4. Remove fuel filler cap and fuel airplane to desired level.

**• NOTE •**

Recommended fueling best practice is to alternate between wings in 75-gallon (284-liter) increments when fueling from a single nozzle. Larger fuel imbalance may lead to fuel venting from low wing if aircraft “leans” towards a heavy wing as fuel reaches maximum tank capacity.

**• NOTE •**

Refer to [Section 2: Limitations, "Fuel"](#) for maximum fuel imbalance information.

5. Remove nozzle, install filler cap, and remove protective cover.
6. Repeat refuel procedure for opposite wing.
7. Remove ground wires.
8. Remove fire extinguisher.

---

## **Fuel Contamination and Sampling**

Typically, fuel contamination results from foreign material such as water, dirt, rust, and fungal or bacterial growth. Additionally, chemicals and additives that are incompatible with fuel or fuel system components are also a source of fuel contamination. To ensure that the proper grade of fuel is used and that contamination is not present, the fuel must be sampled prior to each flight after adding fuel, and sampled prior to the first flight of the day.

Each fuel system drain must be sampled by draining a cupful of fuel into a clear sample cup. Fuel drains are provided for the wing tanks and collector tank drains. A total of four tank drains are located at the low spot in the respective tank.

If sampling reveals contamination, the tank drains must be sampled again repeatedly until all contamination is removed. It is helpful to gently rock the wings and slightly oscillate the pitch attitude to move contaminants to the drain points for sampling. If after repeated samplings (three or more), evidence of significant contamination remains, do not fly the airplane until a mechanic is consulted, the fuel system is drained and purged, and the source of contamination is determined and corrected.

If sampling reveals the airplane has been serviced with an improper fuel grade, do not fly the airplane until the fuel system is drained and refueled with an approved fuel grade.

• NOTE •

Piston fuels such as 100 (Green) or 100LL (Blue) are dyed, while turbine fuels are undyed.

To help reduce the occurrence of contaminated fuel coming from the supplier or fixed based operator, pilots should ensure that the fuel supply has been checked for contamination and that the fuel is properly filtered. Also, between flights, the fuel tanks should be kept as full as operational conditions permit to reduce condensation on the inside of fuel tanks. In extremely humid areas, the fuel supply should be checked frequently and drained of condensation to prevent possible distribution problems due to fuel icing during flight.

## ***De-Fueling***

A portion of the fuel in each tank may be drained from the wing fuel tanks using a siphon hose placed in the cell or tank through the filler neck. Additionally, using the electric fuel pump, the fuel may be pumped out of the bulkhead fitting at the engine firewall using an additional external hose. The remaining fuel may be drained by opening the drain valves. Use the same precautions as when refueling airplane.

• Note •

Refer to AMM Chapter 12: Servicing, Airplane De-Fueling procedures for more information.

## **Ice Protection System (IPS) Servicing**

### ***IPS Fluid Replenishment***

The IPS fluid tank is serviced through a filler cap and neck located on the RH side of the fuselage.

#### **• WARNING •**

**Only use approved IPS fluids. Use of unapproved fluids may cause aircraft damage due to material incompatibility. It may also result in inadequate ice protection which may cause ice accretions that could possibly damage the airframe or engine. Engine damage could lead to loss of thrust resulting in an off-airport landing.**

**Refer to [Section 1: General, "Ice Protection System \(IPS\) Fluid"](#).**

**To prevent IPS fluid contamination, maintain a clean, dedicated measuring container and ensure mouth of fluid container is clean before dispensing. Secure the filler cap immediately after filling. Only reuse IPS fluid drained from the aircraft if it is known to be free of debris, contaminants, and other liquids.**

**Always store fluid in accordance with the manufacturer's recommendations, preferably in the manufacturer's container. In lieu of manufacturer's recommendations, only store fluid in an inert plastic container known to be chemically compatible or in a stainless steel (300 Series) container.**

#### **To add IPS fluid**

1. Dispense approved IPS fluid into clean measuring container.
  2. Remove filler cap.
  3. Use funnel to fill fluid tank to desired level.
  4. Install filler cap.
-

## Battery Service

The aircraft is equipped with rechargeable lithium-ion batteries. The primary start battery (Battery 1) is mounted below the left-hand baggage floor, the battery is accessed via the baggage compartment door. Refer to AMM Chapter 5: Time Limits And Maintenance Checks, Overhaul and Replacement Schedule.

The emergency battery (Battery 2) is mounted on the exterior of the LH parachute bay in the nose, the battery is accessed via the LH access panel in the nose. Refer to AMM Chapter 5: Time Limits And Maintenance Checks, Scheduled Maintenance.

The external power receptacle is located on the LH wing fairing behind the baggage door. Refer to AMM Chapter 24: Electrical Power, External Power for servicing procedures.

## Oxygen System Servicing

### • WARNING •

To preclude the possibility of fire by spontaneous combustion, oil, grease, paint, hydraulic fluid, and other flammable material should be kept away from oxygen equipment.

Service the oxygen system as follows:

### ***Oxygen Supply Cylinder***

- 5-year hydrostatic and regulator overhaul
- 15-year replacement

### ***Crew Mask (Quick Don)***

- 6-year overhaul

### ***Passenger Mask Containment System (MCS)***

- 6-year replacement

### ***Passenger Solenoid***

- Operational check every 200 flight hours

## **Cleaning Exterior Surfaces**

### **• CAUTION •**

Waxing, an unapproved cleaning method, may damage the pneumatic deicers due to material incompatibility. Do not wax leading edge pneumatic deicers.

Prior to cleaning, place the airplane in a shaded area to allow the surfaces to cool.

The airplane should be washed with a mild soap and water. Harsh abrasives or alkaline soaps or detergents could make scratches on painted or plastic surfaces or could cause corrosion of metal. Cover static ports and other areas where cleaning solution could cause damage. Be sure to remove the static port covers before flight.

## **General Exterior Surfaces**

1. Flush away loose dirt with water.
2. Apply cleaning solution with a soft cloth, a sponge, or a soft bristle brush.
3. To remove exhaust stains, allow the solution to remain on the surface longer.
4. To remove stubborn oil and grease, use a cloth dampened with naphtha.
5. Rinse all surfaces thoroughly.

### **• NOTE •**

Any good silicone-free automotive wax may be used to preserve painted surfaces. Soft cleaning cloths or a chamois should be used to prevent scratches when cleaning or polishing. A heavier coating of wax on the leading surfaces will reduce the abrasion problems in these areas.

---

## ***Recommended Exterior Cleaning Products***

| <b>Cleaning Product</b>                   | <b>Cleaning Application</b> | <b>Supplier</b>        |
|-------------------------------------------|-----------------------------|------------------------|
| Pure Carnauba Wax                         | Painted Exterior            | Any Source             |
| Mothers California Gold Pure Carnauba Wax | Painted Exterior            | Wal-Mart Stores        |
| RejeX                                     | Painted Exterior            | Corrosion Technologies |

| Cleaning Product                                                                  | Cleaning Application              | Supplier                    |
|-----------------------------------------------------------------------------------|-----------------------------------|-----------------------------|
| WX/Block System                                                                   | Painted Exterior                  | Wings and Wheels            |
| AeroShell Flight Jacket Plexicoat                                                 | Painted Exterior                  | ShellStore Online           |
| XL-100 Heavy-Duty Cleaner/Degreaser                                               | Painted Exterior and Landing Gear | Buckeye International       |
| Stoddard Solvent PD-680 Type II                                                   | Engine Compartment                | Any Source                  |
| Kerosene                                                                          | Exterior Windshield and Windows   | Any Source                  |
| Klear-To-Land                                                                     | Exterior Windshield and Windows   | D.W. Davies & Co            |
| Prist                                                                             | Exterior Windshield and Windows   | Prist Aerospace             |
| LP Aero Plastics Acrylic Polish & Sealant                                         | Exterior Windshield and Windows   | Aircraft Spruce & Specialty |
| Goodrich Aerospace Cleaner 54023-004                                              | Deicer Exterior                   | Cirrus Aircraft             |
| Goodrich Aerospace Protectant 54023-005                                           | Deicer Exterior                   | Cirrus Aircraft             |
| Goodrich SILVERboot Polish 54023-006                                              | Deicer Exterior                   | Cirrus Aircraft             |
| ICEX® II Ice Adhesion Inhibitor 54023-003 (2 pints), 54023-301 (16 oz pump spray) | Deicer Exterior                   | Cirrus Aircraft             |
| Nuvite NuShine II S Polish                                                        | Engine Inlet Lipskin              | Nuvite Chemical Compounds   |

## Care of Graphics

Graphics require care similar to any fine paint finish. Use high quality products designed specifically for use on automobile finishes. Use products in accordance with the manufacturer's instructions.

## **Exposure to Environmental Conditions**

Graphics, like paint, are degraded by prolonged exposure to sun and atmospheric pollutants. Store the aircraft in a hangar, under a cloth cover, or in a shaded area whenever possible. Protect the aircraft from dew and rain which may contain acidic pollutants (commonly found in large metropolitan areas).

### **• CAUTION •**

If graphics start to discolor or turn brown as a result of exposure to acidic pollution, immediately have a professional remove the graphic from the aircraft to avoid staining the underlying paint.

## **Regular Washing**

Wash graphics whenever the aircraft appears dirty. Contaminants allowed to remain on the exterior may be more difficult to remove.

### **To wash and clean graphics**

1. Rinse off as much dirt and grit as possible with a spray of water.
2. Clean graphic with a wet, non-abrasive detergent such as 3M™ Car Wash Soap 39000, Meguiar's NXT Generation® Car Wash, or Deep Crystal® Car Wash, and a soft, clean cloth or sponge.
3. Rinse thoroughly with clean water.
4. To reduce water spotting, immediately use a silicone squeegee to remove water.
5. Dry with a clean microfiber cloth.

---

## **Pressure Washing**

Although hand washing is preferred, pressure washing may be used when necessary to remove dirt and contaminants. Pressure washing must be performed in accordance with the following procedure:

1. Ensure the water pressure is less than 2000 psi (14 MPa).
2. Ensure water temperature is less than 180 °F (82 °C).
3. Use a spray nozzle with a 40° wide angle spray pattern.

### **• CAUTION •**

Holding the nozzle of a pressure washer at an angle less than 90° to the graphic may lift the edges of the graphic.

4. Keep the spray nozzle perpendicular to the graphic, and at a distance of at least 1 foot (30 cm).
5. To reduce water spotting, immediately use a silicone squeegee to remove water.
6. Dry with a clean microfiber cloth.

## Removing Difficult Contaminants

Difficult contaminants such as bugs, bird droppings, or tree sap may require spot cleaning.

### • CAUTION •

To prevent scratching the graphic, refrain from rough scrubbing and the use of abrasive tools.

## To spot clean difficult contaminants

1. Soften contaminants by soaking with hot, soapy water for several minutes.
2. Rinse thoroughly with clean water.
3. To reduce water spotting, immediately use a silicone squeegee to remove water.
4. Dry with a clean microfiber cloth.

### • CAUTION •

Initially test cleaning products on an inconspicuous area of the graphic to verify they will not cause damage.

5. If further cleaning is needed, one of the following products may be used: Meguiar's Gold Class™ Bug and Tar Remover, 3M™ Citrus Base Cleaner, a mixture of two parts isopropyl alcohol to one part water (mix ratio 2:1), or denatured alcohol.
6. Immediately rinse off all residue with clean water.
7. To reduce water spotting, immediately use a silicone squeegee to remove water.
8. Dry with a clean microfiber cloth.

---

## Cleaning Fuel Spills

### • CAUTION •

Immediately clean fuel spills to avoid degrading the vinyl and adhesive used in the graphic.



### **To clean fuel spills**

1. Wipe off spilled fuel.
  2. Clean graphic with a wet, non-abrasive detergent such as 3M™ Car Wash Soap 39000, Meguiar's NXT Generation® Car Wash, or Deep Crystal® Car Wash, and a soft, clean cloth or sponge.
  3. Rinse thoroughly with clean water.
  4. To reduce water spotting, immediately use a silicone squeegee to remove water.
  5. Dry with a clean microfiber cloth.
- 

### **Graphic Restoration**

If typical cleaning methods fail to produce satisfactory results, refer to the recommended restoration products and mixtures below to help preserve the condition of the graphics on your aircraft.

#### **• CAUTION •**

Do not use abrasive polishes or cutting compounds.

Do not use polish or wax on graphics with a matte or texture finish.

Initially test restoration products and mixtures on an inconspicuous area of the graphic to verify they will not cause damage.

#### **• NOTE •**

Use an all-purpose cleaner to remove wax or wax residue.

RELATED TABLE/FIGURE:

[Refer to “Table 2: Recommended Graphic Restoration Products and Mixtures”.](#)

**Table 2:** Recommended Graphic Restoration Products and Mixtures

| <b>Film or Finish Type</b>                                             | <b>Product or Mixture</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smooth Gloss                                                           | 3M™ Perfect-it™ Show Car Paste Wax 39526;<br>Meguiar's Gold Class™ Carnuaba Plus Premium<br>Liquid Wax                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Matte or Satin Texture                                                 | Mixture of two parts isopropyl alcohol to one part<br>water (mix ratio 2:1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Matte White<br>(1080-M10)<br>Carbon Fiber White<br>Texture (1080-CF10) | Depending on the type and degree of<br>contamination to be removed, use one or more of<br>the following solutions in the order shown:<br><ol style="list-style-type: none"> <li>1. Hot, soapy water solution</li> <li>2. Mixture of two parts isopropyl alcohol to one<br/>part water (mix ratio 2:1)</li> <li>3. Simple Green® All-Purpose Cleaner</li> <li>4. Household chlorine bleach, followed by a<br/>mixture of two parts isopropyl alcohol to one<br/>part water (mix ratio 2:1)</li> <li>5. Mineral spirits, followed by a mixture of two<br/>parts isopropyl alcohol to one part water (mix<br/>ratio 2:1)</li> </ol> |
| Carbon Fiber or<br>Brushed Metal<br>Texture                            | 3M™ Tire Restorer or Meguiar's Natural Shine<br>Protectant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Carbon Fiber Black<br>Texture (1080-CF12)                              | Meguiar's Ultimate Black Plastic Restorer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## **Exterior Windshield and Windows**

Before cleaning an acrylic window, rinse away all dirt particles before applying cloth or chamois. Never rub dry acrylic. Dull or scratched window coverings may be polished using a special acrylic polishing paste.

### **• CAUTION •**

Clean acrylic windows with a solvent-free, non-abrasive, antistatic acrylic cleaner. Do not use gasoline, alcohol, benzene, carbon tetrachloride, thinner, acetone, or glass window cleaning sprays.

Use only a non-abrasive cotton cloth or genuine chamois to clean acrylic windows. Paper towel or newspaper are highly abrasive and will cause hairline scratches.

### ***To clean the Exterior Windshield and Windows***

1. Remove grease or oil using a soft cloth saturated with kerosene then rinse with clean, fresh water.

Wiping with a circular motion can cause glare rings. Use an up and down wiping motion on the windshield in the direction of the airflow to prevent this.

To prevent scratching from dirt that has accumulated on the cloth, fold cloth to expose a clean area after each pass.

2. Using a moist cloth or chamois, gently wipe the windows clean of all contaminants.
  3. Apply acrylic cleaner to one area at a time, then wipe away with a soft, cotton cloth.
  4. Dry the windows using a dry non-abrasive cotton cloth or chamois.
-

## Engine Compartment

The engine exterior and compartment may be cleaned with a dry cleaning solvent, MIL-PRF-680 Type II.

### ***To clean the Engine Compartment***

1. Remove engine nacelle.
  2. Cover fire extinguisher discharge nozzles to ensure no water or debris falls into the discharge tubes during cleaning.
  3. Hand scrub parts with solvent using a soft fiber brush or pad to remove contaminants.
  4. If any residue remains after cleaning, use a stiff fiber brush to remove residue. Repeat previous step.
  5. Using an air compressor, blow components dry with clean, filtered shop air restricted to less than 30 psi.
  6. If solvent is used to clean titanium parts, such as the airframe fire barrier, use clean water for final rinse.
  7. Uncover fire extinguisher discharge nozzles.
  8. Install engine nacelle.
- 

### ***Flight Near/Over Salt Water***

A fresh water rinse of the powerplant must be performed after a flight near/over salt water.

#### **• CAUTION •**

Failure to conduct a fresh water rinse after flight near/over salt water may cause engine damage.

## **Heated Inlet**

### ***To clean the Heated Inlet***

**• NOTE •**

The preferred method of removing bugs is to use a mixture of water and mild soap.

1. Use a clean, lint-free, cotton cloth full wetted in a mixture of water and mild soap or isopropyl alcohol.
  2. Wipe lipskin surface and inner barrel to remove bugs, dirt, and other debris.
  3. Rotate cloth continuously. Use a new cloth as necessary to reduce the chance of any collected debris scratching the lipskin surface.
  4. Dry surface with a new, clean, lint-free, cotton cloth.
- 

### ***To polish the Lipskin (Cosmetic Polish)***

**• CAUTION •**

Hand polish only. Mechanical polishing is prohibited on all areas of the inlet assembly.

1. Use Teflease tape to mask off painted surfaces beyond the trailing edges of the lipskin.
2. Apply a non-abrasive metal polish to the lipskin in small fingertip dabs (0.5 inch diameter) in a 1-inch by 1-inch grid pattern.
3. Using a clean, lint-free, cotton cloth, start polishing the inner surfaces of the lipskin and progress to the outer surfaces. Use a swirl pattern around the lipskin. Ensure the cloth is rotated or folded as necessary.

**• NOTE •**

Black discoloration on the cloth is expected.

4. Gently remove any excess polish with a clean, lint-free, cotton cloth, rotating as necessary until mirror finish is restored.
-

## **Landing Gear**

Before cleaning the landing gear, place a plastic cover or similar material over the wheel and brake assembly.

### ***To clean the Landing Gear***

1. Place a pan under the gear to catch waste.
  2. Spray or brush the gear area with solvent or a mixture of solvent and degreaser, as desired. Where heavy grease and dirt deposits have collected, it may be necessary to brush areas that were sprayed, in order to clean them.
  3. Allow the solvent to remain on the gear from five to ten minutes. Then rinse the gear with additional solvent and allow to dry.
  4. Remove the cover from the wheel and remove the catch pan.
  5. Lubricate the gear in accordance with AMM Chapter 12: Servicing.
- 

## **Deicers**

Before cleaning/maintaining the deicers, place a plastic cover or similar material over the surrounding wing or stabilizer surfaces.

### **• NOTE •**

It is recommended the maintenance process be performed in the following sequence: (i) Clean (ii) Polish (iii) Protect (iv) Apply Ice Adhesion Inhibitor

### ***Preventative Maintenance***

### **• NOTE •**

It is helpful to check for damage with deicer inflated and leak detector fluid. Air leaking damage can be located and marked for repair.

### **To clean/maintain the Deicers**

1. Perform visual inspection of deicers.
    - a Inspect deicer carefully for surface damage: pinholes, cuts, tears, abrasions, scuffs, cracking and/or crazing.
    - b It is permissible to use leak detection fluid or suitable equivalent to check for air leaks in damaged areas.
  2. Wash deicers frequently (per instructions in the Deicer Cleaner Application section) using mild soap and warm water to protect the surface appearance.
-

## ***Deicer Cleaner Application***

Goodrich Aerospace Cleaner is to be used for frequent routine cleaning of the Silver Urethane Deicers. It may be used as needed for removal of bugs, dirt, and grime. Goodrich Aerospace Cleaner may be used as a spot cleaner as well.

**• NOTE •**

Goodrich Aerospace Cleaner does not need to be diluted for use.

Ambient Conditions (Temperatures > 50 °F (10 °C),  
Humidity < 90%).

Dry Time (Use when dry).

### **To apply Deicer Cleaner**

1. Using clean, soft, lint-free cotton cloths, fully wet deicer surface with Goodrich Aerospace Cleaner.  
Ensure cloth is continually wetted with Aerospace Cleaner to avoid abrasion and creating scratches on deicer surfaces.
2. Allow cleaner solution to penetrate surface contamination (approximately 60 to 90 seconds) before gently removing contamination with fully wetted cloths.  
If the deicer surface is stained, allow a longer penetration time for severe cleaning conditions, up to 30 minutes.
3. Rinse deicer as needed with clean water and clean, soft, lint-free, cotton cloths; maintain the cloths fully wetted at all times.  
Ensure cloth is continually wetted with water to avoid abrasion and creating scratches in deicer surface.
4. Dry deicer with clean, soft, lint-free, cotton cloths.  
Avoid excessive pressure when polishing to prevent abrasion and creating scratches in the deicer surface.
5. Repeat cleaner application as needed until deicer surface is clean and free of contamination.

**• NOTE •**

Although not required, the deicer protectant should be used after the cleaner application in order to provide additional UV protection.

---

## **Deicer Polish Application**

Goodrich SILVERboot Polish is a polish designed to reduce cosmetic imperfections and to enhance the silver appearance of the deicers. The product is specially formulated for use on the silver urethane material. The polish is designed and recommended for use after the deicers have been in service for 6 to 12 months. Reapplication is suggested at 3 to 12 month intervals as needed. Polish application on new deicers will not cause damage but will not improve cosmetic appearance. Polish application will not remove or eliminate all cosmetic imperfections.

### **• NOTE •**

Avoid excessive pressure when polishing so as to prevent abrasion and creating scratches in the deicer surface.

Ambient Conditions (Temperatures between 40 °F (4 °C) and 85 °F (29 °C), Humidity < 90%).

Dry Time (Use when dry).

### **To apply Deicer Polish**

1. Clean the deicer using the Goodrich Aerospace Cleaner. Refer to [Section 8: Handling and Servicing, "Deicer Cleaner Application"](#).
  2. Apply a quarter-sized spot of SILVERboot Polish to a clean, soft, lint-free cotton cloth.
  3. Rub polish on the deicer, working in a small area (approximately 6 × 8 inches). A dark gray residue will appear on the deicer and on the polishing cloth. Use medium pressure to wipe deicers, applying a uniform coat of the protectant.
  4. Buff the deicer with a clean, dry rag in the polished area.
  5. Repeat on an adjacent area, blending the polished areas together.  
The polish and buffing cloths may be used on many adjacent areas. Some improvement in polishing action is seen once the polish cloth becomes saturated.
  6. If polish dries prior to buffing, it should be removed by re-polishing.
  7. Deicer protectant may be used immediately on top of polish, but is not required. Refer to [Section 8: Handling and Servicing, "Deicer Protectant Application"](#).
-



## **Deicer Protectant Application**

The Goodrich Aerospace Protectant is designed for frequent routine use. The protectant is resistant to dust, soiling, and staining, and minimizes UV damage.

**• NOTE •**

Ambient Conditions (Temperatures > 50 °F (10 °C),  
Humidity < 90%).

Dry Time (5-15 minutes, use when dry).

### **To apply Deicer Protectant**

1. Clean the deicer using the Goodrich Aerospace Cleaner. Refer to [Section 8: Handling and Servicing, "Deicer Cleaner Application"](#).
2. Agitate the Aerospace Protectant before application.
3. Spray Aerospace Protectant onto a clean, lint-free cloth, saturating the cloth. The protectant may alternately be sprayed directly onto the deicer in addition to spraying the cloth.

Overspray should be wiped off.

Ensure cloth is continually wetted with Aerospace Protectant to avoid abrasion and creating scratches in deicer surface.

4. Use medium pressure to wipe deicers, applying a uniform coat of the protectant.  
Avoid excessive pressure when polishing so as to prevent abrasion and creating scratches in the deicer surface.
5. Wipe deicers one final time with a clean, lint-free cloth to evenly distribute the protectant on the deicer surface.
6. Apply as needed to maintain surface protection and appearance.

**• NOTE •**

Although not required, ICEX II Ice Adhesion Inhibitor should be applied after the deicer protectant in order to provide improved surface finish and to lower ice adhesion strength.

---

## ***Cosmetic Surface Protectant Application***

### **• CAUTION •**

Avoid scratching the unprotected deicer surface. Use clean, soft, lint-free, cotton cloths.

It is important to leave only a thin film of ICEX II. A heavy application can result in a sticky surface that would collect dust and dirt, thereby reducing the product's ice adhesion inhibitor efficiency. The water thins the silicone film, assists in spreading (polishing) the ICEX II to a uniformly even coating, and helps promote the curing and toughening of the film.

### **• NOTE •**

Dry Time (24 hours is recommended. ICEX II will fully harden during this time, but less time can be allowed, if needed.)

## **To apply Cosmetic Surface Protectant**

1. For deicers with previous use and operation, wash prior to application. Refer to [Section 8: Handling and Servicing, "Deicer Cleaner Application"](#).
2. Rinse deicer with clean water and allow to dry.
3. Clean deicer with isopropyl alcohol to remove substances that cannot be removed with soap and water.
4. Apply masking tape on leading edge lined up to the aft edge of the conductive edge sealer. Form neat, straight lines to border the application area.
5. Apply ICEX II to clean, soft, lint-free cotton cloth.

### **• CAUTION •**

Ensure cloth is continually wetted with ICEX II to avoid abrasion and creating scratches on deicer surface.

6. Using cloth wetted with ICEX II and water, apply a light, even coat of ICEX II to entire deicer surface and masked-off region aft of deicer. Wipe cloth in single, continuous, back-and-forth motions, spanwise on deicer.
7. Allow ICEX II to remain undisturbed on deicer surface for at least 5 minutes in order to wet out the surfaces.

### **• CAUTION •**

Ensure cloth is continually wetted with water to avoid abrasion and creating scratches on deicer surface.

8. Using a clean, soft, lint-free cotton cloth wetted with clean water, evenly polish the ICEX II-treated surfaces, turning cloth frequently.

• CAUTION •

Avoid excessive pressure when polishing to prevent abrasion and creating scratches on deicer surface.

9. Using a dry, clean, lint-free cotton cloth, finish polishing the ICEX II-treated surfaces. Lightly polish only to the point where there is a thin, even, oily film on the deicer surface.
- 

## ***Ice Adhesion Inhibitor Application***

When ICEX II is applied to a silver urethane deicer, the resulting surface finish helps protect the deicer from insect impact stains and other forms of surface contamination. The resulting surface finish also helps protect the deicer surface from scratches due to cloth drag during subsequent washing procedures.

• CAUTION •

Avoid scratching the unprotected deicer surface. Use clean, soft, lint-free, cotton cloths.

It is important to leave only a thin film of ICEX II. A heavy application can result in a sticky surface that would collect dust and dirt, thereby reducing the product's ice adhesion inhibitor efficiency. The water thins the silicone film, assists in spreading (polishing) the ICEX II to a uniformly even coating, and helps promote the curing and toughening of the film.

• NOTE •

Ambient Conditions (Temperatures > 50 °F (10 °C),  
Humidity < 90%).

Dry Time (Depending on application, either 1 hour or 24 hours.  
24 hours is recommended.)

## **To apply Ice Adhesion Inhibitor**

1. For deicers with previous use and operation, wash prior to application. Refer to [Section 8: Handling and Servicing, "Deicer Cleaner Application"](#).
2. Rinse deicer with clean water and allow to dry.
3. Clean deicer with isopropyl alcohol to remove substances that cannot be removed with soap and water.
4. Apply masking tape on leading edge lined up to the aft edge of the conductive edge sealer. Form neat, straight lines to border the application area.
5. Apply ICEX II to clean, soft, lint-free cotton cloth.

• CAUTION •

Ensure cloth is continually wetted with ICEX II to avoid abrasion and creating scratches on deicer surface.

6. Using cloth wetted with ICEX II, apply a light, even coat of ICEX II to entire deicer surface by wiping cloth in single, continuous, back-and-forth motions, spanwise on deicer.

• NOTE •

ICEX II used as an ice adhesion inhibitor will never fully dry to hard coating like the cosmetic surface protectant application.

---

## **Cleaning Interior Surfaces**

Seats, carpet, upholstery panels, and headliners should be vacuumed at regular intervals to remove surface dirt and dust. While vacuuming, use a fine bristle nylon brush to help loosen particles.

• CAUTION •

Remove any sharp objects from pockets or clothing to avoid damaging interior panels or upholstery.

## **Interior Windshield and Windows**

Never rub dry acrylic. Dull or scratched window coverings may be polished using a special acrylic polishing paste.

• CAUTION •

Clean acrylic windows with a solvent-free, non-abrasive, antistatic acrylic cleaner. Do not use gasoline, alcohol, benzene, carbon tetrachloride, thinner, acetone, or glass window cleaning sprays.

Use only a non-abrasive cotton cloth or genuine chamois to clean acrylic windows. Paper towel or newspaper are highly abrasive and will cause hairline scratches.

• NOTE •

Wiping with a circular motion can cause glare rings. Use an up and down wiping motion on the windshield in the direction of the airflow to prevent this.

To prevent scratching from dirt that has accumulated on the cloth, fold cloth to expose a clean area after each pass.

### ***To clean Interior Windshield and Windows***

1. Using a moist cloth or chamois, gently wipe the windows clean of all contaminants.
2. Apply acrylic cleaner to one area at a time, then wipe away with a soft, cotton cloth.
3. Dry the windows using a dry, non-abrasive cotton cloth or chamois.

### ***Recommended Interior Cleaning Products***

| <b>Cleaning Product</b>              | <b>Cleaning Application</b>     | <b>Supplier</b> |
|--------------------------------------|---------------------------------|-----------------|
| Prist                                | Interior Windshield and Windows | Prist Aerospace |
| Optimax                              | Display Screens                 | PhotoDon        |
| Mild Dishwasher Soap (abrasive-free) | Cabin Interior                  | Any Source      |
| Leather Care Kit<br>50689-001        | Leather Upholstery              | Cirrus Aircraft |
| Leather Cleaner<br>50684-001         | Leather Upholstery              | Cirrus Aircraft |
| Ink Remover<br>50685-001             | Leather Upholstery              | Cirrus Aircraft |
| Leather Conditioner<br>50686-001     | Leather Upholstery              | Cirrus Aircraft |
| Spot and Stain Remover<br>50687-001  | Leather Upholstery              | Cirrus Aircraft |
| Vinyl Finish Cleaner<br>50688-001    | Vinyl Panels                    | Cirrus Aircraft |
| Vinyl & Leather Cleaner<br>51479-001 | Vinyl and Leather Upholstery    | Cirrus Aircraft |

## **Instrument Panel and Electronic Display Screens**

The instrument panel, control knobs, and plastic trim need only to be wiped clean with a soft, damp cloth. The multifunction display, primary flight display, and other electronic display screens should be cleaned with Optimax - LCD Screen Cleaning Solution as follows:

### **• CAUTION •**

To avoid solution dripping onto display and possibly migrating into component, apply the cleaning solution to cloth first, not directly to the display screen.

Use only a lens cloth or non-abrasive cotton cloth to clean display screens. Paper towels, tissue, or camera lens paper may scratch the display screen.

Clean display screen with power OFF.

### ***To clean Instrument Panel and Electronic Display Screens***

1. Gently wipe the display with a dry, clean, cotton cloth.
  2. Moisten cotton cloth with cleaning solution.
  3. Wipe the soft cotton cloth across the display in one direction, moving from the top of the display to the bottom. Do not rub harshly.
  4. Gently wipe the display with a dry, clean cotton cloth.
- 

## **Headliner and Trim Panels**

The airplane interior can be cleaned with a mild detergent or soap and water. Harsh abrasives or alkaline soaps or detergents should be avoided. Solvents and alcohols may damage or discolor vinyl or urethane parts. Cover areas where cleaning solution could cause damage. Use the following procedure:

### **• CAUTION •**

Solvent cleaners and alcohol should not be used on interior parts. If cleaning solvents are used on cloth, cover areas where cleaning solvents could cause damage.

### ***To clean Headliner and Trim Panels***

1. Clean headliner and side panels with a stiff bristle brush, and vacuum where necessary.
  2. Soiled upholstery may be cleaned with a good upholstery cleaner suitable for the material. Carefully follow the manufacturer's instructions. Avoid soaking or harsh rubbing.
-

## **Leather Upholstery and Seats**

For routine maintenance, occasionally wipe leather upholstery with a soft, damp cloth. For deeper cleaning, start with mix of mild detergent and water. If necessary, work your way up to the products available from Cirrus Aircraft for more stubborn marks and stains. Do not use soaps as they contain alkaline which will alter the leather's pH balance and cause the leather to age prematurely. Cover areas where cleaning solution could cause damage. Use the following procedure:

**• CAUTION •**

Solvent cleaners and alcohol should not be used on leather upholstery.

### ***To clean Leather Upholstery and Seats***

1. Clean leather upholstery with a soft bristle brush, and vacuum where necessary.
  2. Wipe leather upholstery with a soft, damp cloth.
  3. Soiled upholstery may be cleaned with the approved products available from Cirrus Aircraft. Avoid soaking or harsh rubbing.
- 

## **Carpets**

To clean carpets, first remove loose dirt with a whiskbroom or vacuum. For soiled spots and stubborn stains, use a non-flammable, dry cleaning fluid. Floor carpets may be cleaned like any household carpet.

Intentionally Left Blank



## ***Section 9: Log of Supplements***

### **Table of Contents**

As Required

FAA Approved AFM Supplements must be in the airplane for flight operations when the subject optional equipment is installed or the special operations are to be performed.

This Log of Supplements shows all Cirrus Aircraft Supplements available for the aircraft at the corresponding date of the revision level shown in the lower left corner. A check mark in the Part Number column indicates that the supplement is applicable to the AFM. Any installed supplements not applicable to the AFM are provided for reference only.

## ***Section 10: Safety Information***

### **Table of Contents**

|                                                 |    |
|-------------------------------------------------|----|
| Introduction.....                               | 3  |
| Taxiing, Steering, and Braking Practices .....  | 3  |
| Operating Practices .....                       | 3  |
| Brake Maintenance .....                         | 4  |
| Cirrus Airframe Parachute System (CAPS).....    | 5  |
| Deployment Scenarios .....                      | 5  |
| Loss of Thrust.....                             | 5  |
| Pilot Incapacitation .....                      | 6  |
| Other .....                                     | 6  |
| General Deployment Information .....            | 7  |
| Deployment Speed .....                          | 7  |
| Deployment Altitude .....                       | 7  |
| Deployment Attitude .....                       | 7  |
| Landing Considerations.....                     | 8  |
| Emergency Landing Body Position .....           | 8  |
| Emergency Ground Egress .....                   | 8  |
| Water Landings .....                            | 9  |
| Post-Impact Fire .....                          | 9  |
| Ground Gusts .....                              | 9  |
| Safe Return Autoland System (if installed)..... | 10 |
| Activation Scenarios.....                       | 10 |
| Pilot Incapacitation .....                      | 11 |
| Other .....                                     | 11 |
| Landing Considerations.....                     | 11 |

Intentionally Left Blank

## **Introduction**

This aircraft is designed to operate safely and efficiently in a flight environment. However, like any other aircraft, pilots must maintain proficiency to achieve maximum safety, utility, and economy. Cirrus Aircraft strongly recommends that all pilots seek regular recurrent training and that they operate in accordance with the Cirrus Aircraft Flight Operations Manual and Envelope of Safety. In addition, regulations require additional pilot training and proficiency because the SF50 is turbojet-powered (14 CFR 61.58).

As the pilot, you must be thoroughly familiar with the contents of this AFM, the Supplements, Flight Checklist, and operational guides and data provided by manufacturers of equipment installed in this airplane. You must operate the airplane in accordance with the applicable FAA operating rules and within the limitations specified in Section 2 of this AFM.

### **• NOTE •**

Refer to [Section 9: Log of Supplements](#) for applicable FAA operating rules.

The Normal Procedures section of this AFM was designed to provide guidance for day-to-day operation of this airplane. The procedures given are the result of flight testing, FAA certification requirements, and input from pilots with a variety of operational experience. Become fully familiar with the procedures, perform all the required checks, and operate the airplane within the limitations and as outlined in the procedures.

## **Taxiing, Steering, and Braking Practices**

The Vision SF50 uses a full-castering nose wheel and relies on differential braking for directional control while taxiing. Proper braking practices are therefore critical to avoid potential damage to the brakes.

The most common cause of brake damage and/or failure is the creation of excessive heat through improper braking practices. Pilots unaccustomed to full-castering nose wheel steering may be inclined to “ride” the brakes to maintain constant taxi speeds and use the brakes excessively for steering during cross-wing taxiing.

## **Operating Practices**

When taxiing, directional control is accomplished with intermittent braking (toe taps) as necessary. Use only as much power as is necessary to achieve forward movement. Deceleration or taxi speed control using brakes but without a reduction in power will result in increased brake temperature.

“Riding the brakes” while taxiing is similar to driving a car with one foot on the brake and one foot on the gas. This causes a continuous buildup of energy that would otherwise be moving the airplane.

Observe the following operating practices:

- Verify that the parking brake is completely disengaged before taxi.
- Use only the minimum thrust necessary to achieve forward movement. Keep in mind, any additional power added with the thrust will be absorbed in the brakes to maintain constant speed.  
Approximately 38% N1 is the required breakaway thrust at 6000 lb (2722 kg).
- Use minimum necessary inputs of differential braking to achieve directional control.
- Do not “ride the brakes”. Pilots should consciously remove pressure from the brakes while taxiing. Failure to do so results in excessive heat buildup, premature brake wear, and increased possibility of brake failure or fire.
- Avoid unnecessary high-speed taxiing. High-speed taxiing may result in excessive demands on the brakes, increased brake wear, and the possibility of brake failure or fire.
- Brakes have a large energy absorbing capacity; therefore, cooling time should be considered. Energy absorbed during a few seconds of deceleration can take several minutes to dissipate. Always allow adequate cooling time after brake use.
- A cool down period and brake overheat inspection are required after high-energy braking events.

• NOTE •

Refer to [Section 2: Limitations, "Brakes"](#).

## Brake Maintenance

The brakes shall be inspected before every flight for general condition, evidence of overheating, and deterioration.

• NOTE •

Refer to [Section 8: Handling and Servicing, "Brake Servicing"](#) for brake system servicing information and brake overheat inspection procedures.

## **Cirrus Airframe Parachute System (CAPS)**

The Cirrus Airframe Parachute System (CAPS) is designed to lower the aircraft and its passengers to the ground in the event of a life-threatening emergency. CAPS deployment will likely result in damage to, or loss of, the airframe, and possible injury to the aircraft occupants.

The primary purpose of CAPS on the Cirrus Vision SF50 is to bring the aircraft occupants safely to the ground following a loss of thrust event, if the pilot has determined the use of CAPS to be safer than a forced landing. The use of CAPS should also be considered for pilot incapacitation or other emergency situations, if no other recovery options are deemed to exist. CAPS should be used in accordance with the procedures and guidance set forth in this AFM.

The following discussion is meant to guide your thinking about CAPS activation. CAPS activation should be well thought out and mentally practiced by every Cirrus Vision SF50 pilot. Cirrus Aircraft also strongly recommends that pilots complete initial and recurrent factory training programs which focus on the proper use of CAPS.

### **Deployment Scenarios**

This section describes possible scenarios in which CAPS activation is appropriate for the Cirrus Vision SF50. This list is not intended to be exhaustive, but merely illustrative of the type of circumstances when CAPS deployment could be the most appropriate means of saving the aircraft occupants.

#### ***Loss of Thrust***

The primary purpose of CAPS on the Cirrus Vision SF50 is to bring the aircraft occupants safely to the ground following a loss of thrust event or other engine loss, if the pilot has determined the use of CAPS to be safer than a forced landing. CAPS activation is recommended if a forced landing is required on an unprepared surface, over rough or mountainous terrain, over water out of gliding distance to land, over widespread fog, or at night.

In most cases, CAPS is a safer option than performing a forced landing on either an improved or unimproved surface. In some cases, CAPS may be safer than attempting a glide approach in order to land on a runway.

#### **Loss of Thrust during Departure**

If loss of thrust occurs during departure at an altitude of less than 600 feet AGL, it is recommended that the pilot land straight ahead if practicable. If a straight-ahead landing is not practicable, it is recommended that the pilot activate CAPS immediately.

If loss of thrust occurs during departure at an altitude between 600 and 2000 feet AGL, it is recommended that the pilot activate CAPS immediately.

Altitude loss from horizontal flight deployments has been demonstrated to be less than 600 feet.

If loss of thrust occurs during departure at an altitude of greater than 2000 feet AGL, the pilot must determine if a successful forced landing is assured. If not, CAPS should be utilized.

### **Loss of Thrust Enroute**

While attempting to glide to an airfield to perform a power-off landing, the pilot must be continuously aware of altitude and ability to successfully perform the landing. The pilot must determine by 1000 feet AGL if landing is assured or if CAPS will be required.

### **Pilot Incapacitation**

Pilot incapacitation may be the result of anything from a pilot's medical condition to a bird strike that injures the pilot.

*Serials without Safe Return Autoland:* If incapacitation occurs and the passengers are not trained to land the aircraft, CAPS activation by the passengers should be used.

*Serials with Safe Return Autoland:* If incapacitation occurs and the passengers are not trained to land the aircraft, passengers should activate Safe Return Autoland. In the event that Safe Return Autoland is unavailable, as indicated by a red slashed circle (⊗) illuminated on the Safe Return Autoland activation panel or as shown on the respective passenger informational screen while Safe Return Autoland is active, CAPS activation by the passengers should be used.

These scenarios should be discussed with passengers prior to flight and all appropriate passengers should be briefed on CAPS operation so they could effectively activate CAPS if required. The CAPS operation briefing should include the following:

- Reference to the passenger briefing card.
- Use of the LVL button.
- Use of the CAPS activation handle.

### **Other**

CAPS activation should be considered for other life-threatening emergency situations when all other emergency procedures have been unsuccessful or if no other options are deemed to exist.



## **General Deployment Information**

### ***Deployment Speed***

The recommended CAPS activation speed is one that is less than 135 KIAS and 145 KTAS, whichever is lower, which will result in immediate parachute deployment. It is important for the operator to make all reasonable efforts to slow to the minimum possible airspeed prior to CAPS activation. This is necessary to minimize loads on the airframe and aircraft occupants and increase the chances of a successful outcome.

If CAPS is activated at higher speeds, the system is designed to delay deployment of the parachute until the autopilot (triggered by handle pull) slows the airplane to the parachute deployment envelope. The time required for the autopilot to slow the airplane from 250 KIAS (level flight) to the parachute deployment envelope has been demonstrated to be 15-24 seconds. Flying at a slower airspeed when CAPS is activated will reduce this time.

In any event, the system will automatically deploy the parachute approximately 30 seconds after CAPS activation regardless of airspeed.

### ***Deployment Altitude***

No minimum altitude for deployment has been set. This is because the actual altitude loss during a particular deployment depends upon the airplane's airspeed, altitude and attitude at deployment as well as other environmental factors. In all cases, however, the chances of a successful deployment increase with altitude. Altitude loss from horizontal flight deployments has been demonstrated to be less than 600 feet. Deployment at a low altitude leaves little or no time for the aircraft to stabilize under the canopy or for the cabin to be secured. A low altitude deployment increases the risk of injury or death and should be avoided if possible. If circumstances permit, it is advisable to activate the CAPS at or above 1000 feet AGL.

At any altitude, once the CAPS is determined to be the best alternative available for saving the aircraft occupants, activate the system without delay.

### ***Deployment Attitude***

CAPS deployment has been demonstrated from straight and level and vertical descent (nose pointed down) attitudes. These tests showed that as long as the parachute was introduced to the free air by the extraction system, it would successfully recover the aircraft into its level descent attitude under the parachute. Deployment from other attitudes has not been tested and is unknown; however, it may remain the best option.

## Landing Considerations

After a CAPS deployment, the airplane will descend at approximately 1560 feet per minute with a lateral speed equal to the velocity of the surface wind. The CAPS landing touchdown is equivalent to ground impact from a height of approximately 10.5 feet. While the airframe and seats are designed to accommodate the touchdown forces, occupants must be prepared for the landing.

### • WARNING •

**Occupant protection during a CAPS landing has been demonstrated with the landing gear in the UP position. A CAPS landing with the gear in the DOWN position may result in a cracked or punctured fuel tank.**

The overriding consideration in all CAPS-deployed landings is to prepare the occupants for the touchdown in order to protect them as much as possible from injury.

## Emergency Landing Body Position

The most important consideration for a touchdown with CAPS deployed is to protect the occupants from injury, especially back injury. Contacting the ground with the back offset attempting to open a door or secure items increases the likelihood of back injury. All occupants must be in the emergency landing body position well before touchdown. After touchdown, all occupants should maintain the emergency landing body position until the airplane comes to a complete stop.

The emergency landing body position is assumed with tightened seat belt and shoulder harness by placing both hands beside the legs, and holding the upper torso and neck/back of the head erect and against the seat backs. For seats with armrests, the hands can be placed on the armrests. The seat should be back in the upright or second upright notch position. The seat cushions contain an aluminum honeycomb core designed to crush under impact to absorb downward loads and help protect the spine from compression injury.

## Emergency Ground Egress

After CAPS activation, if time permits, the pilot should consider the best option for emergency ground egress once the airplane comes to a complete stop.

### • NOTE •

Refer to [Section 3: Emergency Procedures, "Emergency Ground Egress"](#).

If the cabin door is clear, exit via the cabin door. If the cabin door is blocked, or has jammed on impact, exit via the emergency exit (egress window).

Depending on altitude and condition of the cabin after touchdown, dumping cabin pressure (in accordance with the [CAPS Activation](#) procedure) prior to shutting down electrical systems may be required to allow use of the emergency exit (egress window).

• NOTE •

An emergency exit egress will involve removal of the window from the interior of the airplane. Refer to Chapter 8: Doors, Windows, and Exits of the Pilot's Information Manual (PIM) for necessary instructions.

## **Water Landings**

The ability of the airplane to float after a water landing has not been tested and is unknown. After CAPS activation but before touchdown, if time permits, the pilot should consider the best option for emergency egress should a water landing be likely.

## **Post-Impact Fire**

If there is no fire prior to touchdown and the pilot is able to shut down the engine, fuel, and electrical systems, there is less chance of a post-impact fire. After CAPS activation but before touchdown, if time permits, the pilot should consider the best option for emergency egress in anticipation of the risk of a post-impact fire.

## **Ground Gusts**

If it is known or suspected that ground gusts are present in the landing zone, the parachute could drag the airplane after touchdown, especially if the terrain is flat and without obstacles. In order to ensure that the occupants can escape the airplane in the timeliest manner after the airplane comes to rest, the pilot should consider the best option for emergency ground egress.

• NOTE •

Refer to [Section 3: Emergency Procedures, "Emergency Ground Egress"](#) for more information.

Occupants must be in the emergency landing body position for touchdown. Occupants must not loosen seat belts until the airplane comes to rest. When the airplane comes to rest, the occupants should exit the airplane and immediately move upwind to prevent a sudden gust from dragging the airplane in their direction.

## **Safe Return Autoland System (if installed)**

The Safe Return Autoland system is designed to safely land the aircraft in the event of pilot incapacitation. Once activated, it performs the following:

- Declares an emergency and provides periodic status updates
- Selects the closest suitable airport
- Navigates via the most direct route possible while avoiding terrain and undesirable weather if weather information is available for use
- Lands the aircraft, and brings it to a stop

### **• NOTE •**

Safe Return Autoland evaluates runway length, width, elevation, gradient, terminal area weather, time to destination, and aircraft fuel state to select a destination.

Safe Return Autoland uses METAR data in determining destination suitability. If METAR data is unavailable, the system may select a less suitable destination.

Safe Return Autoland uses NEXRAD weather information to route around weather. If NEXRAD data is unavailable, the system will be unable to route around weather.

### **• WARNING •**

**Safe Return Autoland assumes a fully functional airplane in order to function properly, and is not designed to mitigate system failures. Any system failure that prevents the AFCS or engine from functioning, or the FMS from navigating, will also prevent Safe Return Autoland from functioning. CAPS activation by the passengers should then be used. Other failures including (but not limited to) landing gear, flaps, wheel speed sensors, or radar altimeter, will degrade system performance, and may result in damage to the airframe and possible injury to the occupants.**

Because of these conditions and limitations, Safe Return Autoland should only be activated in a true pilot incapacity emergency situation.

## **Activation Scenarios**

This section describes possible scenarios in which Safe Return Autoland activation is appropriate for the Cirrus Vision SF50. This list is not intended to be exhaustive, but merely illustrative of the type of circumstances when Safe Return Autoland activation could be the most appropriate means of saving the aircraft occupants.

## **Pilot Incapacitation**

Pilot incapacitation may be the result of anything from a pilot's medical condition to a bird strike that injures the pilot such that he/she can not safely fly and land the airplane. If pilot incapacitation occurs, and no other type-rated pilots are on board, passengers should activate the Safe Return Autoland system. If required airplane system functionality is degraded or unavailable, the Safe Return Autoland system will indicate the degraded condition via a red slashed circle (⊗) illuminated on the Safe Return Autoland activation panel or shown on the respective passenger informational screen while Safe Return Autoland is active.

Passengers must be briefed to recognize that, in rare cases, sufficient fuel may be unavailable to reach a suitable runway. This information is provided on the flight display during Safe Return Autoland activation.

In the event that Safe Return Autoland is unavailable, or there is not sufficient fuel to reach the system's target airport once Safe Return Autoland has been activated, CAPS activation by the passengers should then be used. This scenario should be discussed with passengers prior to flight and all appropriate passengers should be briefed on Safe Return Autoland operation so they could effectively activate Safe Return Autoland if required.

The Safe Return Autoland operation briefing should include the following:

- Reference to the passenger briefing card.
- Use of the Safe Return Autoland pushbutton.
- Status indications "system unavailable or degraded" red slashed circle (⊗) to the left of the button, "activated" green landing airplane icon to the right of the button.
- Fuel remaining until landing and estimated time until landing.
- GTC push-to-talk for passenger emergency communications.

## **Other**

Safe Return Autoland activation should be considered any time the pilot is unable to safely fly the aircraft.

## **Landing Considerations**

After a Safe Return Autoland landing, the airplane must be recovered from the runway and inspected for damage due to a potential hard landing.

### **• NOTE •**

The brakes will be locked on until they are reset. Refer to Section 8: Handling and Servicing, "Safe Return Autoland Runway Recovery".

Intentionally Left Blank